

Supporting Information

High-Throughput Synthesis and Optimization of Ionizable Lipids Through A³ Coupling for Efficient mRNA Delivery

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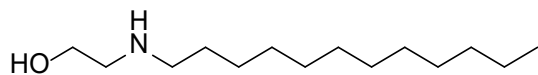
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1. Characterization data

1.1 Characterization data of *A1* – *A28*

2-(Dodecylamino)ethan-1-ol (*A1*): [CAS: 16613-87-9]

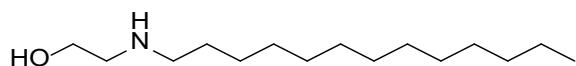


White solid. Yield: 2.00 g (10 mmol scale), 95%.

M.p. 40–41 °C.

^1H NMR (400 MHz, CDCl_3) δ 3.65 – 3.62 (m, 2H), 2.79 – 2.76 (m, 2H), 2.61 (t, J = 7.2 Hz, 2H), 1.88 (br s, 2H), 1.50 – 1.44 (m, 2H), 1.32 – 1.26 (m, 18H), 0.88 (t, J = 6.4 Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 60.8, 51.1, 49.6, 31.9, 30.1, 29.66, 29.63, 29.62, 29.56, 29.3, 27.3, 22.7, 14.1.

2-(Tridecylamino)ethan-1-ol (*A2*): [CAS: 34323-83-6]

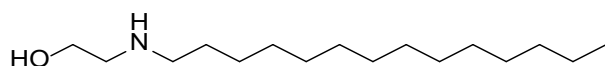


White solid. Yield: 0.38 g (2 mmol scale), 78%.

M.p. 47–49 °C.

^1H NMR (400 MHz, CDCl_3) δ 3.65 – 3.63 (m, 2H), 2.77 – 2.75 (m, 2H), 2.63 – 2.59 (m, 2H), 2.31 (br s, 1H), 1.52 – 1.45 (m, 2H), 1.30 – 1.26 (m, 20H), 0.88 (t, J = 6.4 Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 60.8, 51.1, 49.6, 31.9, 30.1, 29.65, 29.63, 29.62, 29.60, 29.5, 29.3, 27.3, 22.7, 14.1.

2-(Tetradecylamino)ethan-1-ol (*A3*): [CAS: 25737-87-5]



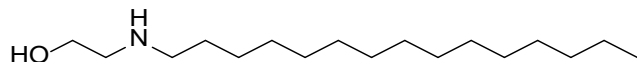
White solid. Yield: 0.99 g (5 mmol scale), 73%.

M.p. 49–50 °C.

^1H NMR (400 MHz, CDCl_3) δ 3.66 – 3.64 (m, 2H), 2.78 – 2.76 (m, 2H), 2.70 (br s, 1H), 2.63

– 2.60 (m, 2H), 1.49 (m, 2H), 1.24 (m, 22H), 0.87 (t, $J = 6.8$ Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 60.5, 51.0, 49.4, 31.9, 29.8, 29.66, 29.65, 29.64, 29.6, 29.5, 29.3, 27.3, 22.7, 14.1.

2-(Pentadecylamino)ethan-1-ol (A4): [CAS: 98389-74-3]

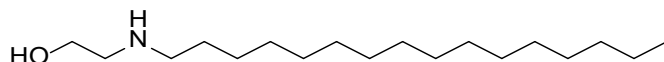


White solid. Yield: 0.40 g (1.7 mmol scale), 87%.

M.p. 57–58 °C.

^1H NMR (400 MHz, CDCl_3) δ 3.65 – 3.62 (m, 2H), 2.78 – 2.75 (m, 2H), 2.62 – 2.59 (m, 2H), 2.21 (br s, 1H), 1.48 – 1.45 (m, 2H), 1.33 – 1.23 (m, 25H), 0.88 (t, $J = 6.8$ Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 60.8, 51.1, 49.5, 31.9, 30.1, 29.7, 29.6, 29.5, 29.3, 27.3, 22.7, 14.1.

2-(Hexadecylamino)ethan-1-ol (A5): [CAS:35902-64-8]

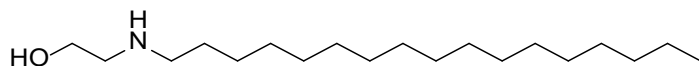


White solid. Yield: 2.15 g (8.5 mmol scale), 89%.

M.p. 48–49 °C.

^1H NMR (400 MHz, CDCl_3) δ 3.62 – 3.60 (m, 2H), 2.87 (br s, 1H), 2.72 – 2.69 (m, 2H), 2.57 (t, $J = 7.2$ Hz, 2H), 1.47 – 1.44 (m, 2H), 1.26 – 1.18 (m, 28H), 0.84 (t, $J = 6.8$ Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 60.6, 51.3, 49.6, 31.9, 30.0, 29.64, 29.60, 29.56, 29.51, 29.3, 27.3, 22.6, 14.0.

2-(Heptadecylamino)ethan-1-ol (A6): [CAS: 13436-74-3]

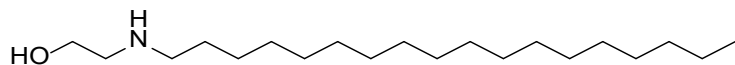


White solid. Yield: 0.35 g (1.6 mmol scale), 74%.

M.p. 57–58 °C.

^1H NMR (400 MHz, CDCl_3) δ 3.66 – 3.63 (m, 2H), 2.78 – 2.76 (m, 2H), 2.63 – 2.59 (m, 2H), 2.32 (br s, 1H), 1.52 – 1.45 (m, 1H), 1.30 – 1.23 (m, 29H), 0.88 (t, $J = 6.4$ Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 60.8, 51.1, 49.5, 31.9, 30.1, 29.7, 29.65, 29.61, 29.5, 29.3, 27.3, 22.7, 14.1.

2-(Nonadecylamino)ethan-1-ol (A7): [CAS: 31314-15-5]

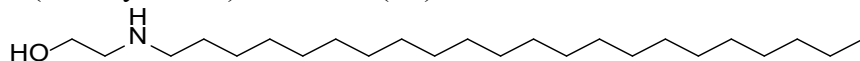


White solid. Yield: 0.70 g (5 mmol scale), 45%.

M.p. 55–56 °C.

^1H NMR (400 MHz, CDCl_3) δ 3.65 – 3.62 (m, 2H), 2.88 – 2.76 (m, 2H), 2.61 (t, J = 7.2 Hz, 2H), 1.91 (br s, 1H), 1.51 – 1.43 (m, 2H), 1.33 – 1.23 (m, 33H), 0.88 (t, J = 6.8 Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 60.9, 51.0, 49.5, 31.9, 30.2, 29.69, 29.66, 29.62, 29.56, 29.3, 27.3, 22.7, 14.1.

2-(Docosylamino)ethan-1-ol (A8)

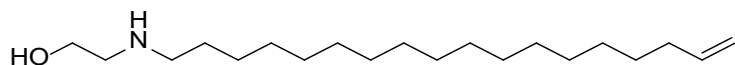


White solid. Yield: 1.32 g (5 mmol scale), 72%.

M.p. 68–70 °C.

^1H NMR (400 MHz, CDCl_3) δ 3.64 – 3.62 (m, 2H), 2.77 – 2.75 (m, 2H), 2.62 – 2.58 (m, 2H), 2.21 (br s, 1H), 1.54 – 1.40 (m, 2H), 1.28 – 1.25 (m, 38H), 0.89 – 0.86 (m, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 60.8, 51.1, 49.5, 31.9, 30.1, 29.7, 29.6, 29.5, 29.3, 27.3, 22.7, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{24}\text{H}_{52}\text{NO}^+$ 370.4043, found 370.4040.

2-(Octadec-17-en-1-ylamino)ethan-1-ol (A9)

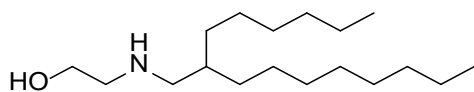


White solid. Yield: 0.3 g (1.5 mmol scale), 64%.

M.p. 50–51 °C.

^1H NMR (400 MHz, CDCl_3) δ 5.81 (ddt, J = 16.9, 10.2, 6.7 Hz, 1H), 5.01 – 4.91 (m, 2H), 3.65 – 3.62 (m, 2H), 2.78 – 2.76 (m, 2H), 2.61 (t, J = 7.2 Hz, 2H), 2.04 (qt, J = 6.8, 1.6 Hz, 2H), 1.96 (br s, 1H), 1.51 – 1.41 (m, 2H), 1.39 – 1.36 (m, 2H), 1.30 – 1.26 (m, 25H). ^{13}C NMR (101 MHz, CDCl_3) δ 139.3, 114.1, 60.8, 51.0, 49.5, 33.8, 30.2, 29.66, 29.61, 29.56, 29.50, 29.1, 28.9, 27.3. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{20}\text{H}_{42}\text{NO}^+$ 312.3261, found 312.3253.

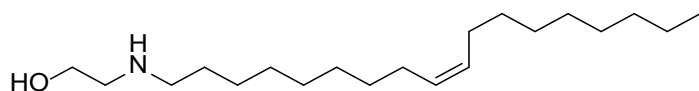
2-((2-Hexyldecyl)amino)ethan-1-ol (A10)



Colorless oily liquid. Yield: 1.36 g (4.9 mmol scale), 98%.

^1H NMR (400 MHz, CDCl_3) δ 3.66 – 3.64 (m, 2H), 2.92 (br s, 1H), 2.80 – 2.77 (m, 2H), 2.54 (d, $J = 6.2$ Hz, 2H), 1.51 – 1.48 (m, 1H), 1.31 – 1.25 (m, 24H), 0.87 (t, $J = 6.6$ Hz, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 60.2, 52.9, 51.2, 37.7, 32.0, 31.9, 30.0, 29.7, 29.6, 29.3, 26.62, 26.58, 22.6, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{18}\text{H}_{40}\text{NO}^+$ 286.3104, found 286.3103.

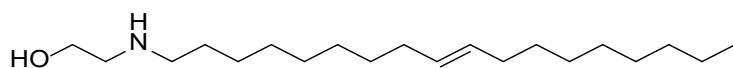
(Z)-2-(octadec-9-en-1-ylamino)ethan-1-ol (A11): [CAS: 31314-16-6]



Colorless oily liquid. Yield: 0.42 g (1.5 mmol scale), 90%.

^1H NMR (400 MHz, CDCl_3) δ 5.36 – 5.33 (m, 2H), 3.64 (t, $J = 5.2$ Hz, 2H), 2.77 (t, $J = 5.2$ Hz, 2H), 2.61 (t, $J = 7.2$ Hz, 2H), 2.04 – 1.99 (m, 6H), 1.52 – 1.45 (m, 2H), 1.33 – 1.27 (m, 22H), 0.88 (t, $J = 6.8$ Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 129.9, 129.8, 60.8, 51.0, 49.5, 31.9, 30.2, 29.7, 29.5, 29.3, 29.2, 27.3, 27.2, 22.7, 14.1.

(E)-2-(octadec-9-en-1-ylamino)ethan-1-ol (A12)

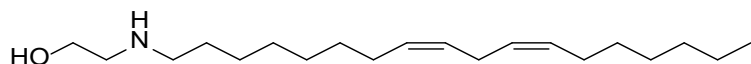


White solid. Yield: 1.428 g (6 mmol scale), 76%.

M.p. 29–31 °C.

^1H NMR (400 MHz, CDCl_3) δ 5.43 – 5.33 (m, 2H), 3.64 (t, $J = 5.2$ Hz, 2H), 2.76 (t, $J = 5.6$ Hz, 2H), 2.61 (t, $J = 7.2$ Hz, 2H), 2.31 (s, 2H), 1.96 (m, 4H), 1.52 – 1.45 (m, 2H), 1.36 – 1.26 (m, 22H), 0.88 (t, $J = 6.4$ Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 130.4, 130.3, 60.8, 51.1, 49.5, 32.6, 31.9, 30.1, 29.63, 29.61, 29.50, 29.46, 29.44, 29.3, 29.2, 29.1, 27.3, 22.7, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{20}\text{H}_{42}\text{NO}^+$ 312.3261, found 312.3263.

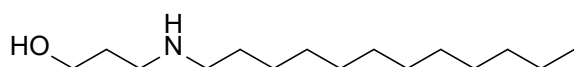
2-(((8Z,11Z)-octadeca-8,11-dien-1-yl)amino) ethan-1-ol (A13): [CAS: 2102622-45-5]



Yellow oily liquid. Yield: 1.177 g (4 mmol scale), 91%.

^1H NMR (400 MHz, CDCl_3) δ 5.40 – 5.30 (m, 2H), 3.64 (t, J = 5.2 Hz, 2H), 2.79 – 2.75 (m, 4H), 2.66 (s, 1H), 2.61 (t, J = 7.2 Hz, 2H), 2.05 (q, J = 6.8 Hz, 4H), 1.51 – 1.45 (m, 2H), 1.38 – 1.25 (m, 17H), 0.89 (t, J = 6.4 Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 130.1, 130.0, 127.93, 127.87, 60.7, 51.1, 49.5, 31.5, 30.0, 29.6, 29.5, 29.3, 29.2, 27.3, 27.2, 27.1, 25.6, 22.5, 14.0.

3-(Dodecylamino)propan-1-ol (A14): [CAS: 52871-79-1]

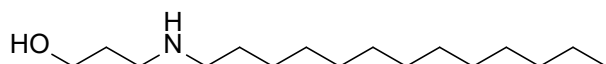


White solid. Yield: 2 g (8.7 mmol scale), 95 %.

M.p. 47–49 °C.

^1H NMR (400 MHz, CDCl_3) δ 3.79 (t, J = 5.2 Hz, 2H), 2.97 (br s, 1H), 2.86 (t, J = 5.6 Hz, 2H), 2.58 (t, J = 7.2 Hz, 2H), 1.70 – 1.64 (m, 2H), 1.46 – 1.41 (m, 2H), 1.28 – 1.24 (m, 19H), 0.86 (t, J = 6.4 Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 64.5, 50.1, 49.9, 31.9, 30.6, 29.9, 29.62, 29.59, 29.55, 29.50, 29.3, 27.2, 22.6, 14.1.

3-(Tridecylamino)propan-1-ol (A15)

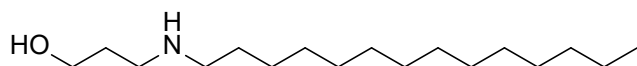


White solid. Yield: 0.38 g (1.6 mmol scale), 92%.

M.p. 54–55 °C.

^1H NMR (400 MHz, CDCl_3) δ 3.81 (t, J = 5.2 Hz, 2H), 3.16 (br s, 1H), 2.87 (t, J = 5.6 Hz, 2H), 2.60 (t, J = 7.2 Hz, 2H), 1.71 – 1.66 (m, 2H), 1.49 – 1.43 (m, 2H), 1.32 – 1.23 (m, 20H), 0.88 (t, J = 6.8 Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 64.5, 50.1, 49.8, 31.9, 30.6, 29.9, 29.64, 29.62, 29.58, 29.55, 29.50, 29.3, 27.2, 22.6, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{16}\text{H}_{36}\text{NO}^+$ 258.2791, found 258.2804.

3-(Tetradecylamino)propan-1-ol (A16): [CAS: 72648-58-9]

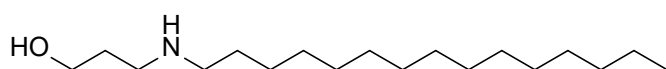


White solid. Yield: 0.96 g (5 mmol scale), 71%.

M.p. 54–55 °C.

^1H NMR (400 MHz, CDCl_3) δ 3.79 (t, $J = 5.2$ Hz, 2H), 3.18 (br s, 1H), 2.86 (t, $J = 5.6$ Hz, 2H), 2.59 (t, $J = 7.2$ Hz, 2H), 1.71 – 1.65 (m, 2H), 1.49 – 1.42 (m, 2H), 1.30 – 1.99 (m, 23H), 0.87 (t, $J = 6.8$ Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 64.4, 50.0, 49.8, 31.9, 30.5, 29.8, 29.66, 29.63, 29.59, 29.55, 29.50, 29.3, 27.2, 22.7, 14.1.

3-(Pentadecylamino)propan-1-ol (A17)

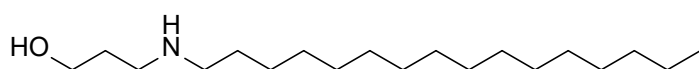


White solid. Yield: 0.4 g (1.6 mmol scale), 87%.

M.p. 61–62 °C.

^1H NMR (400 MHz, CDCl_3) δ 3.81 (t, $J = 5.2$ Hz, 2H), 3.09 (br s, 1H), 2.87 (t, $J = 5.6$ Hz, 2H), 2.59 (t, $J = 7.2$ Hz, 2H), 1.71 – 1.66 (m, 2H), 1.48 – 1.42 (m, 2H), 1.31 – 1.26 (m, 25H), 0.88 (t, $J = 6.8$ Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 64.5, 50.1, 49.9, 31.9, 30.6, 29.9, 29.66, 29.63, 29.59, 29.56, 29.51, 29.3, 27.2, 22.7, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{18}\text{H}_{40}\text{NO}^+$ 286.3104, found 286.3110.

3-(Hexadecylamino)propan-1-ol (A18): [CAS: 72648-60-3]

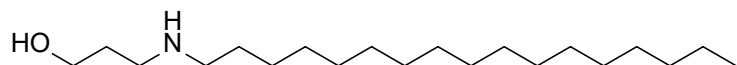


White solid. Yield: 2.3 g (8.5 mmol scale), 88%.

M.p. 52–53 °C.

^1H NMR (400 MHz, CDCl_3) δ 3.80 (t, $J = 5.2$ Hz, 2H), 2.96 (br s, 1H), 2.87 (t, $J = 5.6$ Hz, 2H), 2.59 (t, $J = 7.2$ Hz, 2H), 1.71 – 1.67 (m, 2H), 1.49 – 1.42 (m, 2H), 1.29 – 1.24 (m, 26H), 0.87 (t, $J = 6.8$ Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 64.5, 50.1, 49.8, 31.9, 30.5, 29.9, 29.7, 29.64, 29.60, 29.57, 29.51, 29.3, 27.2, 22.7, 14.1.

3-(Heptadecylamino)propan-1-ol (*A19*)

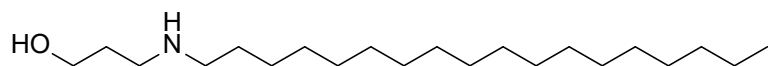


White solid. Yield: 0.5 g (1.6 mmol scale), 99%.

M.p. 68–69 °C.

¹H NMR (400 MHz, CDCl₃) δ 3.81 (t, *J* = 5.2 Hz, 2H), 3.03 (br s, 1H), 2.87 (t, *J* = 5.6 Hz, 2H), 2.60 (t, *J* = 7.2 Hz, 2H), 1.71 – 1.66 (m, 2H), 1.48 – 1.43 (m, 2H), 1.31 – 1.22 (m, 29H), 0.88 (t, *J* = 6.4 Hz, 3H). ¹³C NMR (101 MHz, CDCl₃) δ 64.6, 50.1, 49.9, 31.9, 30.6, 29.9, 29.67, 29.64, 29.60, 29.57, 29.52, 29.3, 27.3, 22.7, 14.1. HRMS (ESI) *m/z*: [M + H]⁺ calcd for C₂₀H₄₄NO⁺ 314.3417, found 314.3427.

3-(Octadecylamino)propan-1-ol (A20): [CAS: 72637-60-6]

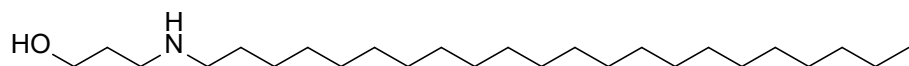


White solid. Yield: 1.25 g (5 mmol scale), 77%.

M.p. 63–64 °C.

¹H NMR (400 MHz, CDCl₃) δ 3.79 (t, *J* = 5.2 Hz, 2H), 3.15 (br s, 1H), 2.86 (t, *J* = 5.6 Hz, 2H), 2.58 (t, *J* = 7.2 Hz, 2H), 1.70 – 1.65 (m, 2H), 1.48 – 1.41 (m, 2H), 1.27 – 1.24 (m, 30H), 0.86 (t, *J* = 6.8 Hz, 3H). ¹³C NMR (101 MHz, CDCl₃) δ 64.4, 50.0, 49.8, 31.9, 30.6, 29.9, 29.7, 29.63, 29.59, 29.55, 29.50, 29.3, 27.2, 22.7, 14.1.

3-(Docosylamino)propan-1-ol (*A21*)



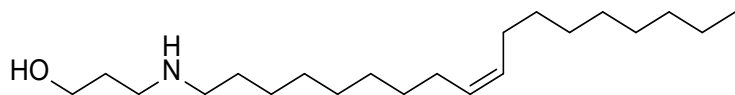
White solid. Yield: 1.59 g (5 mmol scale), 84%.

M.p. 72–73 °C.

¹H NMR (400 MHz, CDCl₃) δ 3.82 – 3.80 (m, 2H), 2.89 – 2.86 (m, 2H), 2.59 (t, *J* = 7.2 Hz, 2H), 1.71 – 1.66 (m, 2H), 1.49 – 1.40 (m, 2H), 1.30 – 1.25 (m, 44H), 0.89 – 0.86 (m, 3H). ¹³C NMR (101 MHz, CDCl₃) δ 64.7, 50.3, 49.9, 31.9, 30.5, 30.0, 29.69, 29.65, 29.62, 29.58, 29.53,

29.4, 27.3, 22.7, 14.1. HRMS (ESI) m/z : $[M + H]^+$ calcd for $C_{25}H_{54}NO^+$ 384.4200, found 384.4206.

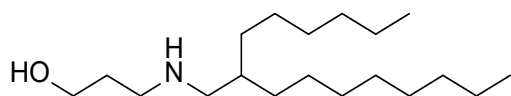
(Z)-3-(octadec-9-en-1-ylamino)propan-1-ol (A22)



Colorless oily liquid. Yield: 0.32 g (1.5 mmol scale), 66%.

1H NMR (400 MHz, $CDCl_3$) δ 5.36 – 5.33 (m, 2H), 3.83 – 3.80 (m, 2H), 2.89 – 2.86 (m, 2H), 2.60 (t, J = 7.2 Hz, 2H), 2.04 – 1.95 (m, 4H), 1.71 – 1.67 (m, 2H), 1.48 – 1.44 (m, 2H), 1.34 – 1.27 (m, 24H), 0.88 (t, J = 6.4 Hz, 3H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 129.9, 129.8, 64.6, 50.2, 49.9, 31.9, 30.5, 29.9, 29.75, 29.74, 29.50, 29.47, 29.3, 29.2, 27.3, 27.2, 22.7, 14.1. HRMS (ESI) m/z : $[M + H]^+$ calcd for $C_{21}H_{44}NO^+$ 326.3417, found 326.3419.

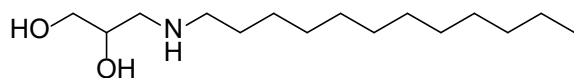
3-((2-Hexyldecyl)amino)propan-1-ol (A23)



Colorless oily liquid. Yield: 1.27 g (5 mmol scale), 85%.

1H NMR (400 MHz, $CDCl_3$) δ 3.80 (t, J = 5.2 Hz, 2H), 2.95 (br s, 1H), 2.85 (t, J = 5.6 Hz, 2H), 2.49 (d, J = 6.0 Hz, 2H), 1.71 – 1.65 (m, 2H), 1.42 (br s, 1H), 1.29 – 1.25 (m, 24H), 0.89 – 0.85 (m, 6H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 64.7, 53.5, 50.7, 37.9, 32.1, 31.87, 31.85, 30.5, 30.0, 29.7, 29.6, 29.3, 26.7, 26.67, 22.7, 14.1. HRMS (ESI) m/z : $[M + H]^+$ calcd for $C_{19}H_{42}NO^+$ 300.3261, found 300.3277.

3-(Dodecylamino)propane-1,2-diol (A24): [CAS: 821-91-0]



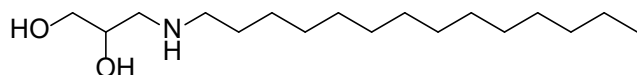
White solid. Yield: 1.80 g (10 mmol scale), 69%.

M.p. 73–74 °C.

1H NMR (400 MHz, $CDCl_3$) δ 3.77 (ddd, J = 8.8, 7.2, 4.0 Hz, 1H), 3.66 (dd, J = 11.6, 4.0 Hz, 1H), 3.56 (dd, J = 11.4, 5.2 Hz, 1H), 3.37 (s, 2H), 2.76 (dd, J = 12.0, 3.6 Hz, 1H), 2.65 (dd, J =

12.0, 7.2 Hz, 1H), 2.63 – 2.54 (m, 2H), 1.49 – 1.43 (m, 2H), 1.30 – 1.22 (m, 18H), 0.87 (t, J = 6.8 Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 69.7, 65.7, 52.3, 49.9, 31.9, 29.8, 29.64, 29.59, 29.50, 29.3, 27.2, 22.6, 14.1.

3-(Tetradecylamino)propane-1,2-diol (A25): [CAS: 111953-19-6]

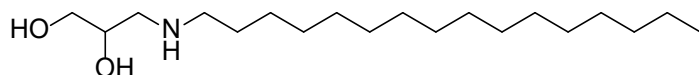


White solid. Yield: 1.14 g (5 mmol scale), 79%.

M.p. 75–76 °C.

^1H NMR (400 MHz, CDCl_3) δ 3.74 – 3.71 (m, 2H), 3.70 – 3.60 (m, 1H), 2.83 – 2.79 (m, 2H), 2.76 (br s, 1H), 2.71 – 2.68 (m, 2H), 2.66 – 2.55 (m, 2H), 1.51 – 1.44 (m, 2H), 1.32 – 1.23 (m, 19H), 0.88 (t, J = 6.8 Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 69.4, 65.9, 52.4, 49.9, 31.9, 30.0, 29.6, 29.6, 29.5, 29.3, 27.2, 22.7, 14.1.

3-(Hexadecylamino)propane-1,2-diol (A26): [CAS: 7517-27-3]

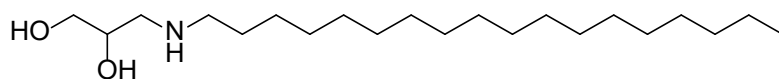


White solid. Yield: 1.30 g (10 mmol scale), 40%.

M.p. 74–75 °C.

^1H NMR (400 MHz, CDCl_3) δ 3.77 – 3.71 (m, 2H), 3.63 – 3.60 (m, 1H), 2.84 – 2.79 (m, 2H), 2.76 (br s, 1H), 2.72 – 2.66 (m, 2H), 2.65 – 2.55 (m, 21H), 1.49 – 1.44 (m, 2H), 1.32 – 1.25 (m, 26H), 0.88 (t, J = 6.4 Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 69.3, 65.9, 52.4, 49.9, 31.9, 30.0, 29.69, 29.66, 29.62, 29.60, 29.5, 29.4, 27.2, 22.7, 14.1.

3-(Octadecylamino)propane-1,2-diol (A27): [CAS: 28900-95-0]



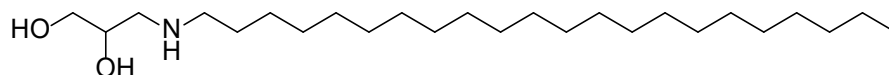
White solid. Yield: 1.64 g (5 mmol scale), 99%.

M.p. 84–86 °C.

^1H NMR (400 MHz, $\text{Methanol-}d_4$) δ 3.77 – 3.71 (m, 1H), 3.49 (dd, J = 5.6, 1.6 Hz, 2H), 3.31 –

3.29 (m, 2H), 2.70 (dd, $J = 12.2, 3.6$ Hz, 1H), 2.65 – 2.53 (m, 3H), 1.54 – 1.48 (m, 2H), 1.33 – 1.28 (m, 34H), 0.90 (t, $J = 6.8$ Hz, 3H). ^{13}C NMR (101 MHz, Methanol- d_4) δ 71.6, 66.1, 53.5, 50.8, 33.1, 30.8, 30.72, 30.67, 30.6, 30.5, 23.7, 14.5.

3-(Docosylamino)propane-1,2-diol (**A28**) [CAS: 2190523-67-0]



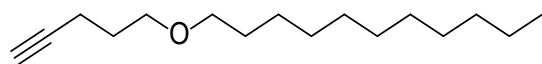
White solid. Yield: 1.64 g (5 mmol scale), 81%.

M.p. 85–87 °C.

^1H NMR (400 MHz, CDCl_3) δ 3.77 – 3.72 (m, 1H), 3.64 – 3.60 (m, 1H), 2.85 – 2.67 (m, 2H), 2.65 – 2.47 (m, 2H), 1.48 – 1.42 (m, 2H), 1.30 – 1.25 (m, 41H), 0.88 (t, $J = 6.6$ Hz, 3H).

1.2 Characterization data of **B8** – **B21**

1-(Pent-4-yn-1-yloxy)undecane (**B8**)

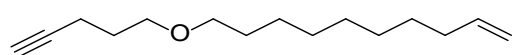


Colorless oil. Yield: 0.28 g (8 mmol scale), 15%.

$R_f = 0.5$ (PE/EtOAc = 50/1, v/v).

^1H NMR (400 MHz, CDCl_3) δ 3.49 (t, $J = 6.4$ Hz, 2H), 3.41 (t, $J = 6.8$ Hz, 2H), 2.29 (td, $J = 7.2, 2.4$ Hz, 2H), 1.93 (t, $J = 2.4$ Hz, 1H), 1.82 – 1.75 (m, 2H), 1.58 – 1.54 (m, 2H), 1.34 – 1.26 (m, 16H), 0.88 (t, $J = 6.8$ Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 84.0, 71.1, 70.0, 68.3, 31.9, 29.1, 29.6, 29.5, 29.3, 28.7, 26.2, 22.7, 15.2, 14.1. HRMS (APCI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{16}\text{H}_{31}\text{O}^+$ 239.2369, found 239.2341.

10-(Pent-4-yn-1-yloxy)dec-1-ene (**B9**)

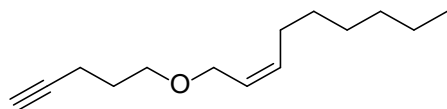


Colorless oil. Yield: 0.29 g (10 mmol scale), 13%.

$R_f = 0.4$ (PE/EtOAc = 50/1, v/v).

^1H NMR (400 MHz, CDCl_3) δ 5.81 (ddt, $J = 16.8, 10.2, 6.8$ Hz, 1H), 5.01 – 4.91 (m, 2H), 3.49 (t, $J = 6.4$ Hz, 2H), 3.41 (t, $J = 6.4$ Hz, 2H), 2.29 (td, $J = 7.2, 2.8$ Hz, 2H), 2.07 – 2.00 (m, 2H), 1.94 (t, $J = 2.4$ Hz, 1H), 1.81 – 1.75 (m, 2H), 1.58 – 1.52 (m, 2H), 1.39 – 1.25 (m, 10H). ^{13}C NMR (101 MHz, CDCl_3) δ 139.2, 114.1, 84.0, 71.0, 69.0, 68.3, 33.8, 29.7, 29.4, 29.1, 28.9, 28.6, 26.1, 15.2. HRMS (APCI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{15}\text{H}_{27}\text{O}^+$ 223.2056, found 223.2053.

(Z)-1-(pent-4-yn-1-yloxy)non-2-ene (B10)

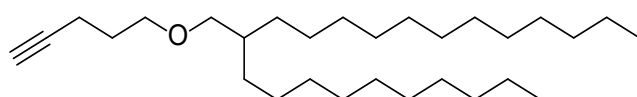


Colorless oil. Yield: 0.4 g (8 mmol scale), 24%.

$R_f = 0.4$ (PE/EtOAc = 50/1, v/v).

^1H NMR (400 MHz, CDCl_3) δ 5.62 – 5.49 (m, 2H), 4.02 (d, $J = 5.6$ Hz, 2H), 3.51 (t, $J = 6.2$ Hz, 2H), 2.29 (td, $J = 7.2, 2.8$ Hz, 2H), 2.06 (q, $J = 6.8$ Hz, 2H), 1.94 (t, $J = 2.8$ Hz, 1H), 1.83 – 1.75 (m, 2H), 1.39 – 1.25 (m, 8H), 0.90 – 0.87 (m, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 133.7, 126.0, 84.0, 68.4, 68.3, 66.4, 31.7, 29.5, 28.9, 28.7, 27.6, 22.6, 15.2, 14.0. HRMS (APCI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{14}\text{H}_{25}\text{O}^+$ 209.1900, found 209.1879.

11-((Pent-4-yn-1-yloxy)methyl) tricosane (B11): [CAS: 1353749-82-2]

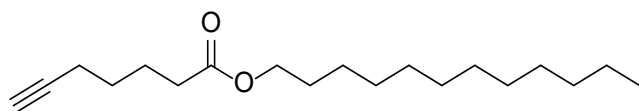


Colorless oil. Yield: 2.1 g (10 mmol scale), 50%.

$R_f = 0.6$ (PE/EtOAc = 50/1, v/v).

^1H NMR (400 MHz, CDCl_3) δ 3.47 (t, $J = 6.0$ Hz, 2H), 3.27 (d, $J = 6.0$ Hz, 2H), 2.28 (td, $J = 7.2, 2.8$ Hz, 2H), 1.93 (t, $J = 2.8$ Hz, 1H), 1.81 – 1.74 (m, 2H), 1.56 – 1.50 (m, 1H), 1.32 – 1.22 (m, 40H), 0.90 – 0.86 (m, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 84.2, 74.3, 69.2, 68.2, 38.2, 31.9, 31.4, 30.1, 29.69, 29.68, 29.4, 28.7, 26.8, 22.7, 15.3, 14.1.

Dodecyl hept-6-ynoate (B12)

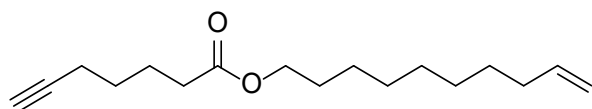


Colorless oily. Yield: 2.02 g (8 mmol scale), 86%.

$R_f = 0.7$ (PE/EtOAc = 10/1, v/v).

^1H NMR (400 MHz, CDCl_3) δ 4.06 (t, $J = 6.8$ Hz, 2H), 2.33 (t, $J = 7.4$ Hz, 2H), 2.21 (td, $J = 7.2, 2.8$ Hz, 2H), 1.95 (t, $J = 2.8$ Hz, 1H), 1.79 – 1.72 (m, 3H), 1.65 – 1.53 (m, 5H), 1.36 – 1.26 (m, 16H), 0.88 (t, $J = 6.4$ Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 173.4, 83.9, 68.5, 64.5, 33.7, 31.9, 29.6, 29.52, 29.48, 29.3, 29.2, 28.6, 27.8, 25.9, 24.0, 22.6, 18.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{19}\text{H}_{35}\text{O}_2^+$ 295.2632, found 295.2652.

Dec-9-en-1-yl hept-6-ynoate (**B13**)

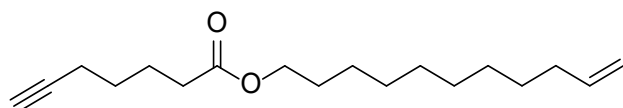


Colorless oily. Yield: 2.06 g (8 mmol scale), 98%.

$R_f = 0.6$ (PE/EtOAc = 10/1, v/v).

^1H NMR (400 MHz, CDCl_3) δ 5.81 (ddt, $J = 16.8, 10.2, 6.8$ Hz, 1H), 5.02 – 4.92 (m, 2H), 4.06 (t, $J = 6.8$ Hz, 2H), 2.33 (t, $J = 7.4$ Hz, 2H), 2.21 (td, $J = 7.0, 2.8$ Hz, 2H), 2.07 – 2.01 (m, 2H), 1.95 (t, $J = 2.8$ Hz, 1H), 1.79 – 1.71 (m, 2H), 1.66 – 1.53 (m, 4H), 1.41 – 1.29 (m, 10H). ^{13}C NMR (101 MHz, CDCl_3) δ 173.5, 139.1, 114.1, 83.9, 68.5, 64.5, 33.8, 29.3, 29.1, 29.0, 28.8, 28.6, 27.8, 25.8, 24.0, 18.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{17}\text{H}_{29}\text{O}_2^+$ 265.2162, found 265.2184.

Undec-10-en-1-yl hept-6-ynoate (**B14**)

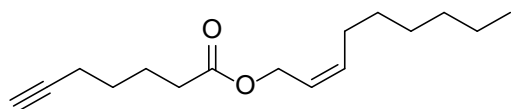


Colorless oily. Yield: 0.862 g (5 mmol scale), 62%.

$R_f = 0.7$ (PE/EtOAc = 10/1, v/v).

^1H NMR (400 MHz, CDCl_3) δ 5.81 (ddt, $J = 16.8, 10.2, 6.8$ Hz, 1H), 5.02 – 4.91 (m, 2H), 4.06 (t, $J = 6.8$ Hz, 2H), 2.33 (t, $J = 7.2$ Hz, 2H), 2.21 (td, $J = 7.2, 2.8$ Hz, 2H), 2.07 – 2.01 (m, 2H), 1.95 (t, $J = 2.8$ Hz, 1H), 1.78 – 1.71 (m, 3H), 1.66 – 1.53 (m, 5H), 1.40 – 1.28 (m, 10H). ^{13}C NMR (101 MHz, CDCl_3) δ 173.4, 139.1, 114.1, 83.9, 68.5, 64.5, 33.8, 29.4, 29.3, 29.2, 29.1, 28.9, 28.6, 27.8, 25.9, 24.0, 18.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{18}\text{H}_{31}\text{O}_2^+$ 279.2319, found 279.2338.

(Z)-non-2-en-1-yl hept-6-ynoate (B15)

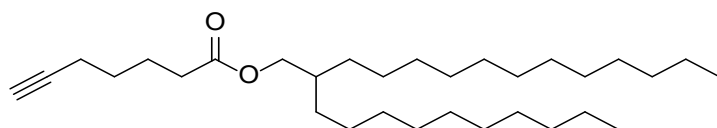


Colorless oily. Yield: 1.536 g (8 mmol scale), 77%.

$R_f = 0.7$ (PE/EtOAc = 10/1, v/v).

^1H NMR (400 MHz, CDCl_3) δ 5.66 – 5.61 (m, 1H), 5.55 – 5.51 (m, 1H), 4.63 (dd, $J = 6.8, 1.2$ Hz, 2H), 2.34 (t, $J = 7.2$ Hz, 2H), 2.21 (td, $J = 7.2, 2.8$ Hz, 2H), 2.10 (qd, $J = 7.2, 1.6$ Hz, 2H), 1.95 (t, $J = 2.8$ Hz, 1H), 1.78 – 1.72 (m, 2H), 1.67 – 1.53 (m, 2H), 1.41 – 1.23 (m, 9H), 0.88 (t, $J = 6.8$ Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 173.3, 135.5, 123.2, 83.9, 68.5, 60.3, 33.7, 31.6, 29.3, 28.8, 27.8, 27.5, 24.0, 22.6, 18.1, 14.0. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{15}\text{H}_{25}\text{O}_2^+$ 251.2006, found 251.2017.

2-Decyltetradecyl hept-6-ynoate (B16)



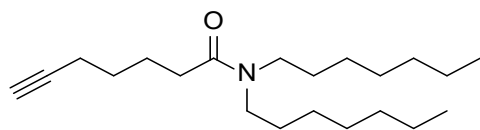
Colorless oily. Yield: 2.3 g (5 mmol scale), 99%.

$R_f = 0.8$ (PE/EtOAc = 10/1, v/v).

^1H NMR (400 MHz, CDCl_3) δ 3.97 (d, $J = 6.0$ Hz, 2H), 2.33 (t, $J = 7.2$ Hz, 2H), 2.21 (td, $J = 7.2, 2.8$ Hz, 2H), 1.94 (t, $J = 2.8$ Hz, 1H), 1.79 – 1.71 (m, 2H), 1.63 – 1.53 (m, 2H), 1.29 – 1.26 (m, 41H), 0.90 – 0.86 (m, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 173.6, 83.9, 68.5, 67.2, 37.3, 33.8, 31.9, 31.3, 29.9, 29.7, 29.65, 29.60, 29.4, 27.9, 26.7, 24.1, 22.7, 18.1, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{31}\text{H}_{59}\text{O}_2^+$

463.4510, found 463.4501.

***N,N*-diheptylhept-6-ynamide (*B17*)**

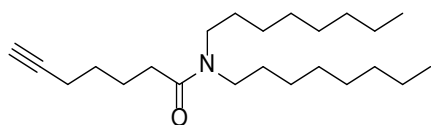


Colorless oily liquid. Yield: 1.33 g (4.8 mmol scale), 87%.

$R_f = 0.5$ (PE/EtOAc = 5/1, v/v).

^1H NMR (400 MHz, CDCl_3) δ 3.30 – 3.26 (m, 2H), 3.22 – 3.18 (m, 2H), 2.31 (t, $J = 7.4$ Hz, 2H), 2.22 (td, $J = 7.2, 2.8$ Hz, 2H), 1.94 (t, $J = 2.8$ Hz, 1H), 1.78 – 1.73 (m, 2H), 1.62 – 1.46 (m, 6H), 1.34 – 1.24 (m, 16H), 0.91 – 0.86 (m, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 172.0, 84.2, 68.3, 47.9, 45.9, 32.5, 31.8, 31.7, 29.13, 29.06, 28.99, 28.2, 27.8, 27.0, 26.8, 24.5, 22.6, 22.5, 18.3, 14.0. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{21}\text{H}_{40}\text{NO}^+$ 322.3104, found 322.3113.

***N,N*-dioctylhept-6-ynamide (*B18*)**

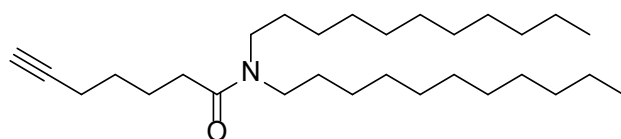


Colorless oily liquid. Yield: 1.7 g (5 mmol scale), 99%.

$R_f = 0.6$ (PE/EtOAc = 5/1, v/v).

^1H NMR (400 MHz, CDCl_3) δ 3.30 – 3.26 (m, 2H), 3.22 – 3.18 (m, 2H), 2.31 (t, $J = 7.2$ Hz, 2H), 2.22 (td, $J = 7.2, 2.8$ Hz, 2H), 1.94 (t, $J = 2.4$ Hz, 1H), 1.80 – 1.73 (m, 2H), 1.62 – 1.48 (m, 6H), 1.31 – 1.25 (m, 20H), 0.88 (q, $J = 6.8$ Hz, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 172.0, 84.2, 68.3, 47.9, 45.9, 32.5, 31.8, 31.7, 29.4, 29.3, 29.22, 29.17, 29.12, 28.2, 27.8, 27.0, 26.9, 24.5, 22.6, 18.3, 14.0. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{23}\text{H}_{44}\text{NO}^+$ 350.3417, found 350.3430.

***N,N*-diundecylhept-6-ynamide (*B19*)**

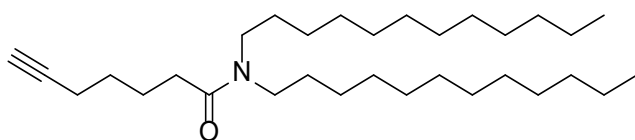


Colorless oily liquid. Yield: 2.1 g (5 mmol scale), 99%.

$R_f = 0.6$ (PE/EtOAc = 5/1).

^1H NMR (400 MHz, CDCl_3) δ 3.30 – 3.26 (m, 2H), 3.21 – 3.14 (m, 2H), 2.31 (t, $J = 7.2$ Hz, 2H), 2.22 (td, $J = 7.2, 2.8$ Hz, 2H), 1.94 (t, $J = 2.8$ Hz, 1H), 1.80 – 1.72 (m, 2H), 1.63 – 1.46 (m, 6H), 1.30 – 1.25 (m, 32H), 0.88 (td, $J = 6.8, 2.4$ Hz, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 172.0, 84.3, 68.4, 48.0, 45.9, 32.5, 31.9, 29.59, 29.55, 29.43, 29.35, 29.32, 29.2, 28.2, 27.8, 27.1, 26.9, 24.6, 22.7, 18.3, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{29}\text{H}_{56}\text{NO}^+$ 434.4356, found 434.4365.

***N,N*-didodecylhept-6-ynamide (*B20*)**

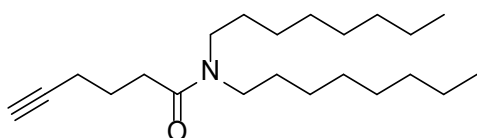


Colorless oily liquid. Yield: 1.89 g (4.5 mmol scale), 91%.

$R_f = 0.7$ (PE/EtOAc = 5/1, v/v).

^1H NMR (400 MHz, CDCl_3) δ 3.30 – 3.26 (m, 2H), 3.21 – 3.17 (m, 2H), 2.31 (t, $J = 7.6$ Hz, 2H), 2.22 (td, $J = 7.2, 2.8$ Hz, 2H), 1.94 (t, $J = 2.4$ Hz, 1H), 1.80 – 1.69 (m, 2H), 1.62 – 1.46 (m, 6H), 1.30 – 1.24 (m, 36H), 0.88 (td, $J = 6.8, 1.6$ Hz, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 172.0, 84.3, 68.4, 47.9, 45.9, 32.5, 31.9, 29.61, 29.59, 29.54, 29.4, 29.3, 29.1, 28.2, 27.8, 27.1, 26.9, 24.6, 22.7, 18.3, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{31}\text{H}_{60}\text{NO}^+$ 462.4669, found 462.4676.

***N,N*-dioctylhex-5-ynamide (*B21*)**



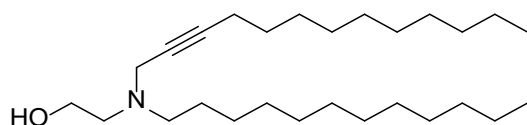
Colorless oily liquid. Yield: 1.6 g (4.8 mmol scale), 99%.

$R_f = 0.6$ (PE/EtOAc = 5/1, v/v).

^1H NMR (400 MHz, CDCl_3) δ 3.31 – 3.27 (m, 2H), 3.24 – 3.20 (m, 2H), 2.44 (t, J = 7.2 Hz, 2H), 2.28 (td, J = 6.8, 2.4 Hz, 2H), 1.95 (t, J = 2.6 Hz, 1H), 1.91 – 1.84 (m, 2H), 1.58 – 1.48 (m, 4H), 1.31 – 1.25 (m, 20H), 0.90 – 0.86 (m, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 171.6, 84.0, 68.7, 47.9, 45.9, 31.8, 31.7, 31.4, 29.4, 29.3, 29.22, 29.18, 29.1, 27.8, 27.0, 26.9, 24.0, 22.6, 17.9, 14.0. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{22}\text{H}_{42}\text{NO}^+$ 336.3261, found 336.3280.

1.3 Characterization data of 1a, A11B2, A11B4, A11B7, A11B9 and A28B1

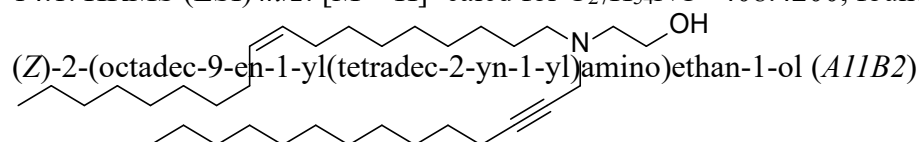
2-(Dodecyl(tridec-2-yn-1-yl)amino)ethan-1-ol (1a)



Colorless oily liquid. Yield: 127 mg (0.5 mmol scale), 63%.

R_f = 0.3 ($\text{CH}_2\text{Cl}_2/\text{MeOH}$ = 25:1, v/v).

^1H NMR (400 MHz, CDCl_3) δ 3.57 (t, J = 5.2 Hz, 2H), 3.38 (t, J = 2.4 Hz, 2H), 2.66 (t, J = 5.2 Hz, 2H), 2.50 (dd, J = 8.8, 7.2 Hz, 2H), 2.18 (tt, J = 6.8, 2.4 Hz, 2H), 1.53 - 1.43 (m, 4H), 1.40 - 1.36 (m, 2H), 1.30 - 1.26 (m, 30H), 0.90 - 0.85 (m, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 85.4, 74.2, 58.3, 55.0, 53.2, 41.9, 31.9, 29.63, 29.56, 29.3, 29.1, 28.9, 28.8, 27.6, 27.4, 22.7, 18.6, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{27}\text{H}_{54}\text{NO}^+$ 408.4200, found 408.4219.

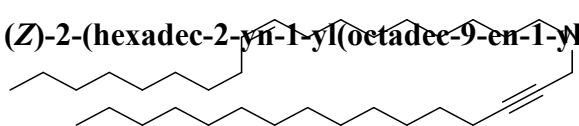


Colorless oily liquid. Yield: 153 mg (0.5 mmol scale), 61%.

R_f = 0.30 ($\text{CH}_2\text{Cl}_2/\text{MeOH}$ = 50/1, v/v).

^1H NMR (400 MHz, CDCl_3) δ 5.39 – 5.30 (m, 2H), 3.56 (t, J = 5.4 Hz, 2H), 3.37 (t, J = 2.2 Hz, 2H), 2.65 (t, J = 5.4 Hz, 2H), 2.56 – 2.45 (m, 2H), 2.18 (tt, J = 7.2, 2.2 Hz, 2H), 2.03 – 1.98 (m, 4H), 1.52 – 1.42 (m, 6H), 1.39 – 1.25 (m, 36H), 0.89 – 0.86 (m, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 129.9, 129.8, 85.4, 74.3, 58.4, 55.0, 53.2, 41.9, 31.9, 29.8, 29.7, 29.63, 29.56, 29.5, 29.34, 29.31, 29.1, 28.9, 28.8, 27.6, 27.4, 27.2, 22.7, 18.6, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{34}\text{H}_{66}\text{NO}^+$ 504.5139, found 504.5171.

(Z)-2-(hexadec-2-yn-1-yl(octadec-9-en-1-yl)amino)ethan-1-ol (A11B4)

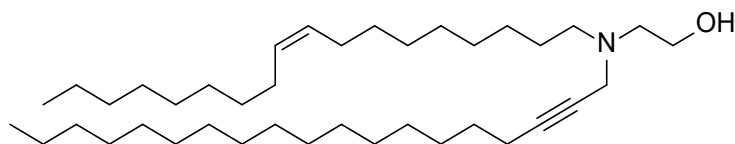


Colorless oily liquid. Yield: 240 mg (0.5 mmol scale), 90%.

R_f = 0.34 ($\text{CH}_2\text{Cl}_2/\text{MeOH}$ = 50/1, v/v).

^1H NMR (400 MHz, CDCl_3) δ 5.38 – 5.31 (m, 2H), 3.57 (t, J = 5.2 Hz, 2H), 3.37 (t, J = 2.4 Hz, 2H), 2.66 (t, J = 5.2 Hz, 2H), 2.52 – 2.48 (m, 2H), 2.20 – 2.15 (m, 2H), 2.01 (q, J = 6.4 Hz, 4H), 1.61 (s, 4H), 1.52 – 1.41 (m, 6H), 1.39 – 1.21 (m, 37H), 0.88 (t, J = 6.4 Hz, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 129.9, 129.8, 85.4, 74.3, 58.4, 55.0, 53.2, 41.9, 31.9, 29.8, 29.69, 29.65, 29.56, 29.53, 29.35, 29.31, 29.1, 29.0, 28.8, 27.6, 27.3, 27.2, 22.7, 18.6, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{36}\text{H}_{70}\text{NO}^+$ 532.5452, found 532.5479.

(Z)-2-(nonadec-2-yn-1-yl(octadec-9-en-1-yl)amino)ethan-1-ol (A11B7)

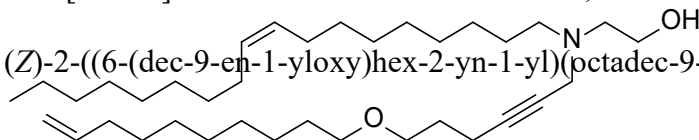


Colorless oily liquid. Yield: 198 mg (0.4 mmol scale), 86%.

R_f = 0.30 ($\text{CH}_2\text{Cl}_2/\text{MeOH}$ = 50/1, v/v).

^1H NMR (400 MHz, CDCl_3) δ 5.36 – 5.33 (m, 2H), 3.56 (t, J = 5.4 Hz, 2H), 3.37 (t, J = 2.2 Hz, 2H), 2.66 (t, J = 5.6 Hz, 2H), 2.52 – 2.48 (m, 2H), 2.18 (tt, J = 6.8, 2.2 Hz, 2H), 2.01 (q, J = 6.6 Hz, 4H), 1.57 (s, 4H), 1.51 – 1.42 (m, 6H), 1.40 – 1.26 (m, 42H), 0.88 (t, J = 6.8 Hz, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 129.9, 129.8, 85.4, 74.3, 58.4, 55.0, 53.2, 41.9, 31.9, 29.8, 29.7, 29.65, 29.57, 29.5, 29.4, 29.3, 29.1, 28.9, 28.8, 27.6, 37.4, 27.2, 22.7, 18.6, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{39}\text{H}_{76}\text{NO}^+$ 574.5921, found 574.5967.

(Z)-2-(((6-(dec-9-en-1-yloxy)hex-2-yn-1-yl)(octadec-9-en-1-yl)amino)ethan-1-ol (A11B9)



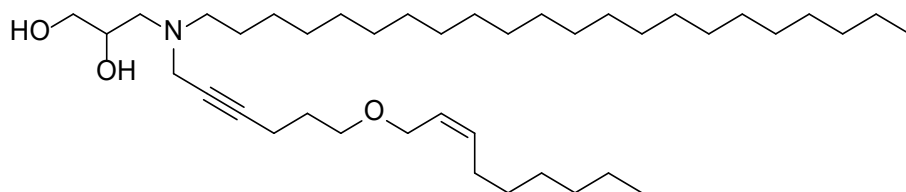
Colorless oily liquid. Yield: 114 mg (0.5 mmol scale), 42%.

R_f = 0.32 ($\text{CH}_2\text{Cl}_2/\text{MeOH}$ = 50/1, v/v).

^1H NMR (400 MHz, CDCl_3) δ 5.80 (ddt, J = 13.2, 10.2, 6.4 Hz, 1H), 5.39 – 5.30 (m, 2H), 5.01

– 4.90 (m, 2H), 3.56 (t, $J = 5.4$ Hz, 2H), 3.48 (t, $J = 6.4$ Hz, 2H), 3.39 (t, $J = 6.8$ Hz, 2H), 3.36 (t, $J = 2.0$ Hz, 2H), 2.65 (t, $J = 5.4$ Hz, 2H), 2.51 – 2.47 (m, 2H), 2.30 – 2.26 (m, 2H), 2.06 – 1.98 (m, 6H), 1.81 – 1.72 (m, 2H), 1.60 – 1.53 (m, 2H), 1.45 – 1.42 (m, 2H), 1.39 – 1.25 (m, 33H), 0.87 (t, $J = 6.8$ Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 139.2, 129.9, 129.8, 114.1, 84.6, 74.5, 71.1, 69.2, 58.3, 55.0, 53.2, 41.9, 33.8, 31.9, 29.8, 29.71, 29.68, 29.6, 29.52, 29.51, 29.4, 29.30, 29.28, 29.13, 29.06, 28.9, 27.6, 27.3, 27.2, 26.1, 22.7, 15.5, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{36}\text{H}_{68}\text{NO}_2^+$ 546.5245, found 546.5264.

(Z)-3-(docosyl(6-(non-2-en-1-yloxy)hex-2-yn-1-yl)amino)propane-1,2-diol (A28B10)



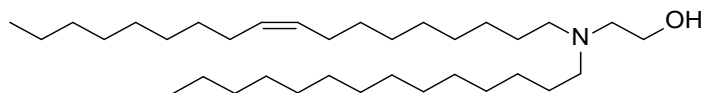
Colorless oily liquid. Yield: 56 mg (0.6 mmol scale), 15%.

$R_f = 0.25$ ($\text{CH}_2\text{Cl}_2/\text{MeOH} = 50/1$, v/v).

^1H NMR (400 MHz, CDCl_3) δ 5.60 – 5.48 (m, 2H), 4.01 (d, $J = 5.6$ Hz, 2H), 3.74 (dd, $J = 8.8$, 3.6 Hz, 2H), 3.51 – 3.46 (m, 3H), 3.38 (qt, $J = 17.2$, 2.4 Hz, 2H), 2.60 – 2.51 (m, 3H), 2.50 – 2.43 (m, 1H), 2.28 (td, $J = 6.8$, 2.4 Hz, 2H), 2.05 (q, $J = 6.8$ Hz, 2H), 1.76 (quint, $J = 6.8$ Hz, 2H), 1.44 – 1.41 (m, 2H), 1.37 – 1.25 (m, 48H), 0.89 – 0.85 (m, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 133.7, 126.0, 84.9, 74.3, 68.7, 67.3, 66.5, 64.8, 56.1, 53.8, 42.5, 31.9, 31.7, 29.7, 29.6, 29.54, 29.49, 29.3, 29.1, 28.9, 27.6, 27.5, 27.3, 22.7, 22.6, 15.5, 14.08, 14.05. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{40}\text{H}_{78}\text{NO}_3^+$ 620.5976, found 620.5980.

1.4 Characterization data of R-A11B2, R-A11B4, R-A11B7, R-A11B9 and R-A28B10, 2a

(Z)-2-(octadec-9-en-1-yl(tetradecyl)amino)ethan-1-ol (R-A11B2)



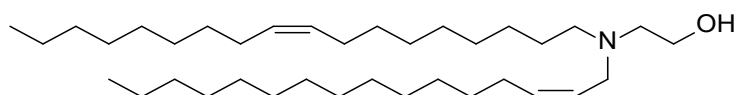
Colorless oily liquid. Yield: 15 mg (0.14 mmol scale), 21%.

$R_f = 0.35$ ($\text{CH}_2\text{Cl}_2/\text{MeOH} = 25/1$, v/v).

^1H NMR (400 MHz, CDCl_3) δ 5.39 – 5.33 (m, 2H), 3.55 (t, $J = 5.2$ Hz, 2H), 2.60 (t, $J = 5.2$ Hz,

2H), 2.49 – 2.45 (m, 4H), 2.03 – 1.94 (m, 4H), 1.47 – 1.41 (m, 4H), 1.30 – 1.25 (m, 44H), 0.88 (t, $J = 6.4$ Hz, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 129.9, 129.8, 58.2, 55.6, 53.8, 32.6, 31.9, 29.8, 29.75, 29.69, 29.65, 29.63, 29.55, 29.53, 29.5, 29.4, 29.31, 29.26, 27.4, 27.2, 26.9, 22.7, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{34}\text{H}_{70}\text{NO}^+$ 508.5452, found 508.5464.

2-(((*Z*)-hexadec-2-en-1-yl)((*Z*)-octadec-9-en-1-yl)amino)ethan-1-ol (*R'*-A11B4)

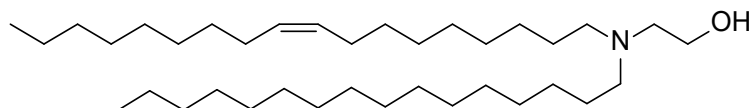


Colorless oily liquid. Yield: 21 mg (0.05 mmol scale), 79%.

$R_f = 0.3$ ($\text{CH}_2\text{Cl}_2/\text{MeOH} = 25/1$, v/v).

^1H NMR (400 MHz, CDCl_3) δ 5.58 – 5.52 (m, 1H), 5.45 – 5.40 (m, 1H), 5.38 – 5.30 (m, 2H), 3.55 (t, $J = 5.2$ Hz, 2H), 3.15 (d, $J = 6.4$ Hz, 2H), 2.59 (t, $J = 5.4$ Hz, 2H), 2.49 – 2.42 (m, 2H), 2.06 – 1.95 (m, 6H), 1.49 – 1.42 (m, 2H), 1.36 – 1.26 (m, 44H), 0.88 (t, $J = 6.4$ Hz, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 133.4, 129.9, 129.8, 125.9, 58.2, 55.0, 53.6, 50.1, 31.9, 29.8, 29.67, 29.66, 29.63, 29.60, 29.5, 29.4, 29.31, 29.28, 29.2, 27.5, 27.4, 27.2, 22.7, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{36}\text{H}_{72}\text{NO}^+$ 534.5608, found 534.5613.

(*Z*)-2-(nonadecyl(octadec-9-en-1-yl)amino)ethan-1-ol (*R*-A11B4)

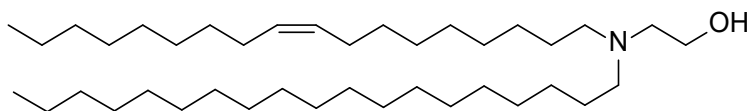


Colorless oily liquid. Yield: 26 mg (0.15 mmol scale), 33%.

$R_f = 0.28$ ($\text{CH}_2\text{Cl}_2/\text{MeOH} = 25/1$, v/v).

^1H NMR (400 MHz, CDCl_3) δ 5.39 – 5.33 (m, 2H), 3.76 – 3.74 (m, 2H), 2.88 – 2.86 (m, 2H), 2.77 – 2.73 (m, 4H), 2.01 – 1.94 (m, 4H), 1.60 (t, $J = 8.0$ Hz, 4H), 1.31 – 1.25 (m, 48H), 0.87 (t, $J = 6.8$ Hz, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 130.0, 129.7, 57.4, 56.9, 56.6, 54.0, 31.9, 29.8, 29.68, 29.66, 29.61, 29.53, 29.51, 29.4, 29.35, 29.30, 29.2, 27.20, 27.16, 27.08, 25.0, 22.7, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{36}\text{H}_{74}\text{NO}^+$ 536.5765, found 536.5789.

(*Z*)-2-(hexadecyl(octadec-9-en-1-yl)amino)ethan-1-ol (*R*-A11B7)

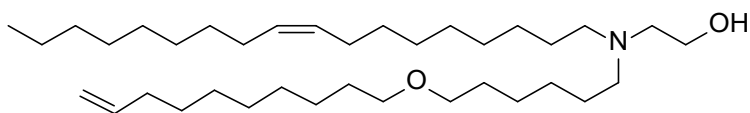


Colorless oily liquid. Yield: 21 mg (0.1 mmol scale), 36%.

$R_f = 0.30$ ($\text{CH}_2\text{Cl}_2/\text{MeOH} = 25/1$, v/v).

^1H NMR (400 MHz, CDCl_3) δ 5.40 – 5.33 (m, 2H), 3.54 (t, $J = 5.2$ Hz, 2H), 2.59 (t, $J = 5.2$ Hz, 2H), 2.46 (t, $J = 7.2$ Hz, 4H), 2.03 – 1.95 (m, 4H), 1.46 – 1.42 (m, 4H), 1.34 – 1.23 (m, 54H), 0.88 (t, $J = 6.4$ Hz, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 129.9, 129.8, 58.2, 55.6, 53.8, 32.6, 31.9, 29.7, 29.6, 29.5, 29.36, 29.31, 29.27, 27.4, 27.2, 27.0, 22.7, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{39}\text{H}_{80}\text{NO}^+$ 578.6234, found 578.6242.

(Z)-2-(((6-(dec-9-en-1-yloxy)hexyl)(octadec-9-en-1-yl)amino)ethan-1-ol (R-A11B9)

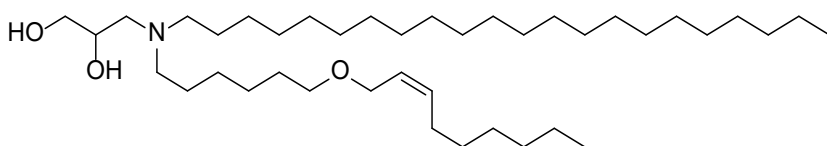


Colorless oily liquid. Yield: 32 mg (0.1 mmol scale), 55%.

$R_f = 0.35$ ($\text{CH}_2\text{Cl}_2/\text{MeOH} = 25/1$, v/v).

^1H NMR (400 MHz, CDCl_3) δ 5.58 – 5.38 (m, 3H), 5.37 – 5.33 (m, 2H), 3.54 (t, $J = 5.4$ Hz, 2H), 3.388 (t, $J = 6.4$, 2H), 3.379 (t, $J = 6.8$, 2H), 3.15 (d, $J = 6.8$ Hz, 2H), 2.59 (t, $J = 5.6$ Hz, 2H), 2.45 (t, $J = 7.2$ Hz, 2H), 2.12 (q, $J = 7.6$ Hz, 2H), 2.04 – 1.98 (m, 4H), 1.66 – 1.56 (m, 4H), 1.46 – 1.42 (m, 2H), 1.29 – 1.25 (m, 38H), 0.87 (t, $J = 6.4$ Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 132.4, 129.9, 129.8, 126.7, 71.0, 70.0, 58.3, 55.0, 53.7, 50.1, 31.9, 29.77, 29.76, 29.61, 29.55, 29.54, 29.52, 29.31, 29.27, 27.4, 27.2, 26.2, 24.1, 22.7, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{36}\text{H}_{72}\text{NO}_2^+$ 550.5558, found 550.5577.

(Z)-3-(docosyl(6-(non-2-en-1-yloxy)hexyl)amino)propane-1,2-diol (R-A28B10)

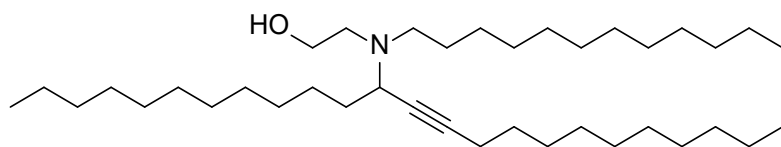


Colorless oily liquid. Yield: 10 mg (0.06 mmol scale), 27%.

$R_f = 0.30$ ($\text{CH}_2\text{Cl}_2/\text{MeOH} = 25/1$, v/v).

^1H NMR (400 MHz, CDCl_3) δ 5.61 – 5.50 (m, 2H), 4.02 (d, J = 5.6 Hz, 2H), 3.80 – 3.73 (m, 2H), 3.52 – 3.40 (m, 4H), 2.64 – 2.58 (m, 2H), 2.55 – 2.51 (m, 2H), 2.30 (tt, J = 7.2, 2.2 Hz, 2H), 2.08 – 1.99 (m, 4H), 1.81 – 1.73 (m, 4H), 1.51 – 1.43 (m, 2H), 1.30 – 1.25 (m, 49H), 0.90 – 0.86 (m, 6H). ^{13}C NMR (176 MHz, CDCl_3) δ 133.7, 126.0, 68.7, 67.2, 66.5, 64.7, 56.2, 53.9, 42.6, 31.9, 31.7, 29.70, 29.68, 29.65, 29.643, 29.638, 29.62, 29.52, 29.51, 29.4, 29.1, 28.9, 27.6, 27.3, 22.7, 22.6, 15.6, 14.12, 14.08. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{40}\text{H}_{82}\text{NO}_3^+$ 624.6289, found 624.6210.

2-(Dodecyl(tetracos-13-yn-12-yl)amino)ethan-1-ol (2a)



Colorless oily liquid. Yield: 28 mg (0.2 mmol scale), 25%.

R_f = 0.3 (PE/EtOAc = 20/1, v/v).

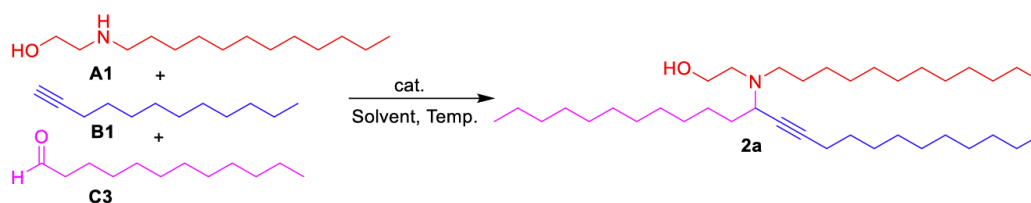
^1H NMR (400 MHz, CDCl_3) δ 3.57 (td, J = 10.8, 3.6 Hz, 1H), 3.53 – 3.45 (m, 1H), 3.43 – 3.36 (m, 1H), 2.86 – 2.64 (m, 2H), 2.56 – 2.46 (m, 2H), 2.46 – 2.32 (m, 1H), 2.18 (td, J = 6.8, 2 Hz, 2H), 1.63 – 1.517 (m, 2H), 1.516 – 1.43 (m, 3H), 1.428 – 1.35 (m, 4H), 1.34 – 1.14 (m, 46H), 0.88 (t, J = 6.4 Hz, 9H). ^{13}C NMR (101 MHz, CDCl_3) δ 84.5, 58.5, 53.3, 52.5, 51.2, 31.9, 29.7, 29.6, 29.4, 29.12, 29.1, 28.8, 27.5, 26.8, 22.7, 18.6, 14.1. HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{38}\text{H}_{76}\text{NO}^+$ 562.5873, found 562.5969.

2. Supplementary figures and tables

Table S1. Optimization for A³ coupling reaction with formaldehyde as the substrate. (^aNMR yield; ^bisolation yield)

Entry	Catalyst (mol %)	Temperature	Time	Solvent	Yield %
1	CuCl (30)	50 °C	5 h	PhMe	82
2	CuCl (30)	50 °C	5 h	EtOH	75
3	CuCl (30)	50 °C	8 h	EtOH	85 ^a (73 ^b)

Table S2. Optimization for A³ coupling reaction conditions with dodecanal as the substrate (^aN₂ atmosphere; ^bNMR yield).



Entry	Catalyst (mol %)	Temperature	Time	Solvent	Yield %
1	CuCl (15)	100 °C	12h	EtOH	complicated
2	Zn(OTf) ₂ (15)	100 °C	12h	EtOH	complicated
3	Zn(OTf) ₂ (15)	100 °C	12h	MeCN	complicated
4	Zn(OTf) ₂ (15)	100 °C	12h	DCE	complicated
5	Zn(OTf) ₂ (15)	100 °C	12h	TFA	complicated
6	Zn(OTf) ₂ (15)	100 °C	12h	THF	complicated
7	Zn(OTf) ₂ (15)	100 °C	12h	H ₂ O	complicated
8	Zn(OTf) ₂ (15)	100 °C	12h	PhMe	complicated
9	Zn(OTf) ₂ (15)	120 °C	12h	PhMe	complicated
10 ^a	Zn(OTf) ₂ (15)	120 °C	12h	PhMe	27
11 ^a	Zn(OTf) ₂ (15)	120 °C	36h	PhMe	65 ^b

Table S3. Physicochemical characterizations of LNPs.

LNPs	Size (nm)	PDI	Zeta (mV)	EE (%)	pKa
A11B2	164.0 ± 2.6	0.10	-0.7 ± 0.1	67.3 ± 0.8	4.54
A11B4	170.5 ± 2.2	0.10	-5.3 ± 1.3	51.5 ± 3.1	4.49
A11B7	139.5 ± 5.5	0.12	-6.2 ± 1.2	54.4 ± 1.6	4.20
A11B9	179.8 ± 1.8	0.10	-9.2 ± 0.9	51.7 ± 3.6	4.48
A28B10	146.2 ± 0.4	0.13	-12.5 ± 1.7	80.9 ± 1.4	4.43
R-A11B2	148.5 ± 0.2	0.09	-8.9 ± 2.2	64.5 ± 1.8	5.62
R-A11B4	161.1 ± 1.6	0.09	-9.9 ± 2.7	64.1 ± 3.3	5.65
R-A11B7	161.8 ± 1.2	0.15	-10.4 ± 0.6	64.5 ± 0.6	4.83
R-A11B9	188.2 ± 1.2	0.14	-11.7 ± 2.5	48.5 ± 2.2	5.11
R-A28B10	158.0 ± 0.6	0.07	-14.6 ± 2.9	45.4 ± 2.5	5.11

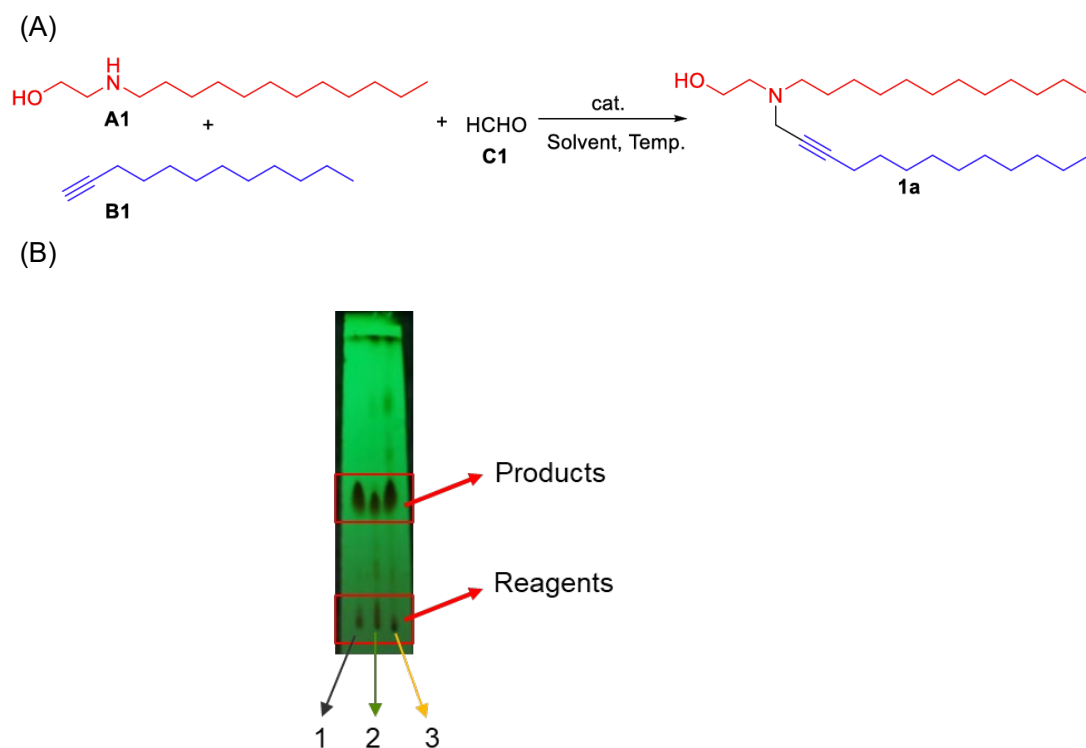
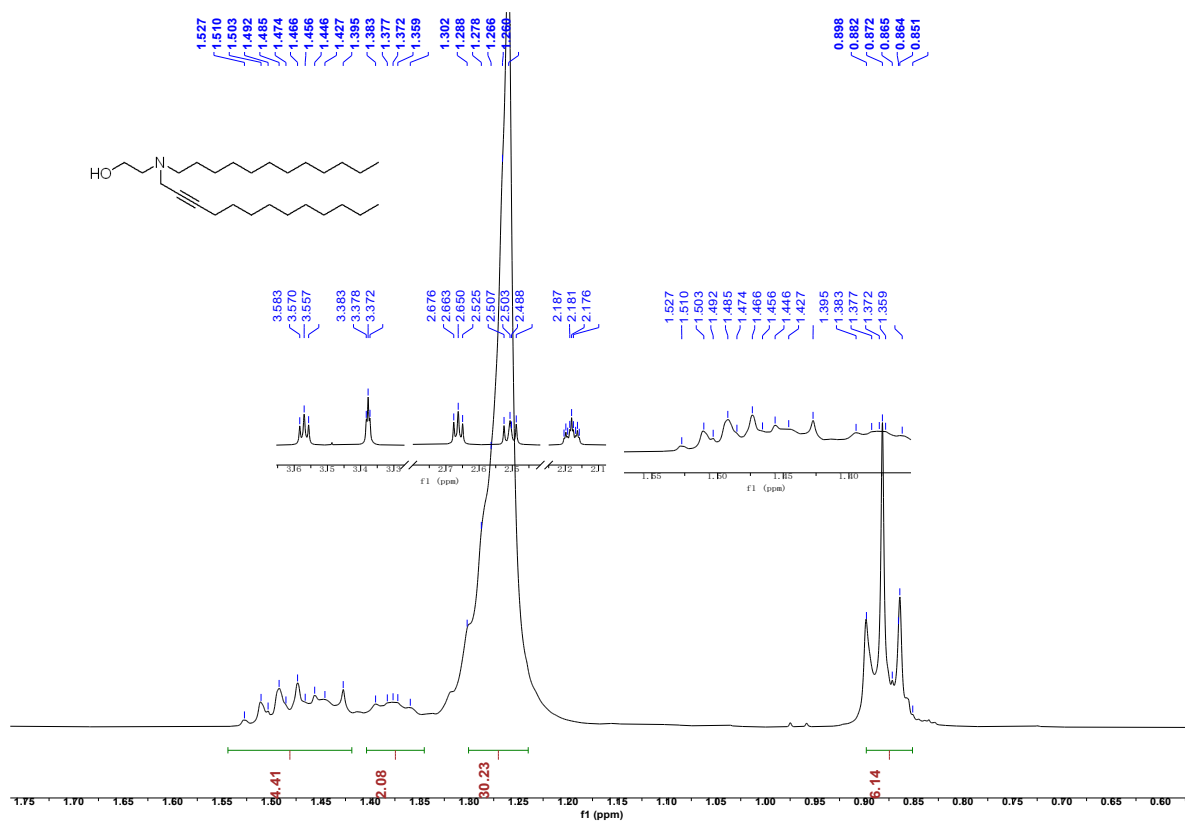
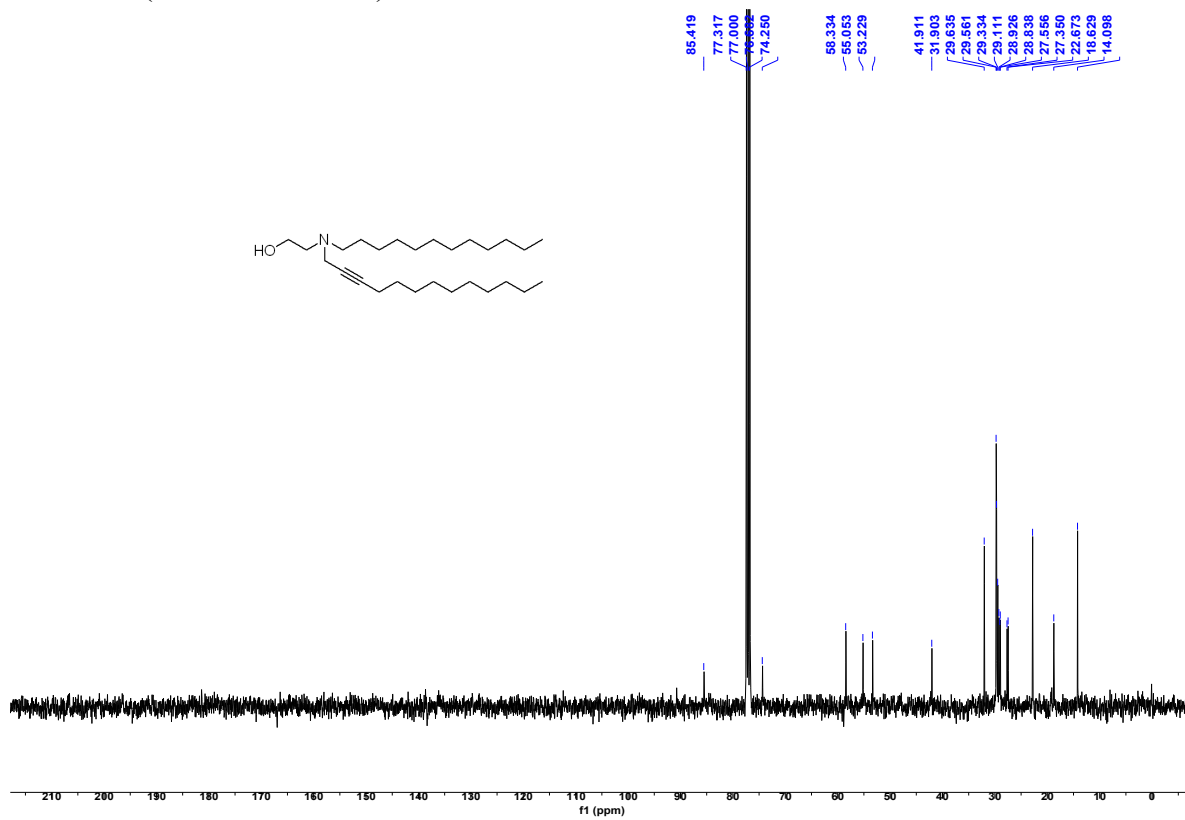


Figure S1. (A) A³ coupling reaction using formaldehyde. (B) Thin layer chromatography (TLC) analysis of each reaction condition listed in Table S1.

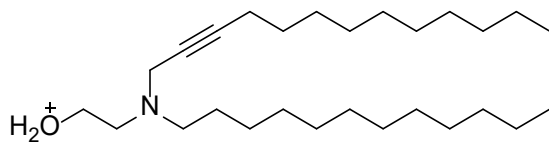
¹H NMR (400 MHz, CDCl₃)



¹³C NMR (101 MHz, CDCl₃)



HRMS (ESI-positive)



Chemical Formula: $C_{27}H_{54}NO^+$

m/z: 408.4200 (100.0%), 409.4234 (29.2%), 410.4268 (4.1%)

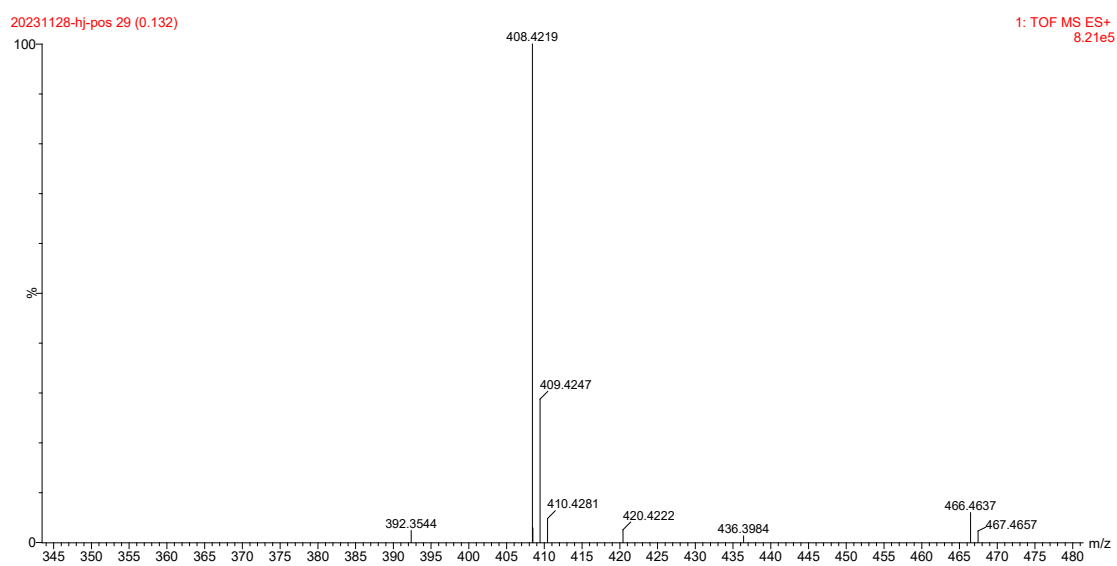
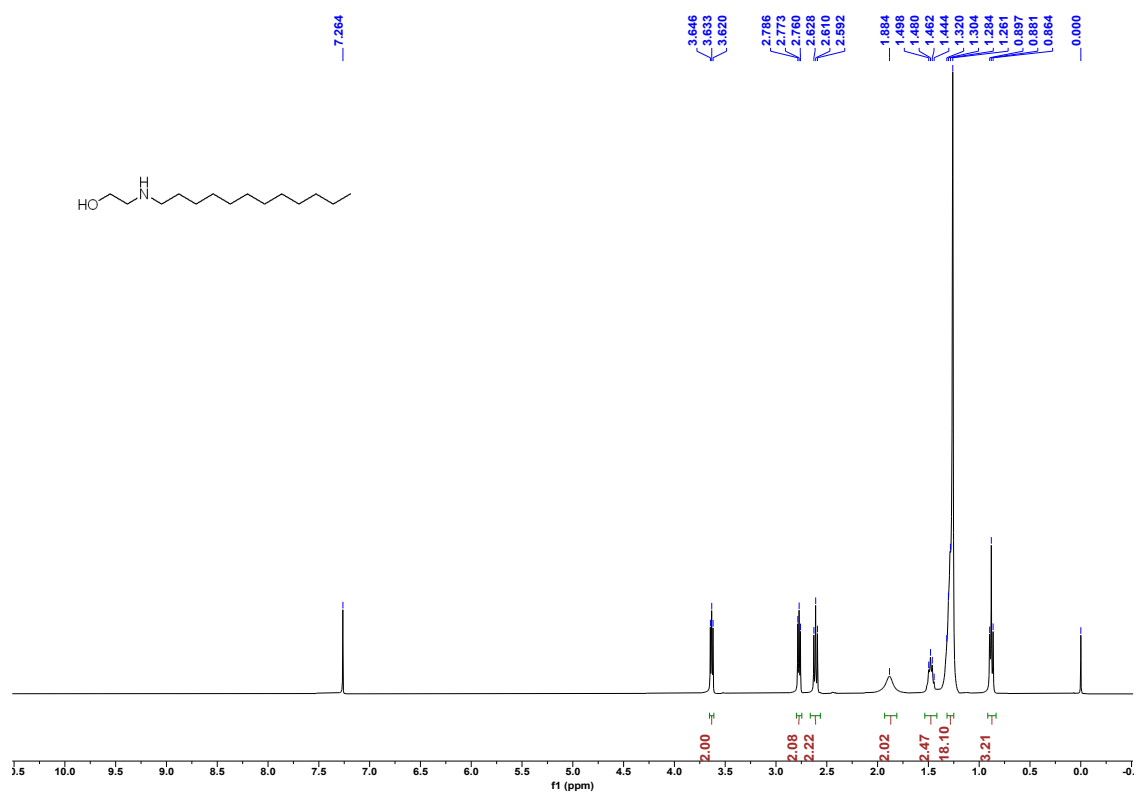


Figure S2. 1H , ^{13}C NMR spectra and HRMS of 2-(Dodecyl(tridec-2-yn-1-yl)amino)ethan-1-ol (*1a*)



¹³C NMR (101 MHz, CDCl₃)

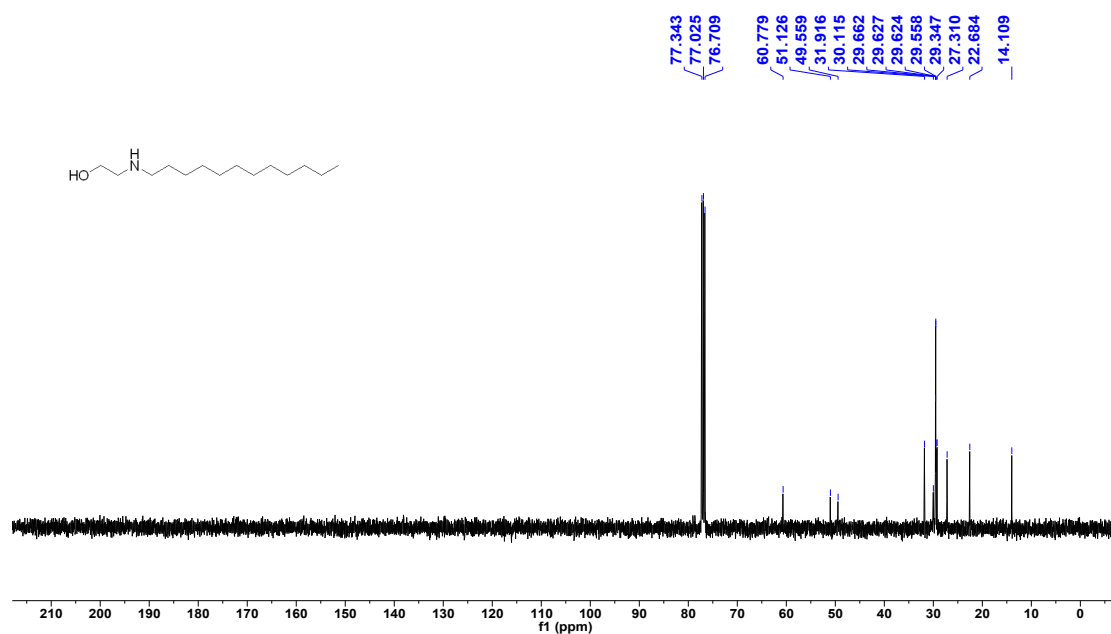
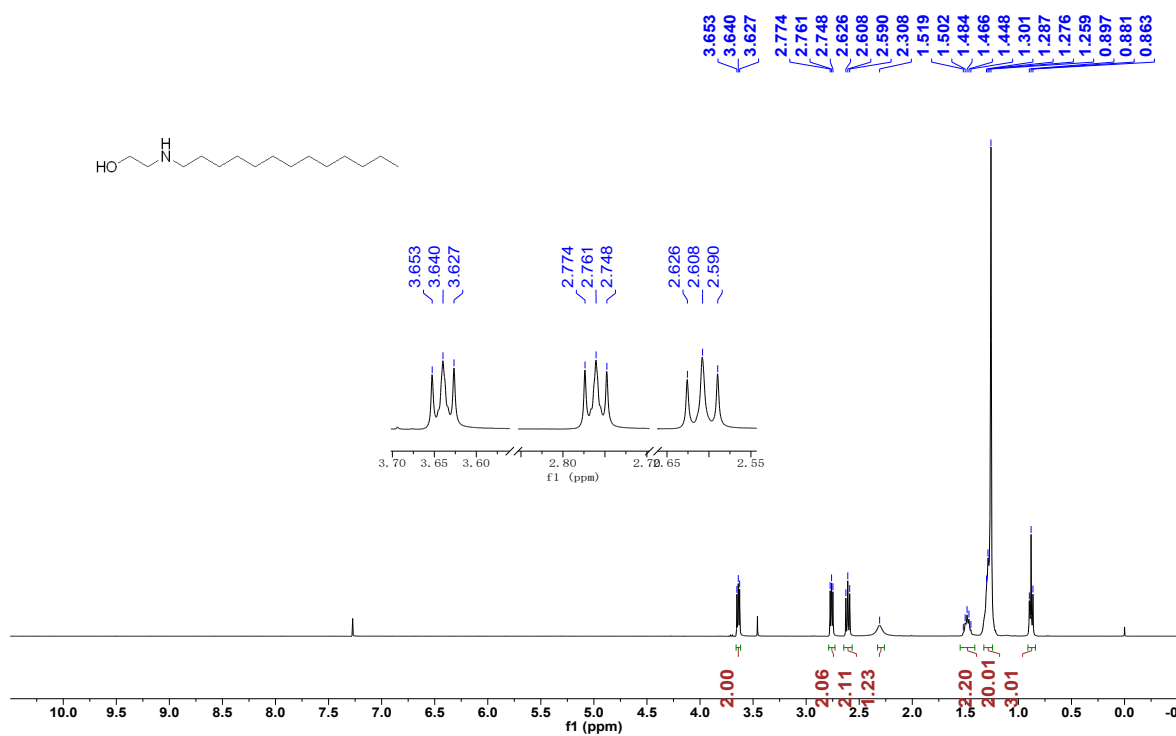


Figure S3. ¹H and ¹³C NMR spectra of 2-(Dodecylamino)ethan-1-ol (*AI*)

^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)

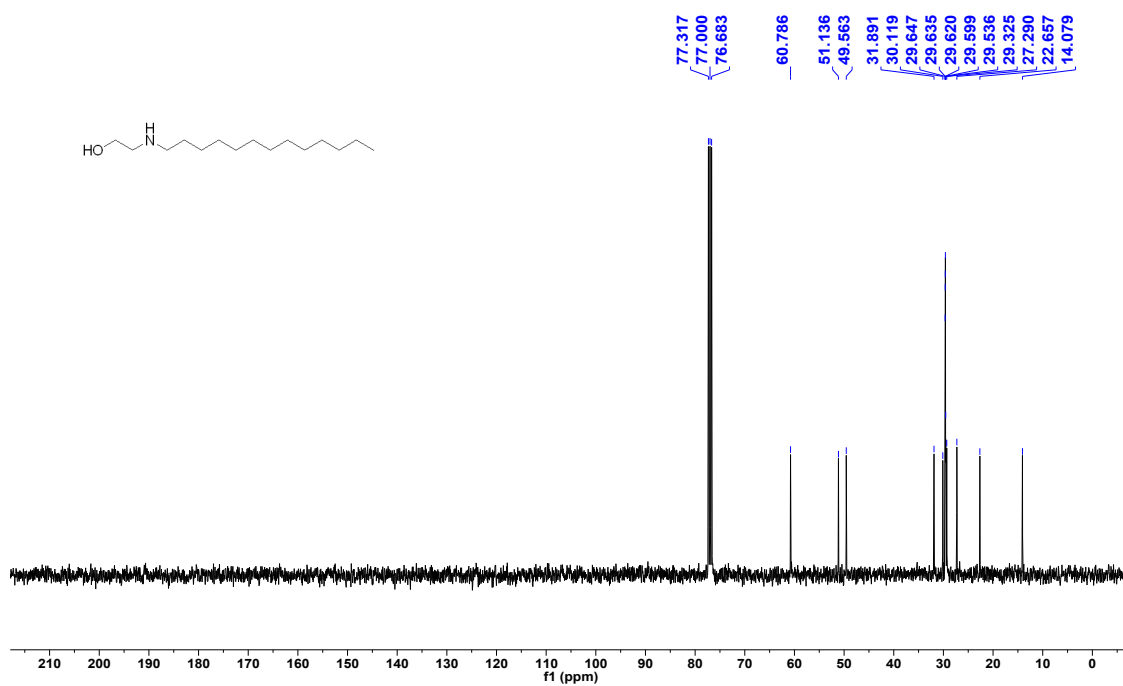
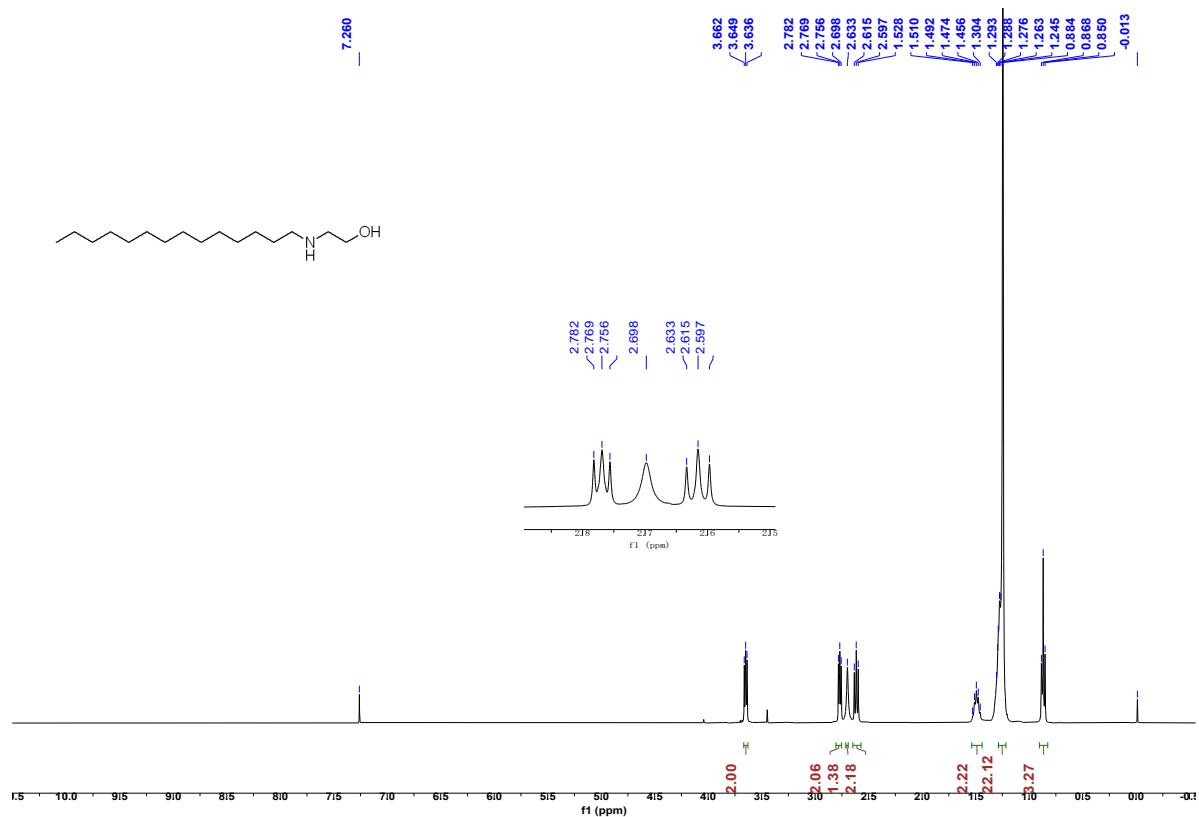


Figure S4. ^1H and ^{13}C NMR spectra of 2-(Tridecylamino)ethan-1-ol (A2)

^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)

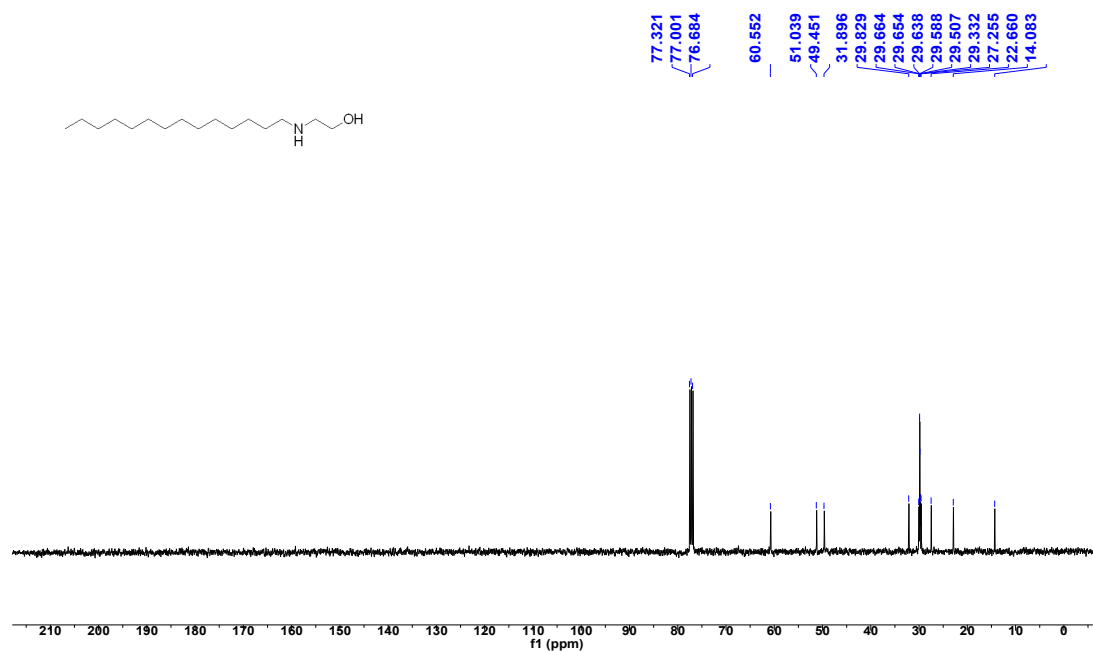
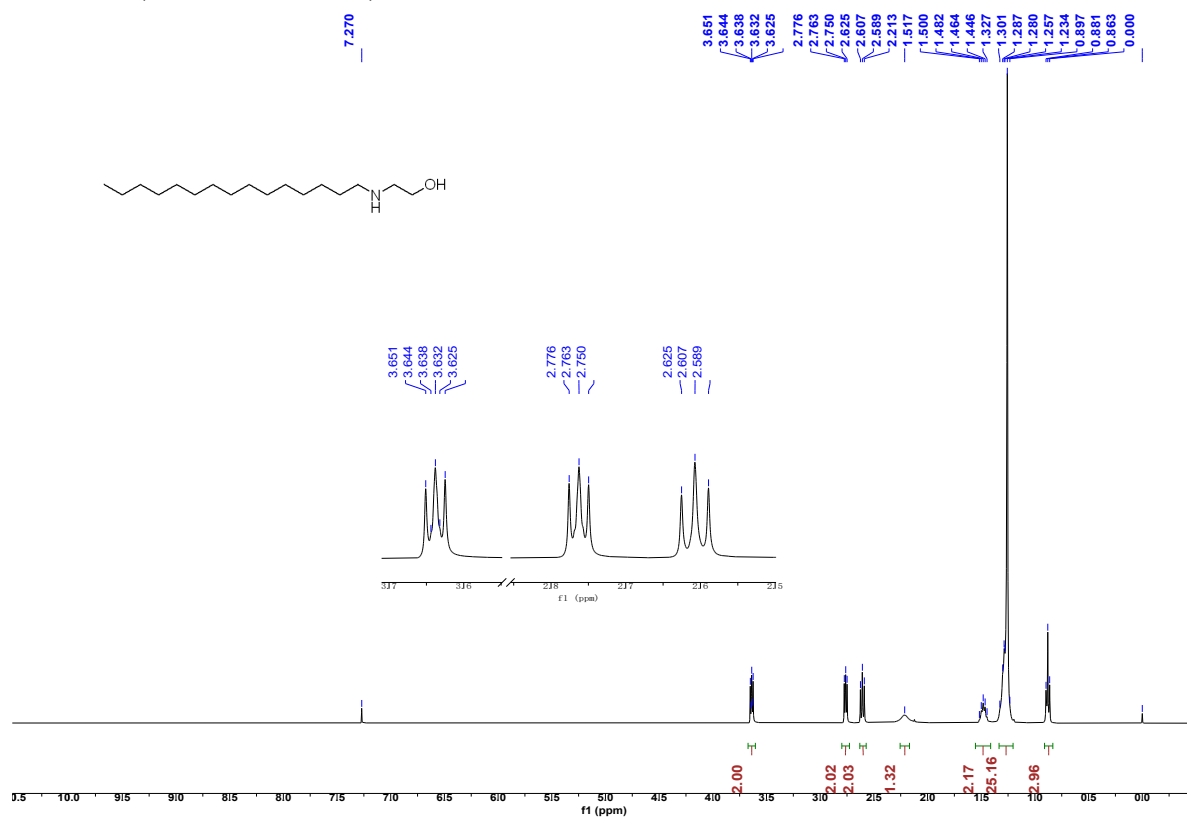


Figure S5. ^1H and ^{13}C NMR spectra of 2-(Tetradecylamino)ethan-1-ol (*A3*)

^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)

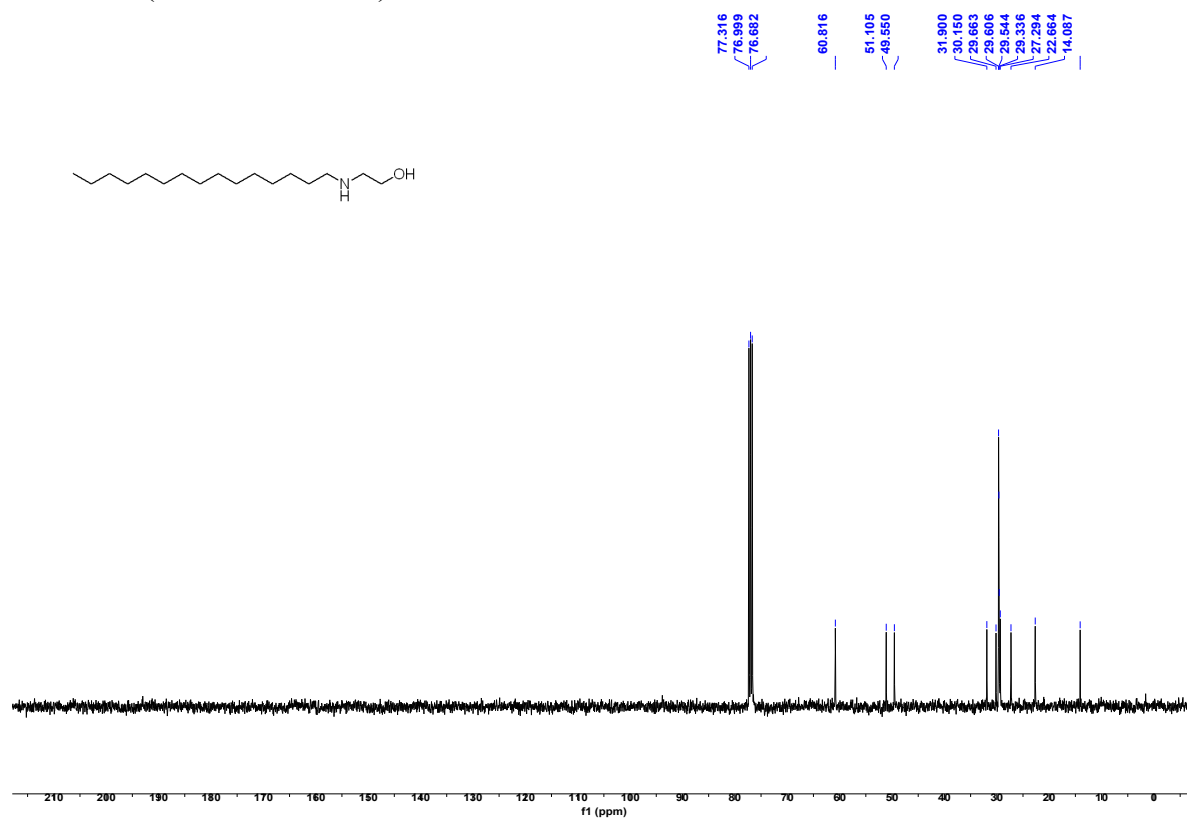
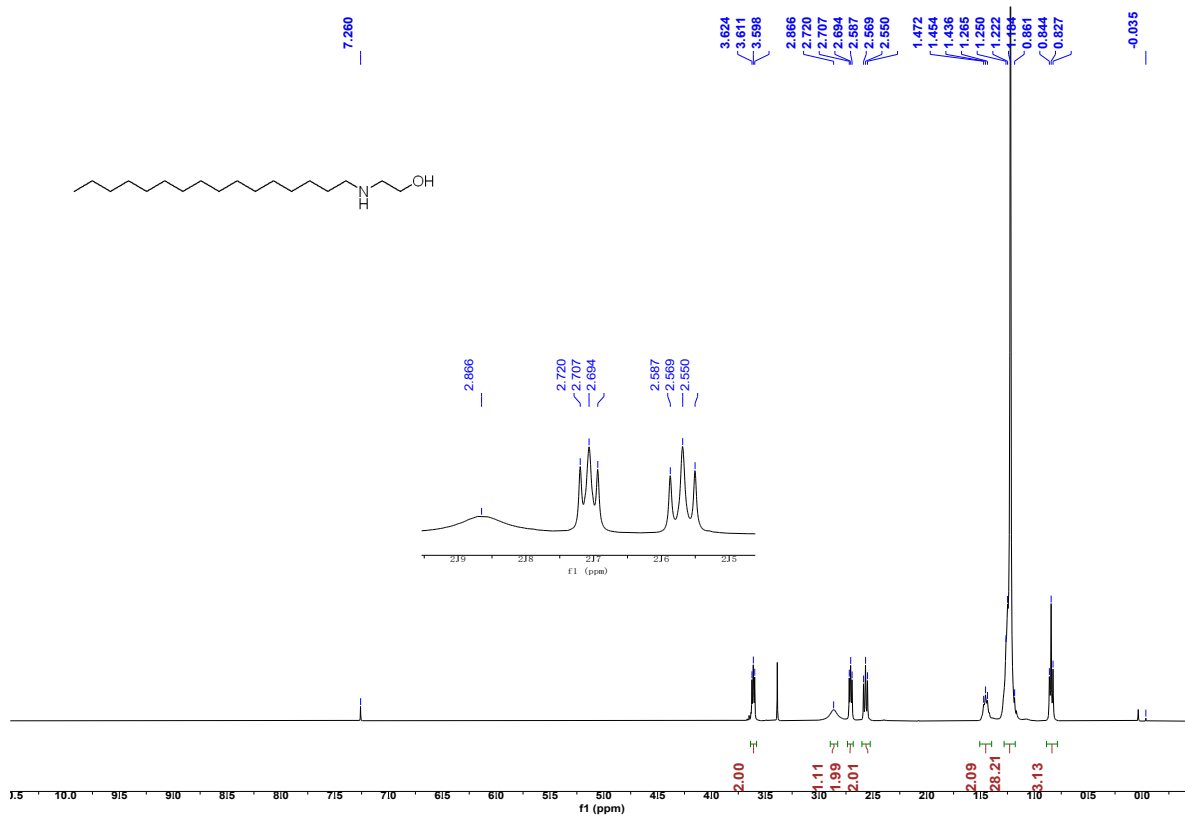


Figure S6. ^1H and ^{13}C NMR spectra of 2-(Pentadecylamino)ethan-1-ol (A4)

^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)

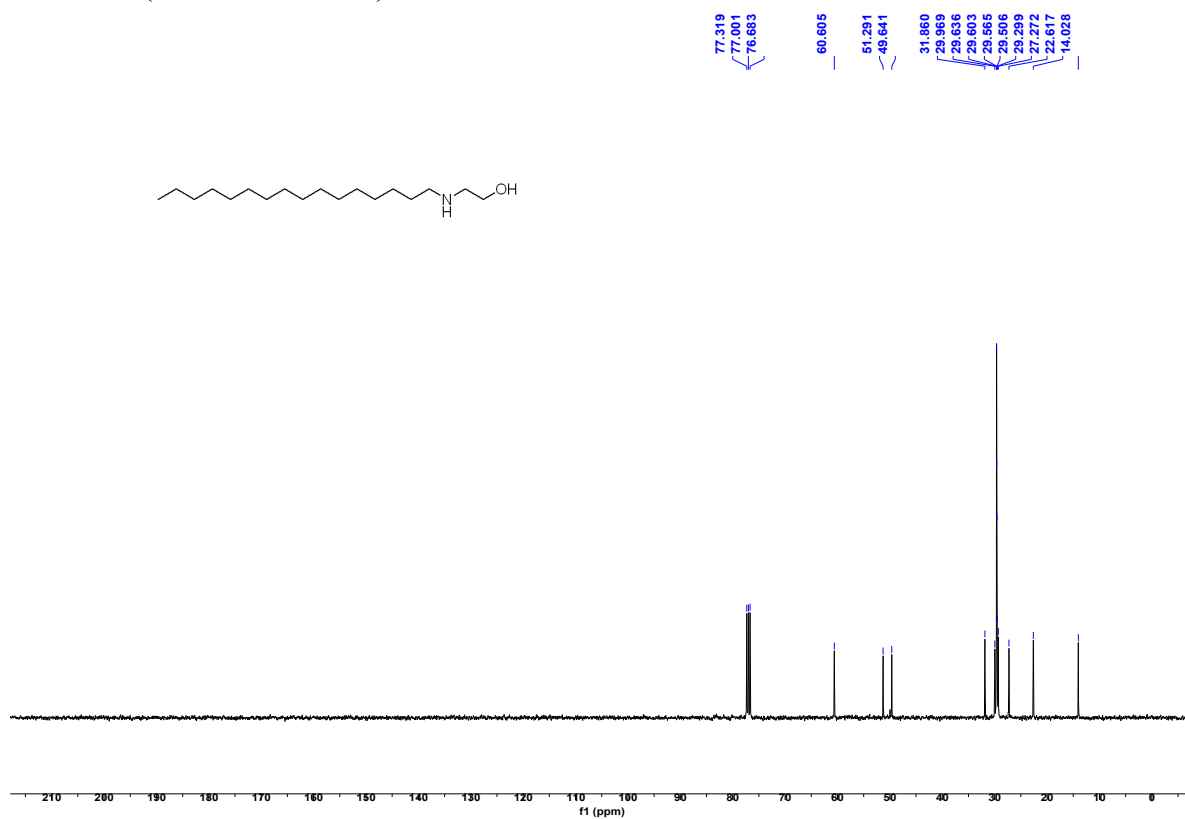
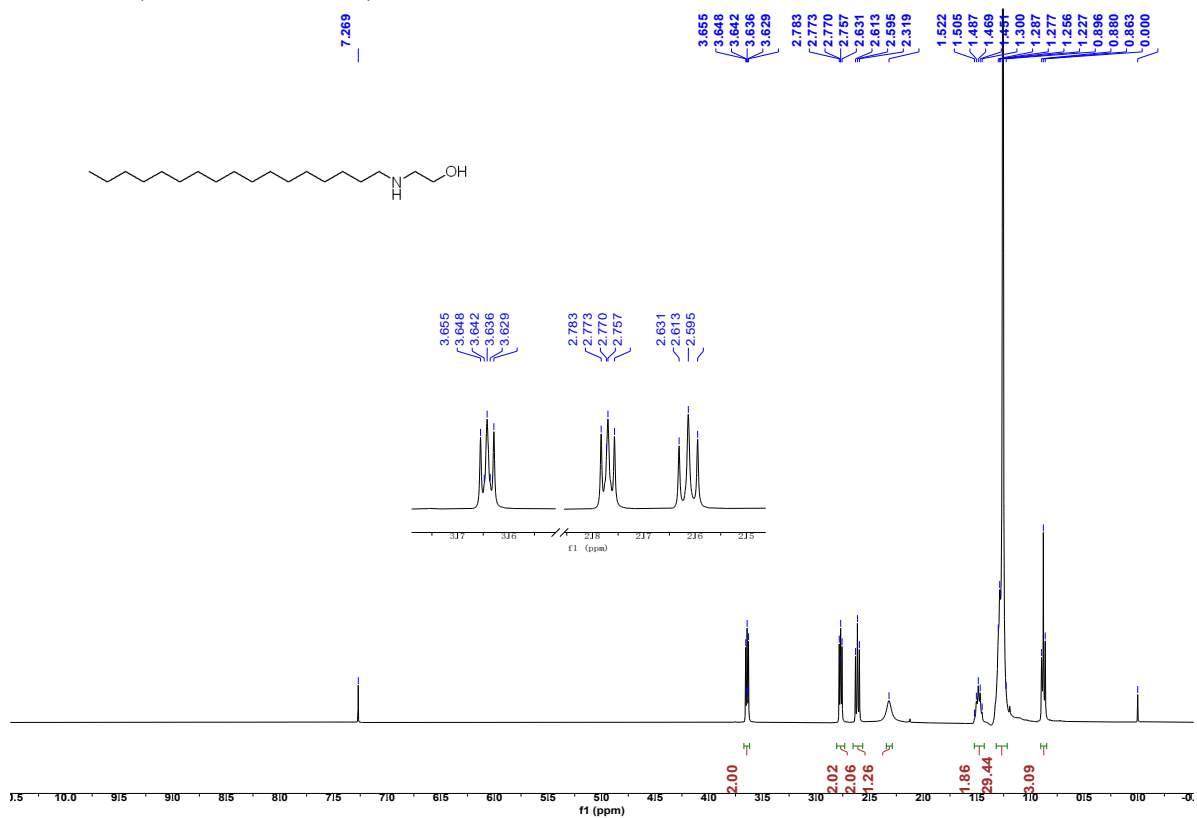


Figure S7. ^1H and ^{13}C NMR spectra of 2-(Hexadecylamino)ethan-1-ol (*A5*)

^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)

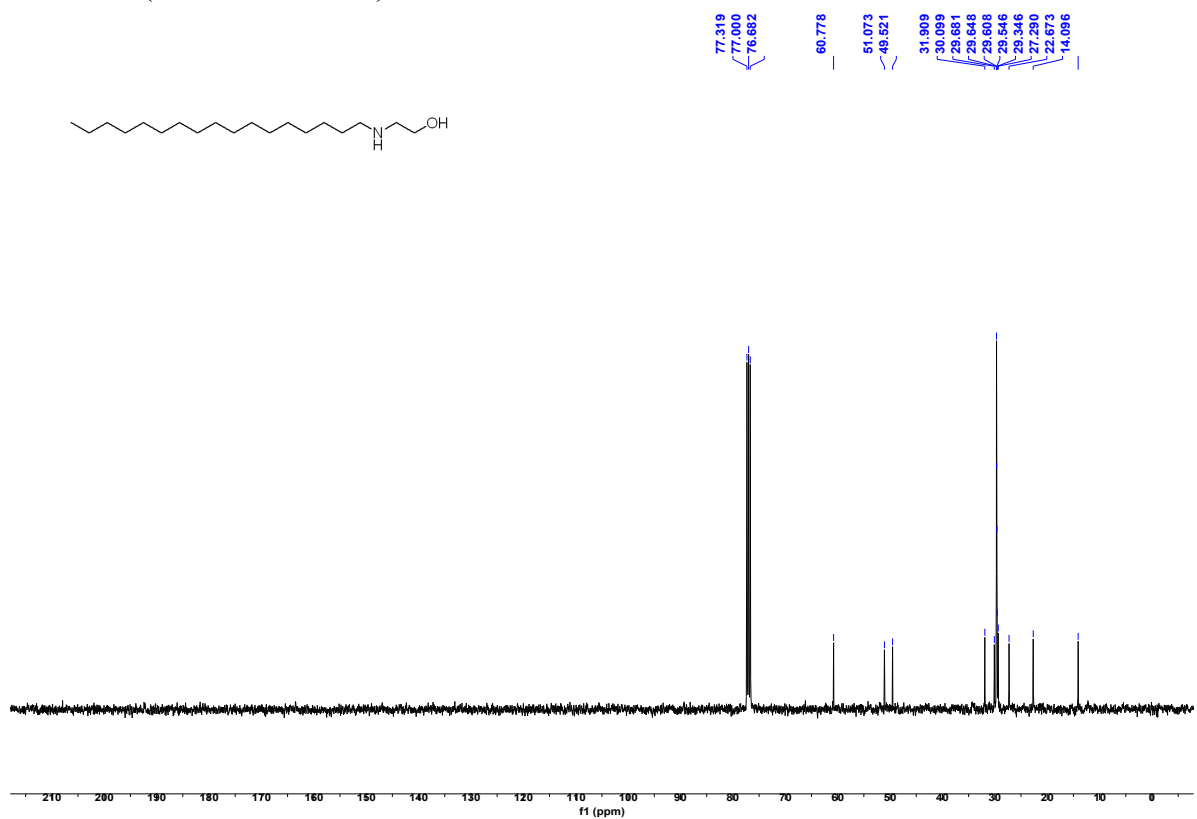
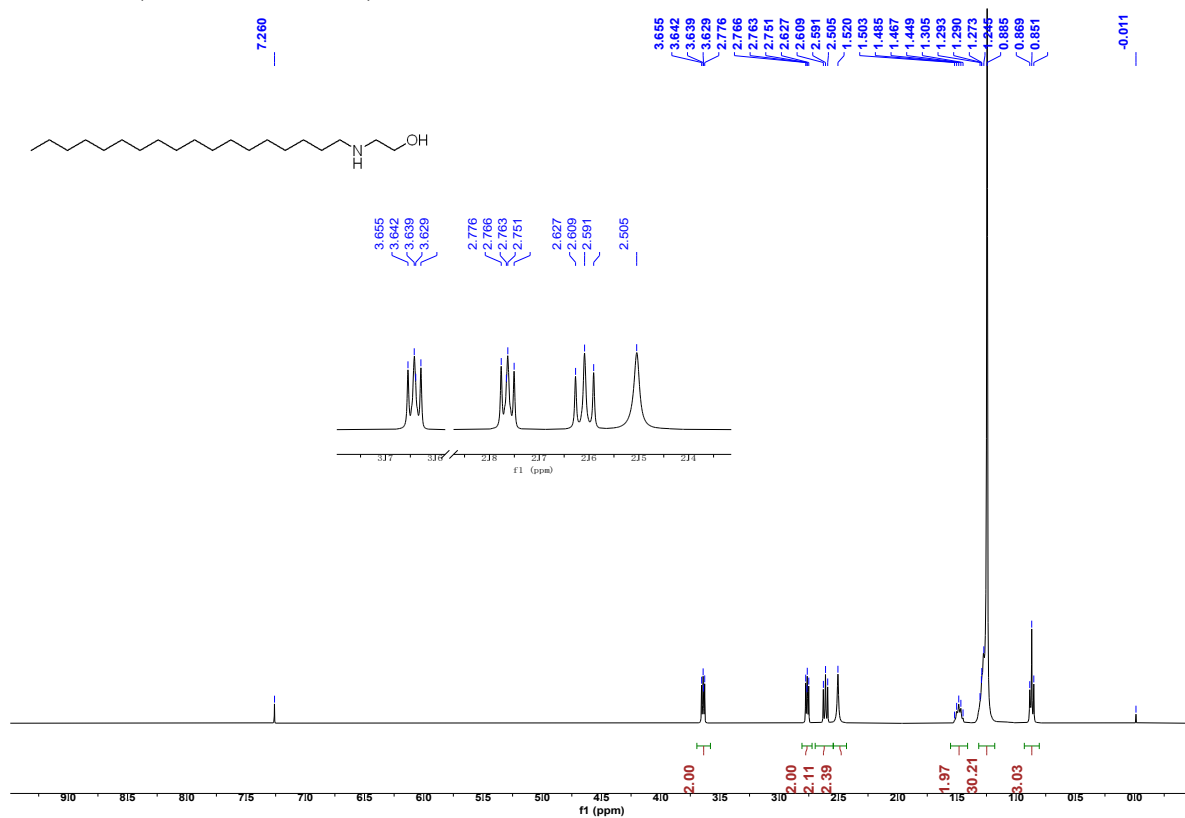


Figure S8. ^1H and ^{13}C NMR spectra of 2-(Heptadecylamino)ethan-1-ol (A6)

^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)

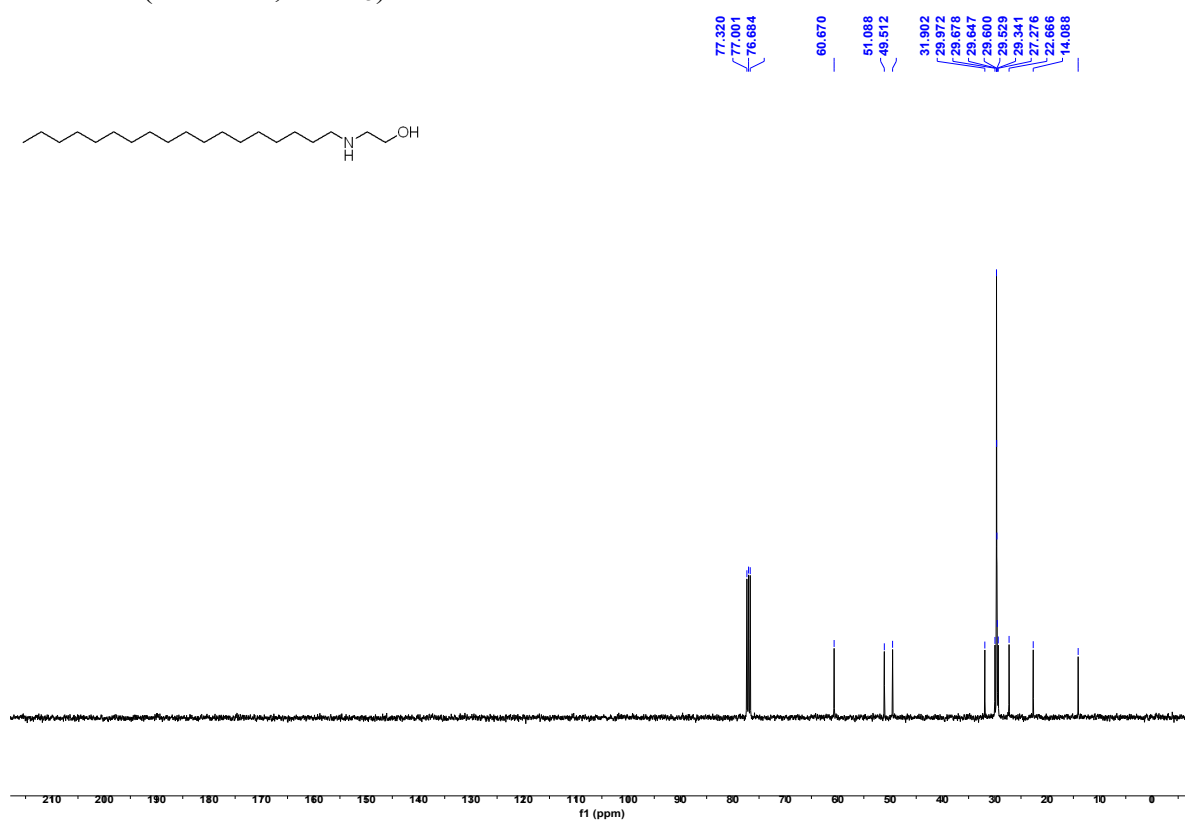
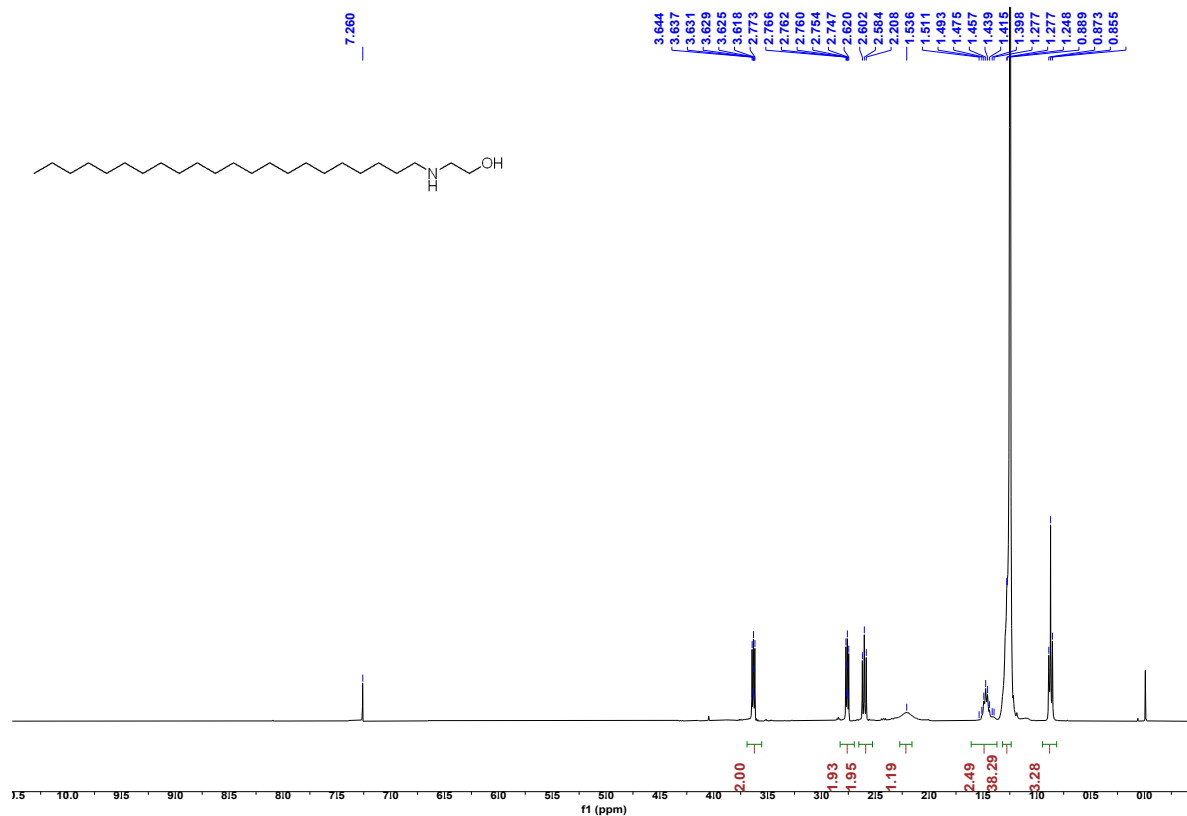
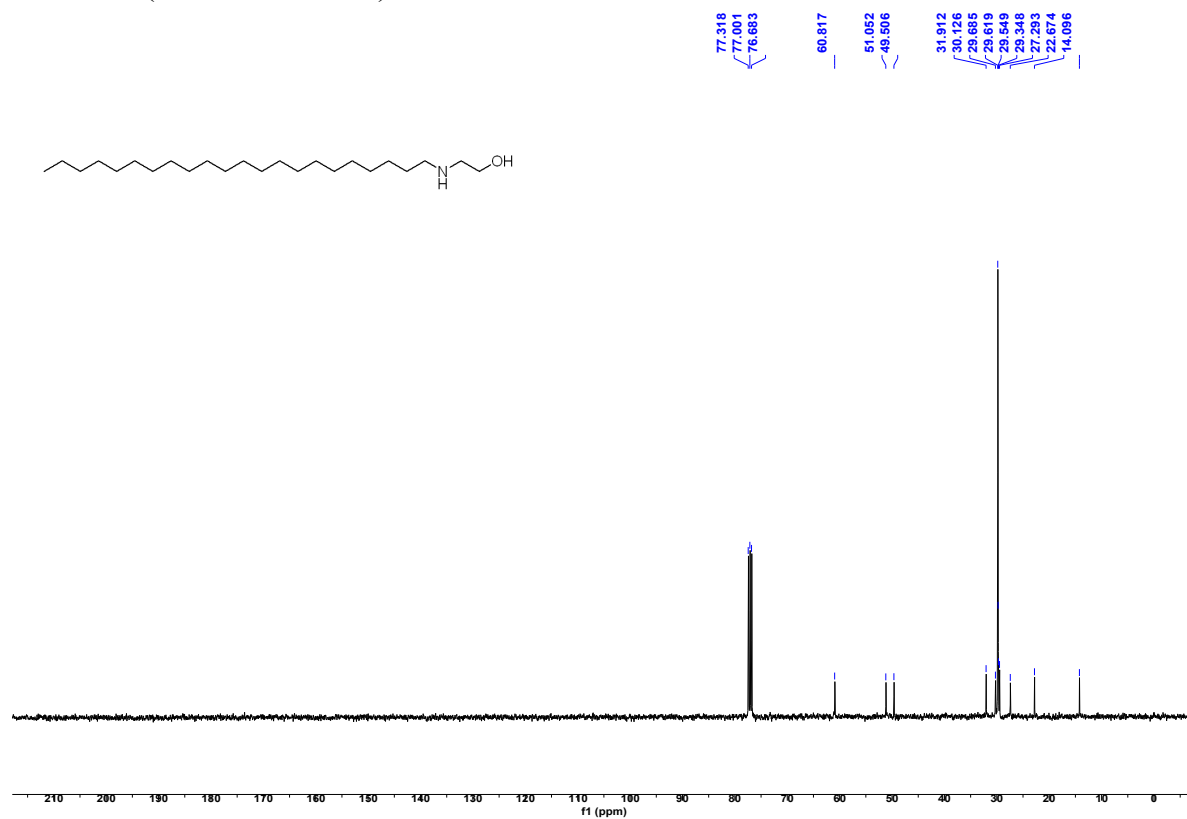


Figure S9. ^1H and ^{13}C NMR spectra of 2-(Nonadecylamino)ethan-1-ol (*A7*)

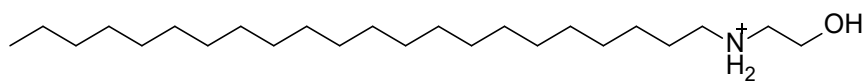
^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)



HRMS (ESI-positive)



Chemical Formula: $C_{24}H_{52}NO^+$

m/z: 370.4044 (100.0%), 371.4077 (26.0%), 372.4111 (3.2%)

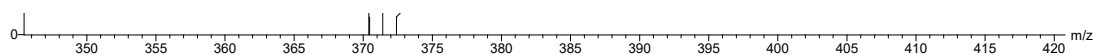
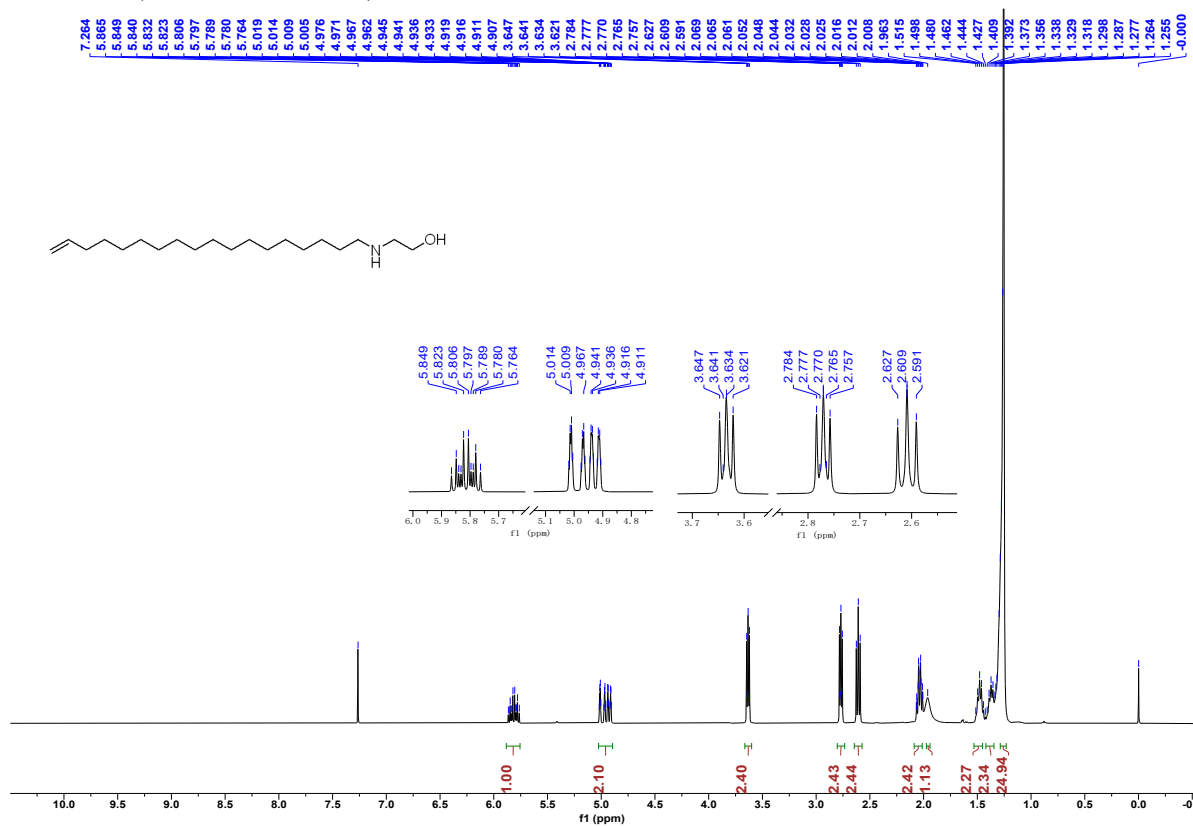
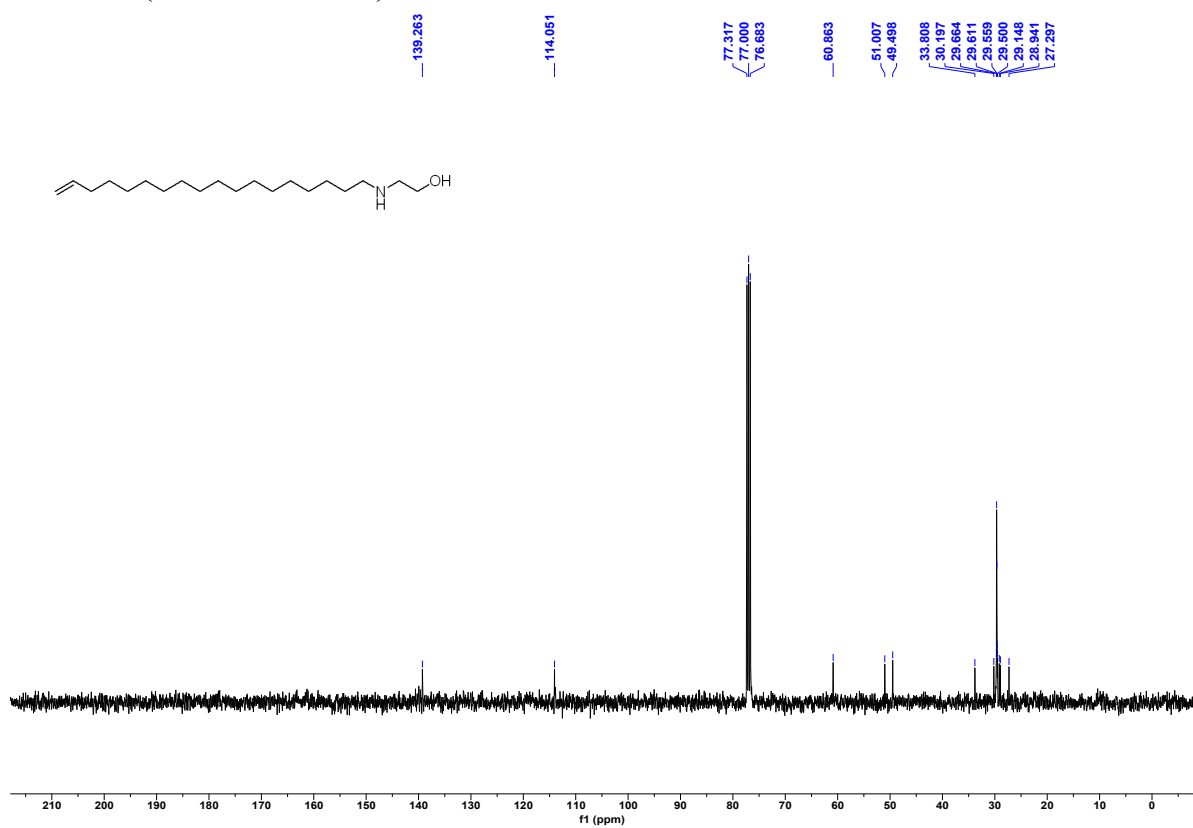


Figure S10. 1H , ^{13}C NMR spectra and HRMS of 2-(Docosylamino)ethan-1-ol (*A8*)

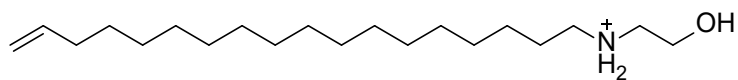
¹H NMR (400 MHz, CDCl₃)



¹³C NMR (101 MHz, CDCl₃)



HRMS (ESI-positive)



Chemical Formula: $\text{C}_{20}\text{H}_{42}\text{NO}^+$

m/z: 312.3261 (100.0%), 313.3295 (21.6%), 314.3328 (2.2%)

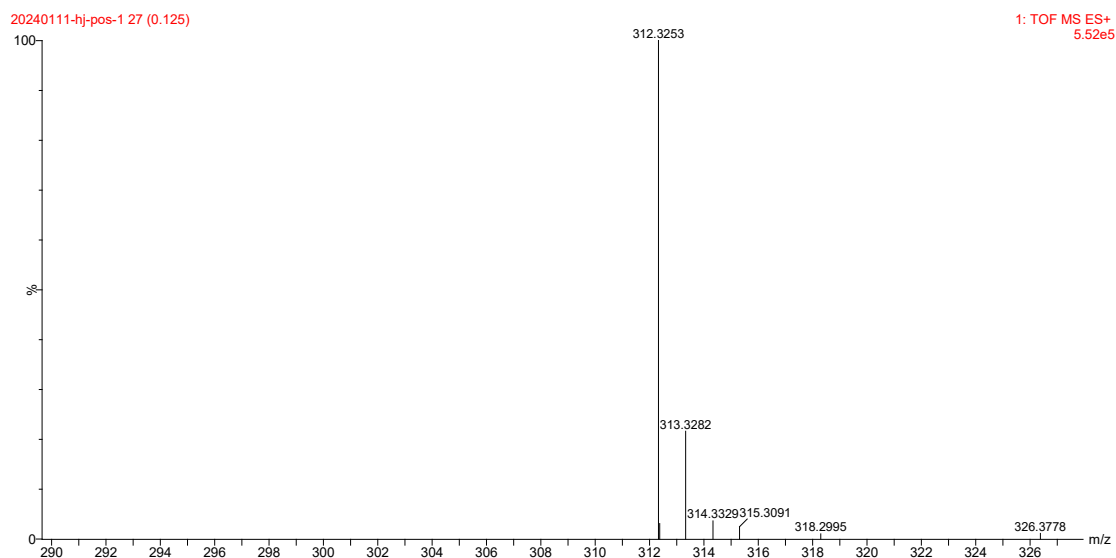
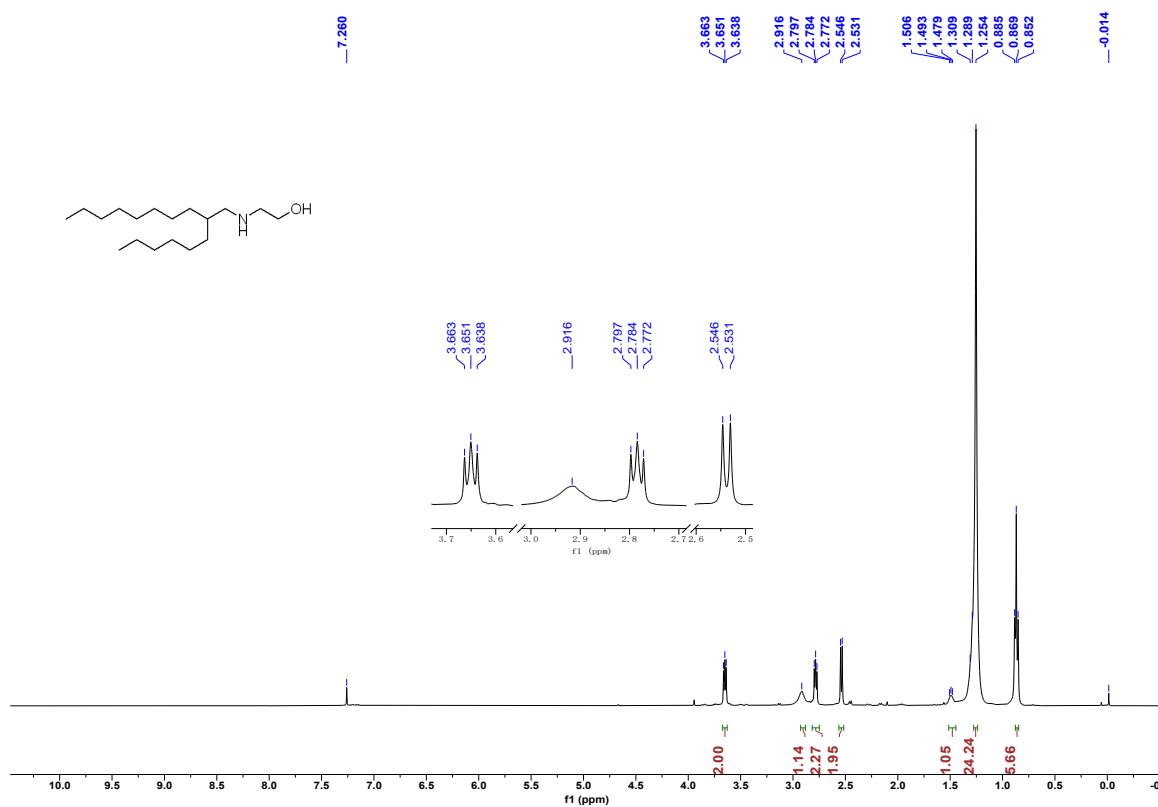
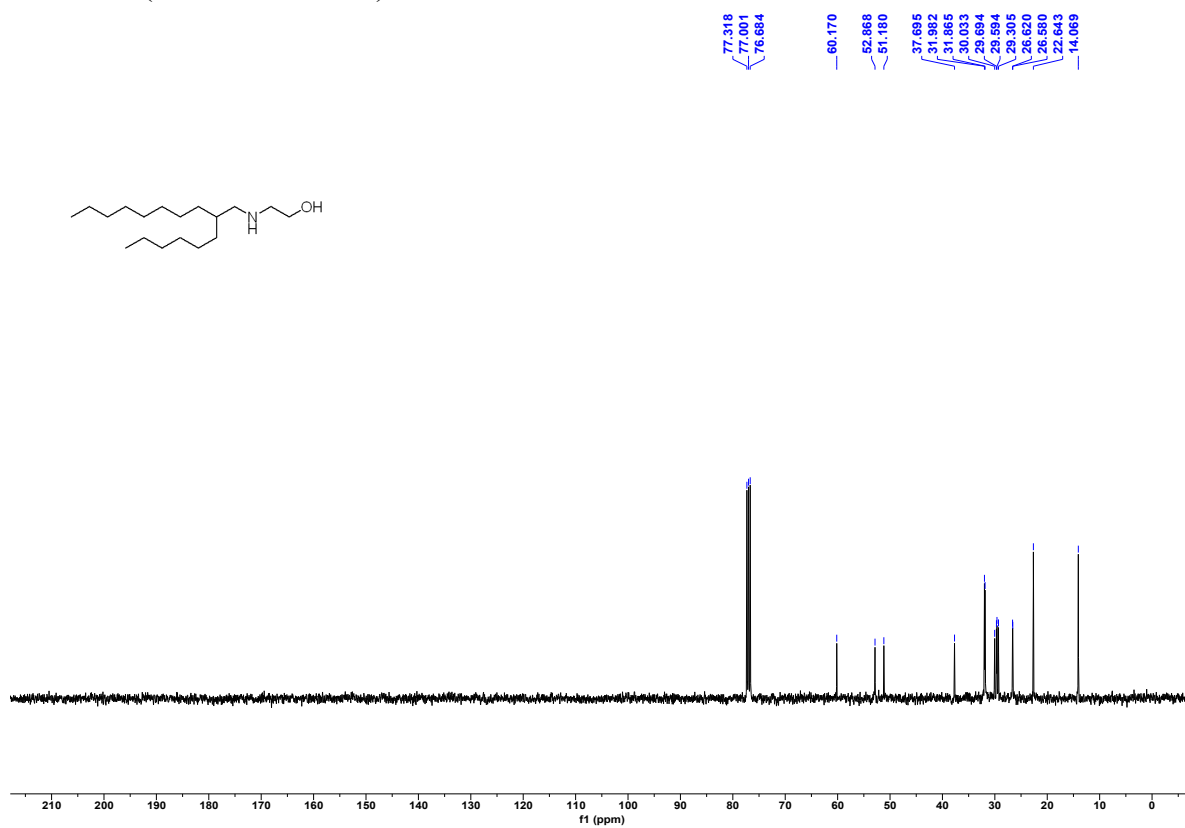


Figure S11. ^1H , ^{13}C NMR spectra and HRMS of 2-(Octadec-17-en-1-ylamino)ethan-1-ol (*49*)

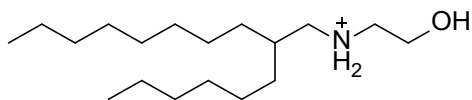
^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)



HRMS (ESI-positive)



Chemical Formula: $\text{C}_{18}\text{H}_{40}\text{NO}^+$

m/z : 286.3105 (100.0%), 287.3138 (19.5%), 288.3172 (1.8%)

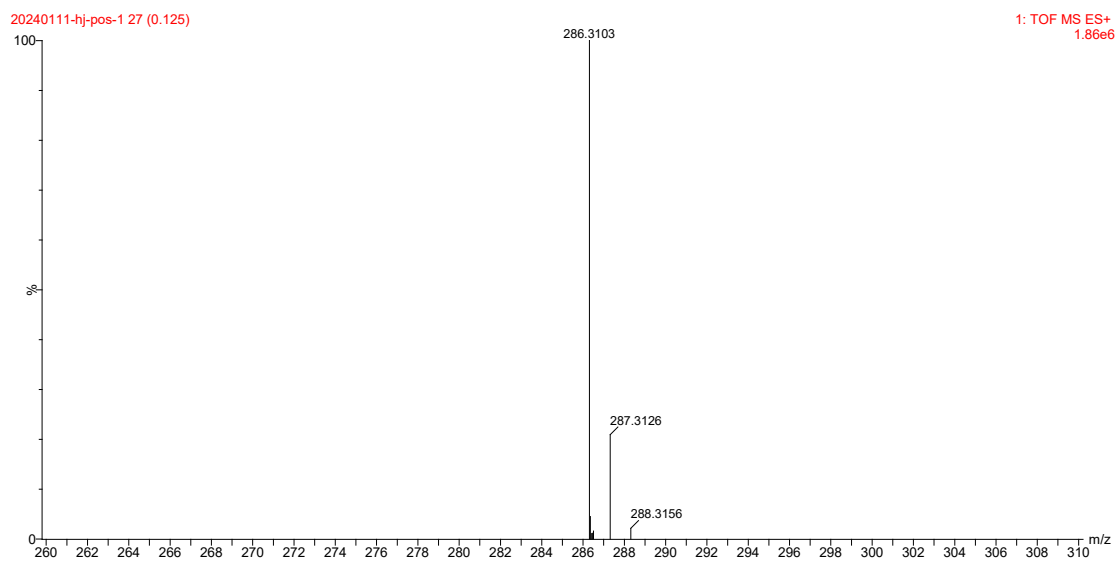
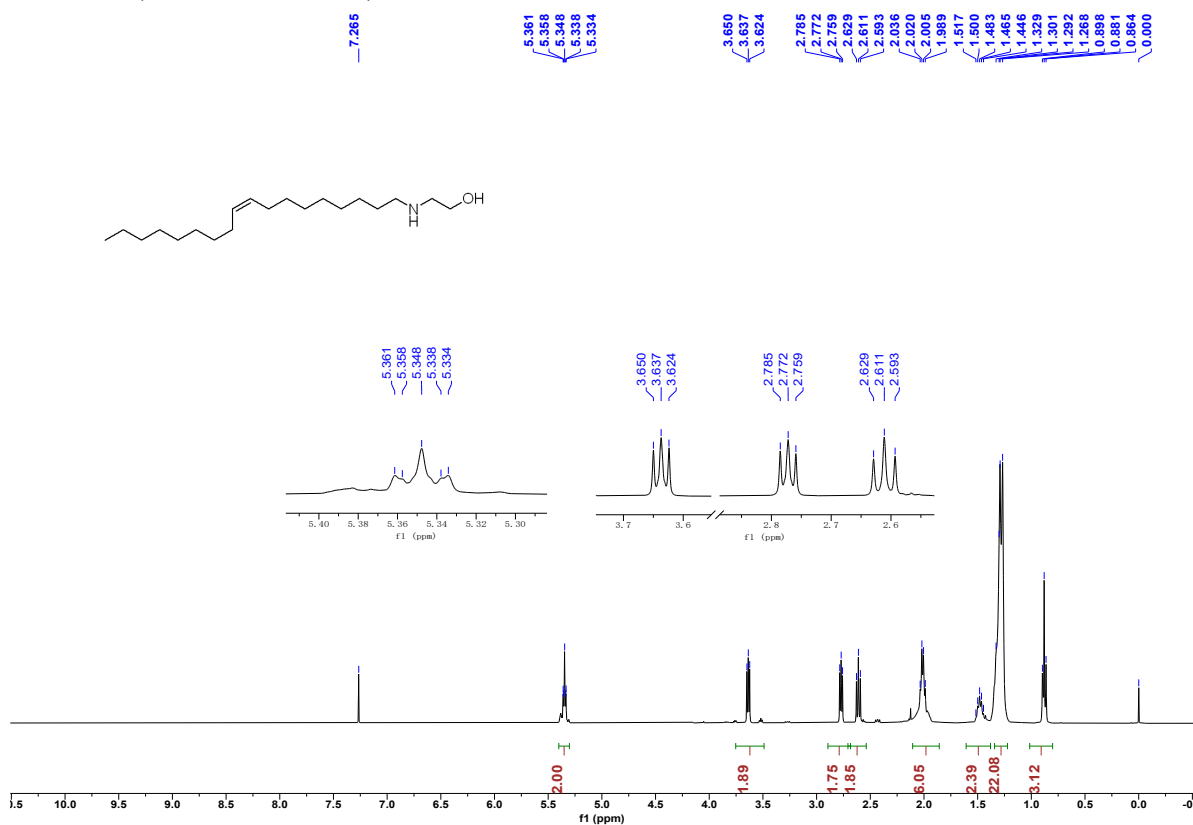


Figure S12. ^1H , ^{13}C NMR spectra and HRMS of 2-((2-Hexyldecyl)amino)ethan-1-ol (*110*)

^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)

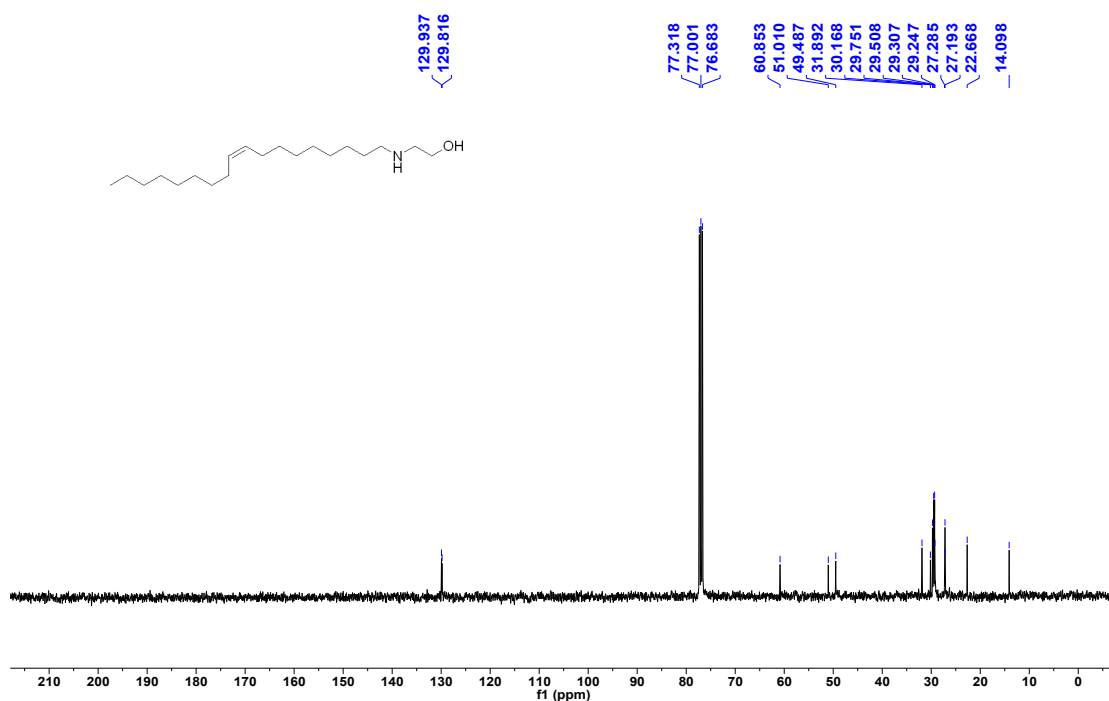
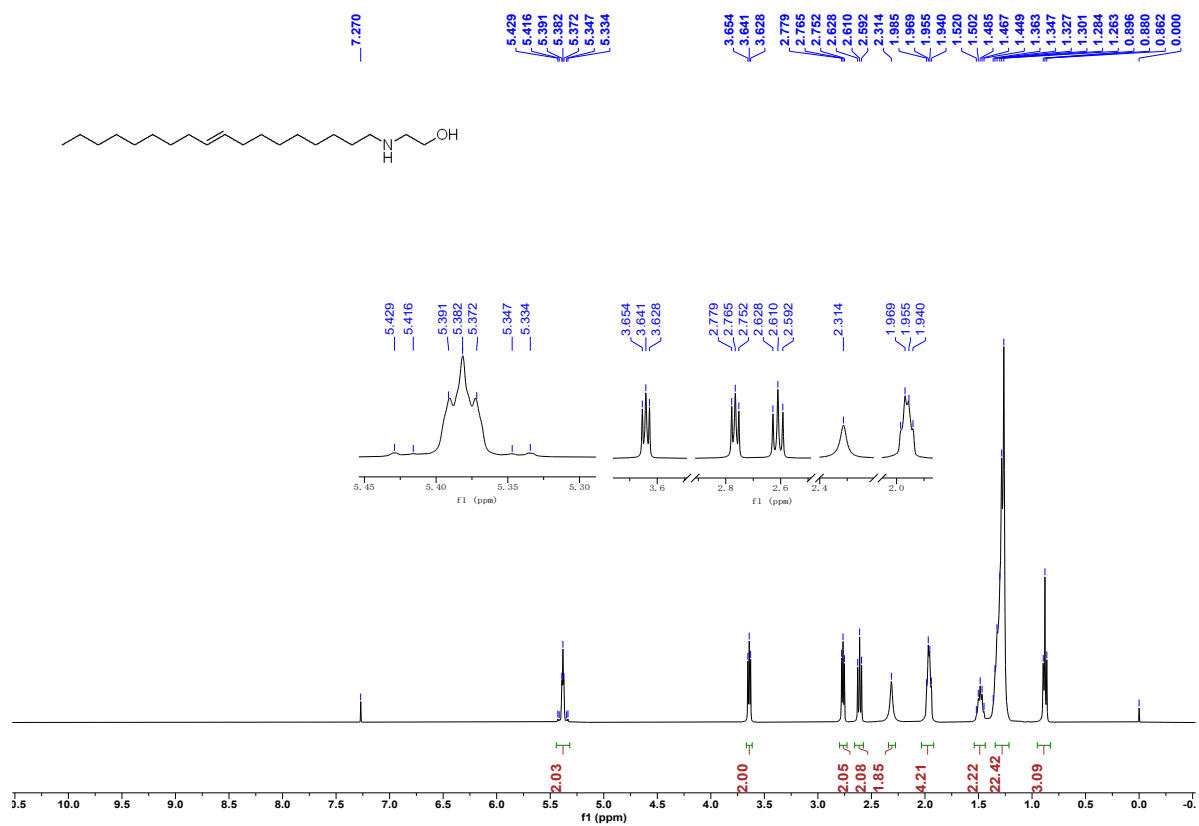
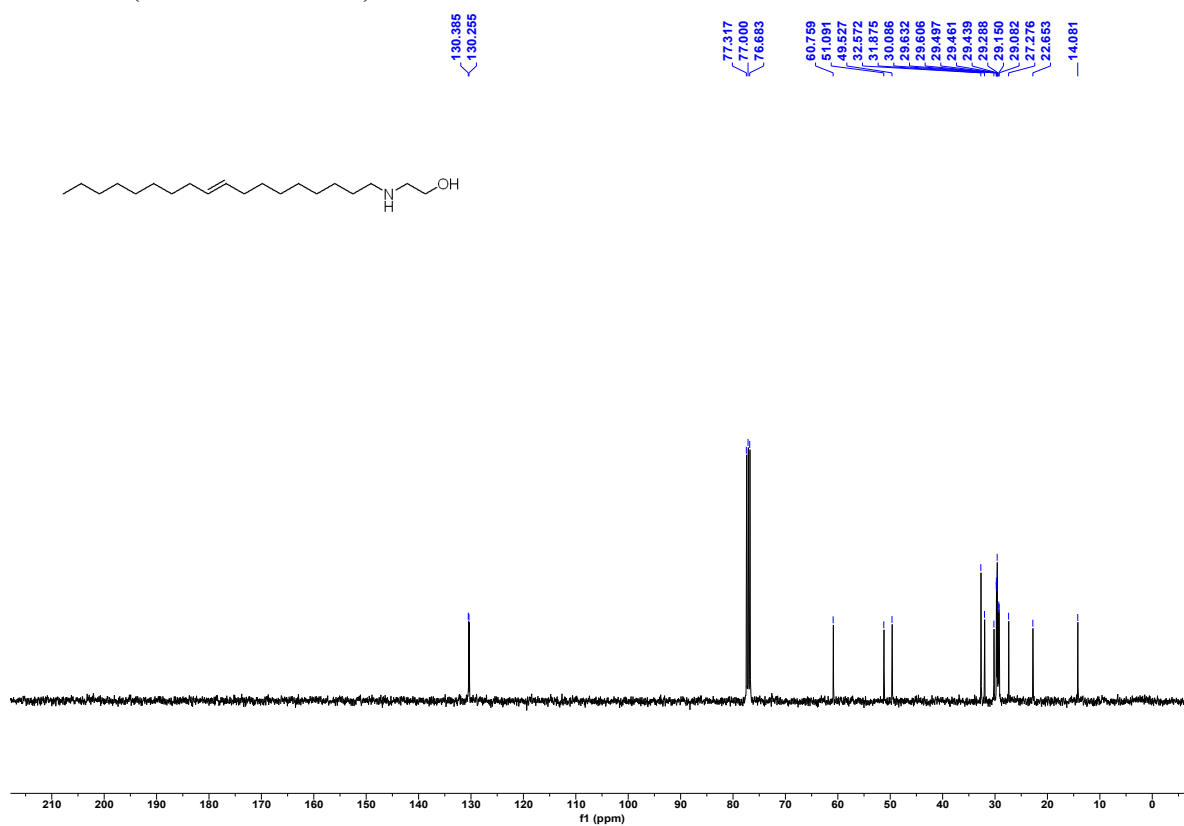


Figure S13. ^1H and ^{13}C NMR spectra of (Z)-2-(octadec-9-en-1-ylamino)ethan-1-ol (*All*)

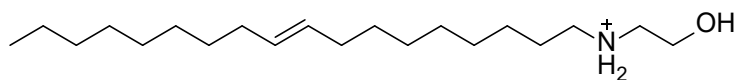
^1H NMR (400 MHz, CDCl_3)



¹³C NMR (101 MHz, CDCl₃)



HRMS (ESI-positive)



Chemical Formula: $\text{C}_{20}\text{H}_{42}\text{NO}^+$

m/z: 312.3261 (100.0%), 313.3295 (21.6%), 314.3328 (2.2%)

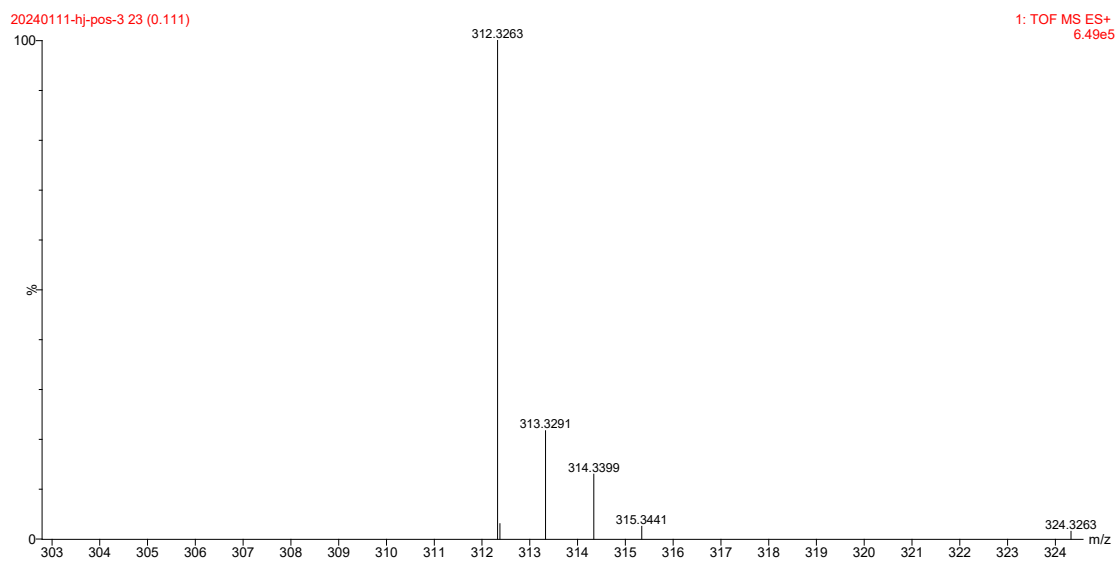


Figure S14. ^1H , ^{13}C NMR spectra and HRMS of (*E*)-2-(octadec-9-en-1-ylamino)ethan-1-ol (A12)

[illegible]CCCCC/C=C/C=C\CCCCCNCO

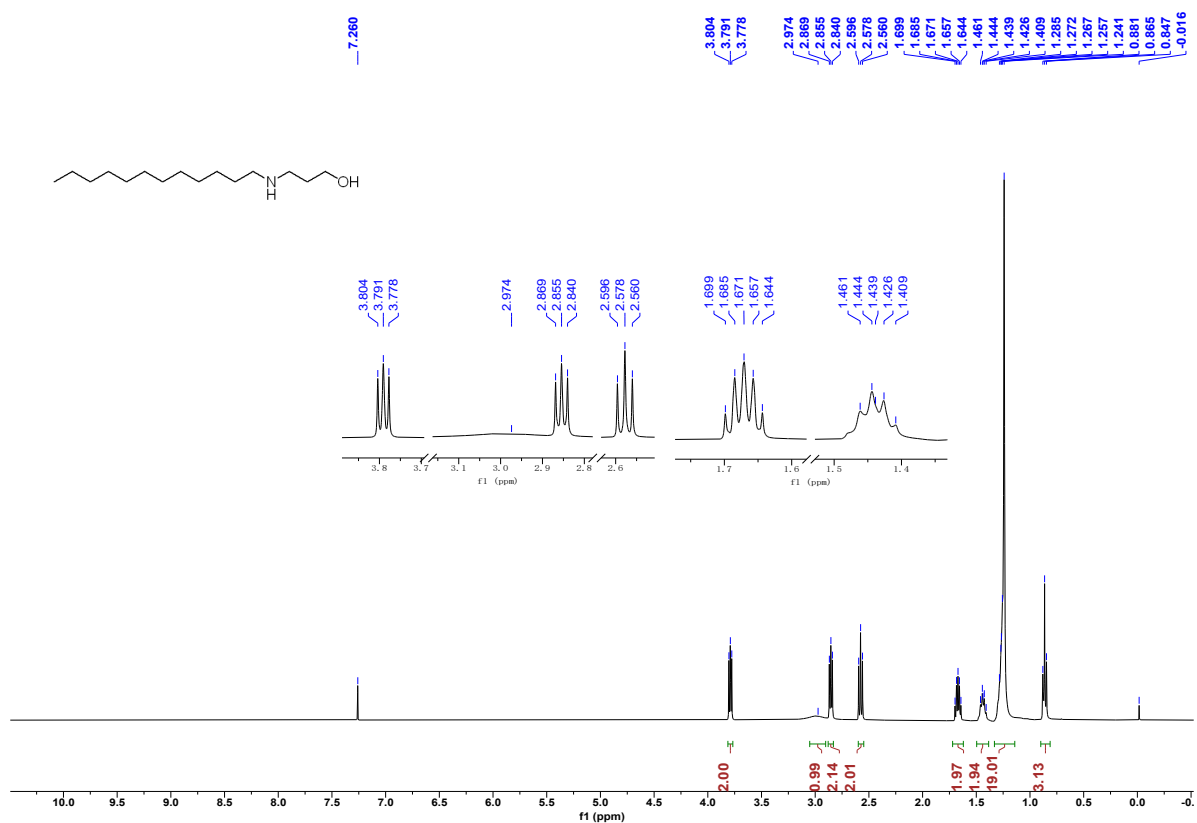
Chemical structure: CCCCC/C=C/C=C\CCCCCNCO

¹³C NMR peaks (ppm):

- 130.130
- 130.044
- 127.834
- 127.869
- 77.318
- 77.000
- 76.683
- 60.664
- 51.152
- 49.525
- 49.473
- 29.996
- 29.608
- 29.463
- 29.295
- 29.212
- 27.258
- 27.169
- 27.145
- 26.975
- 22.519
- 14.015

S43 / S110

^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)

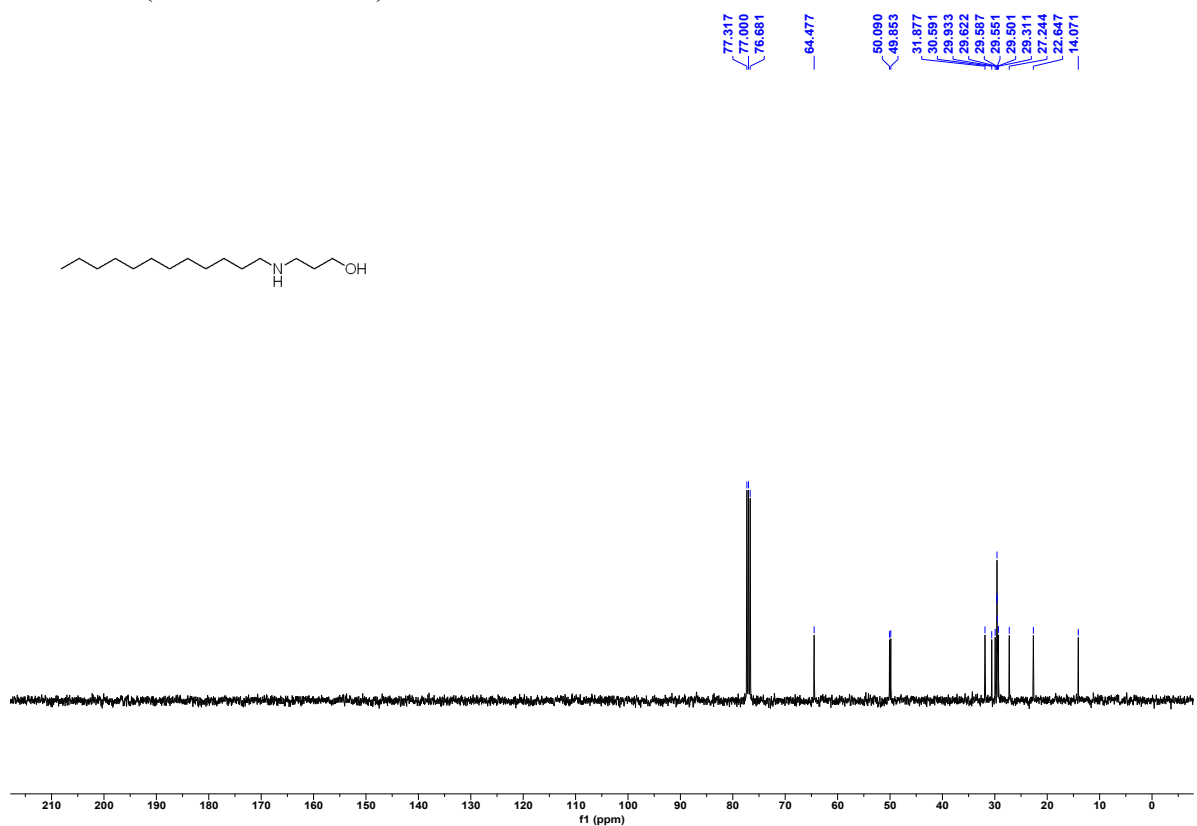
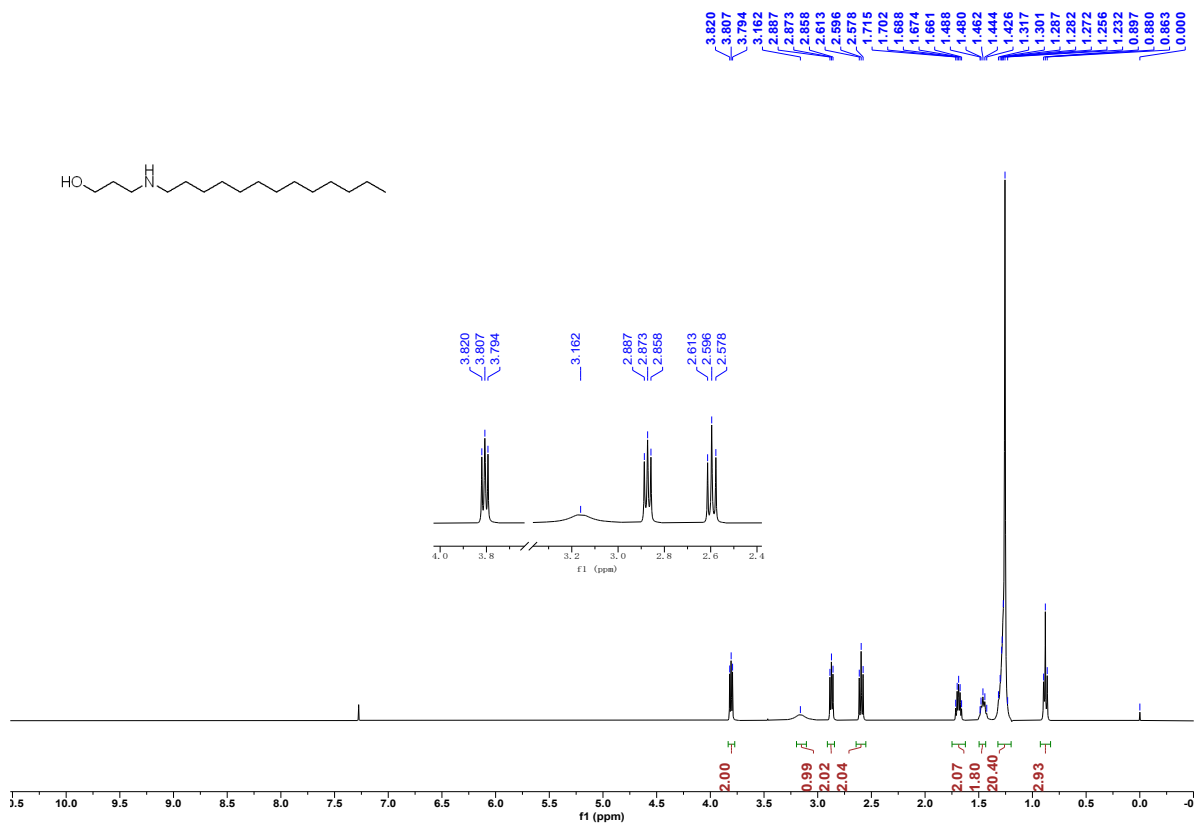
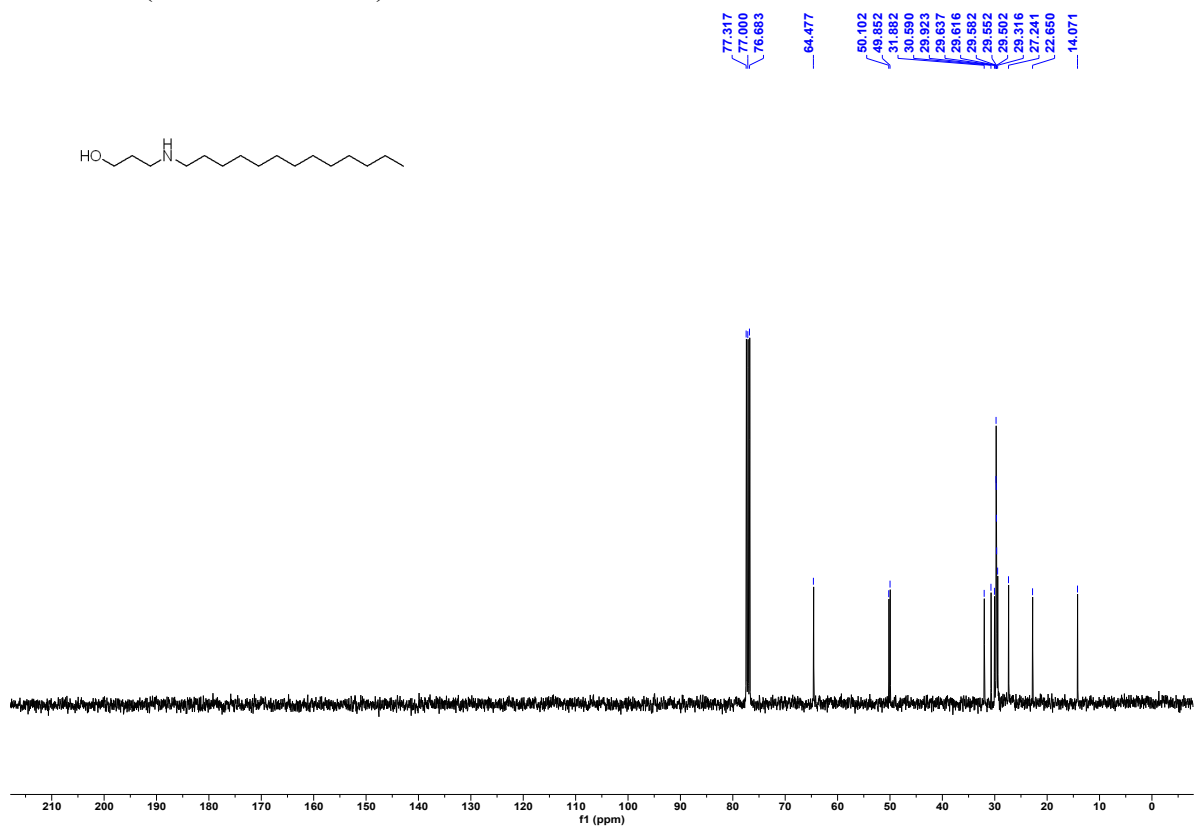


Figure S16. ^1H and ^{13}C NMR spectra of 3-(Dodecylamino)propan-1-ol (*A14*)

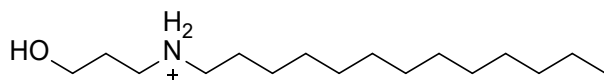
^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)



HRMS (ESI-positive)



Chemical Formula: $\text{C}_{16}\text{H}_{36}\text{NO}^+$

m/z: 258.2791 (100.0%), 259.2825 (17.3%), 260.2859 (1.4%)

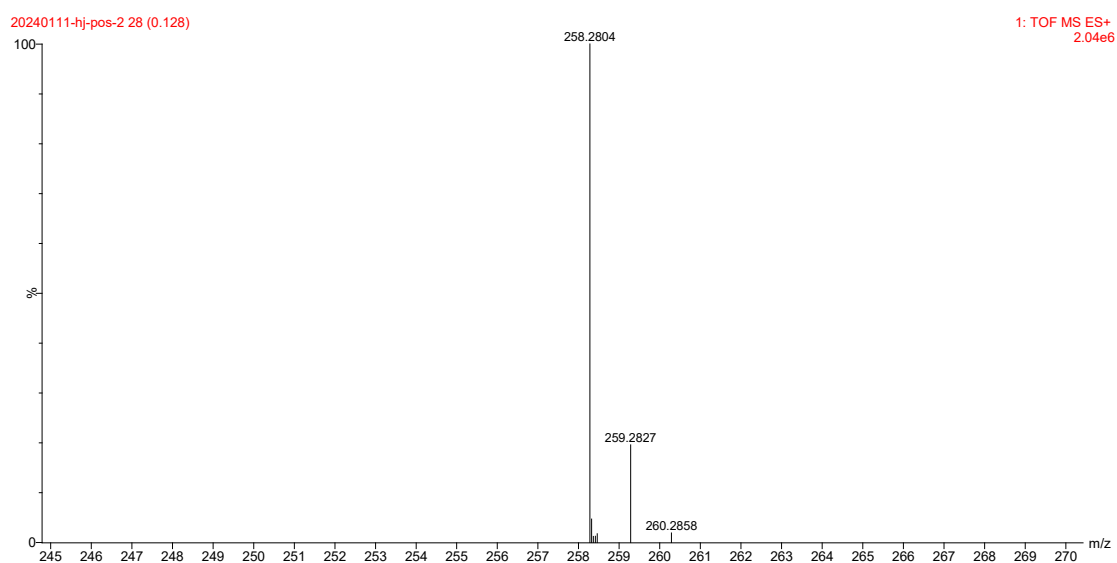
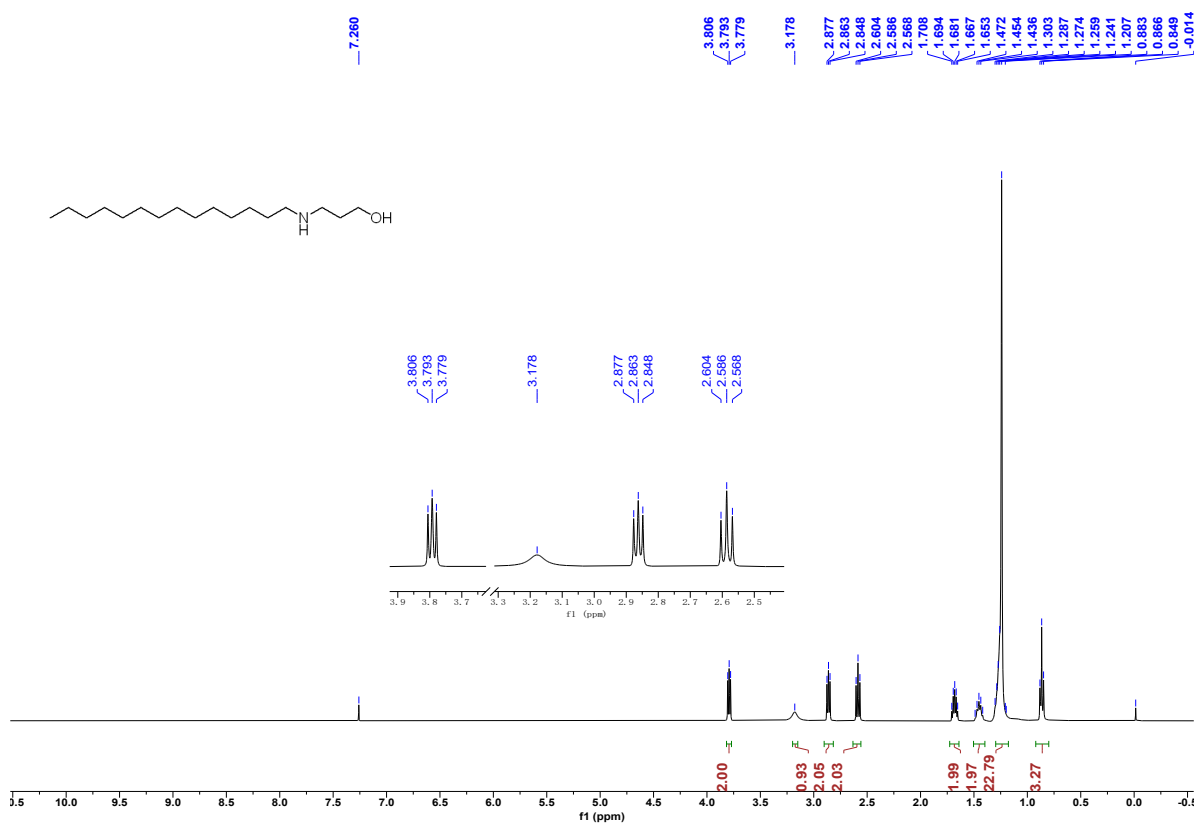


Figure S17. ^1H , ^{13}C NMR spectra and HRMS of 3-(Tridecylamino)propan-1-ol (*A15*)

^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)

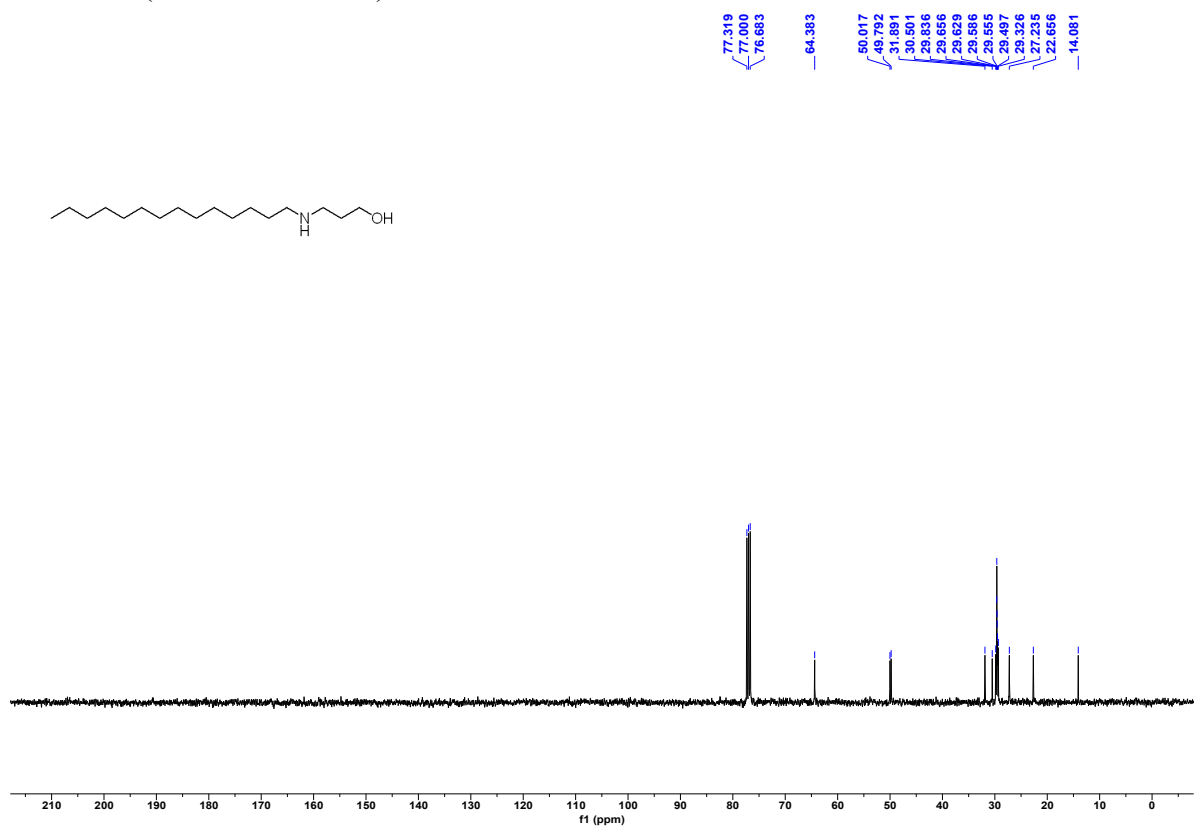
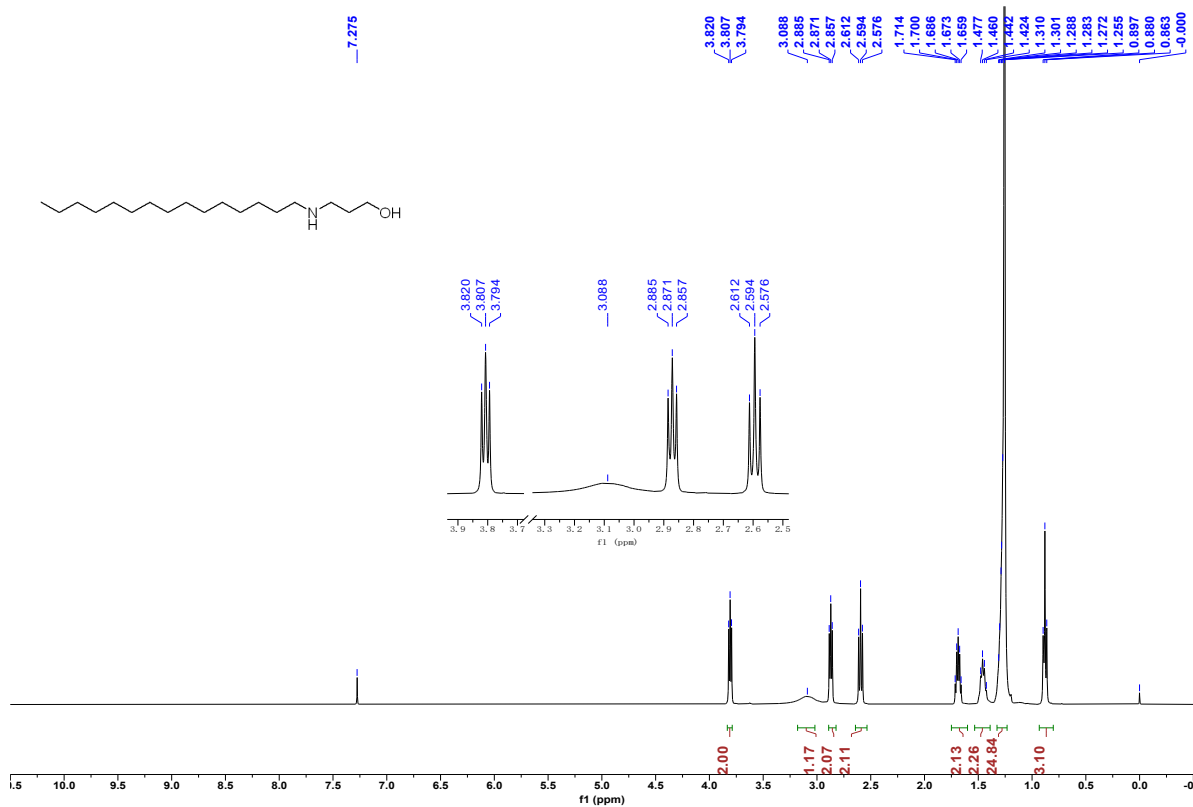
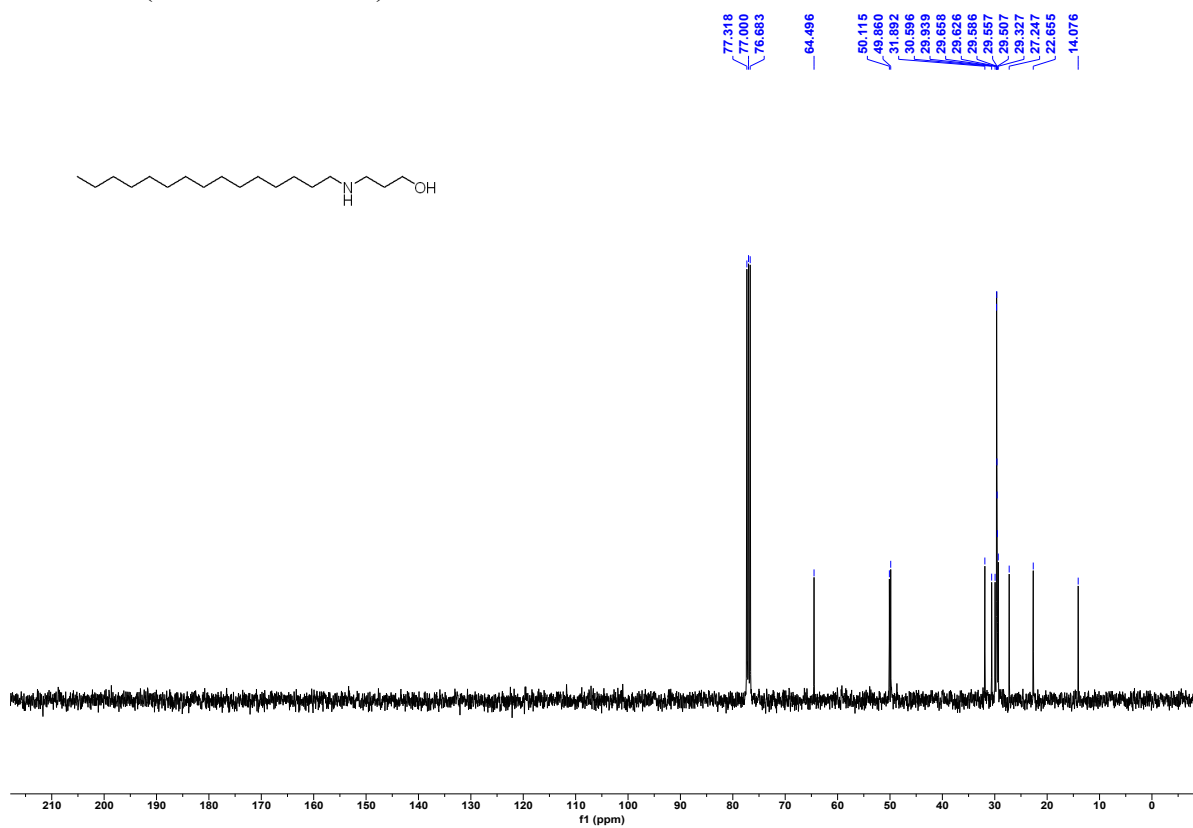


Figure S18. ^1H and ^{13}C NMR spectra of 3-(Tetradecylamino)propan-1-ol (*A16*)

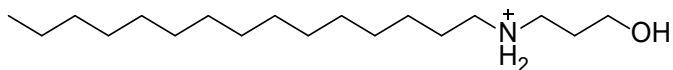
^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)



HRMS (ESI-positive)



Chemical Formula: $C_{18}H_{40}NO^+$

m/z: 286.3104 (100.0%), 287.3138 (19.5%), 288.3172 (1.8%)

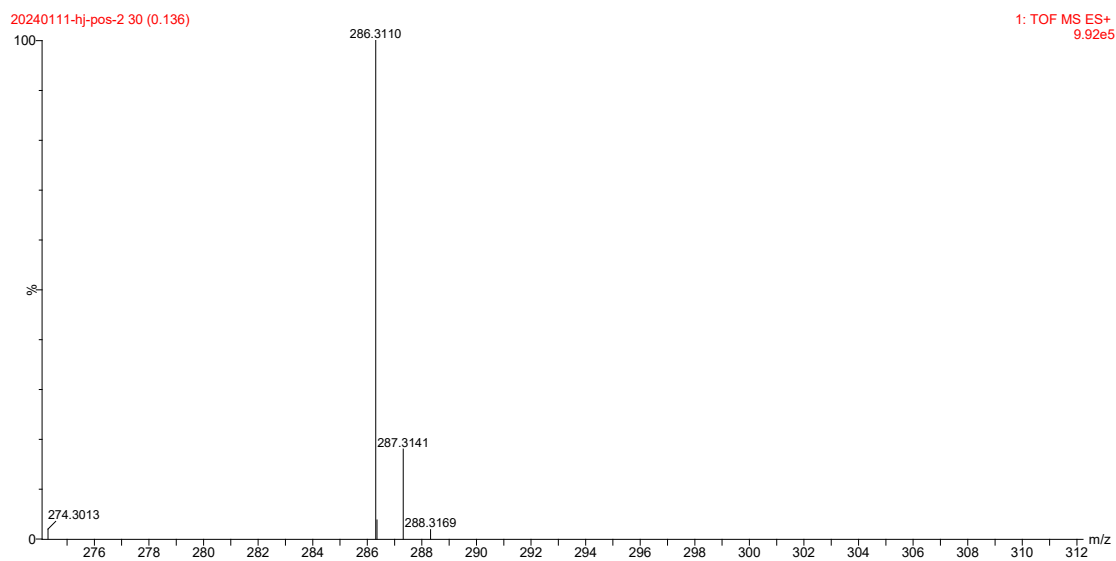
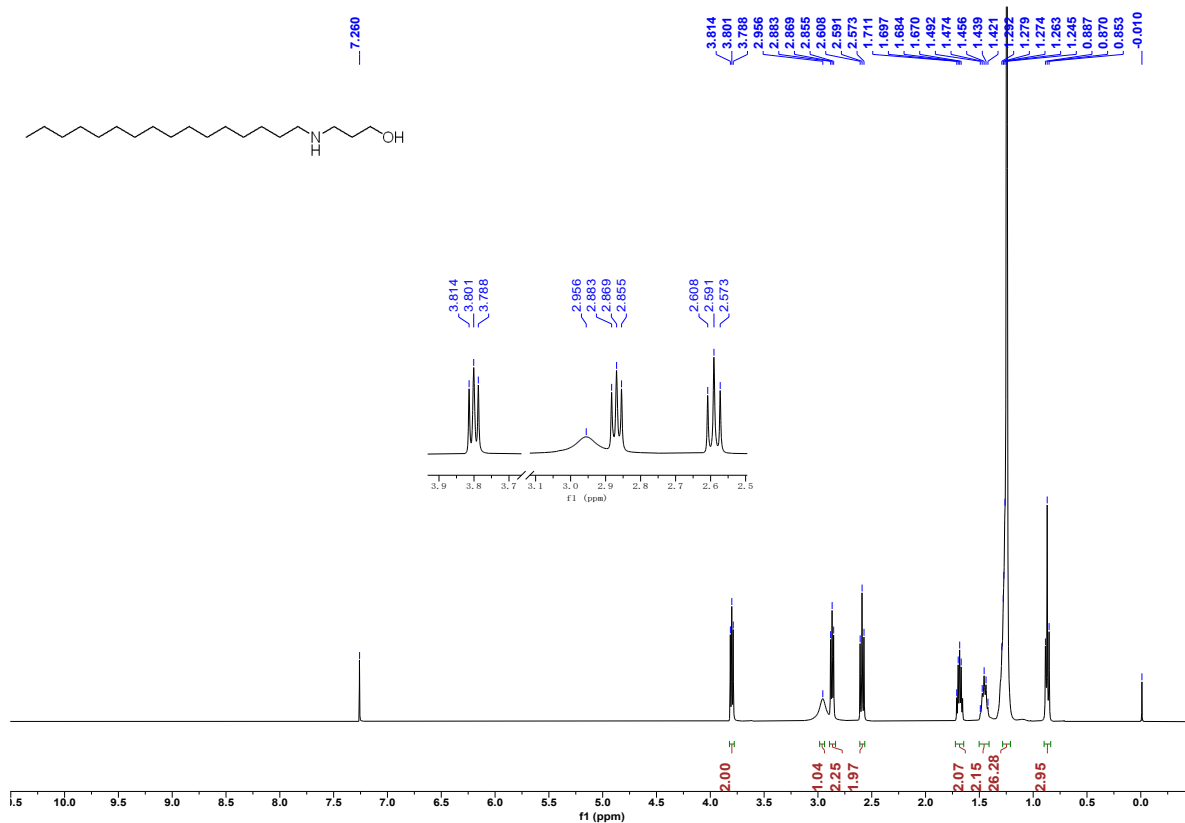


Figure S19. 1H , ^{13}C NMR spectra and HRMS of 3-(Pentadecylamino)propan-1-ol (*A17*)

^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)

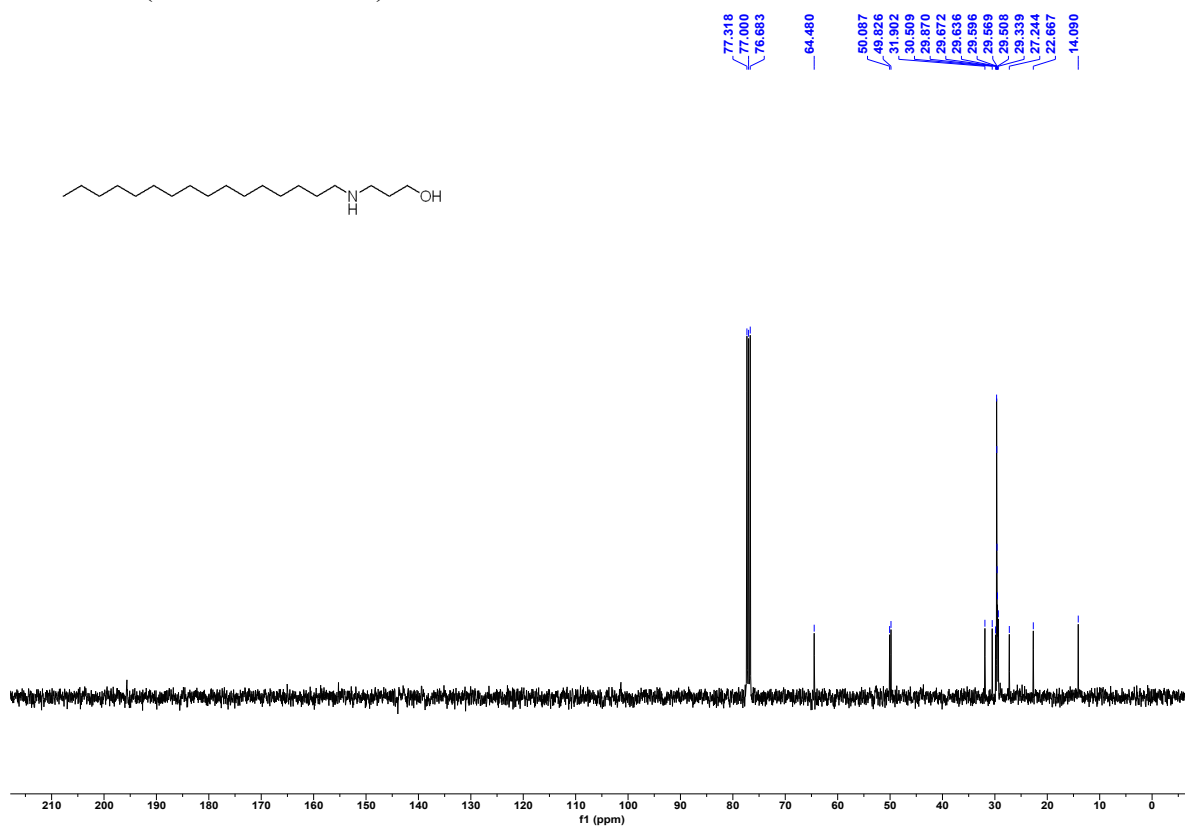
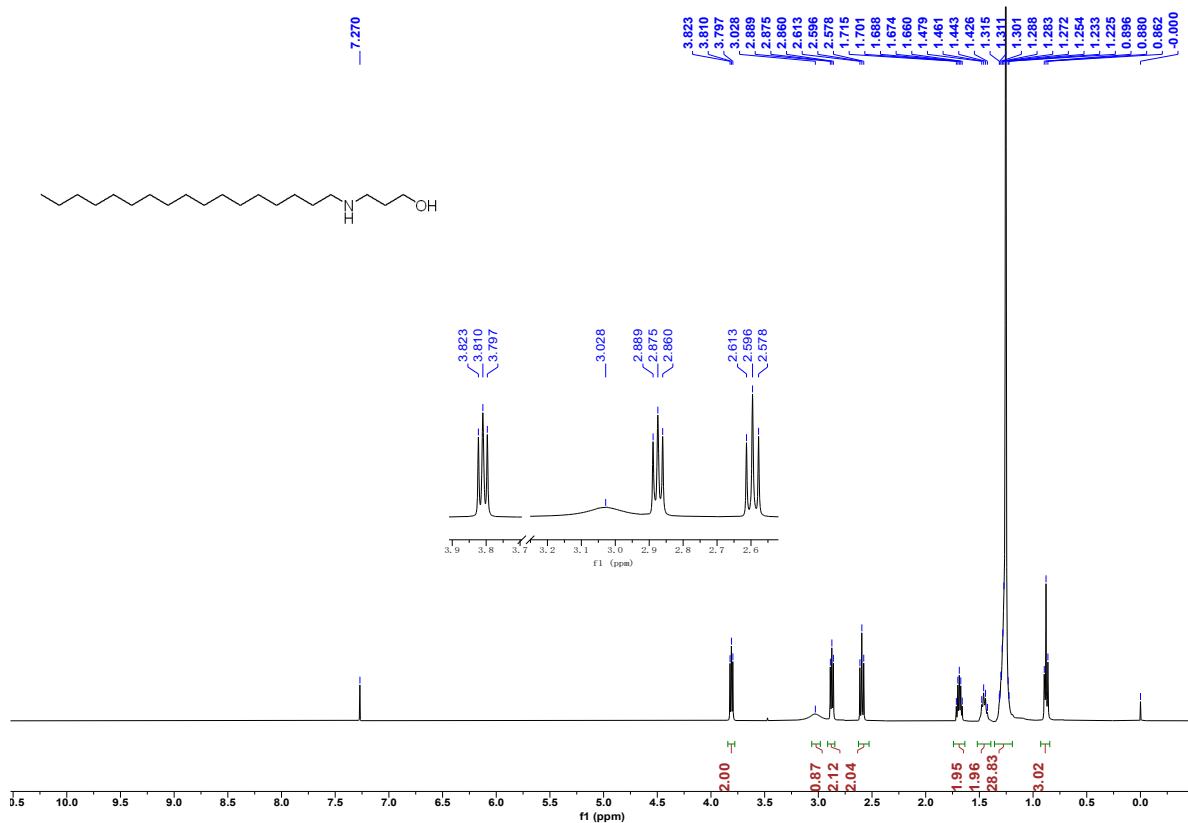
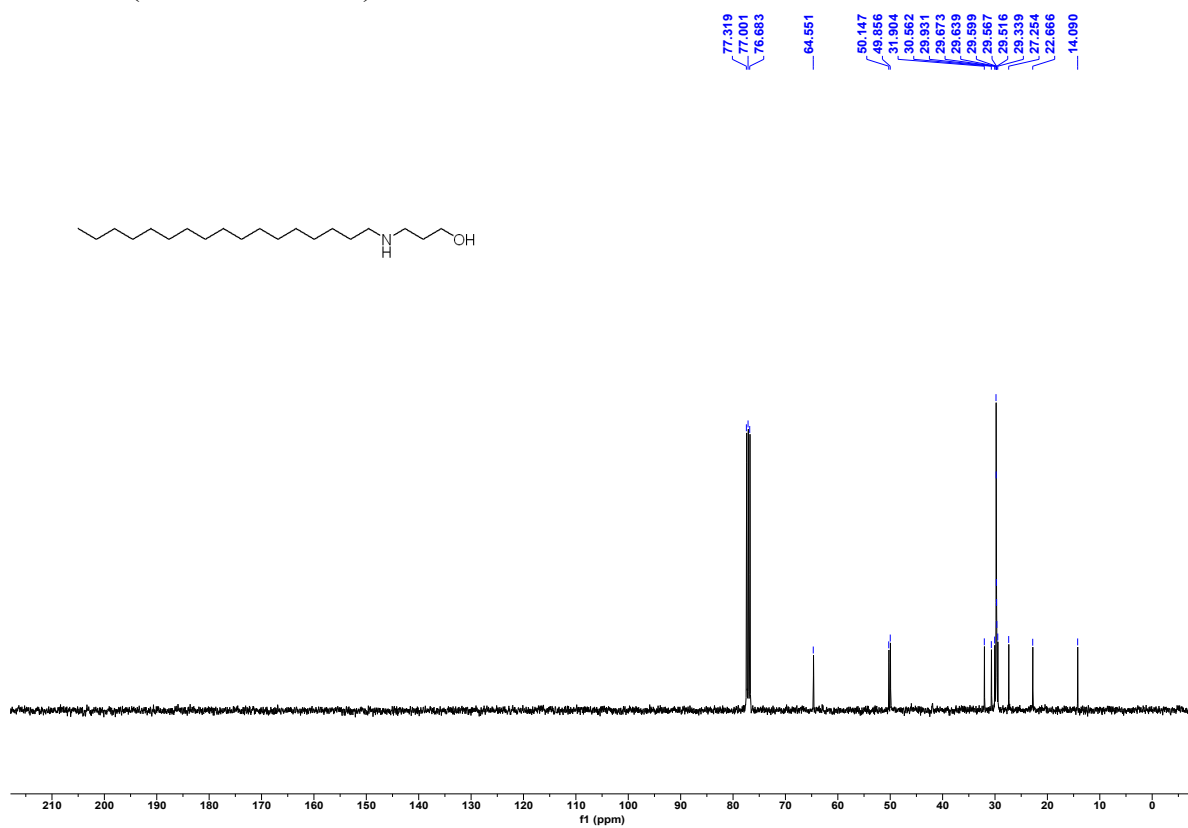


Figure S20. ^1H and ^{13}C NMR spectra of 3-(Hexadecylamino)propan-1-ol (*A18*)

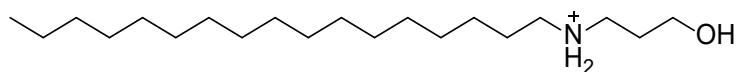
^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)



HRMS (ESI-positive)



Chemical Formula: C₂₀H₄₄NO⁺

m/z: 314.3418 (100.0%), 315.3451 (21.6%), 316.3485 (2.2%)

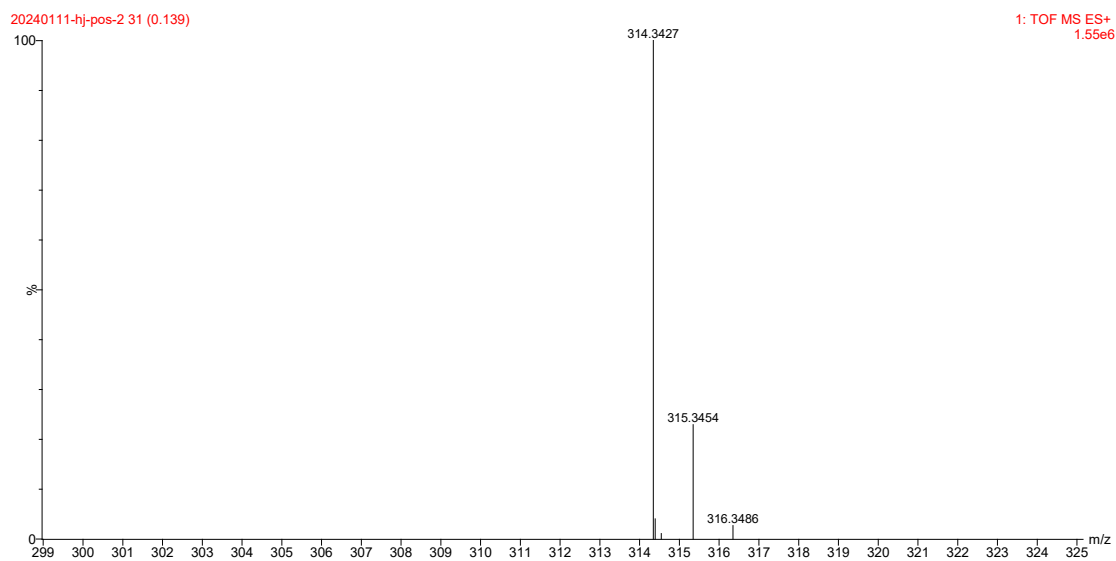
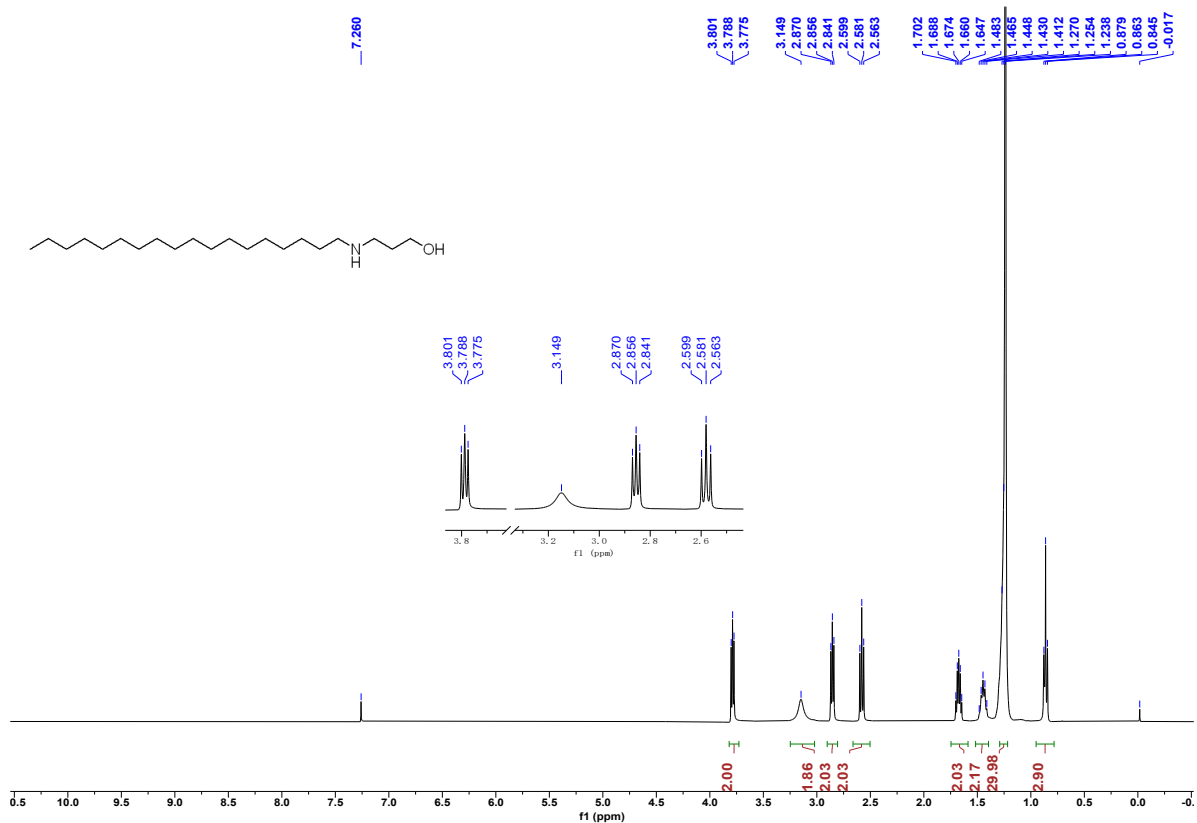


Figure S21. ¹H, ¹³C NMR spectra and HRMS of 3-(Heptadecylamino)propan-1-ol (*A19*)

^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)

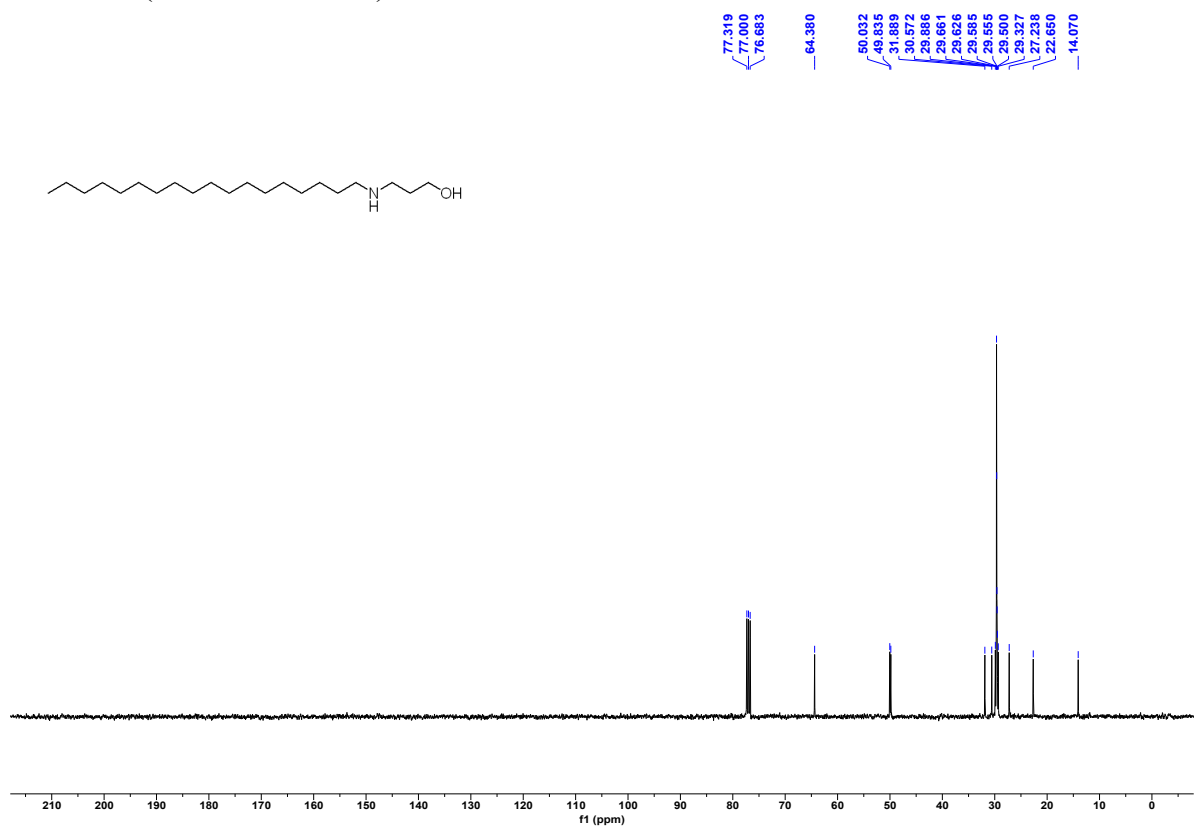
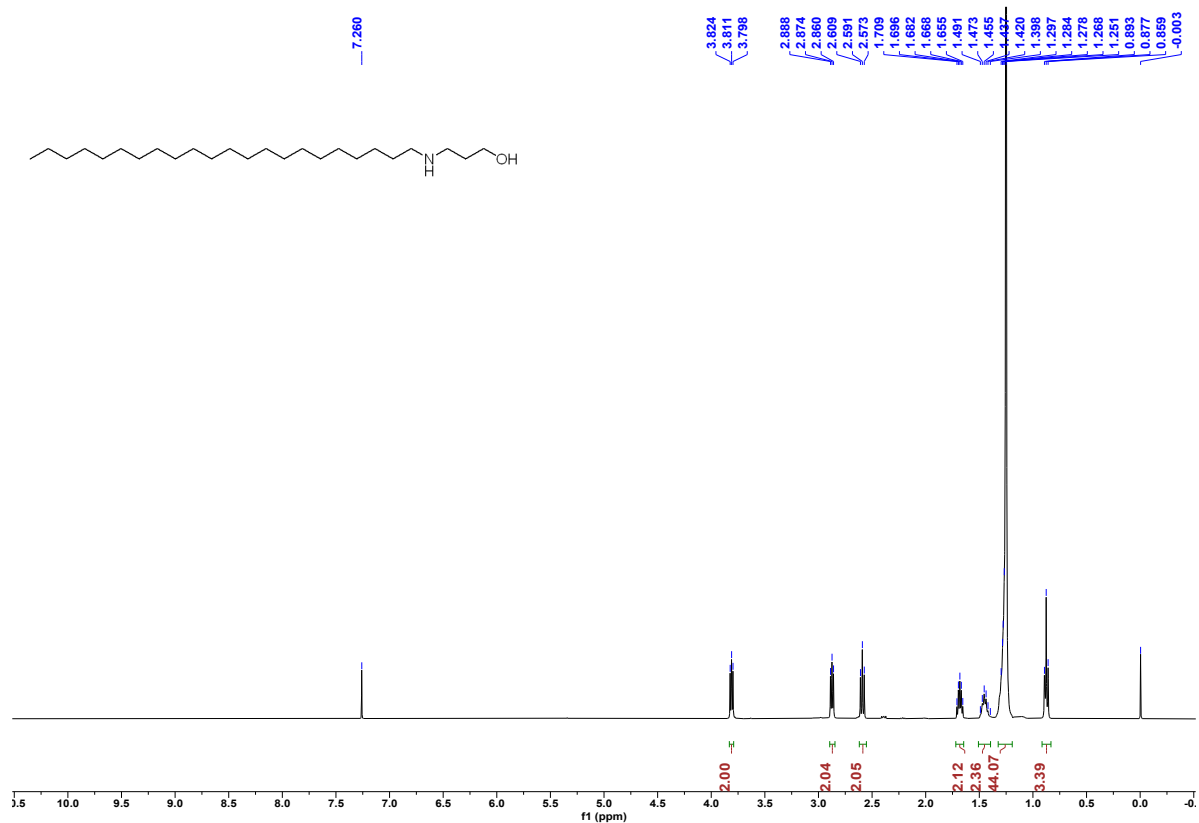
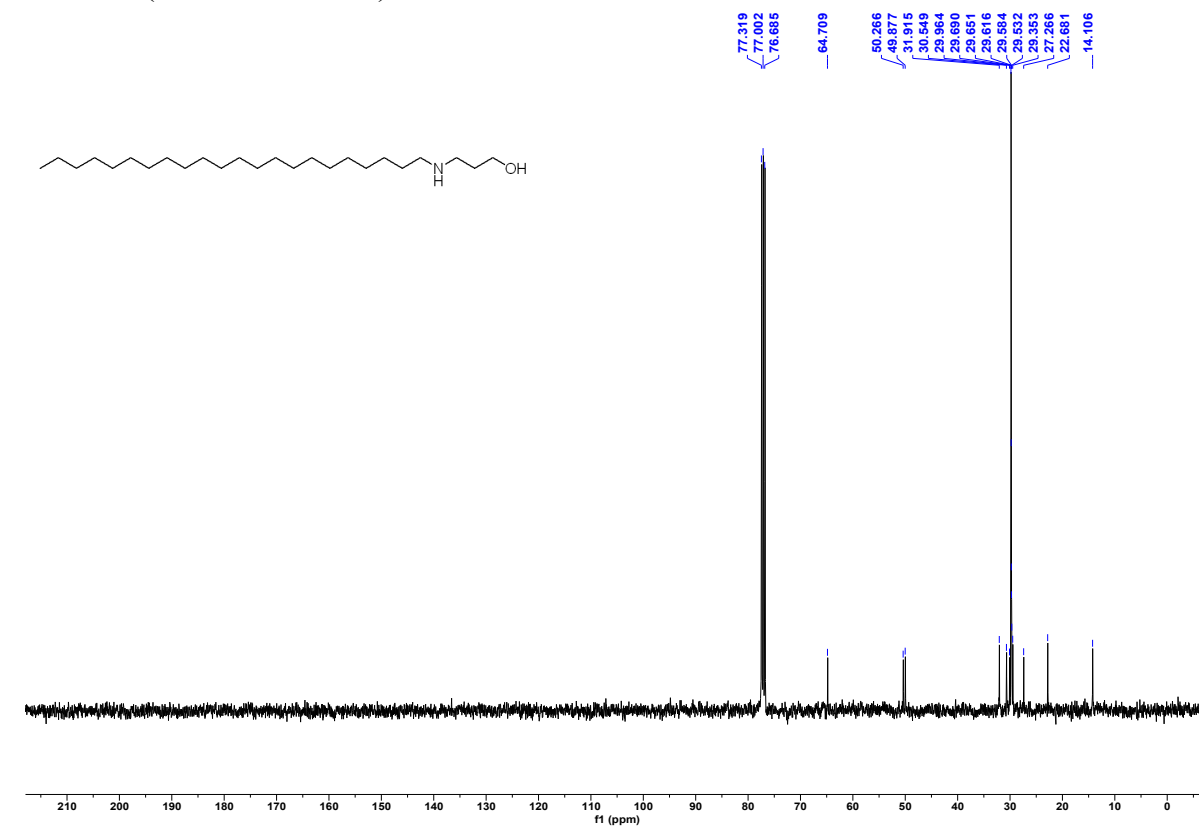


Figure S22. ^1H and ^{13}C NMR spectra of 3-(Octadecylamino)propan-1-ol (*A20*)

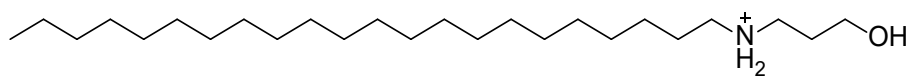
^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)



HRMS (ESI-positive)



Chemical Formula: C₂₅H₅₄NO⁺
m/z: 384.4200 (100.0%), 385.4234 (27.0%), 386.4268 (3.5%)

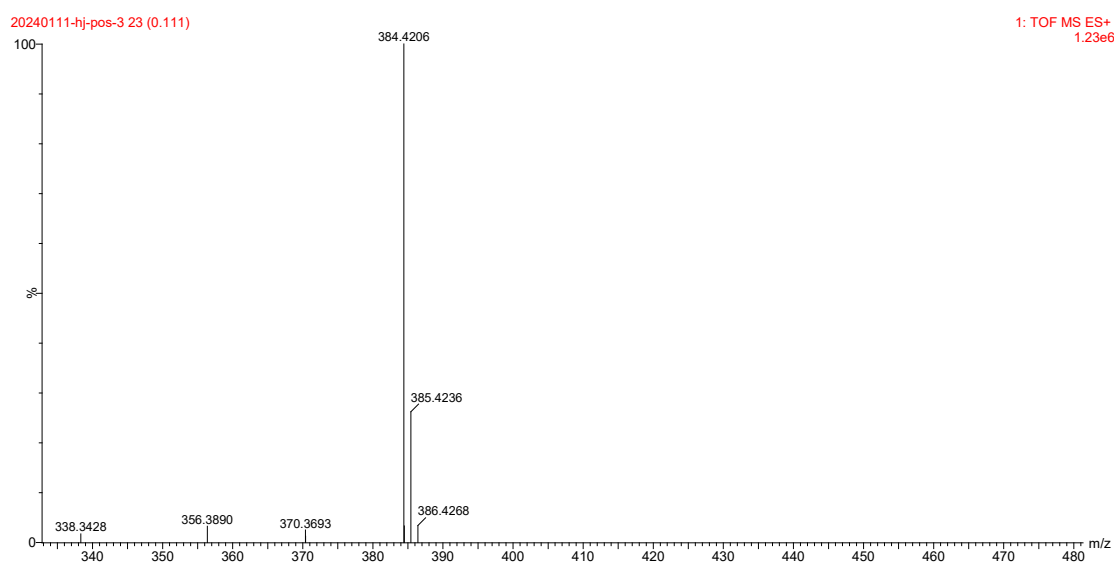
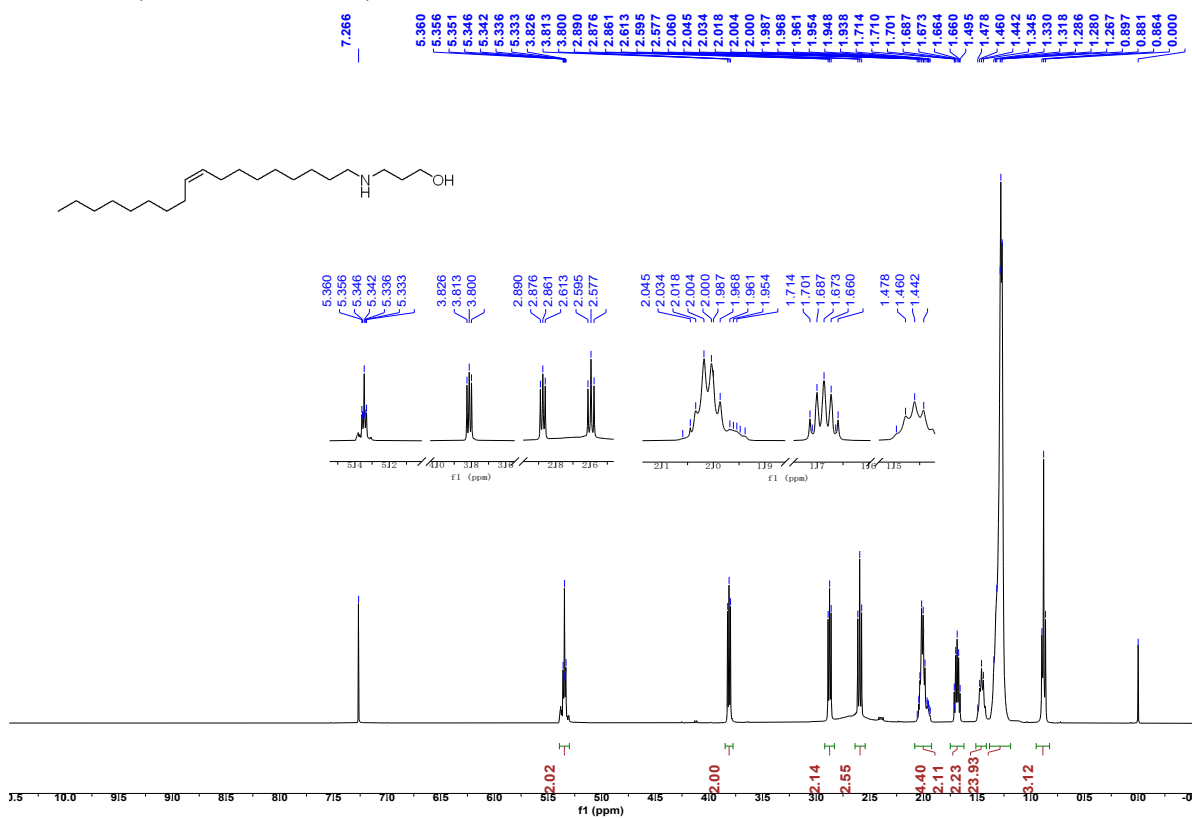
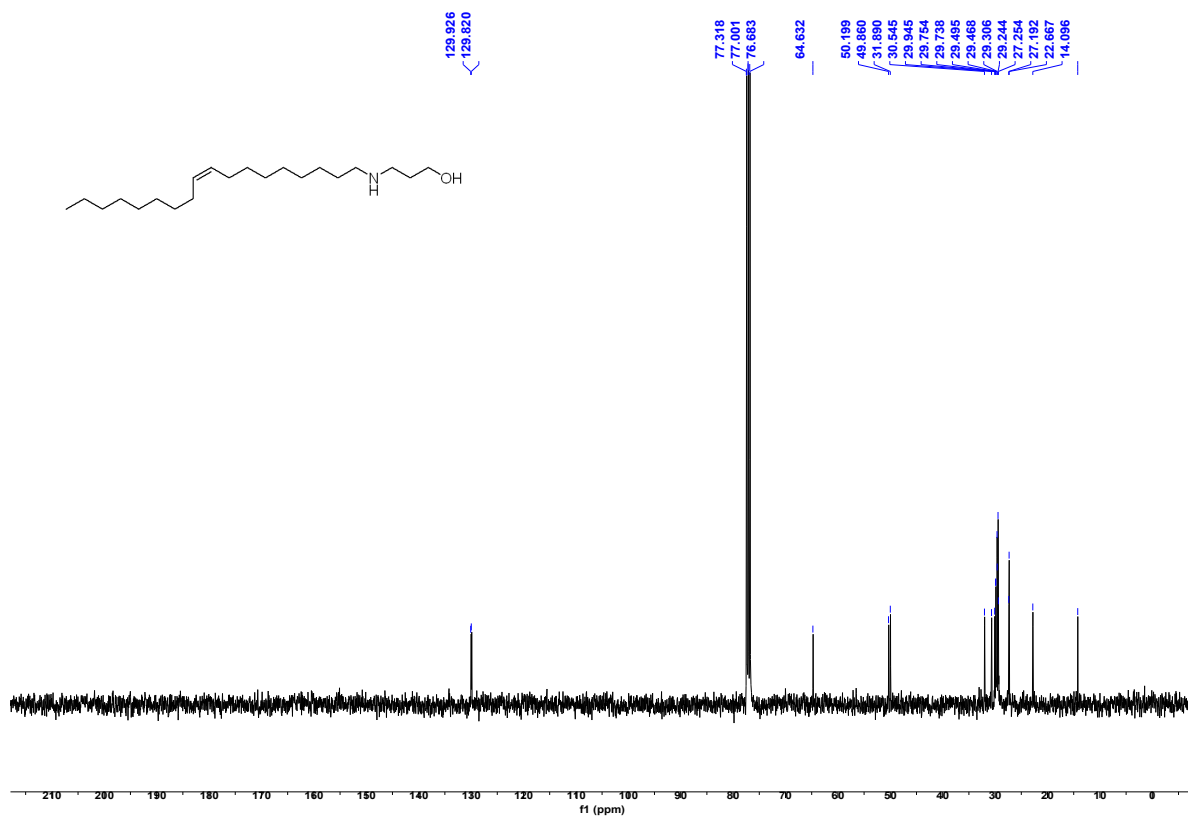


Figure S23. ¹H, ¹³C NMR spectra and HRMS of 3-(Docosylamino)propan-1-ol (*A2I*)

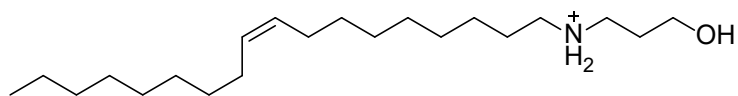
¹H NMR (400 MHz, CDCl₃)



¹³C NMR (101 MHz, CDCl₃)



HRMS (ESI-positive)



Chemical Formula: C₂₁H₄₄NO⁺

m/z: 326.3418 (100.0%), 327.3451 (22.7%), 328.3485 (2.5%)

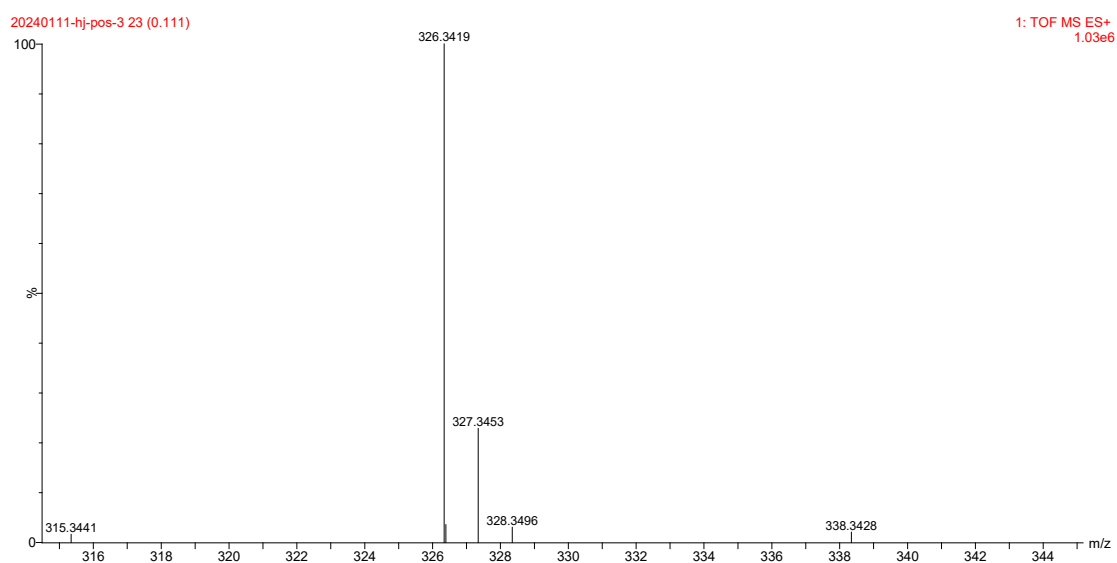
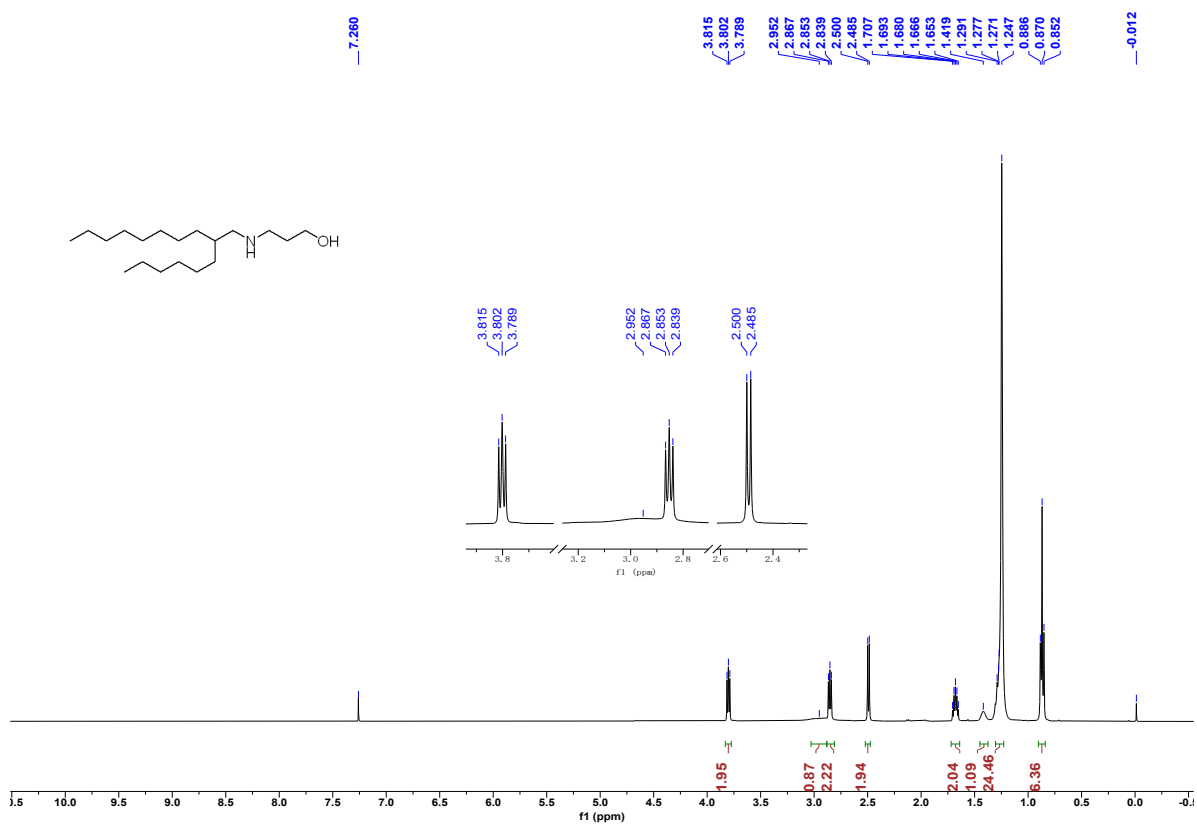
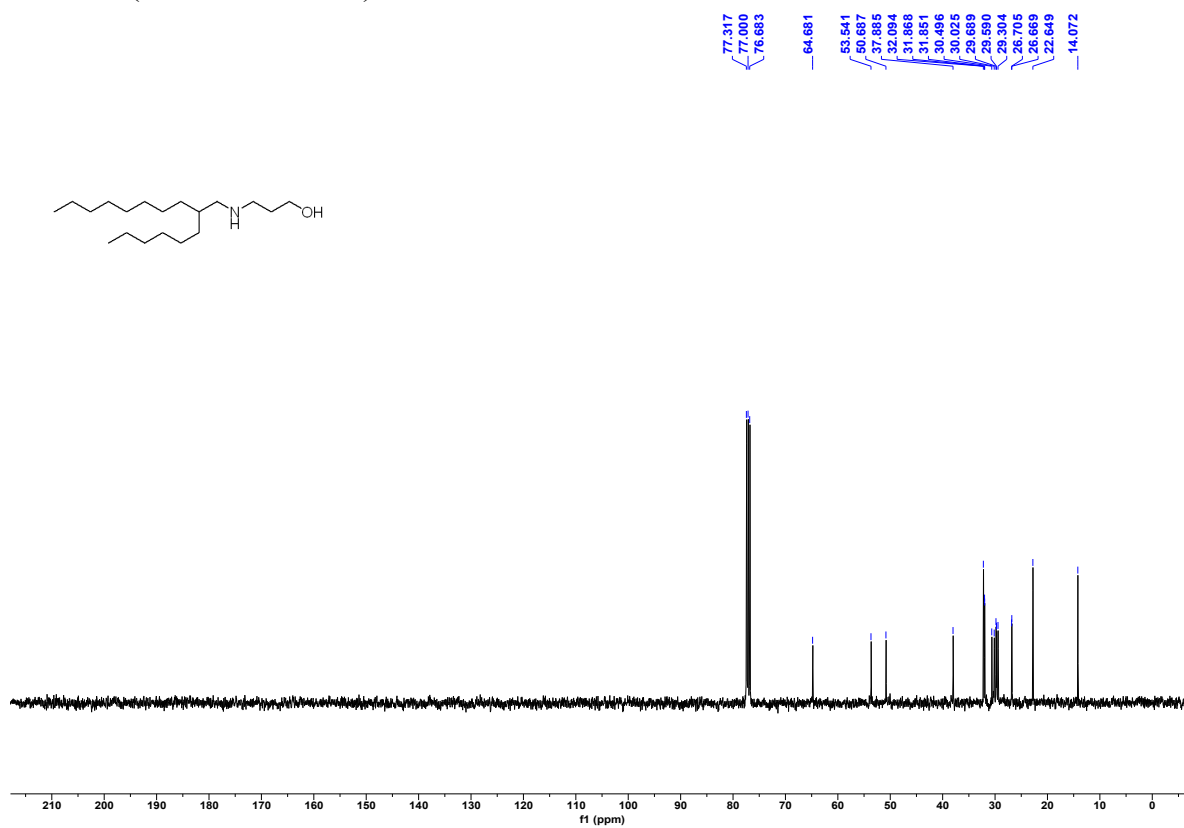


Figure S24. ¹H, ¹³C NMR spectra and HRMS of (Z)-3-(octadec-9-en-1-ylamino)propan-1-ol (A22)

^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)



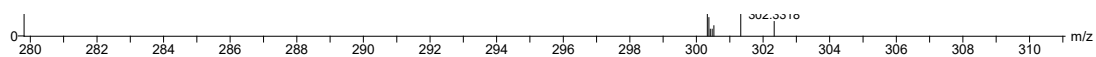
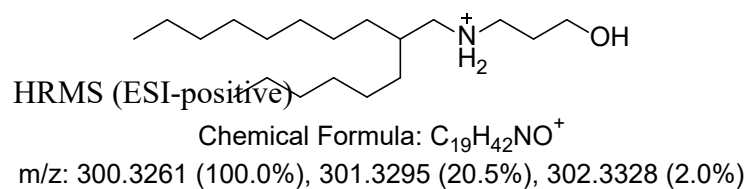
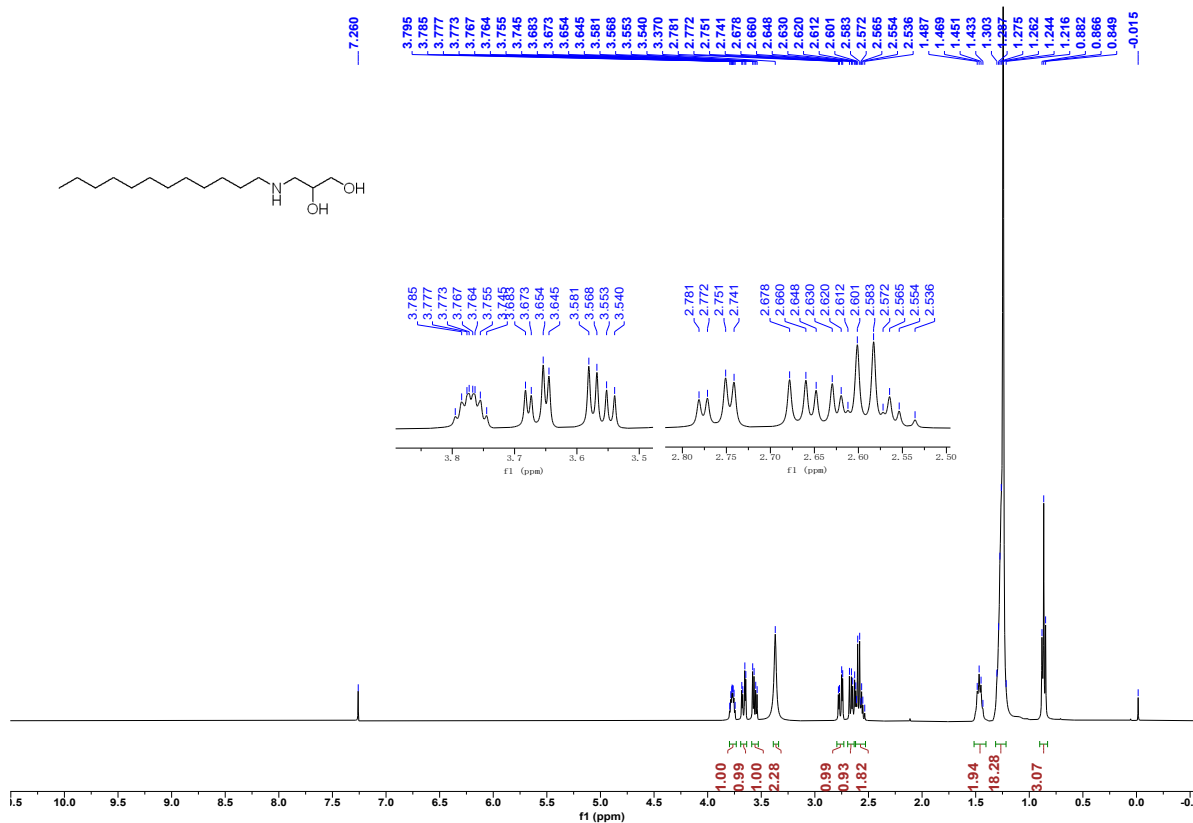


Figure S25. ^1H , ^{13}C NMR spectra and HRMS of 3-((2-Hexyldecyl)amino)propan-1-ol (*A23*)

^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)

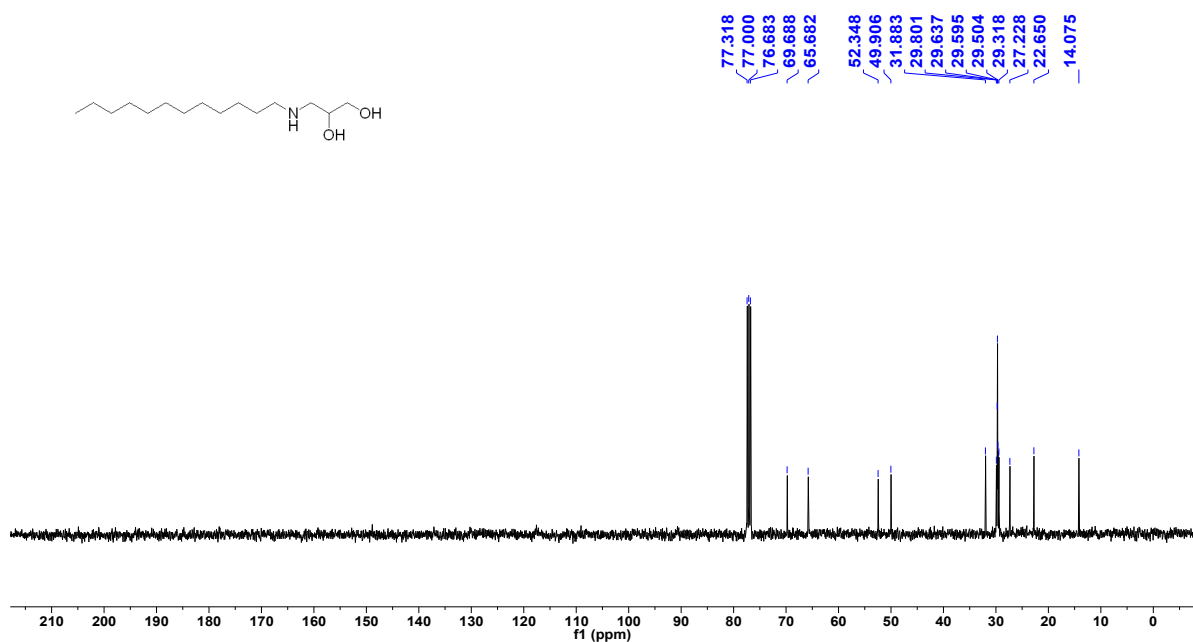
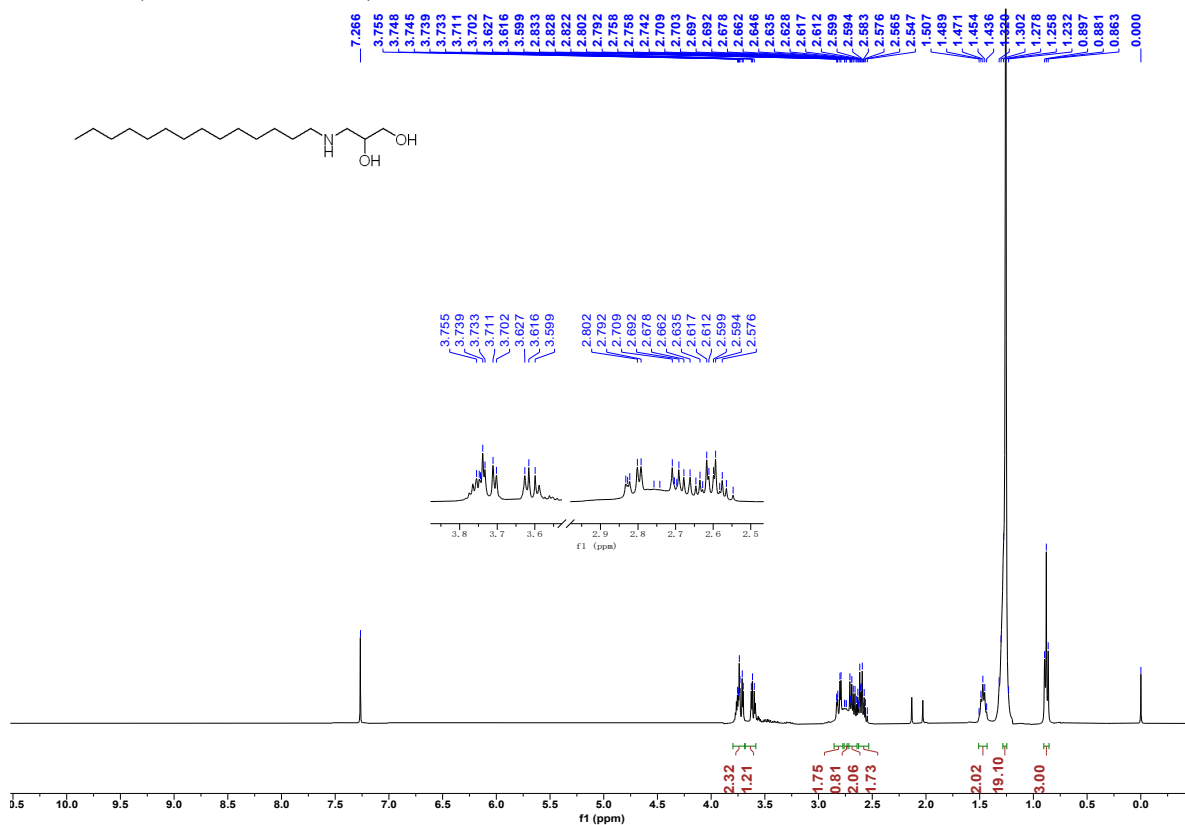


Figure S26. ^1H and ^{13}C NMR spectra of 3-(Dodecylamino)propane-1,2-diol (*A24*)

^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)

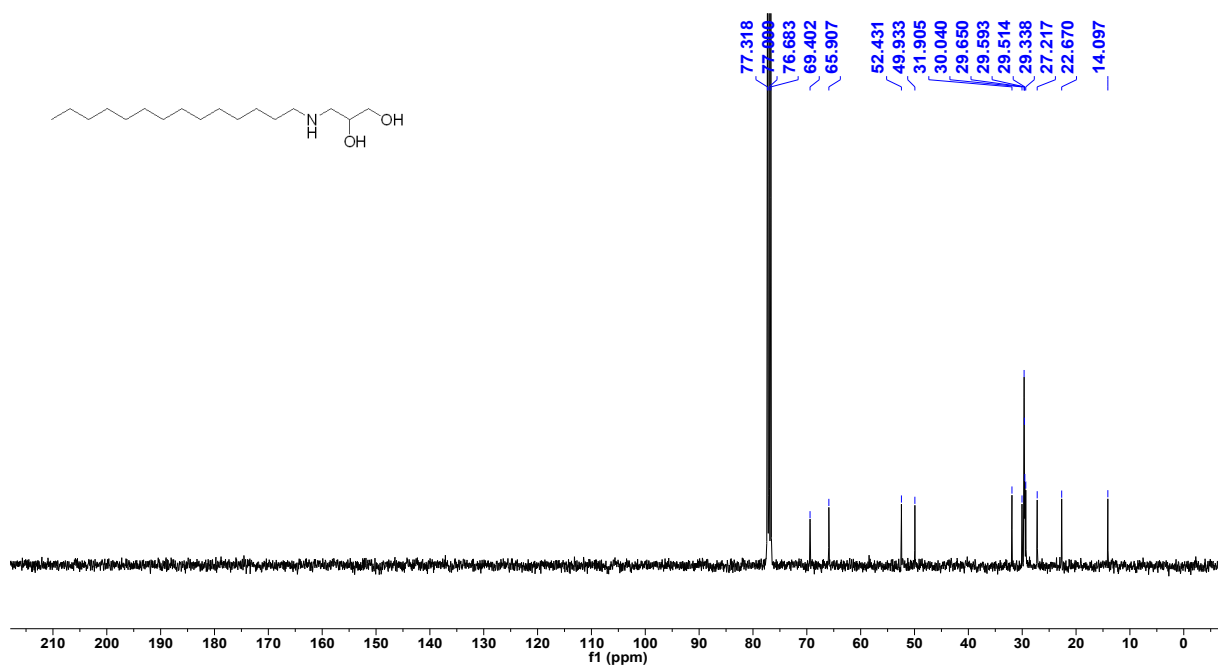
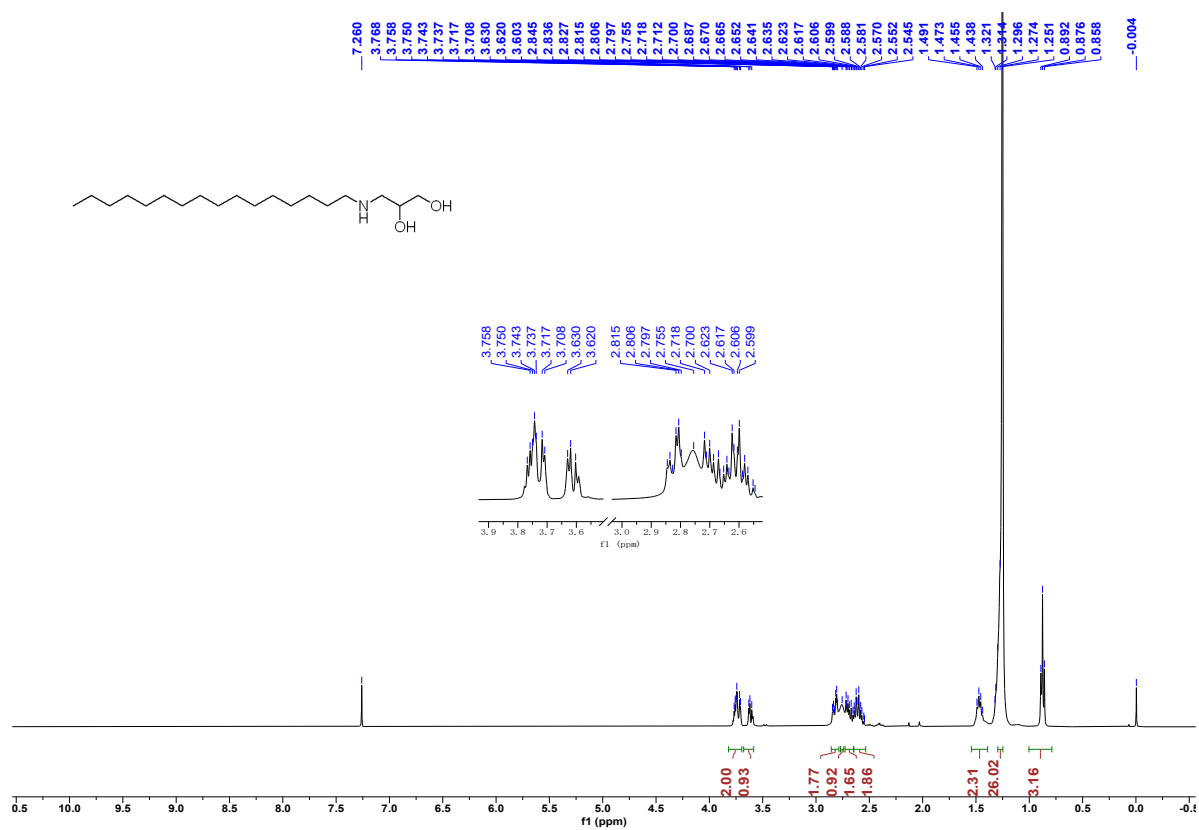


Figure S27. ^1H and ^{13}C NMR spectra of 3-(Tetradecylamino)propane-1,2-diol (A25)

^1H NMR (400 MHz, CDCl_3)



¹³C NMR (101 MHz, CDCl₃)

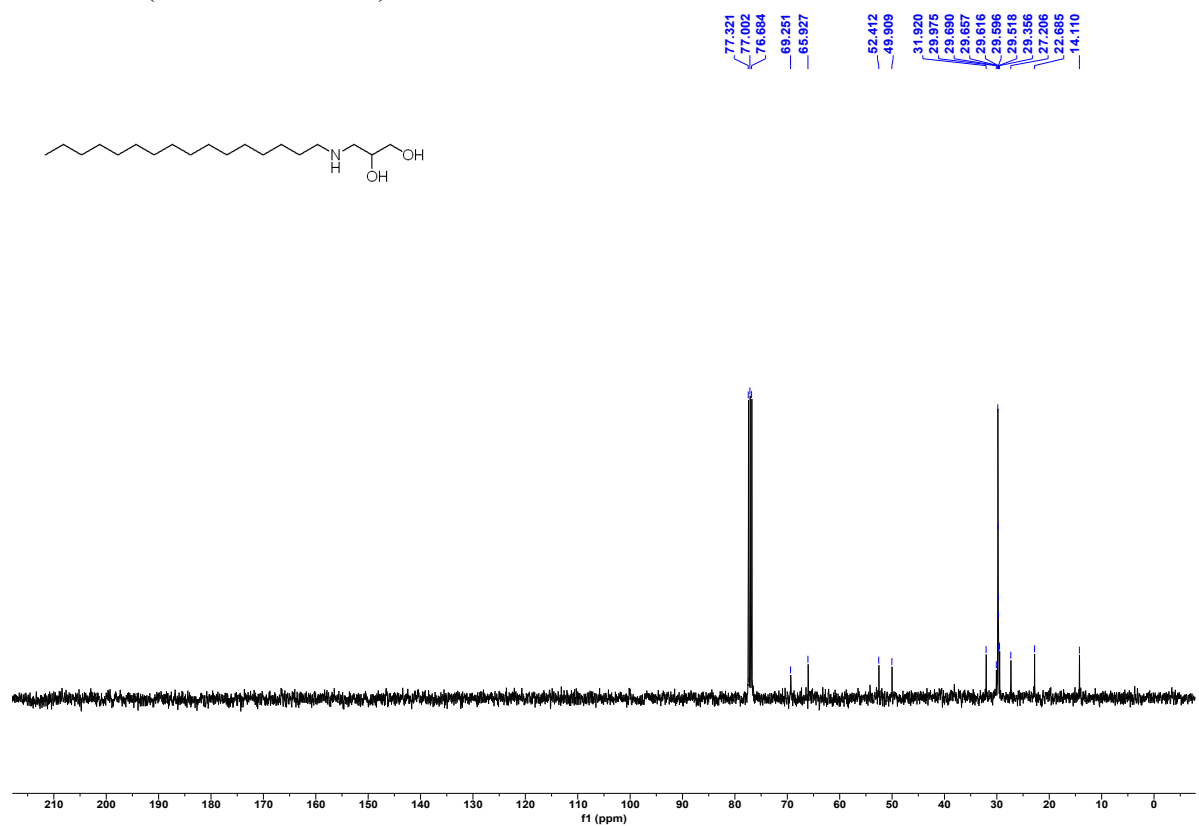
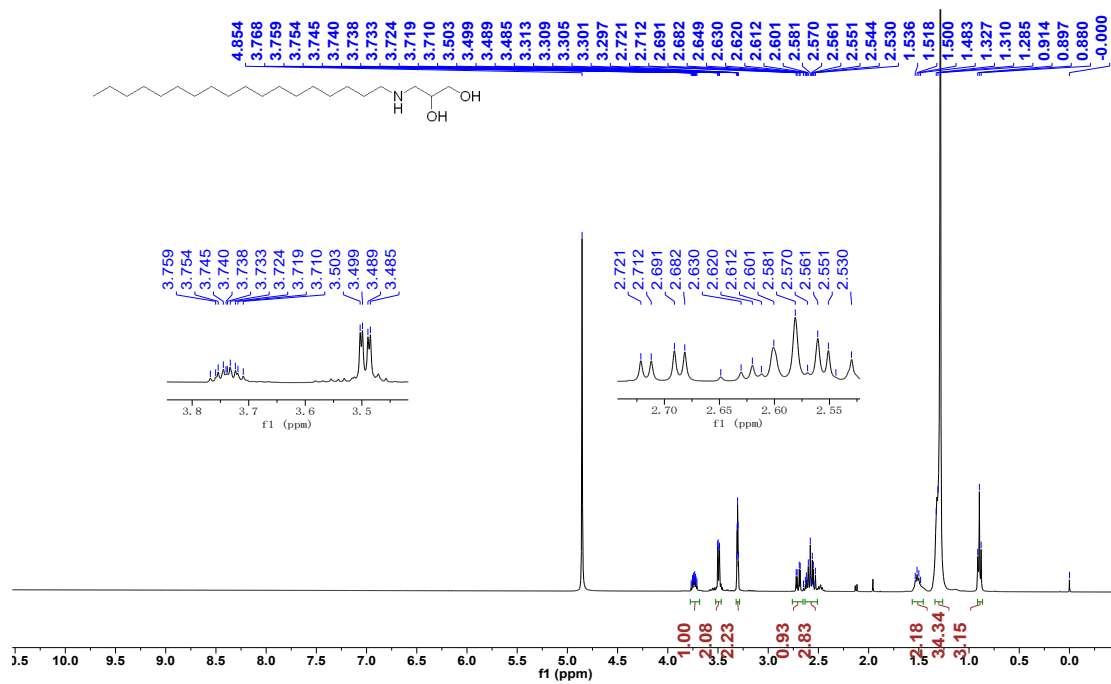


Figure S28. ¹H and ¹³C NMR spectra of 3-(Hexadecylamino)propane-1,2-diol **A26**)

¹H NMR (400 MHz, Methanol-*d*₄)



¹³C NMR (101 MHz, Methanol-*d*₄)

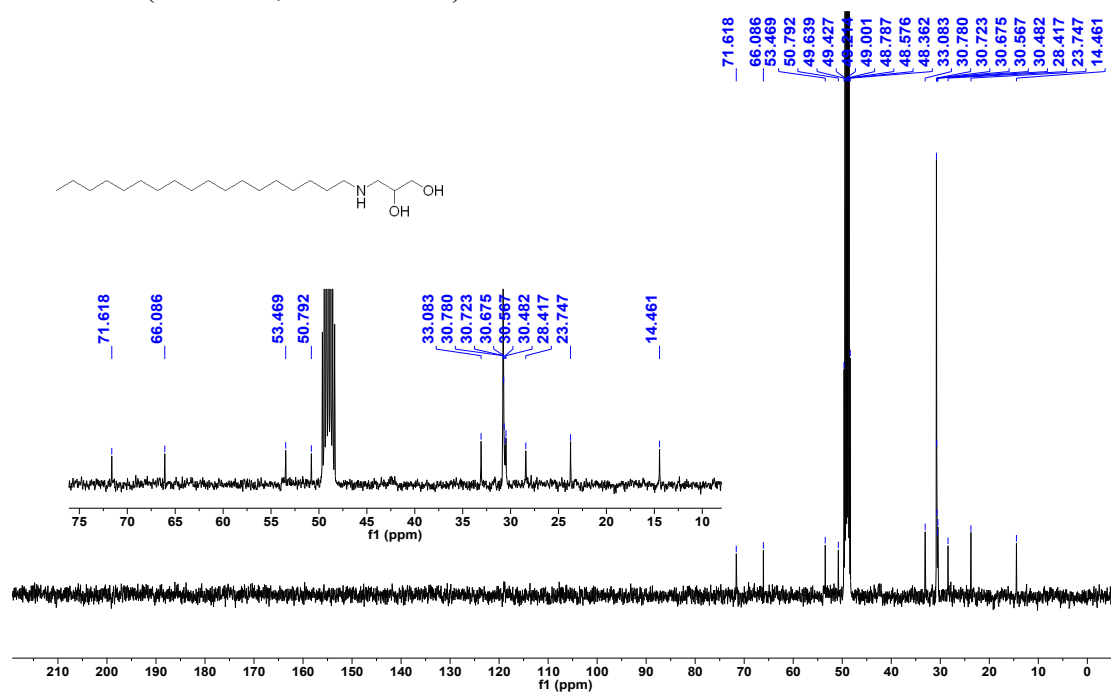


Figure S29. ¹H and ¹³C NMR spectra of 3-(Octadecylamino)propane-1,2-diol (427)

^1H NMR (400 MHz, CDCl_3)

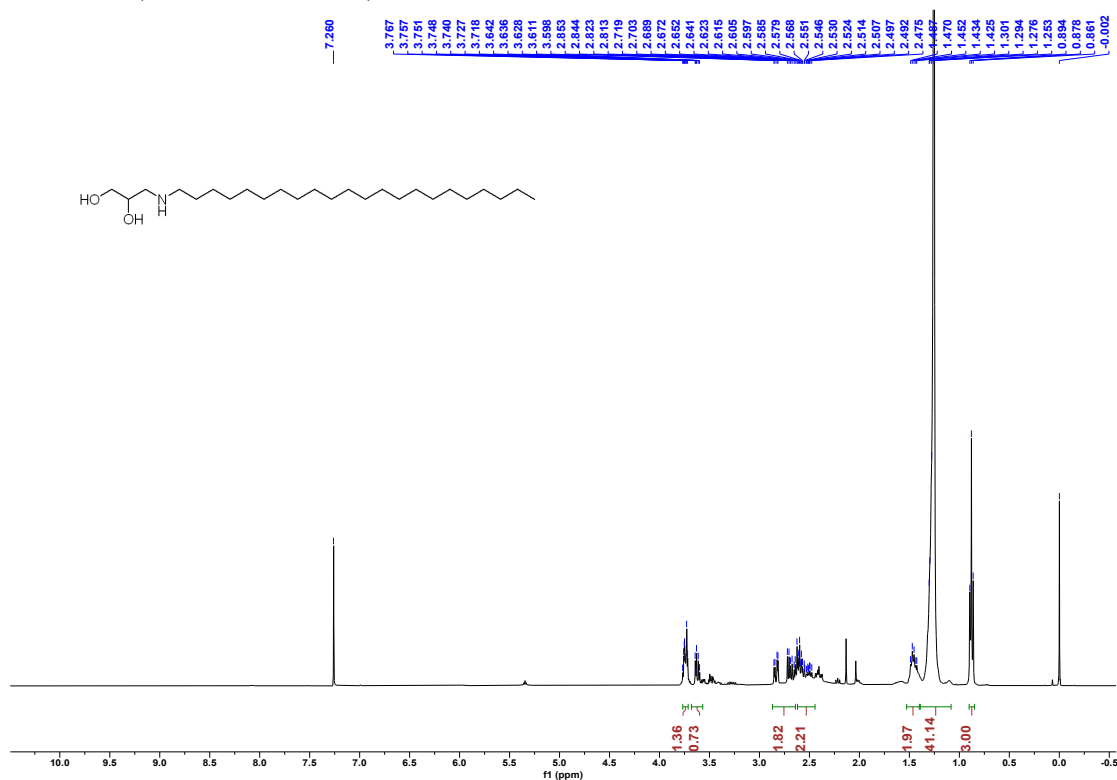
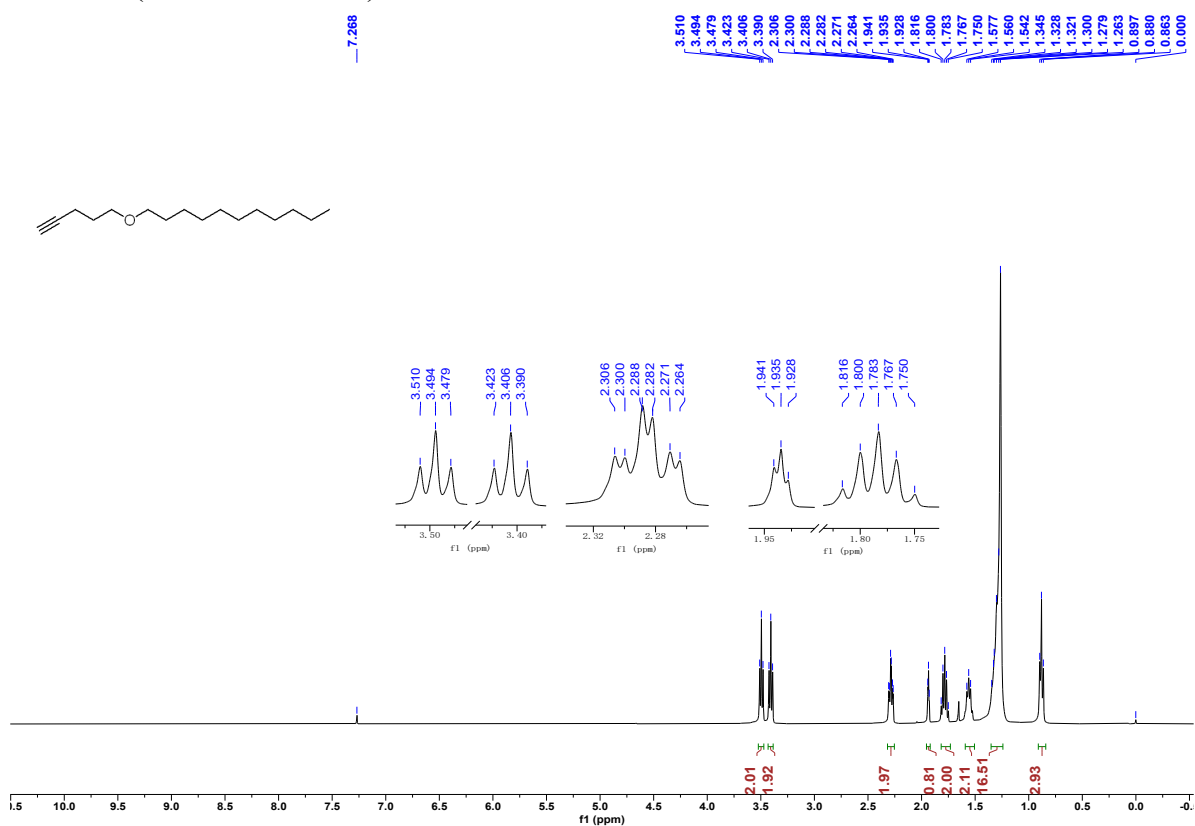
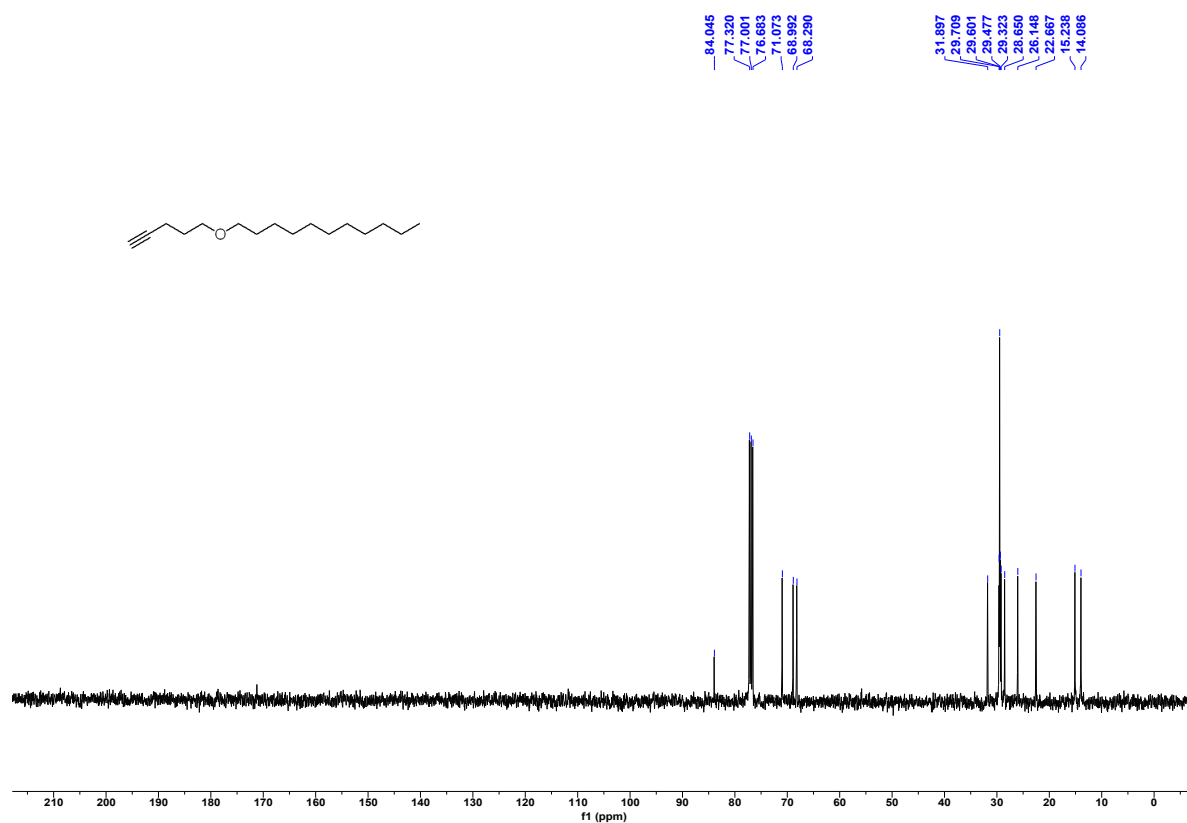


Figure S30. ^1H NMR spectra of 3-(docosylamino)propane-1,2-diol (A28)

^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)



HRMS (APCI-positive)

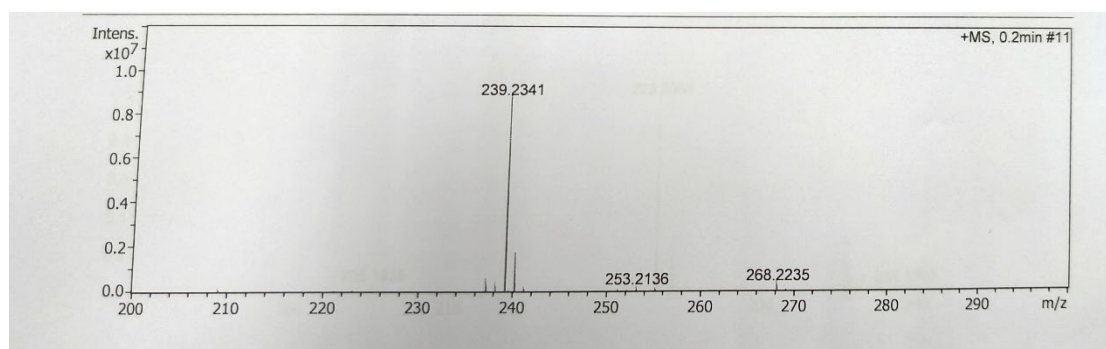
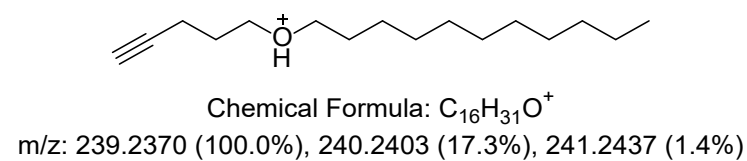
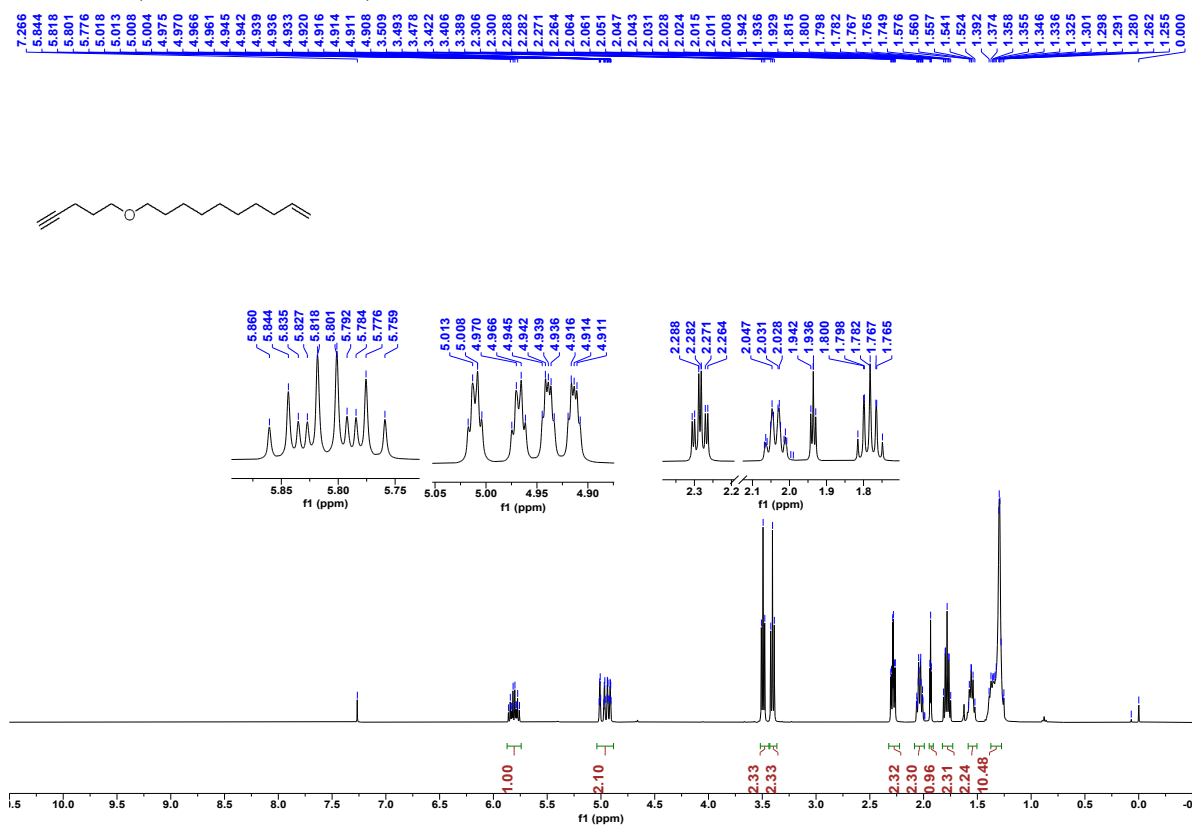
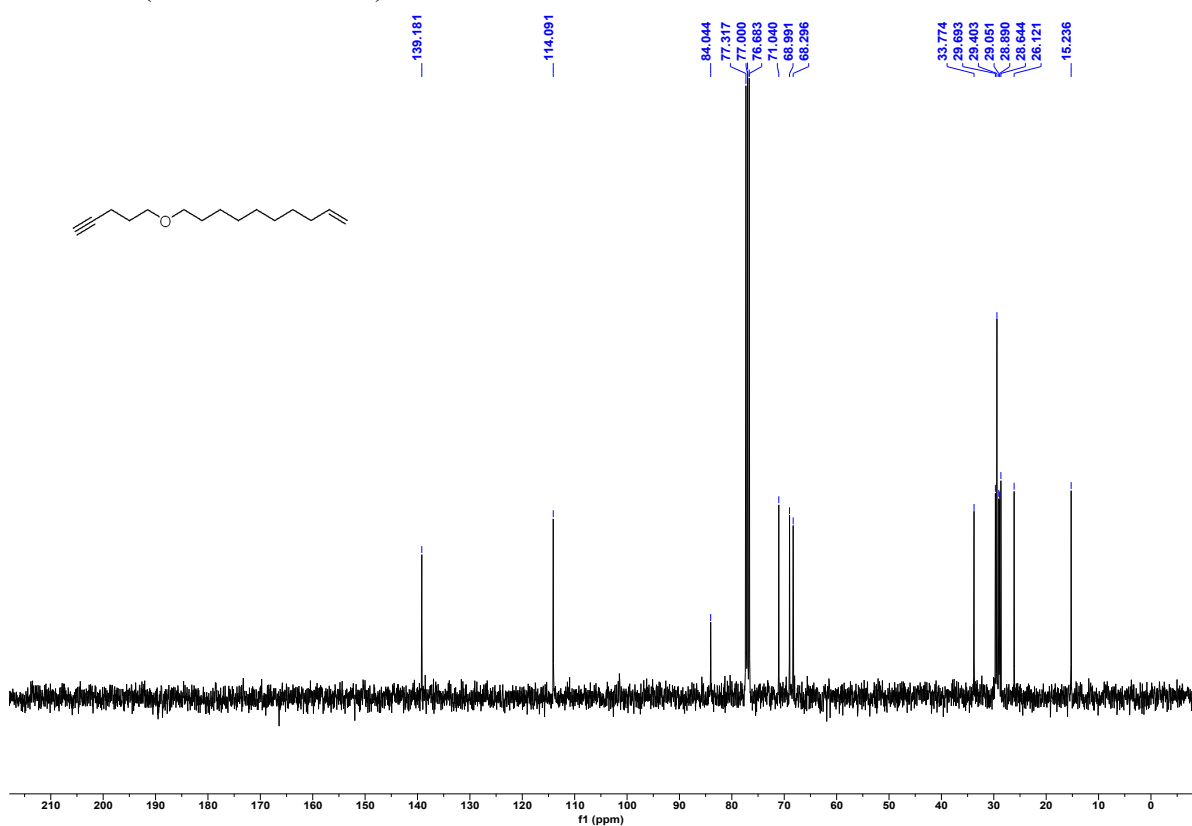


Figure S31. ^1H , ^{13}C NMR spectra and HRMS of 1-(Pent-4-yn-1-yloxy)undecane (*B8*)

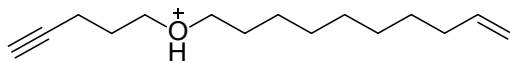
¹H NMR (400 MHz, CDCl₃)



¹³C NMR (101 MHz, CDCl₃)



HRMS (APCI-positive)



Chemical Formula: $C_{15}H_{27}O^+$

m/z: 223.2057 (100.0%), 224.2090 (16.2%), 225.2124 (1.2%)

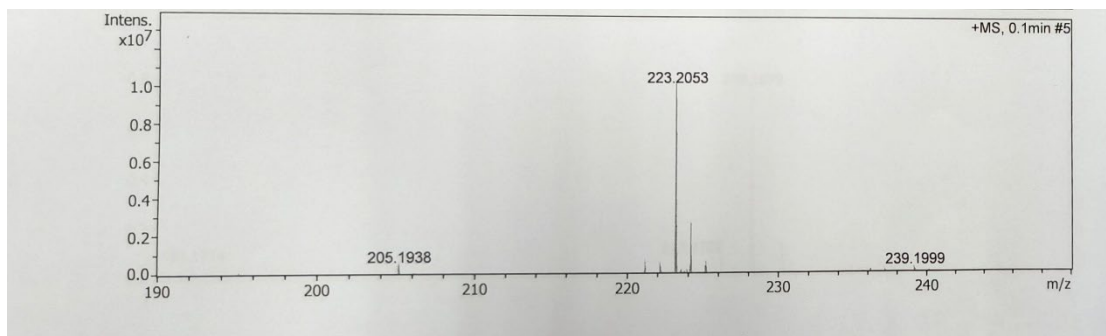
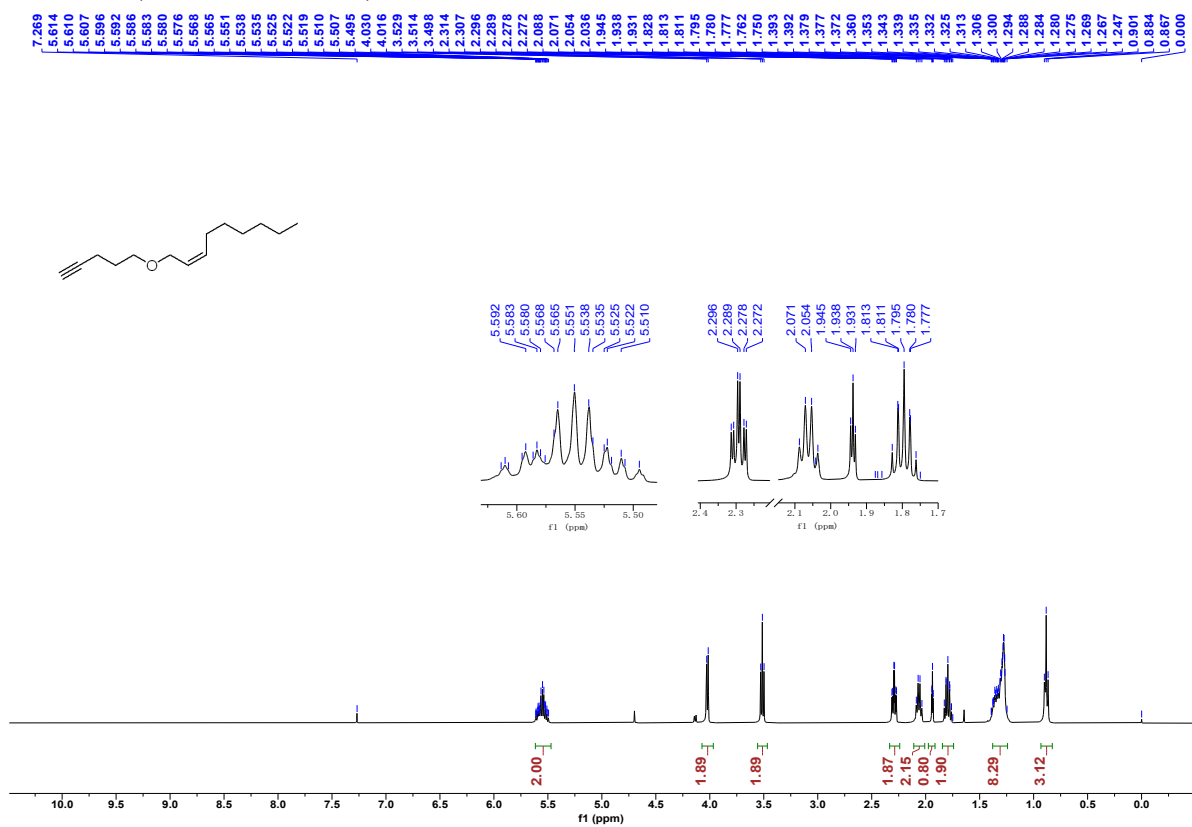
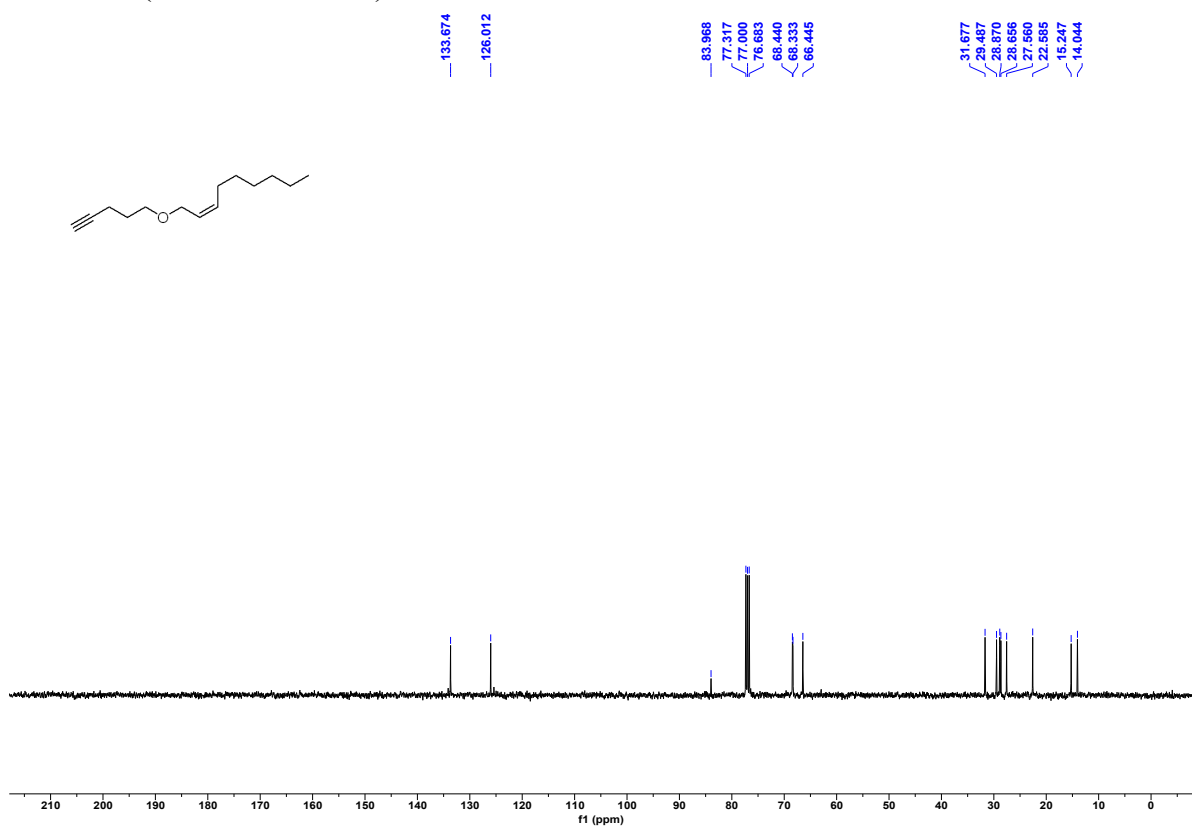


Figure S32. 1H , ^{13}C NMR spectra and HRMS of 10-(Pent-4-yn-1-yloxy)dec-1-ene (*B9*)

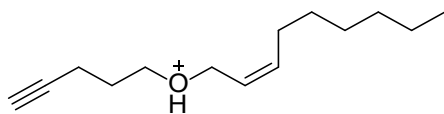
¹H NMR (400 MHz, CDCl₃)



¹³C NMR (101 MHz, CDCl₃)



HRMS (APCI-positive)



Chemical Formula: $C_{14}H_{25}O^+$

m/z: 209.1900 (100.0%), 210.1934 (15.1%), 211.1968 (1.1%)

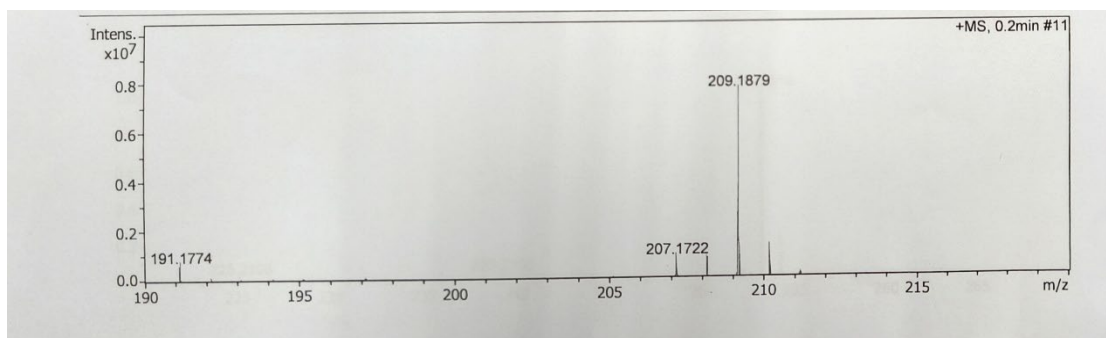
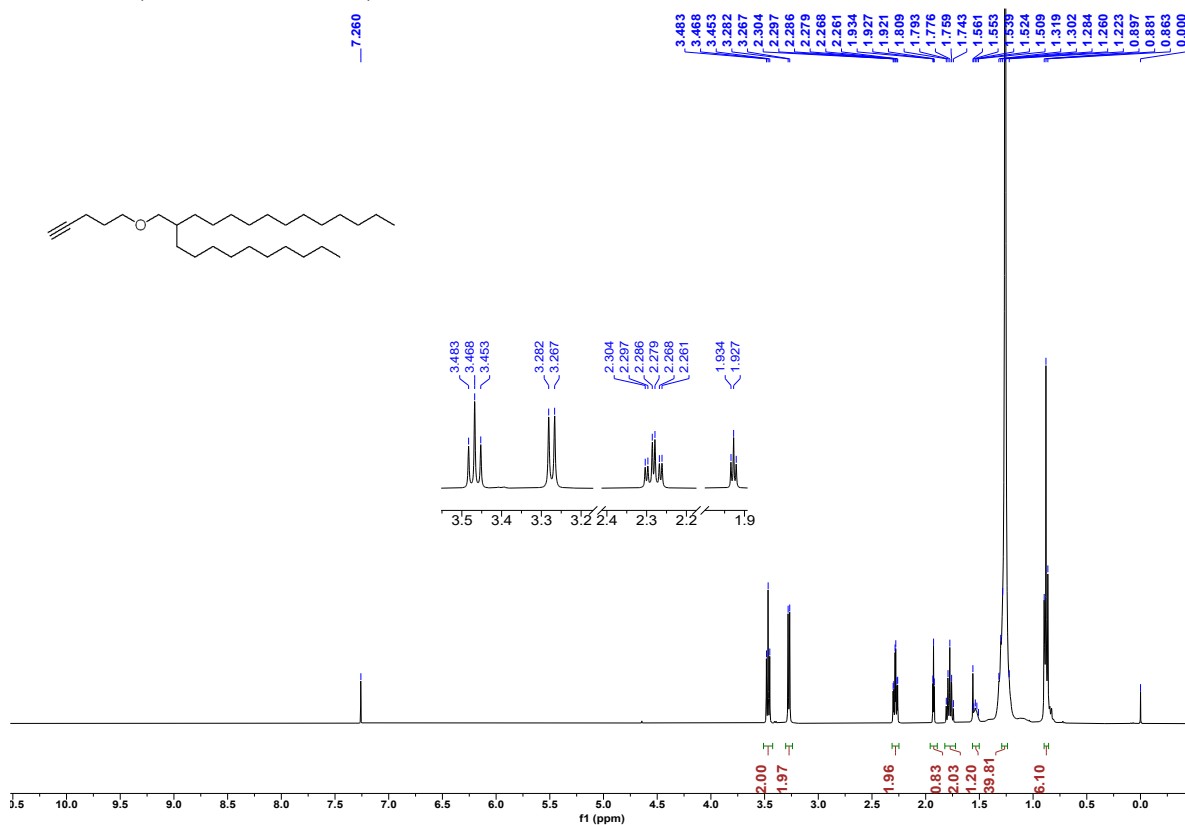


Figure S33. 1H , ^{13}C NMR spectra and HRMS of (Z)-1-(pent-4-yn-1-yloxy)non-2-ene (*B10*)

^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)

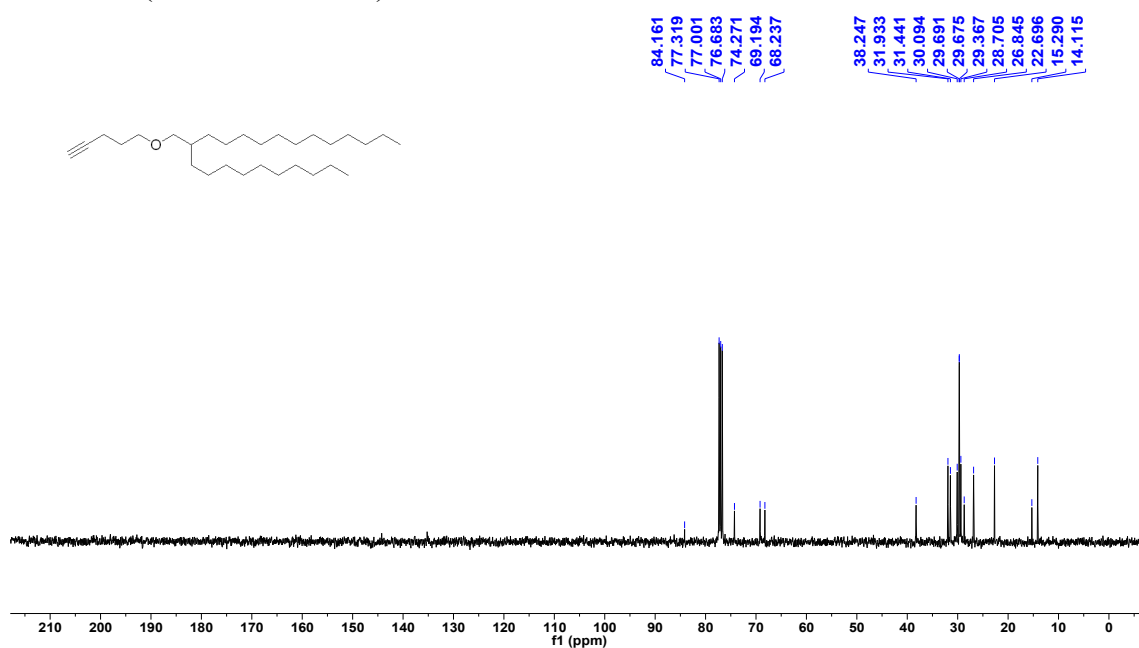
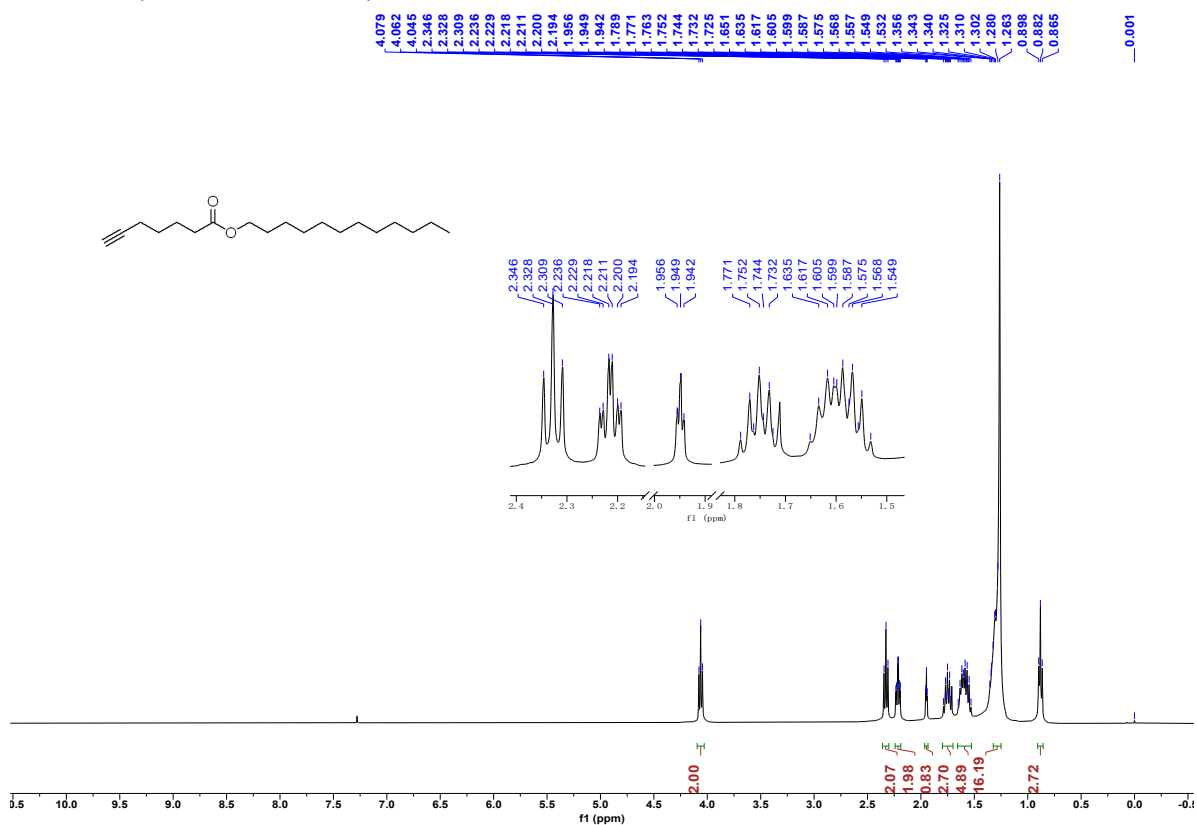
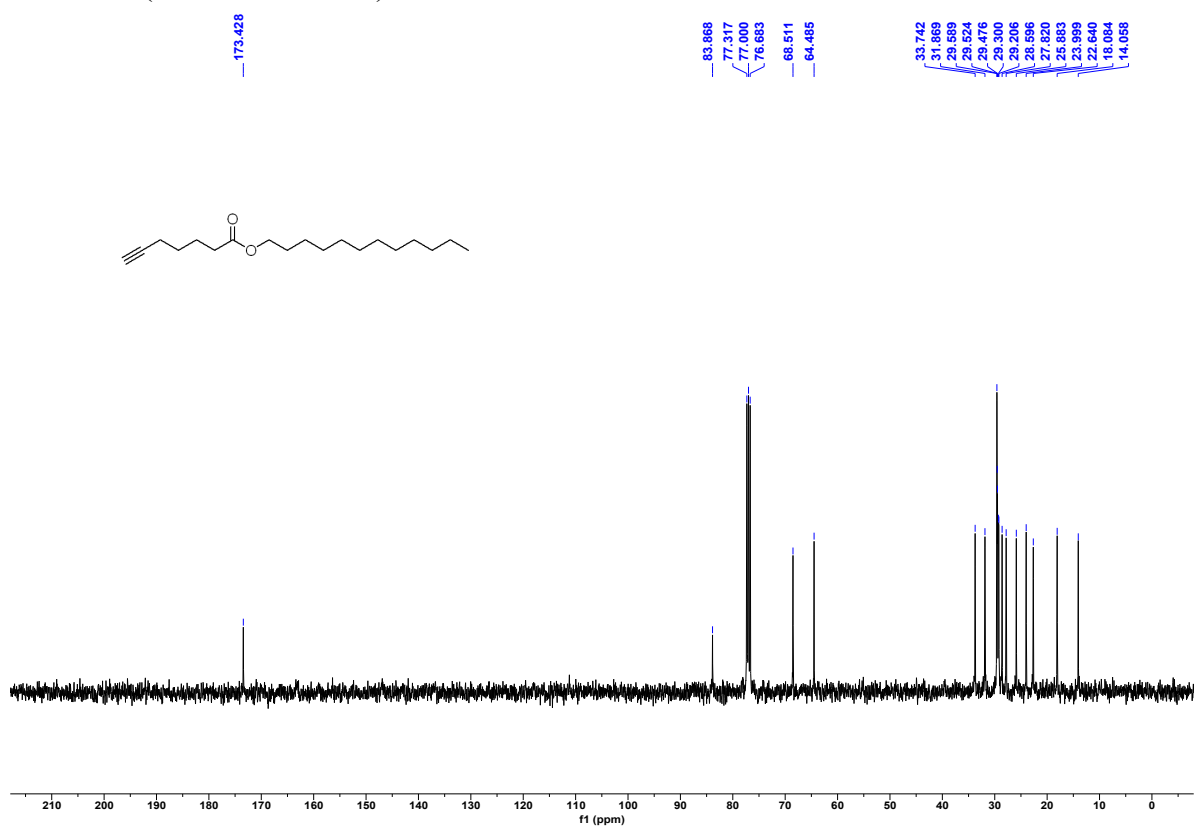


Figure S34. ^1H and ^{13}C NMR spectra of 11-((Pent-4-yn-1-yloxy)methyl)tricosane (*B11*)

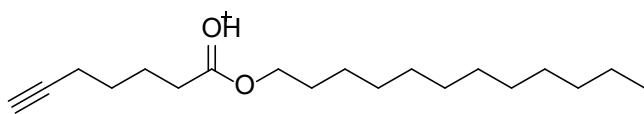
^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)



HRMS (ESI-positive)



Chemical Formula: $\text{C}_{19}\text{H}_{35}\text{O}_2^+$

m/z: 295.2632 (100.0%), 296.2666 (20.5%), 297.2699 (2.0%)

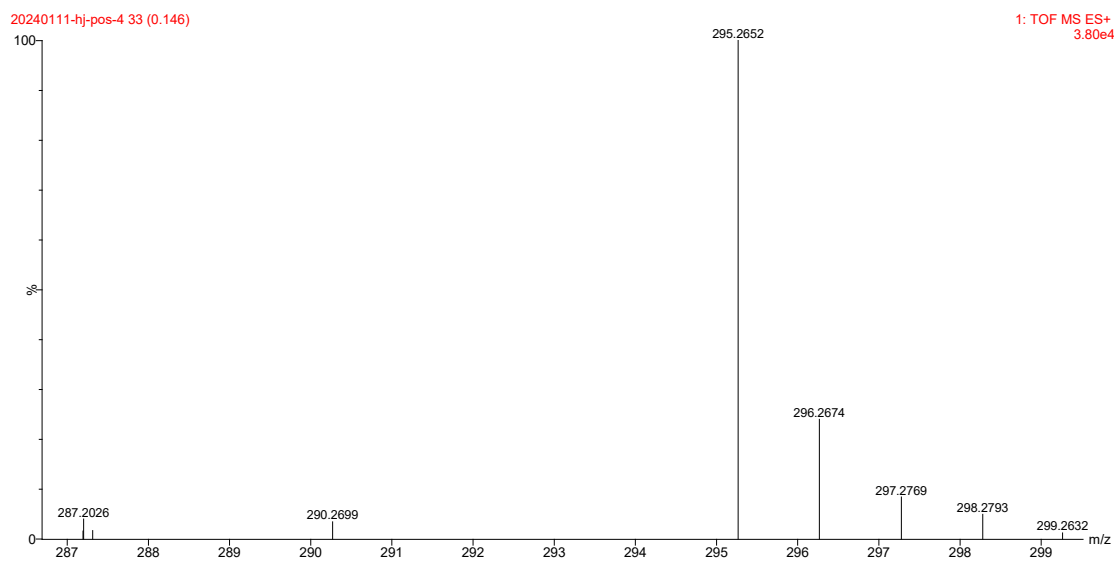
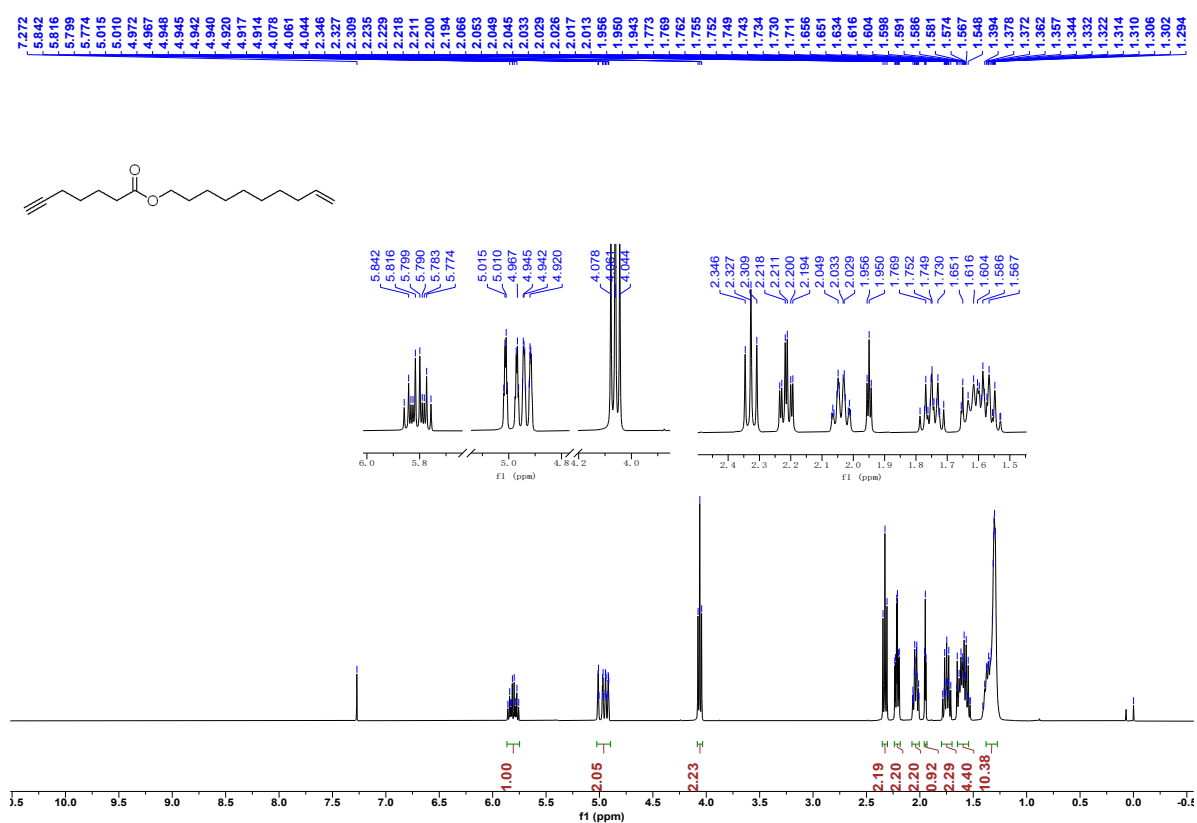
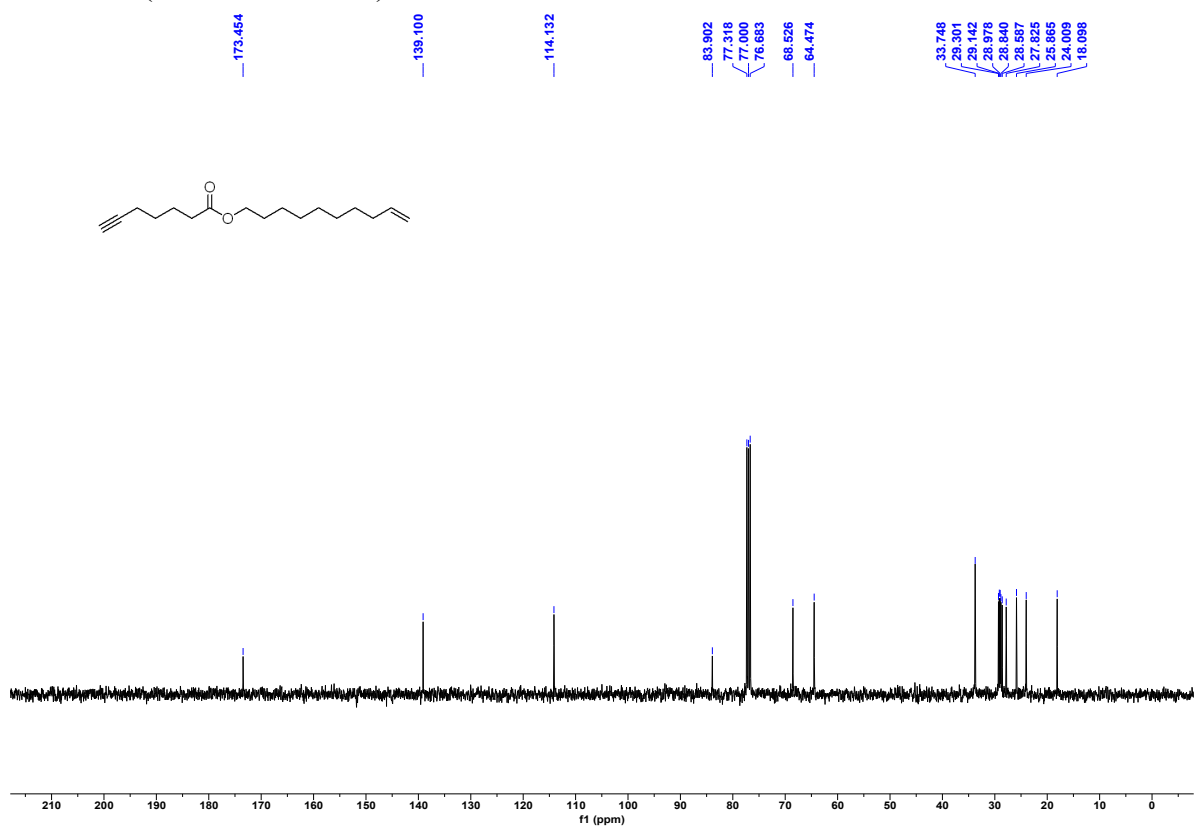


Figure S35 ^1H , ^{13}C NMR spectra and HRMS of Dodecyl hept-6-ynoate (*B12*)

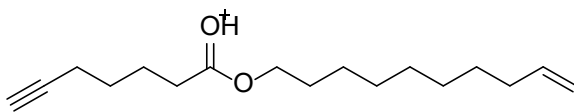
¹H NMR (400 MHz, CDCl₃)



¹³C NMR (101 MHz, CDCl₃)



HRMS (ESI-positive)



Chemical Formula: $C_{17}H_{29}O_2^+$

m/z: 265.2163 (100.0%), 266.2196 (18.4%), 267.2230 (1.6%)

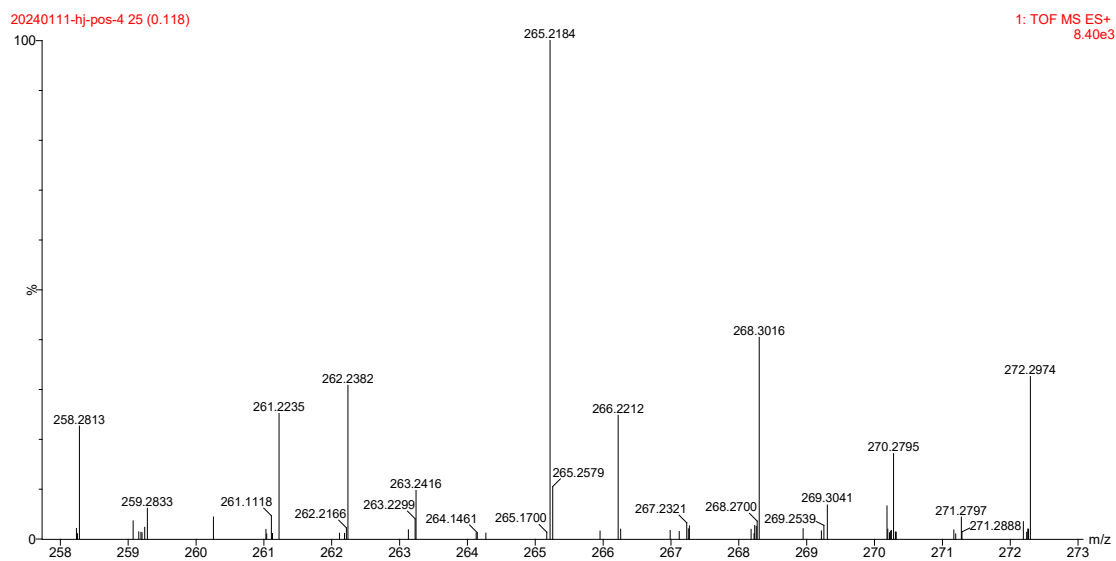
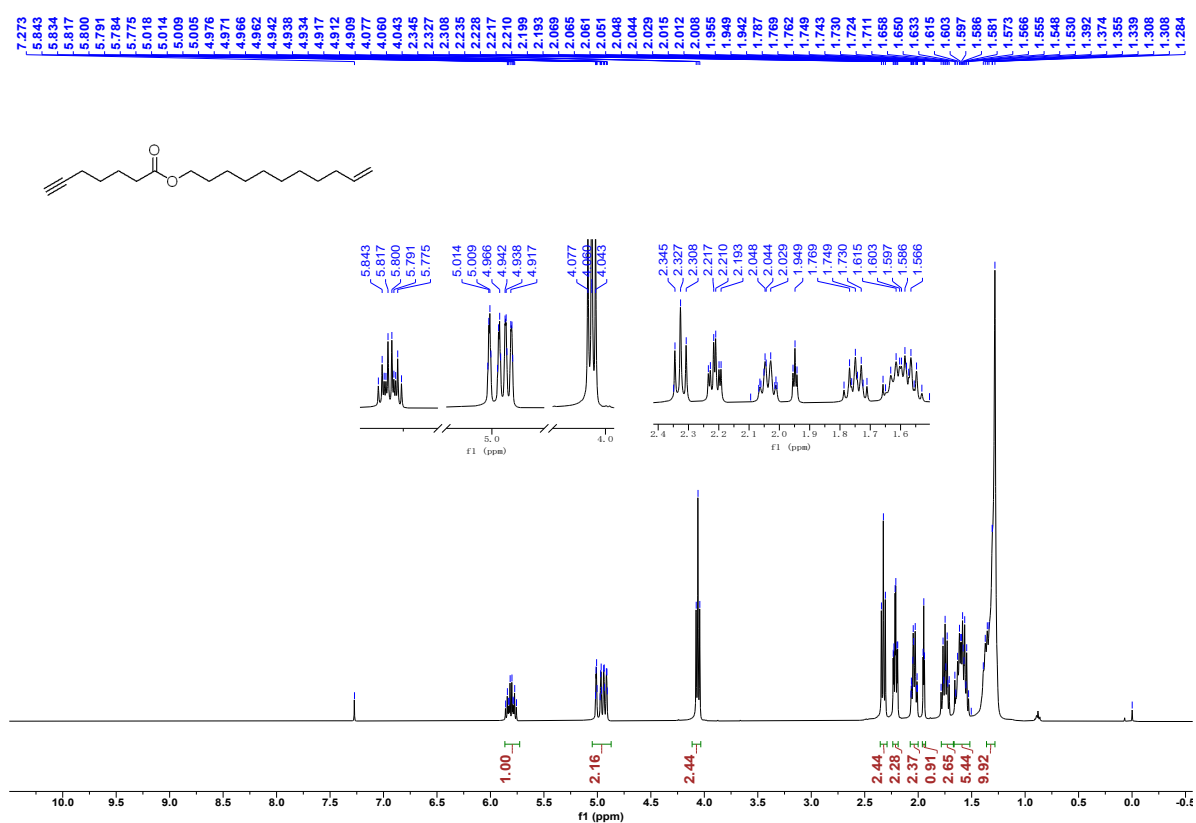
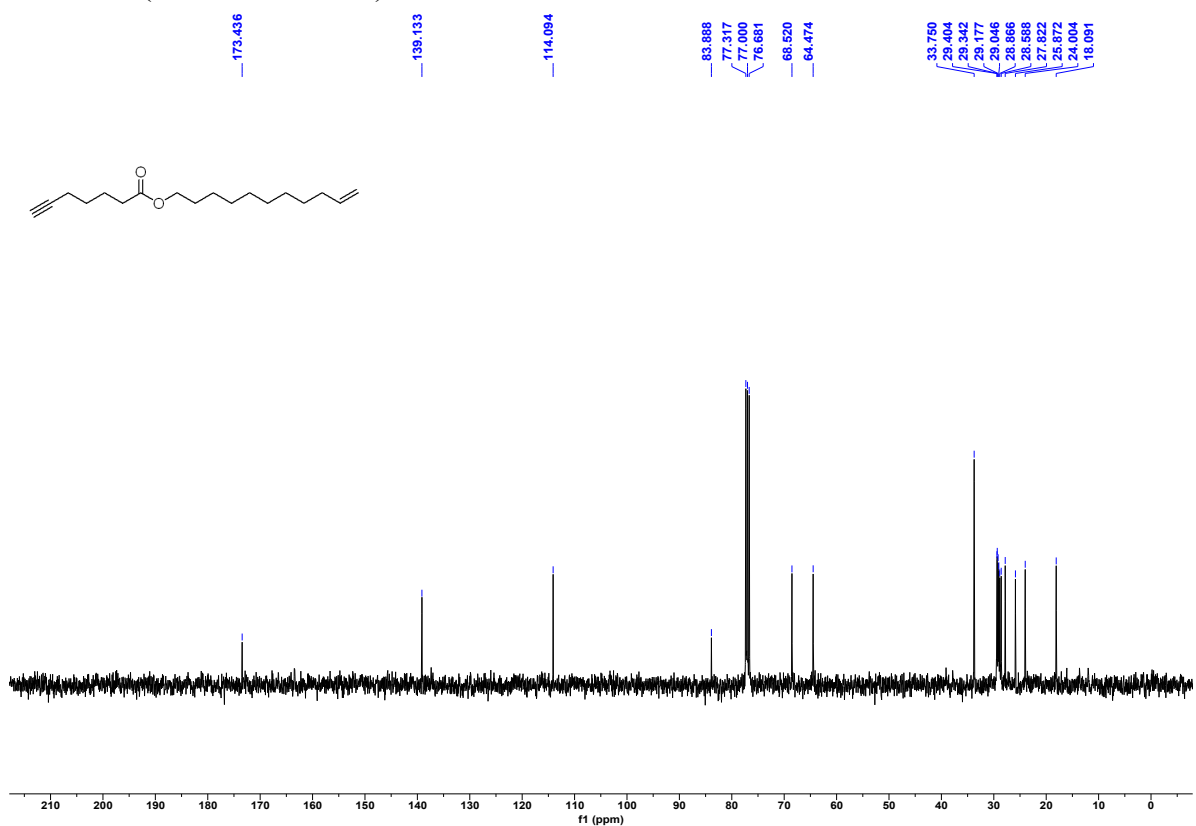


Figure S36. 1H , ^{13}C NMR spectra and HRMS of Dec-9-en-1-yl hept-6-ynoate (*B13*)

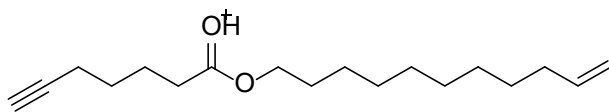
¹H NMR (400 MHz, CDCl₃)



¹³C NMR (101 MHz, CDCl₃)



HRMS (ESI-positive)



Chemical Formula: $C_{18}H_{31}O_2^+$

m/z: 279.2319 (100.0%), 280.2353 (19.5%), 281.2386 (1.8%)

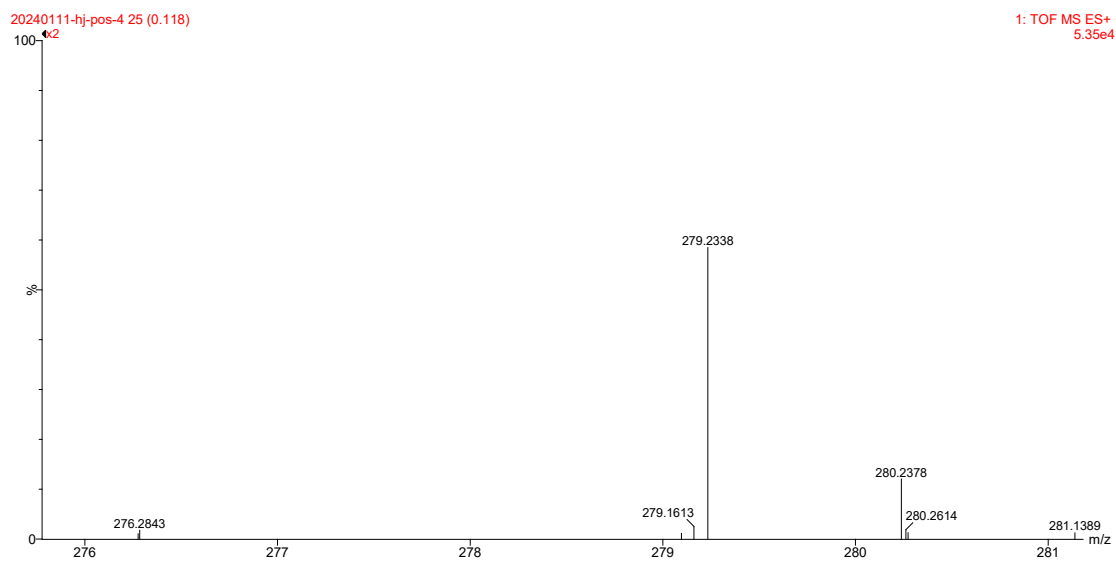
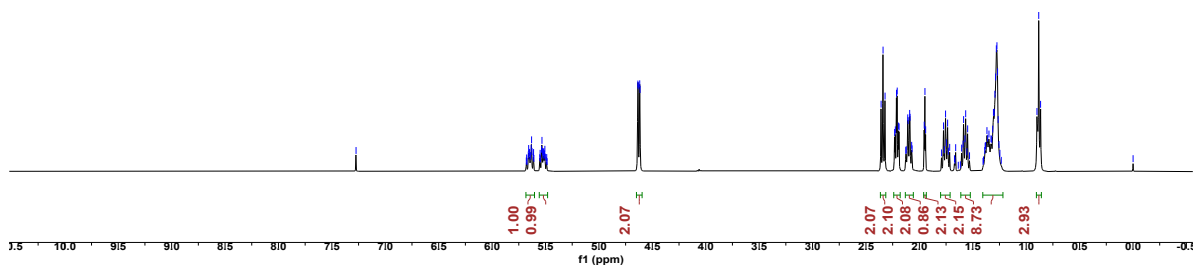
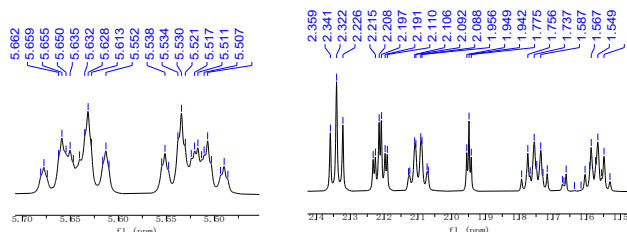
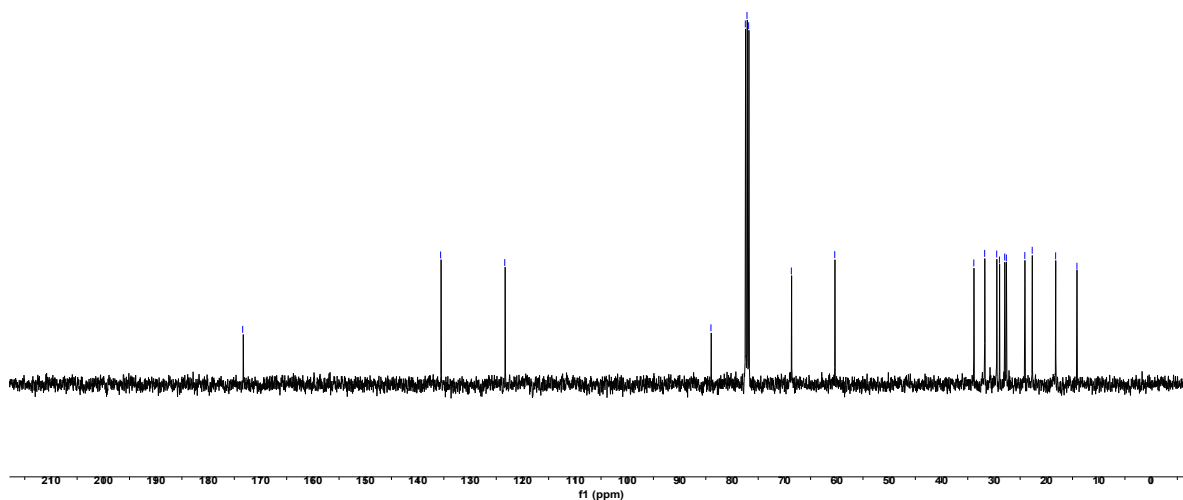


Figure S37. 1H , ^{13}C NMR spectra and HRMS of Undec-10-en-1-yl hept-6-ynoate (*B14*)

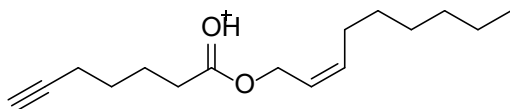
7.275	7.275
5.659	5.659
5.650	5.650
5.635	5.635
5.632	5.632
5.628	5.628
5.613	5.613
5.552	5.552
5.538	5.538
5.534	5.534
5.530	5.530
5.521	5.521
5.517	5.517
5.507	5.507
4.636	4.636
4.633	4.633
4.619	4.619
4.616	4.616
2.359	2.359
2.341	2.341
2.322	2.322
2.233	2.233
2.226	2.226
2.215	2.215
2.208	2.208
1.949	1.949
1.942	1.942
1.793	1.793
1.775	1.775
1.768	1.768
1.758	1.758
1.748	1.748
1.710	1.710
1.707	1.707
1.717	1.717
1.710	1.710
1.661	1.661
1.605	1.605
1.591	1.591
1.587	1.587
1.574	1.574
1.567	1.567
1.556	1.556
1.549	1.549
1.385	1.385
1.389	1.389
1.381	1.381
1.368	1.368
1.361	1.361
1.349	1.349
1.331	1.331
1.324	1.324
1.321	1.321
1.317	1.317
1.307	1.307
1.300	1.300
1.294	1.294
1.287	1.287
1.280	1.280
1.275	1.275
1.269	1.269
1.263	1.263
1.255	1.255
0.900	0.900
0.883	0.883
0.866	0.866



173.256	83.880	33.704
	77.319	31.633
	77.001	29.337
	76.684	28.811
	68.532	27.810
		27.498
		23.972
		22.561
		18.090
		14.024
135.475	60.252	
123.238		



HRMS (ESI-positive)



Chemical Formula: $C_{16}H_{27}O_2^+$

m/z: 251.2006 (100.0%), 252.2040 (17.3%), 253.2073 (1.4%)

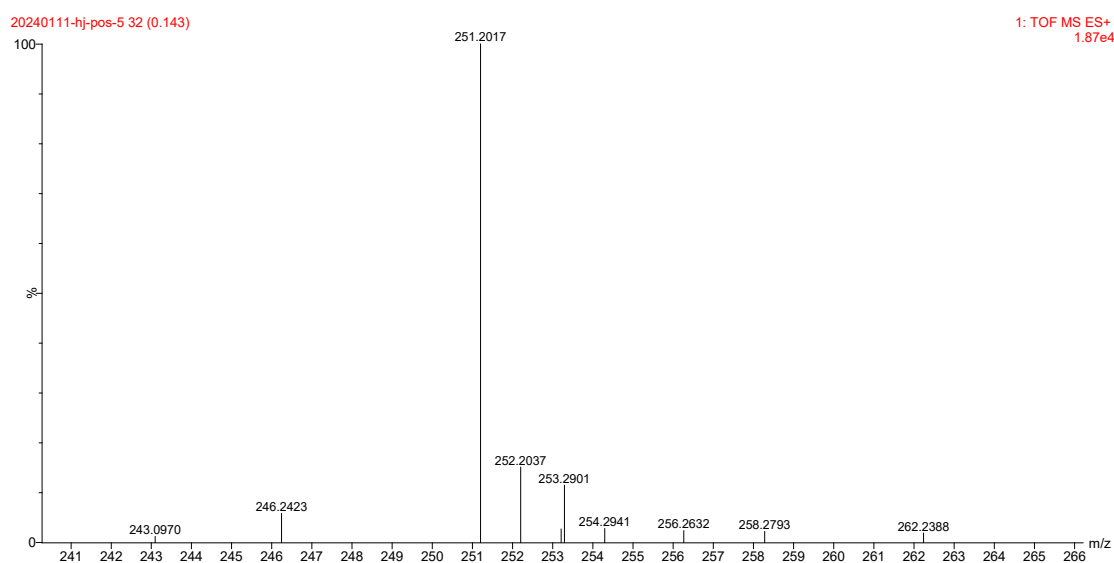


Figure S38. 1H , ^{13}C NMR spectra and HRMS of (*Z*)-non-2-en-1-yl hept-6-ynoate (*B15*)

CCCCCCCCCCCCCCCCCOC(=O)CCCC#C

Chemical Structure: 1-octadecyl 4-pentynoate

¹H NMR Data:

Chemical Shift (ppm)	Integration
7.264	2.00
3.980	1.97
2.364	1.96
2.352	0.78
2.334	2.03
2.315	2.27
2.236	41.35
2.229	5.93
2.218	
2.212	
2.201	
2.194	
1.949	
1.942	
1.936	
1.930	
1.922	
1.905	
1.892	
1.886	
1.884	
1.875	
1.868	
1.857	
1.848	
1.832	
1.829	
1.825	
1.822	
1.800	
0.886	
0.881	
0.864	
0.000	

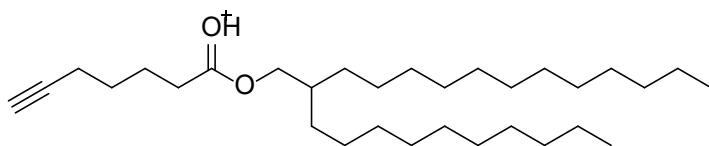
CCCCC#CCCC(=O)OCC(CCCCCCCCCCCCCCCCCC)CCCCCCCCCCCCCCCC

Chemical structure: CCCCC#CCCC(=O)OCC(CCCCCCCCCCCCCCCCCC)CCCCCCCCCCCCCCCC

¹³C NMR peaks (ppm):

- 173.580
- 83.890
- 77.317
- 77.000
- 76.683
- 68.537
- 67.183
- 37.276
- 33.835
- 31.912
- 31.257
- 29.948
- 29.682
- 29.647
- 29.596
- 29.346
- 27.857
- 26.691
- 24.063
- 22.679
- 18.125
- 14.102

HRMS (ESI-positive)



Chemical Formula: $C_{31}H_{59}O_2^+$

m/z: 463.4510 (100.0%), 464.4544 (33.5%), 465.4577 (5.4%)

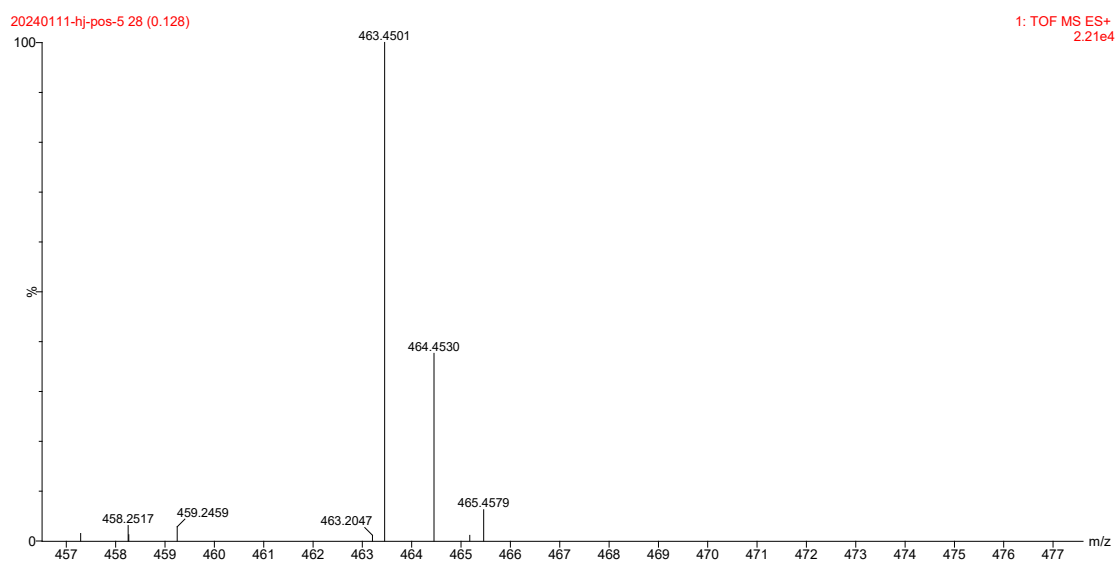
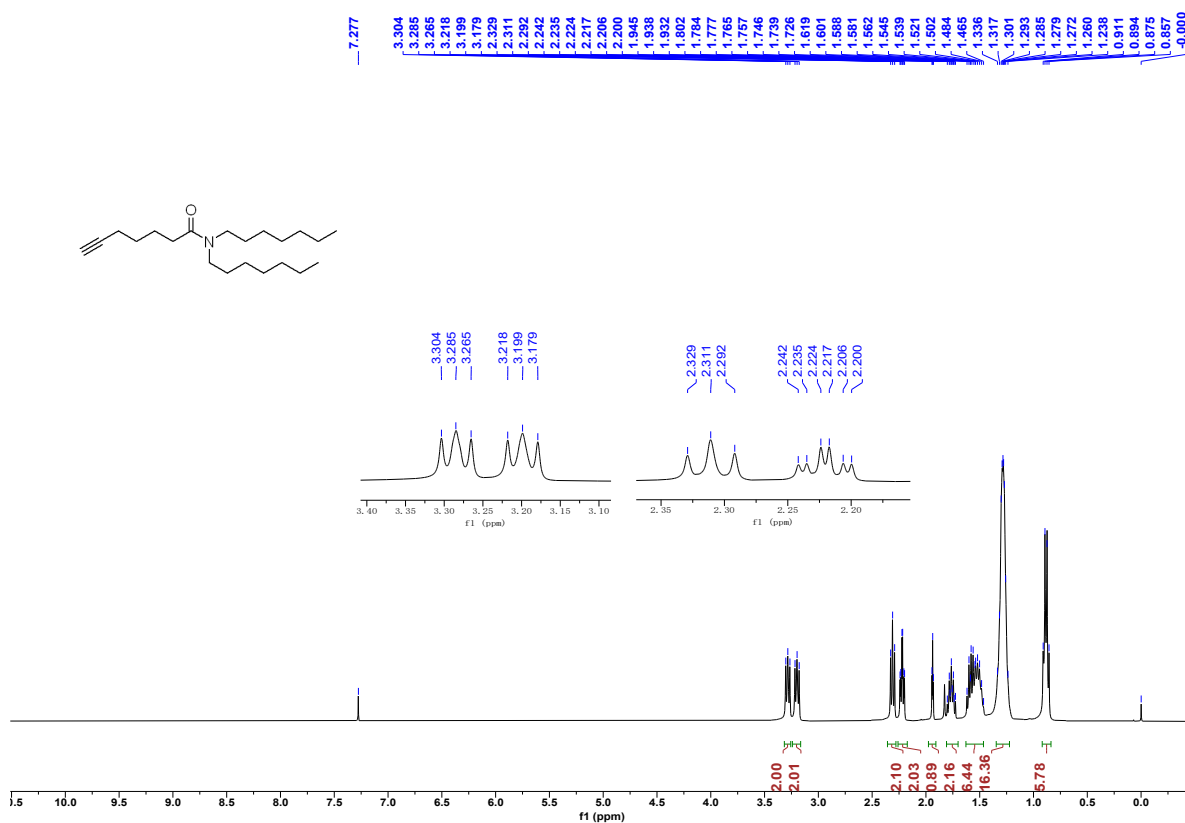
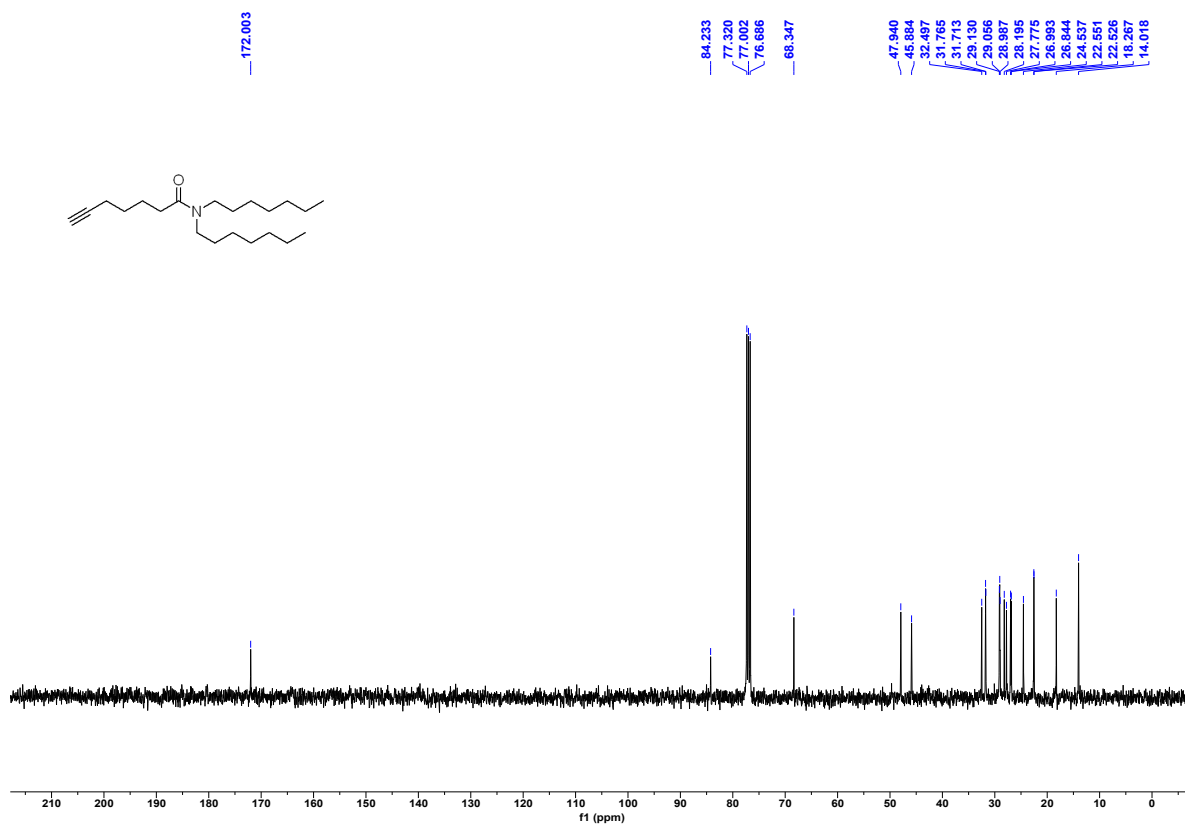


Figure S39. 1H , ^{13}C NMR spectra and HRMS of 2-Decyltetradecyl hept-6-ynoate (*B16*)

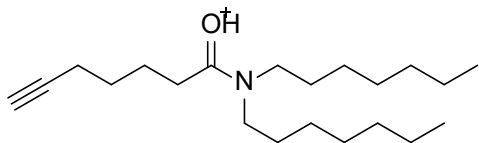
^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)



HRMS (ESI-positive)



Chemical Formula: $C_{21}H_{40}NO^+$

m/z: 322.3105 (100.0%), 323.3138 (22.7%), 324.3172 (2.5%)

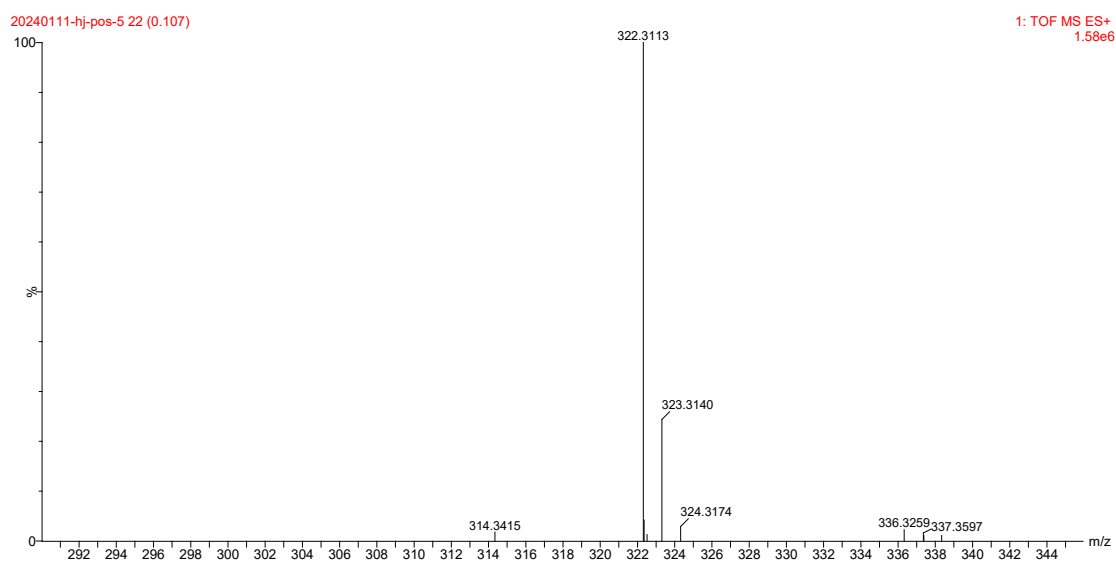
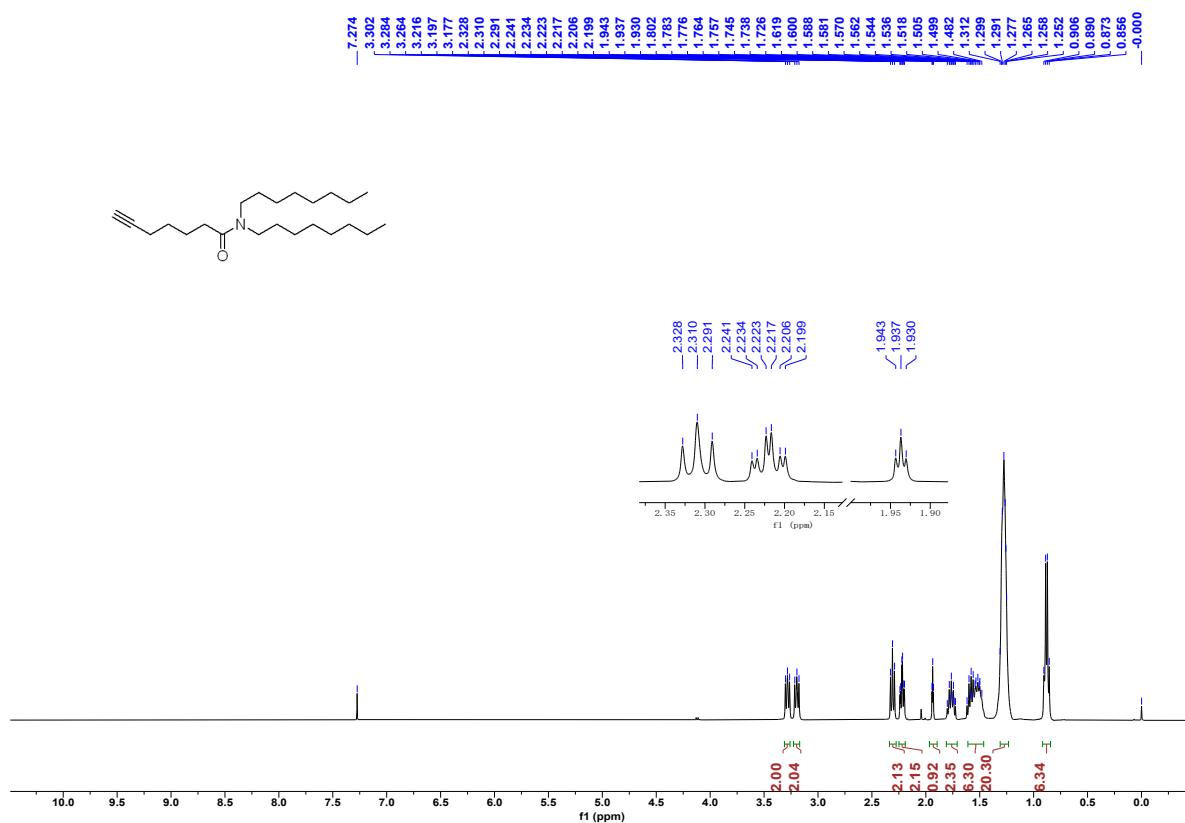
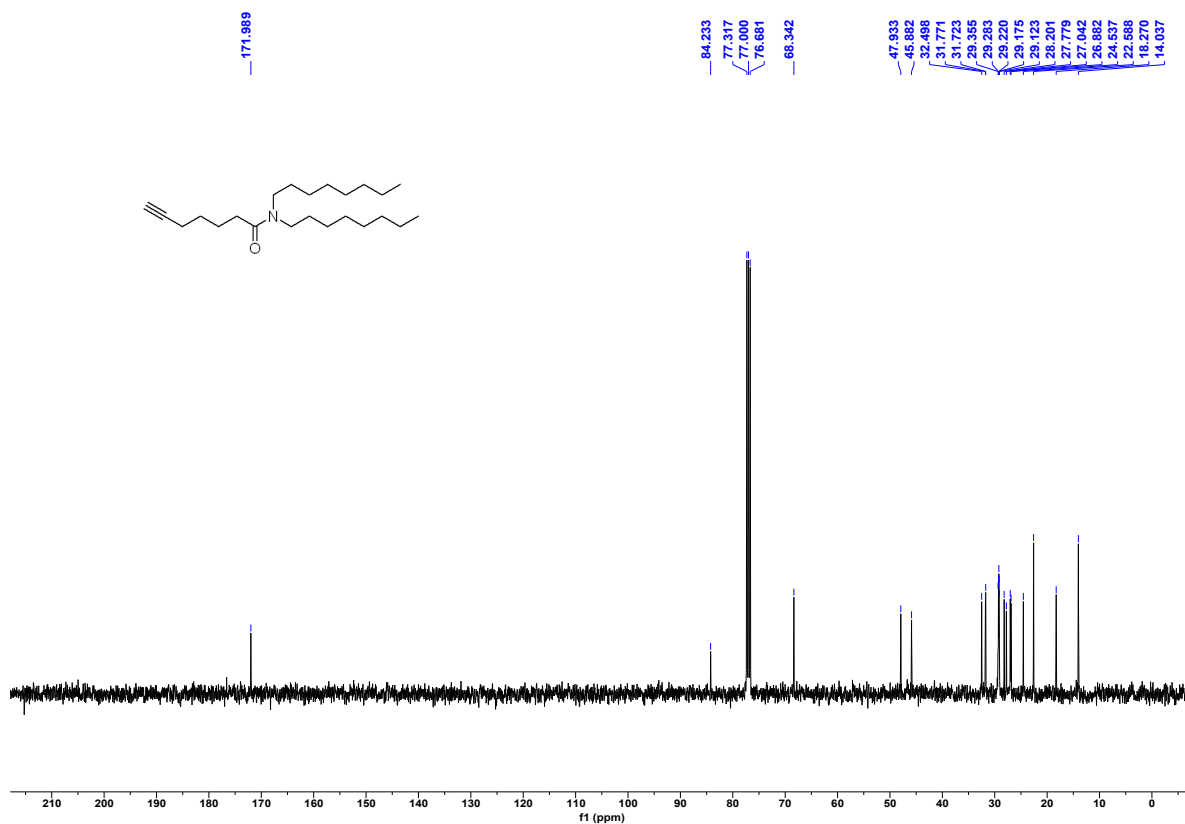


Figure S40. 1H , ^{13}C NMR spectra and HRMS of *N,N*-diheptylhept-6-ynamide (*B17*)

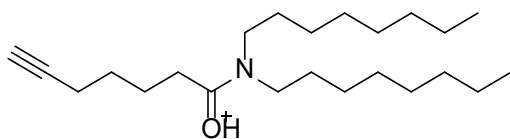
^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)



HRMS (ESI-positive)



Chemical Formula: $C_{23}H_{44}NO^+$

m/z: 350.3418 (100.0%), 351.3451 (24.9%), 352.3485 (3.0%)

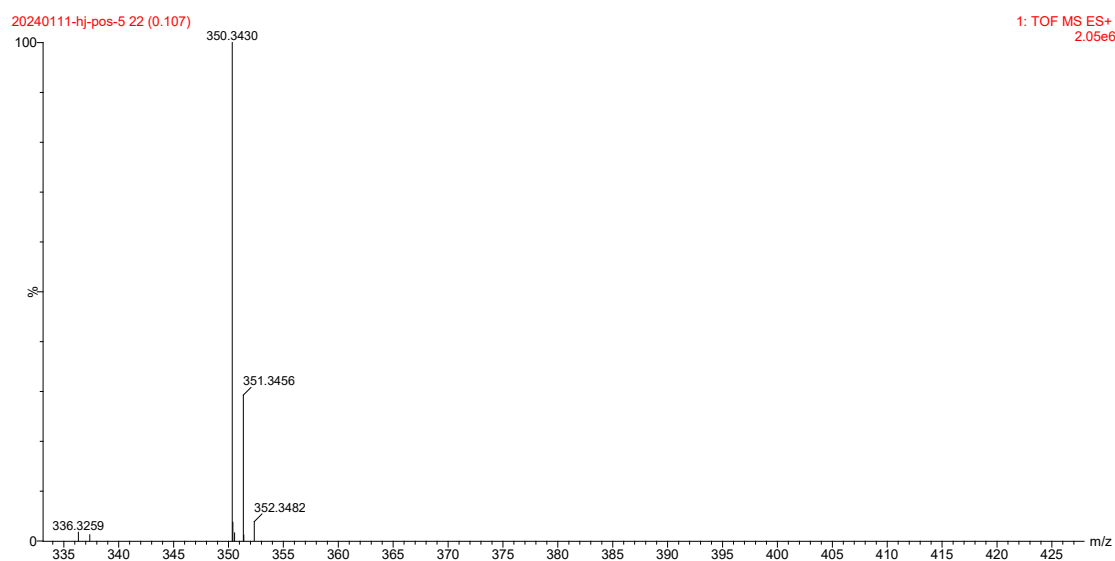
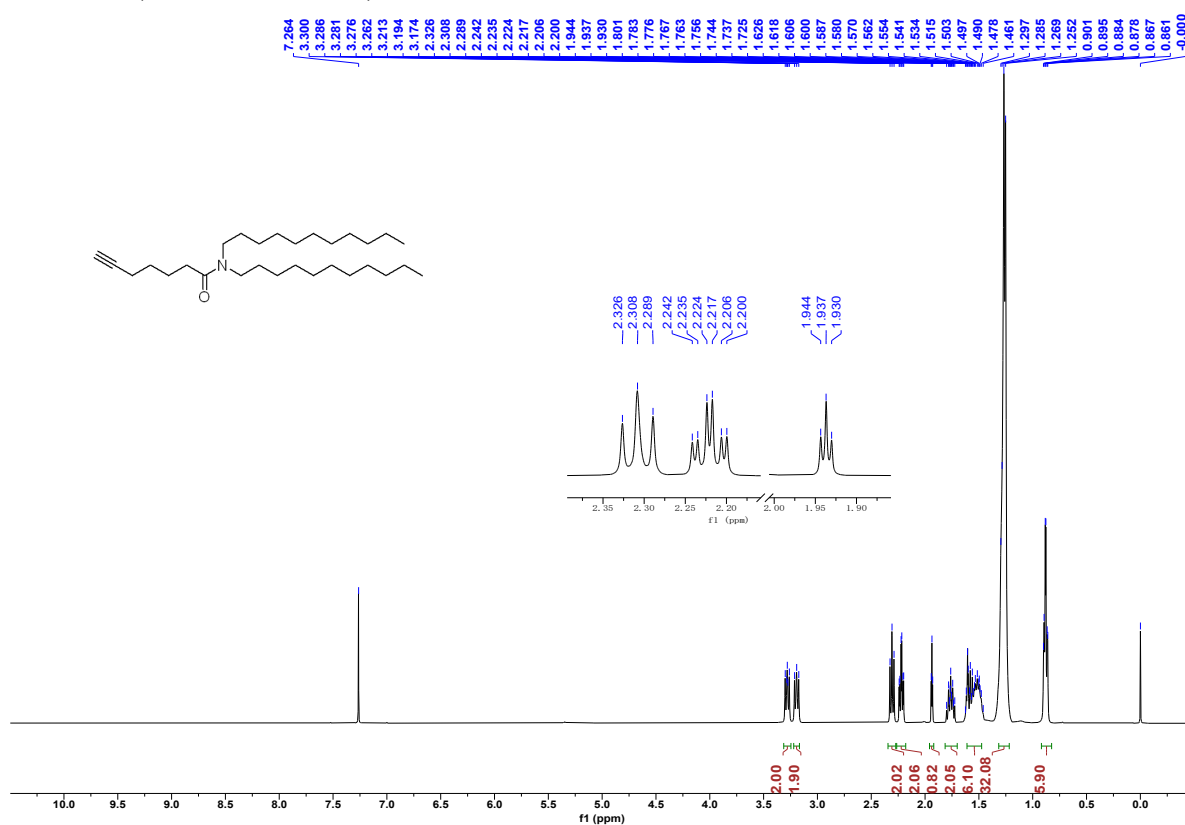
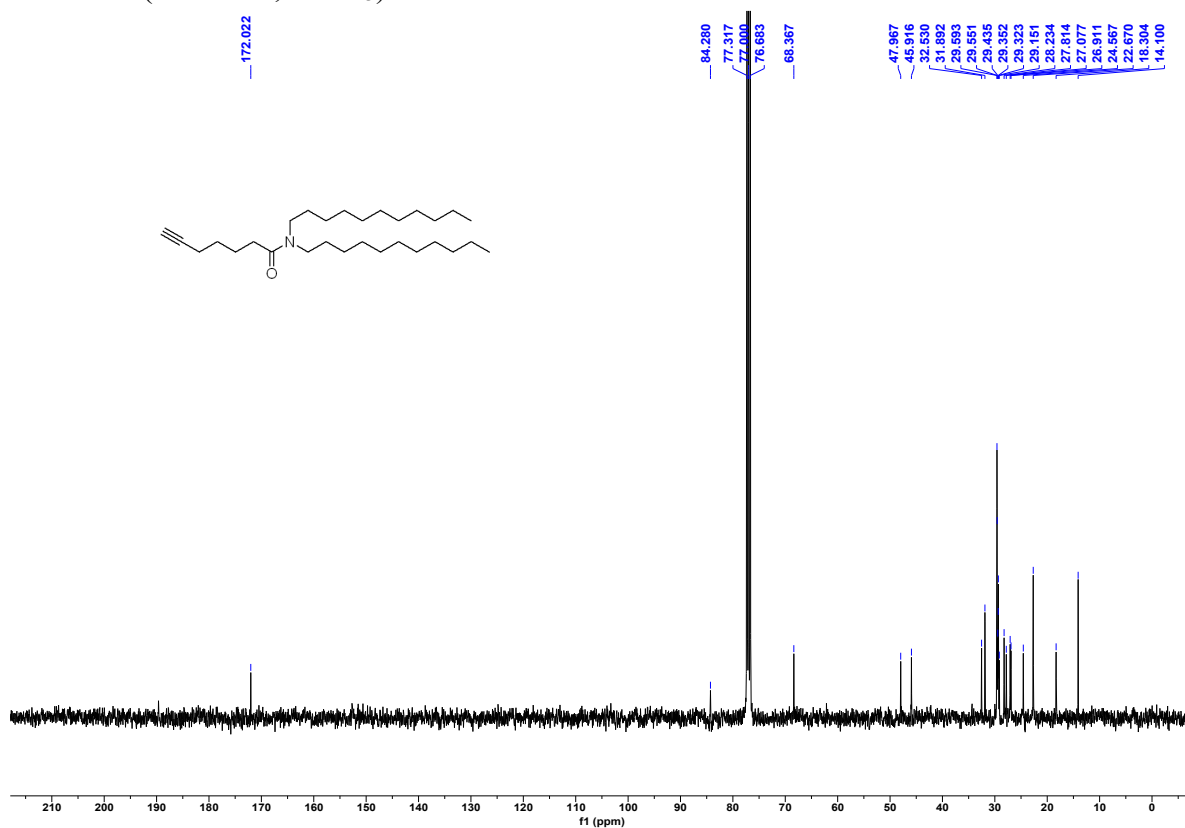


Figure S41. 1H , ^{13}C NMR spectra and HRMS of *N,N*-dioctylhept-6-ynamide (*B18*)

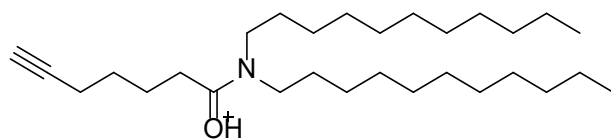
^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)



HRMS (ESI-positive)



Chemical Formula: $\text{C}_{29}\text{H}_{56}\text{NO}^+$

m/z: 434.4357 (100.0%), 435.4390 (31.4%), 436.4424 (4.7%)

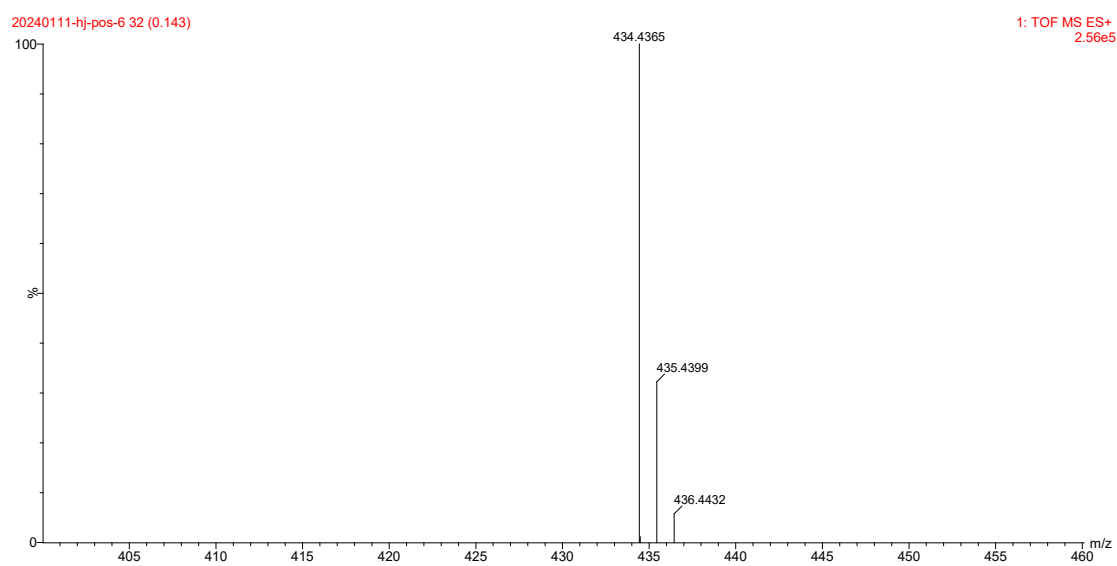
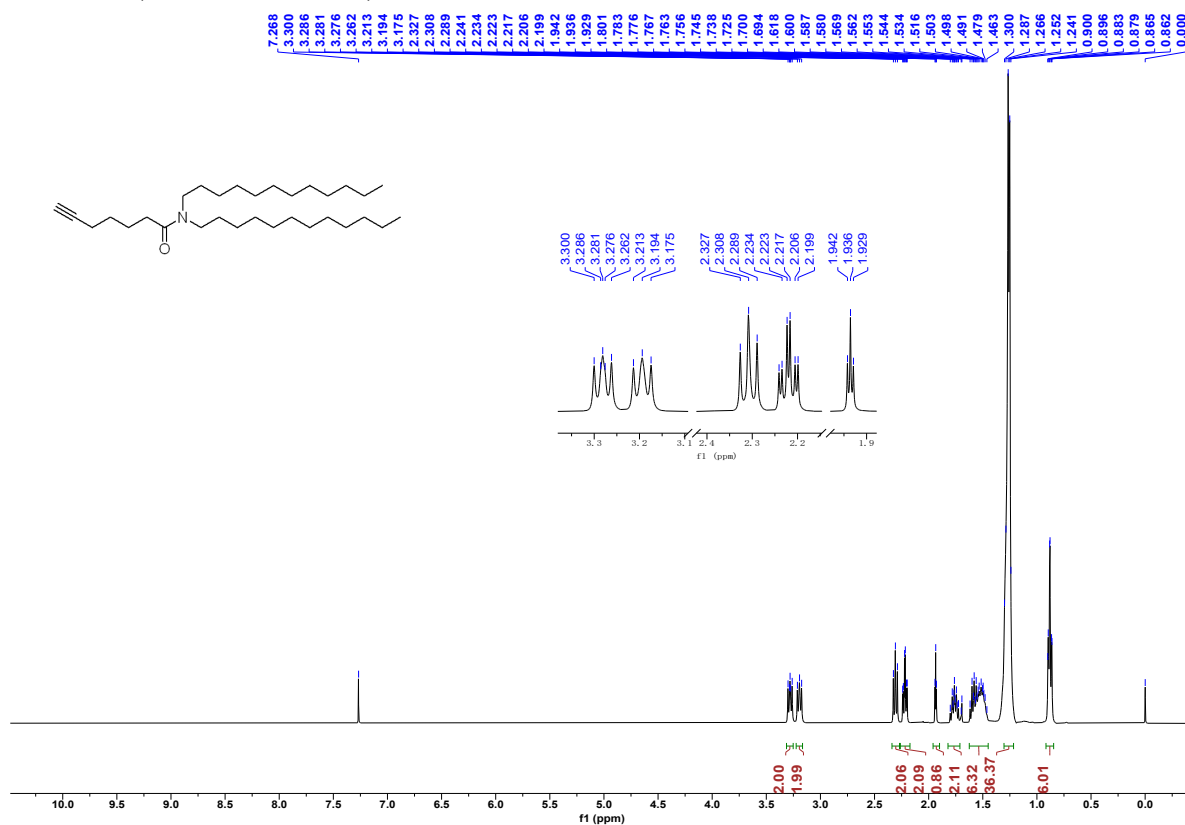
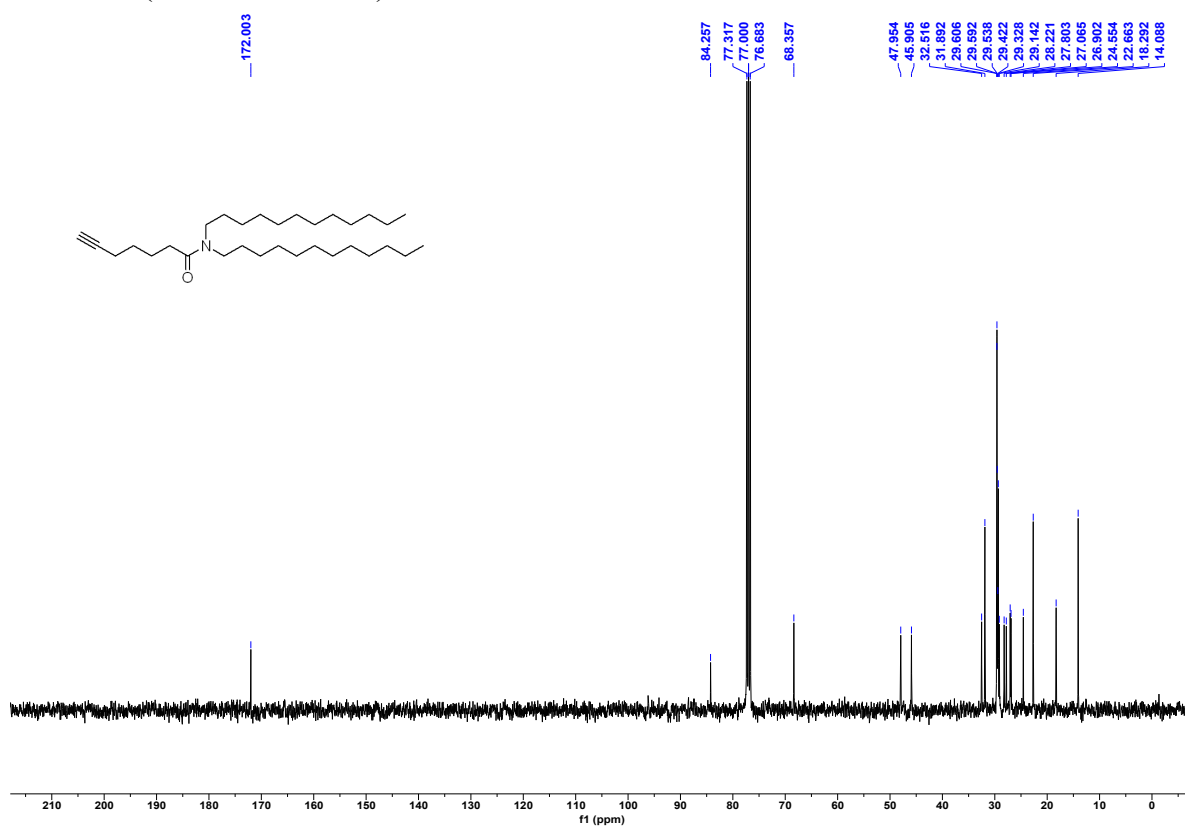


Figure S42. ^1H , ^{13}C NMR spectra and HRMS of *N,N*-diundecylhept-6-ynamide (*B19*)

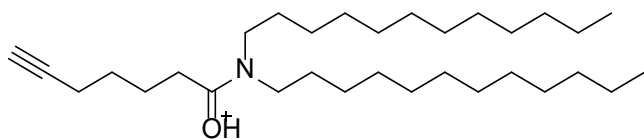
^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)



HRMS (ESI-positive)



Chemical Formula: $\text{C}_{31}\text{H}_{60}\text{NO}^+$

m/z : 462.4670 (100.0%), 463.4703 (33.5%), 464.4737 (5.4%)

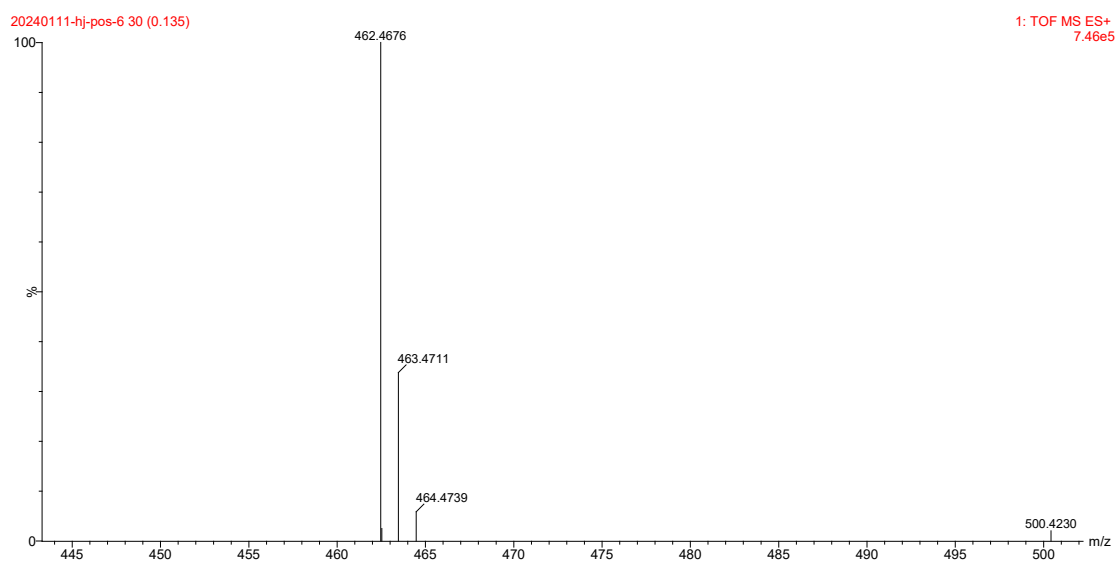
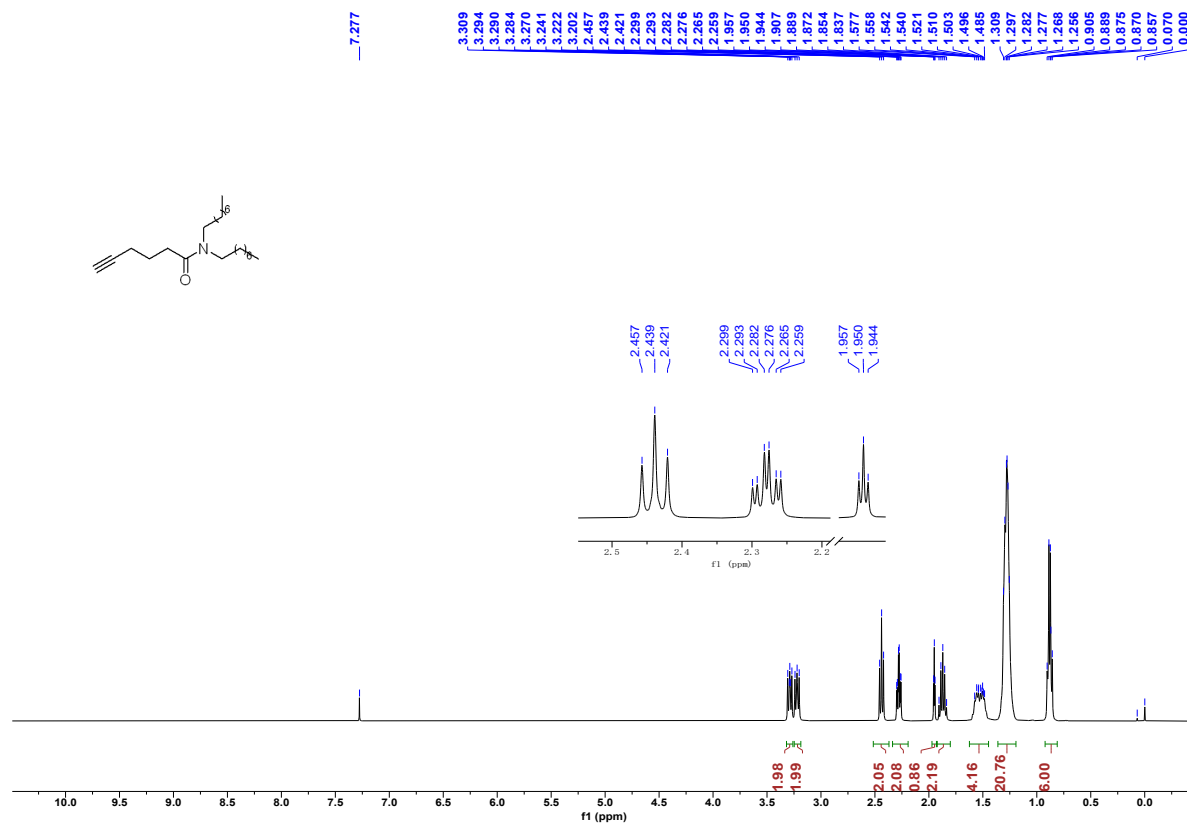
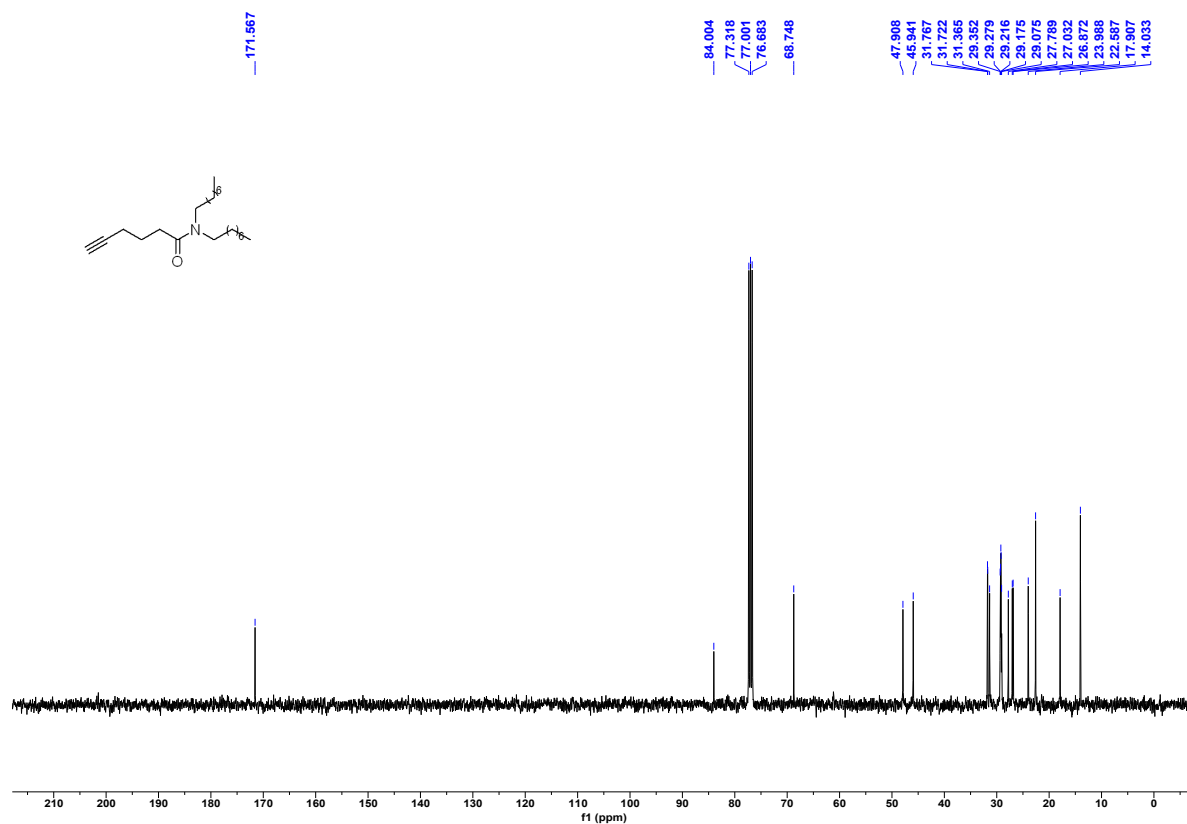


Figure S43. ^1H , ^{13}C NMR spectra and HRMS of *N,N*-didodecylhept-6-ynamide (*B20*)

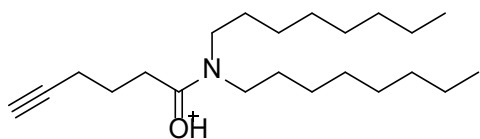
¹H NMR (400 MHz, CDCl₃)



¹³C NMR (101 MHz, CDCl₃)



HRMS (ESI-positive)



Chemical Formula: $C_{22}H_{42}NO^+$

m/z: 336.3261 (100.0%), 337.3295 (23.8%), 338.3328 (2.7%)

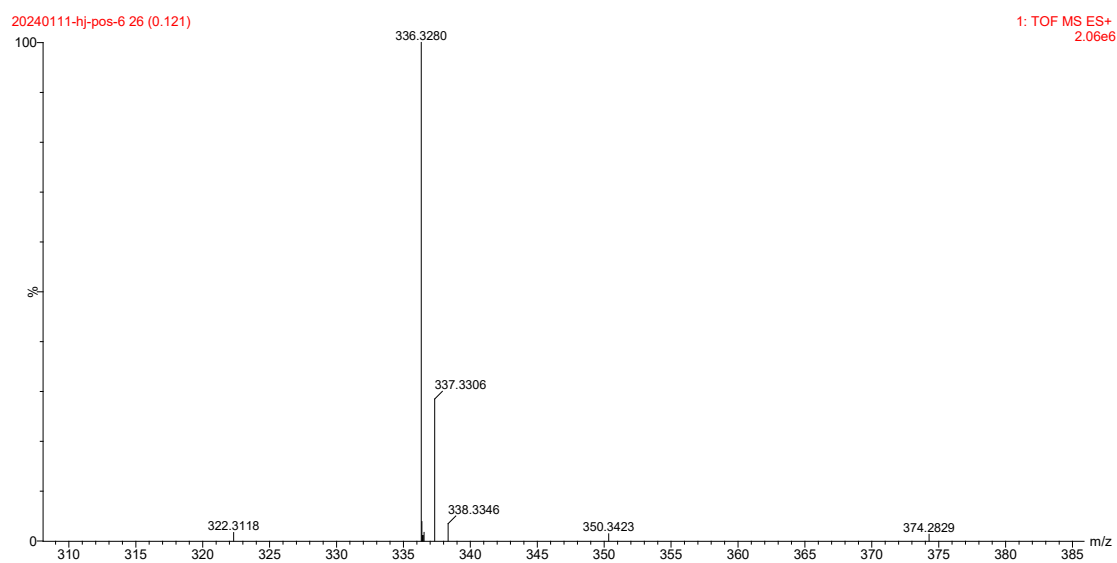


Figure S44. 1H , ^{13}C NMR spectra and HRMS of *N,N*-dioctylhex-5-ynamide (*B2I*)

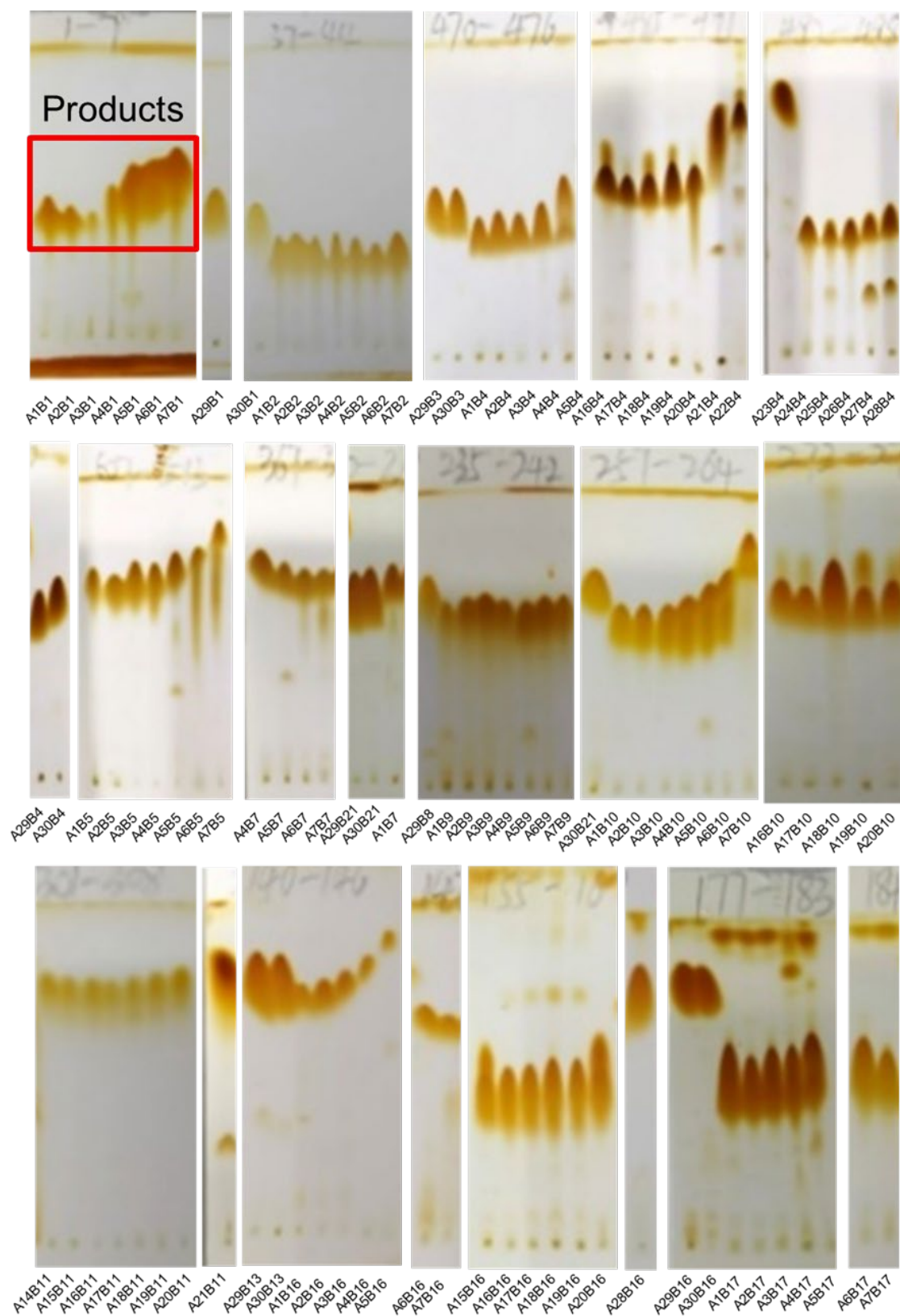
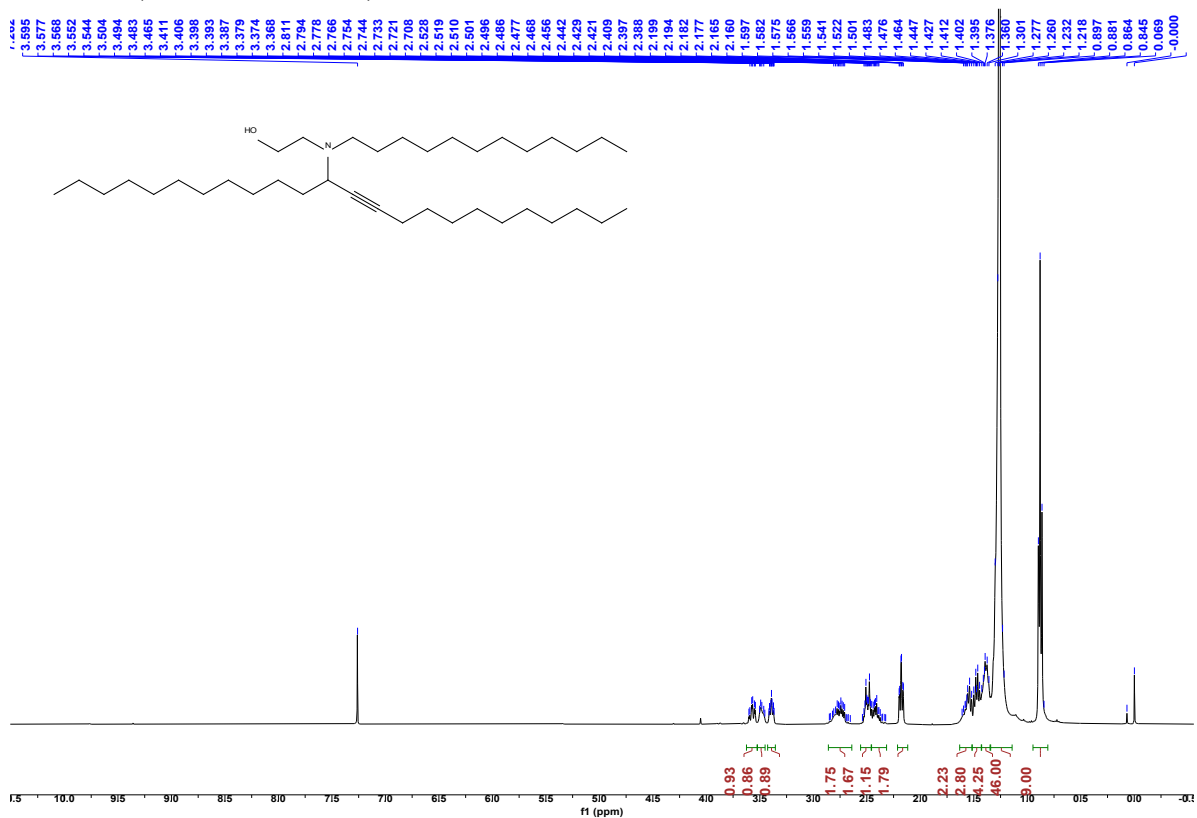
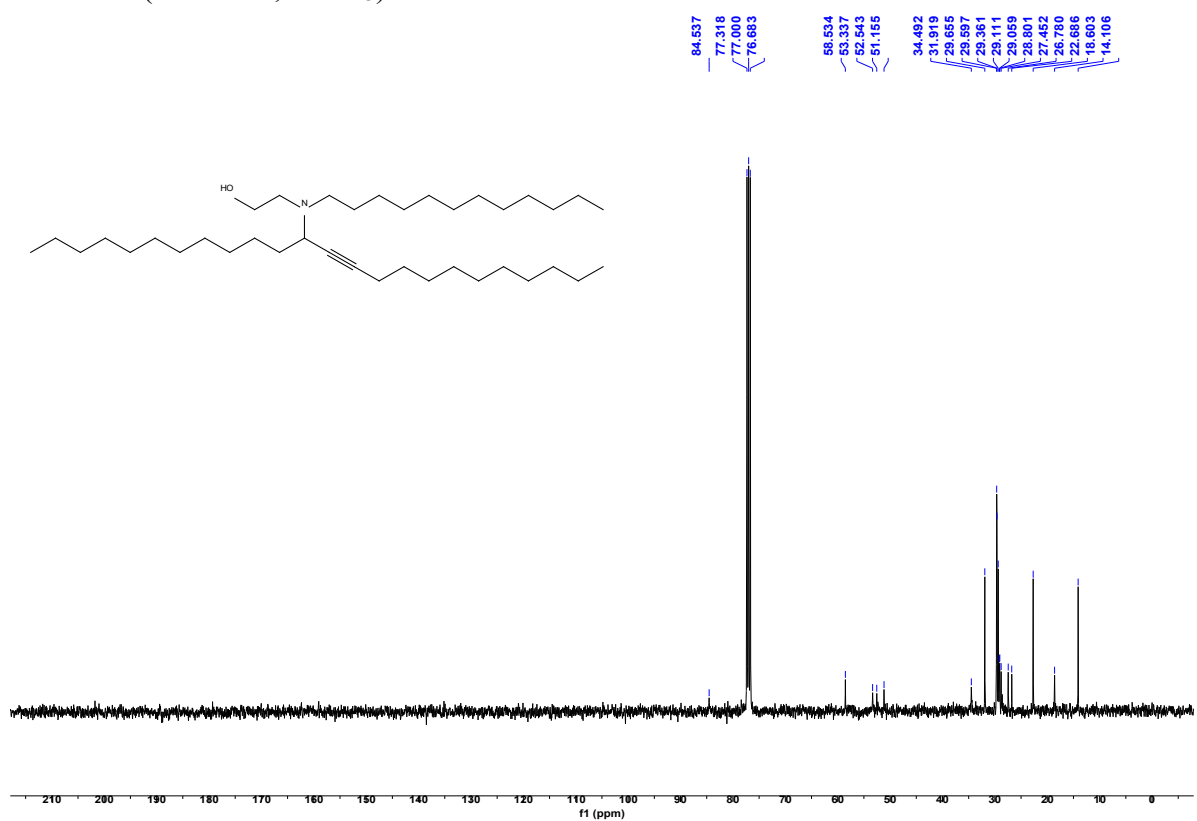


Figure S45. Representative thin layer chromatography (TLC) plates of A^3 coupling reactions.

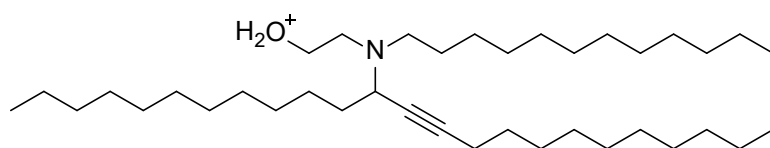
¹H NMR (400 MHz, CDCl₃)



¹³C NMR (101 MHz, CDCl₃)



HRMS (ESI-positive)



Chemical Formula: $\text{C}_{38}\text{H}_{76}\text{NO}^+$

m/z : 562.5922 (100.0%), 563.5955 (41.1%), 564.5989 (8.2%), 565.6023 (1.1%)

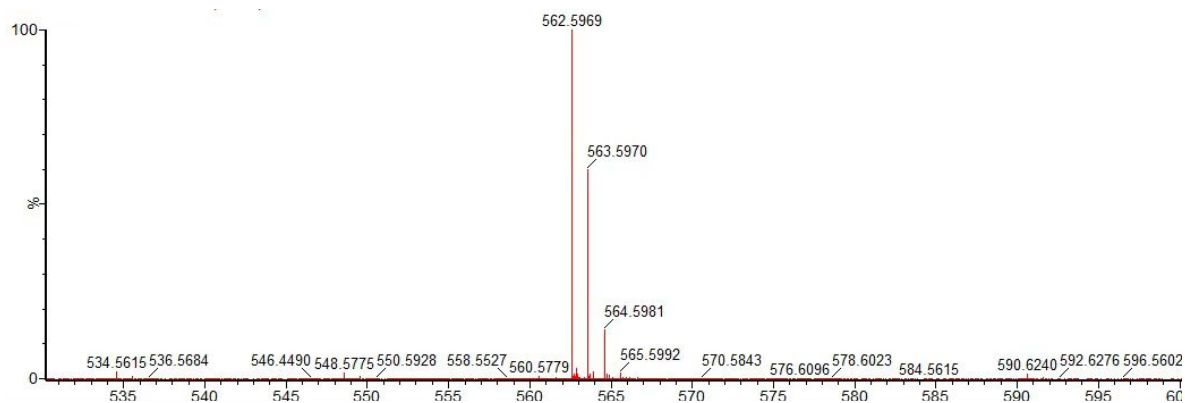


Figure S46. ^1H , ^{13}C NMR spectra and HRMS of 2-(Dodecyl(tetracos-13-yn-12-yl)amino)ethan-1-ol (*2a*)

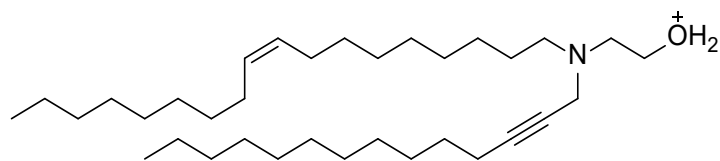
Chemical structure: CCCCC/C=C\CCCCC/C#C/CN(CO)CCCCC#C

¹H NMR spectrum (400 MHz, CDCl₃) showing peaks from 0.0 to 7.3 ppm. The spectrum includes an inset zoom of the 5.28-5.42 ppm region. Integration values are provided below the peaks.

Chemical Shift (ppm)	Integration
7.260	2.00
5.392	2.31
5.388	2.29
5.383	2.38
5.379	2.47
5.369	2.12
5.356	3.92
5.343	6.14
5.330	36.30
5.319	8.26
5.304	
3.575	
3.562	
3.549	
3.373	
3.367	
3.362	
2.667	
2.653	
2.640	
2.497	
2.478	
2.175	
2.170	
2.158	
2.016	
2.001	
1.985	
1.522	
1.506	
1.487	
1.481	
1.451	
1.437	
1.421	
1.390	
1.377	
1.369	
1.354	
1.344	
1.337	
1.324	
1.296	
1.284	
1.260	
1.253	
1.117	
0.894	
0.877	
0.660	
0.004	

[illegible]

HRMS (ESI-positive)



Chemical Formula: $C_{34}H_{66}NO^+$

m/z: 504.5139 (100.0%), 505.5173 (36.8%), 506.5207 (6.6%)

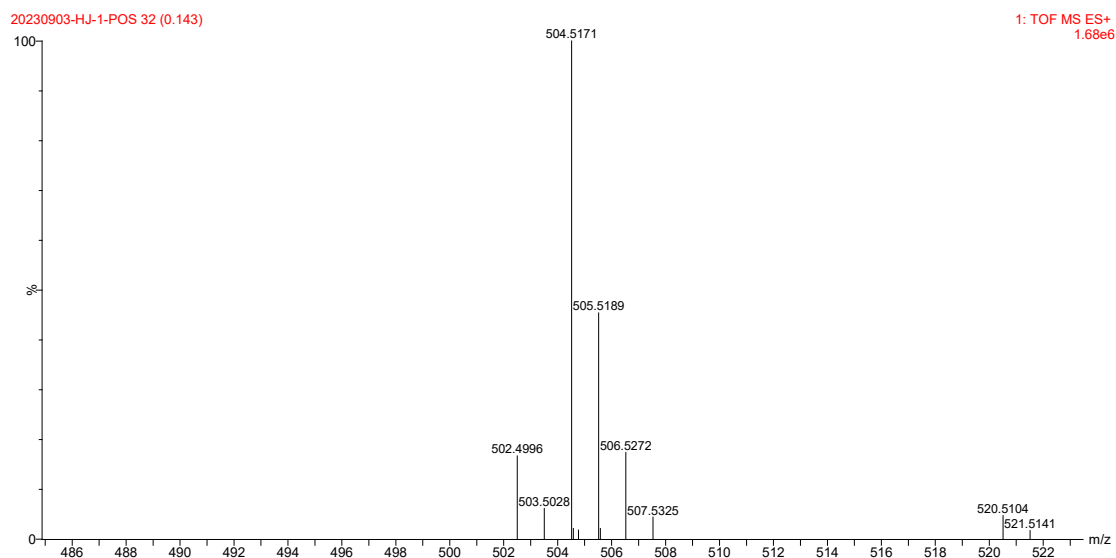
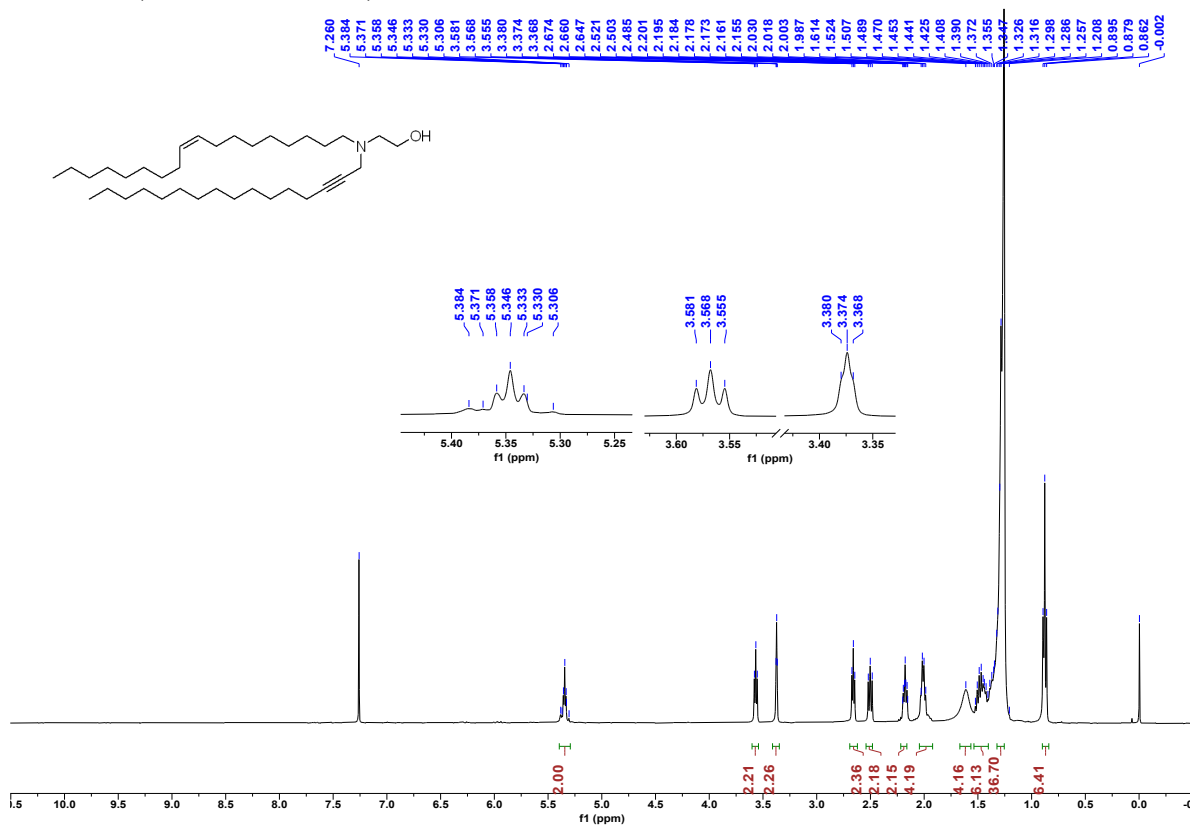
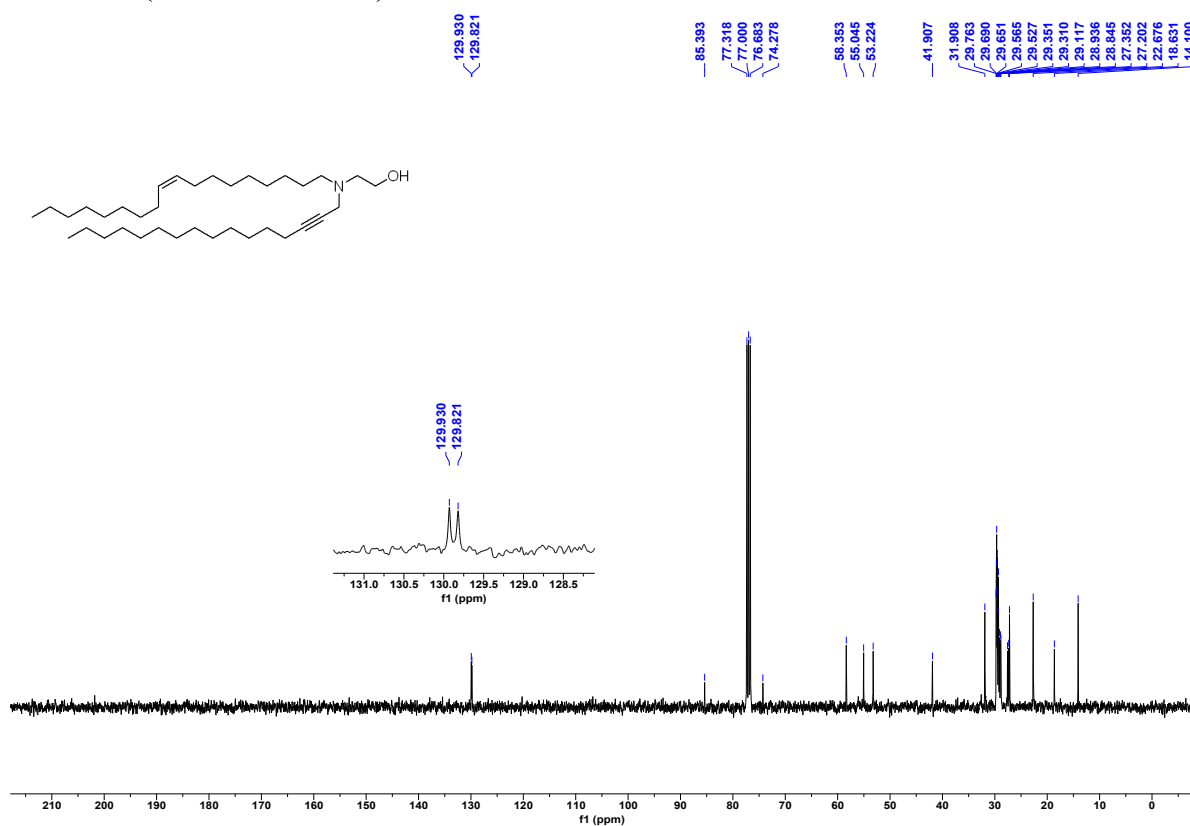


Figure S47. 1H , ^{13}C NMR spectra and HRMS of (Z)-2-(octadec-9-en-1-yl)(tetradec-2-yn-1-yl)aminoethan-1-ol (*A11B2*)

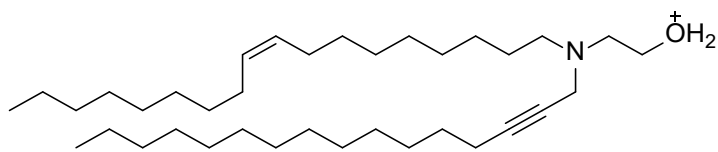
^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)



HRMS (ESI-positive)



Chemical Formula: $C_{36}H_{70}NO^+$

m/z: 532.5452 (100.0%), 533.5486 (38.9%), 534.5520 (7.4%)

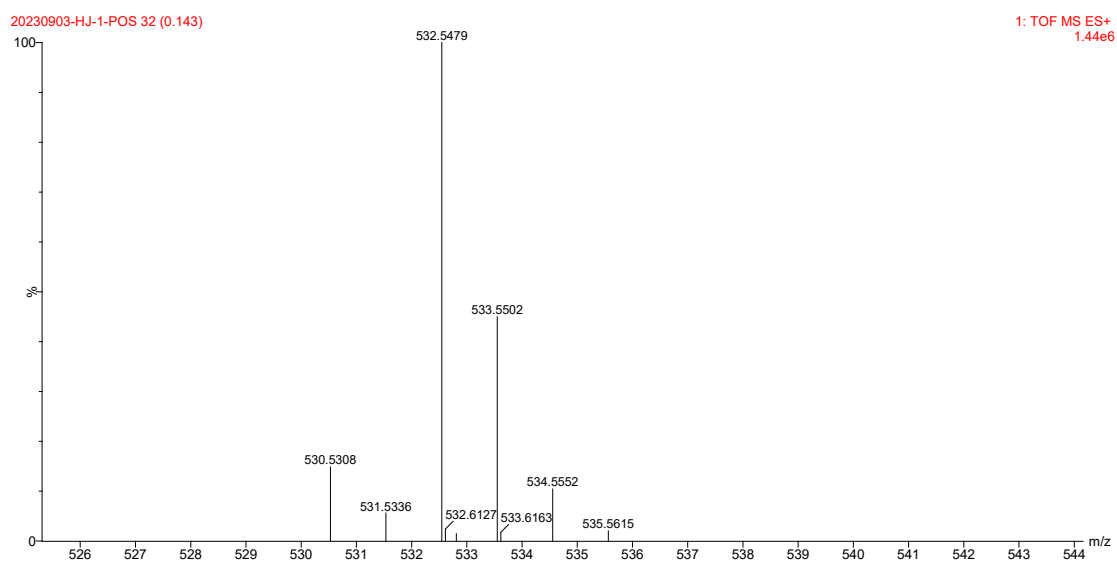
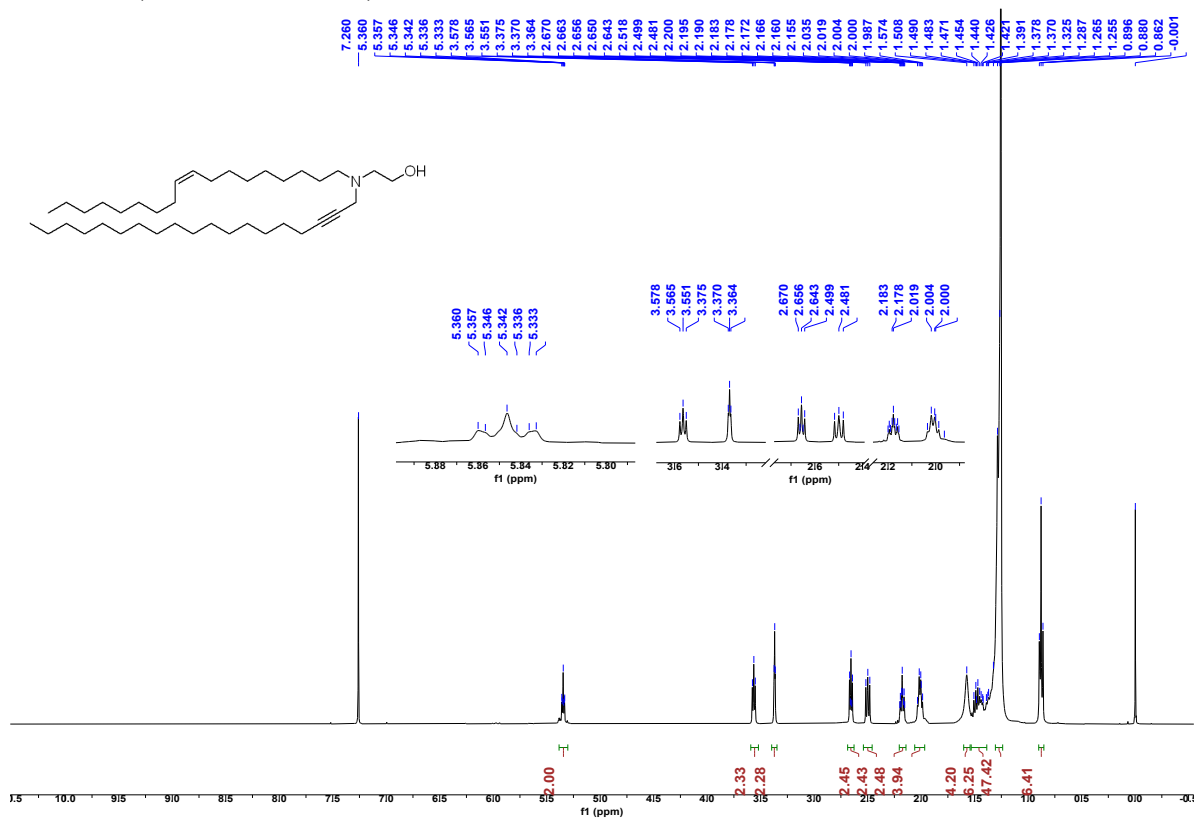
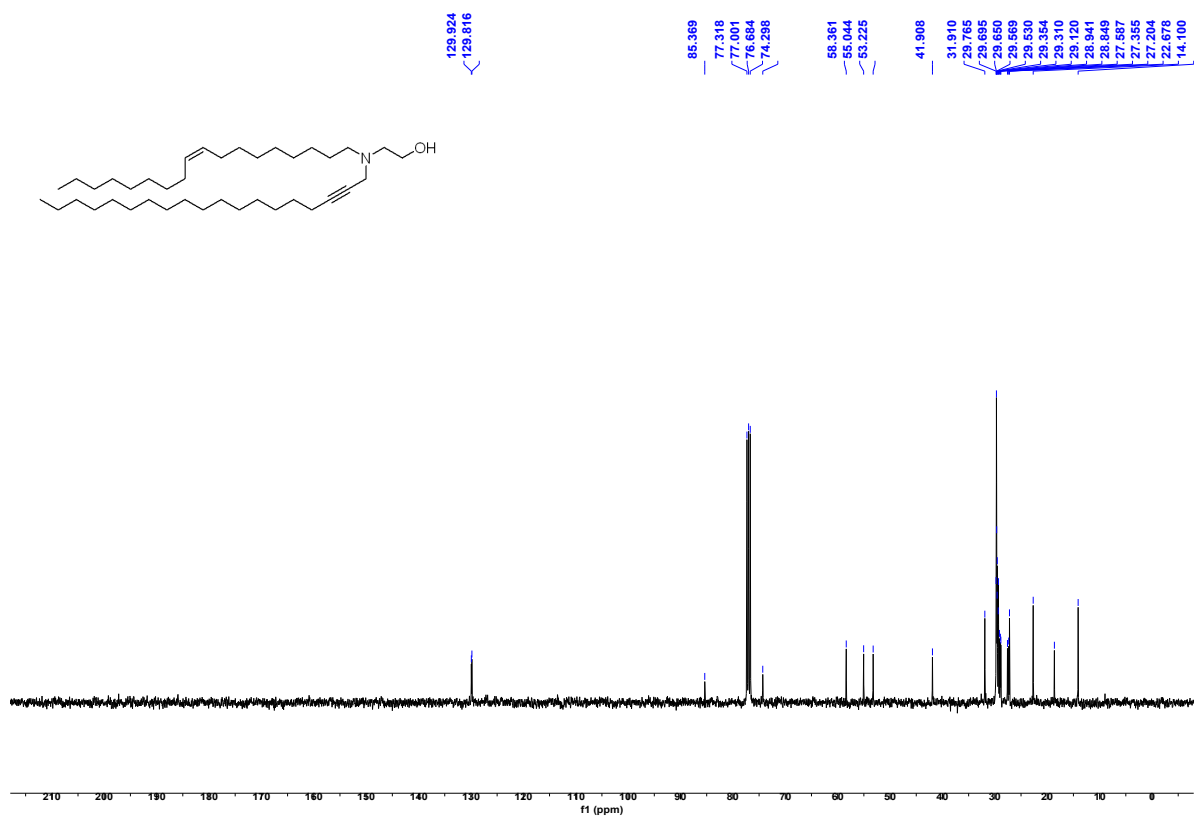


Figure S48. 1H , ^{13}C NMR spectra and HRMS of (Z)-2-(hexadec-2-yn-1-yl)(octadec-9-en-1-yl)aminoethan-1-ol (*A11B4*)

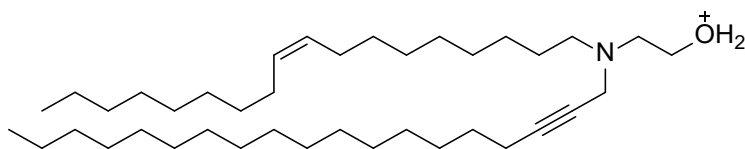
^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)



HRMS (ESI-positive)



Chemical Formula: $C_{39}H_{76}NO^+$

m/z: 574.5922 (100.0%), 575.5955 (42.2%), 576.5989 (8.7%), 577.6023 (1.2%)

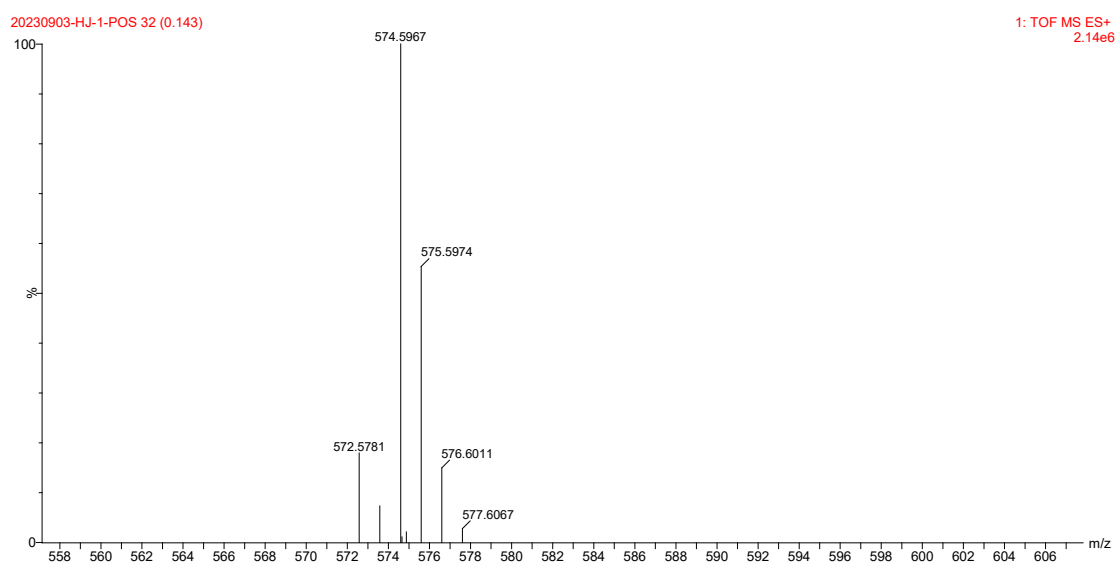
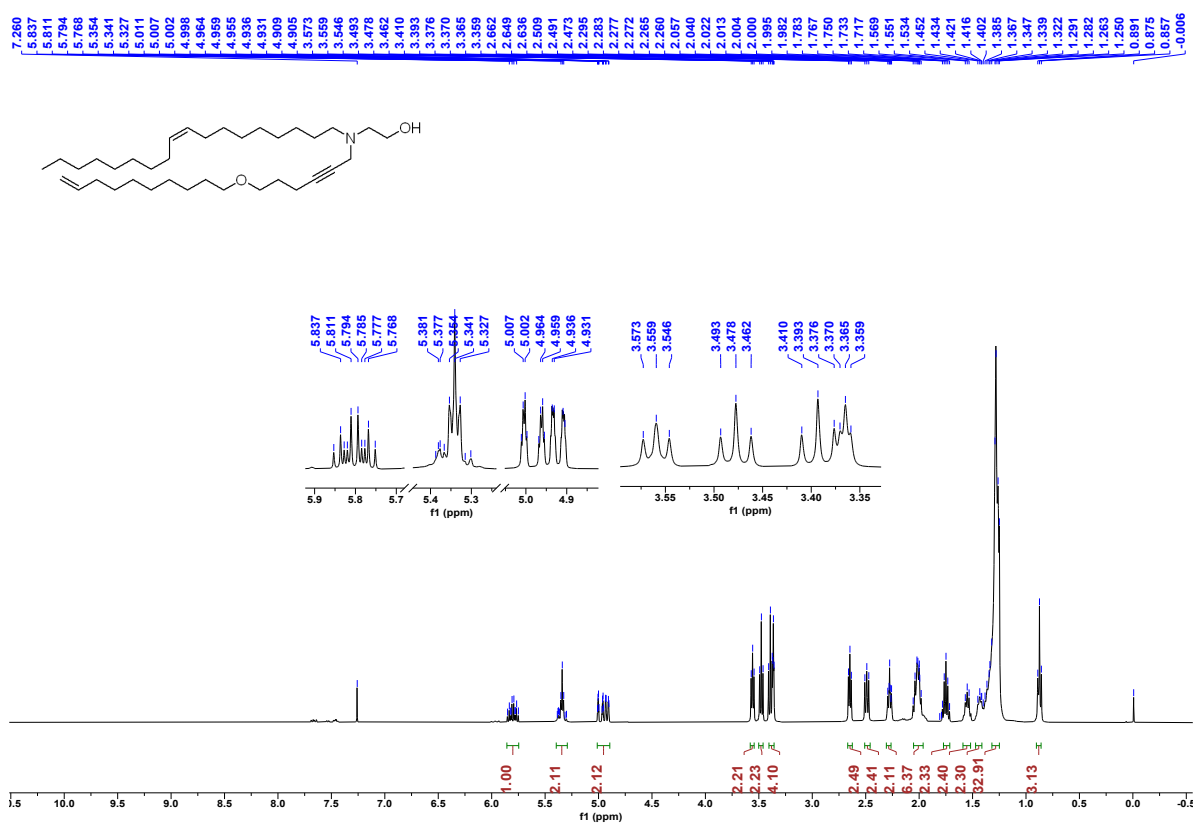
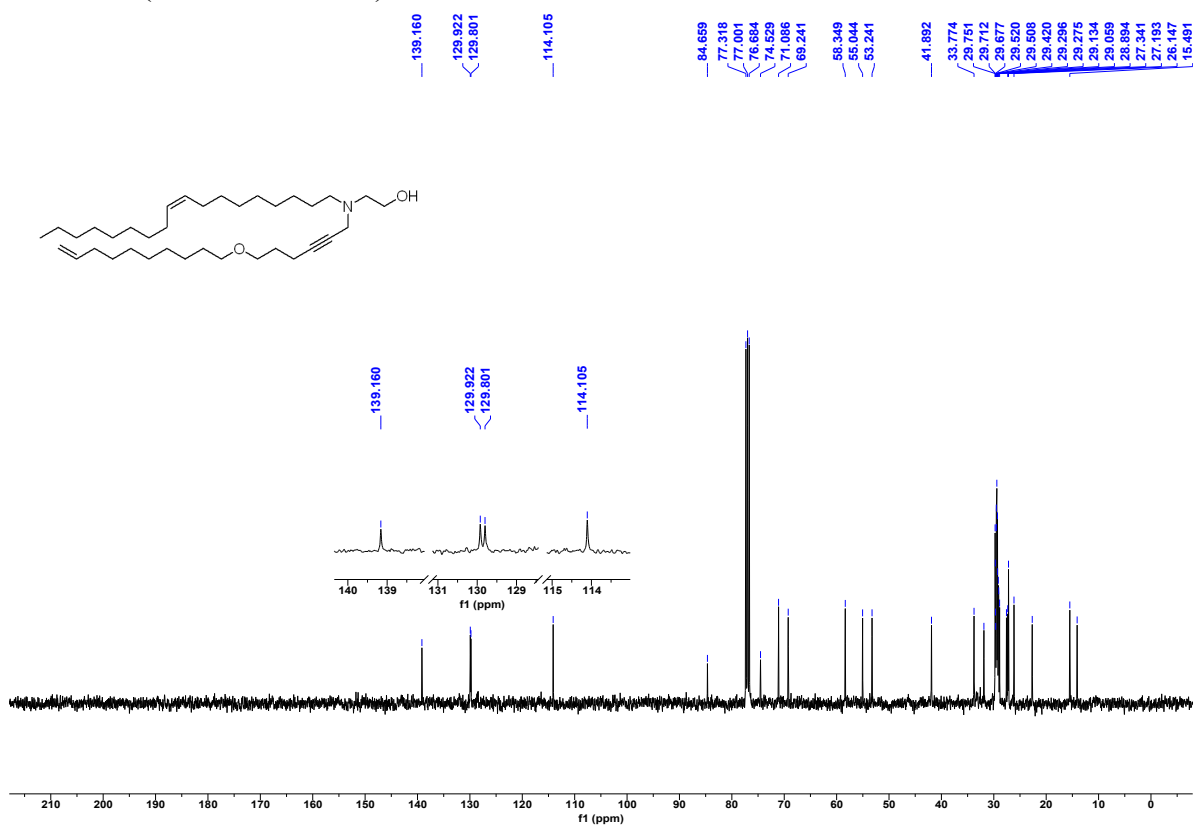


Figure S49. 1H , ^{13}C NMR spectra and HRMS of (Z)-2-(nonadec-2-yn-1-yl)(octadec-9-en-1-yl)aminoethan-1-ol (*A11B7*)

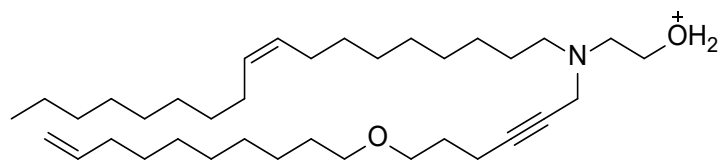
¹H NMR (400 MHz, CDCl₃)



¹³C NMR (101 MHz, CDCl₃)



HRMS (ESI-positive)



Chemical Formula: $C_{36}H_{68}NO_2^+$

m/z: 546.5245 (100.0%), 547.5279 (38.9%), 548.5312 (7.4%)

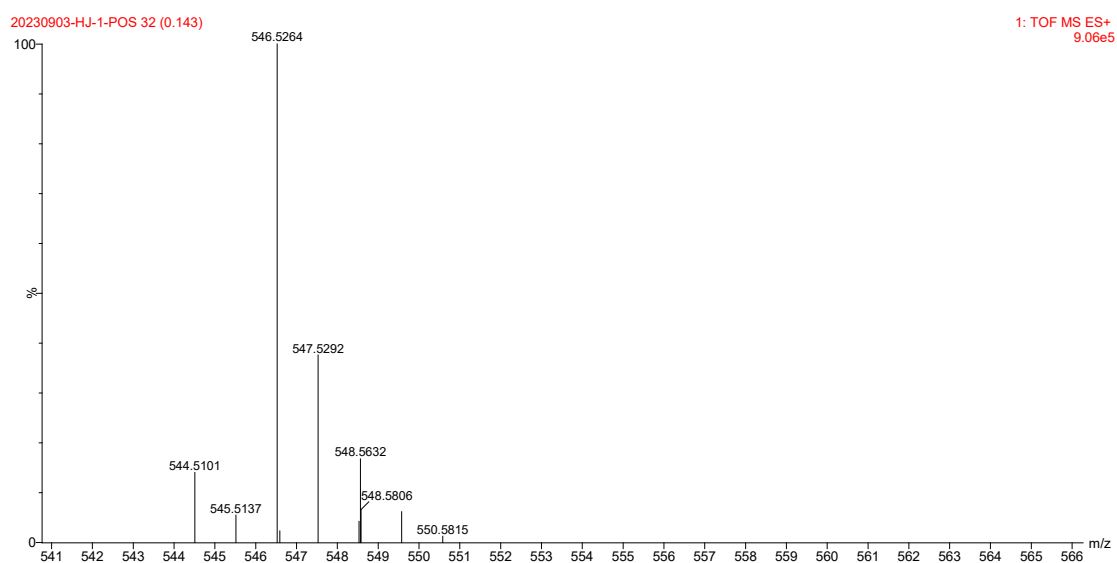
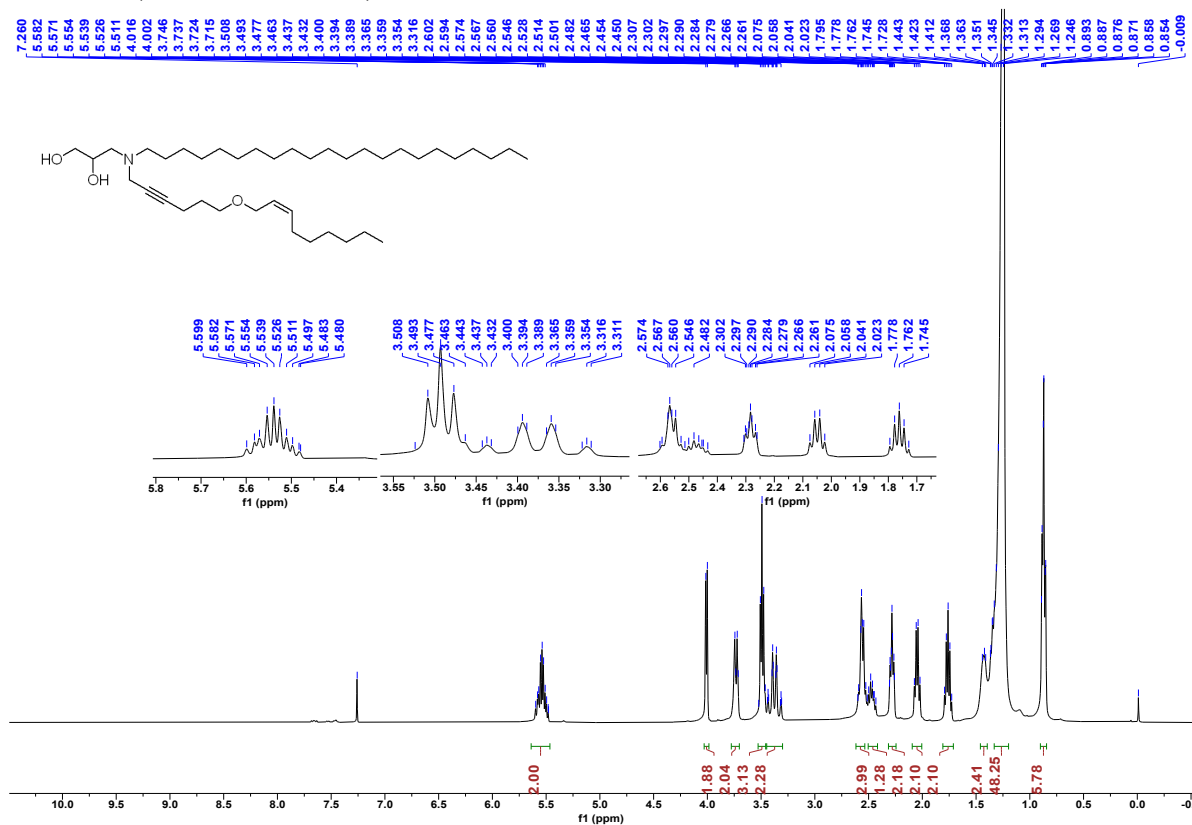


Figure S50. 1H , ^{13}C NMR spectra and HRMS of (Z)-2-((6-(dec-9-en-1-yloxy)hex-2-yn-1-yl)(octadec-9-en-1-yl)amino)ethan-1-ol (*A11B9*)

¹H NMR (400 MHz, CDCl₃)



¹³C NMR (101 MHz, CDCl₃)

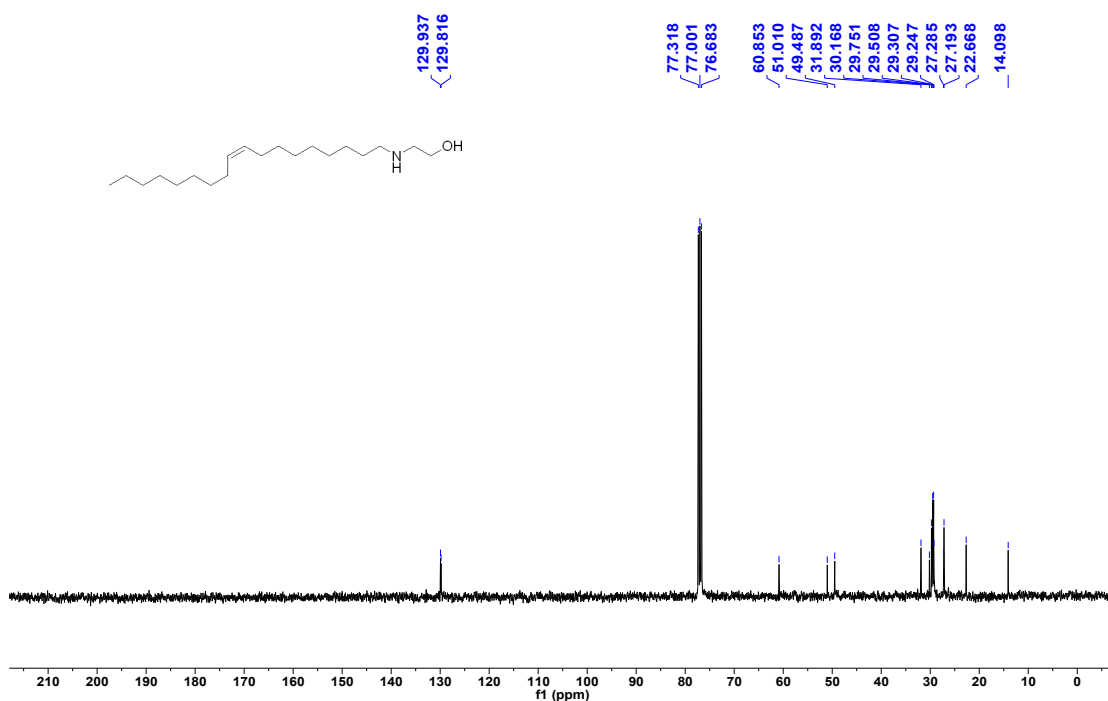


Figure S51. ¹H, ¹³C NMR spectra and HRMS of (Z)-3-(docosyl(6-(non-2-en-1-yloxy)hex-2-yn-1-yl)amino)propane-1,2-diol (*A28B10*)

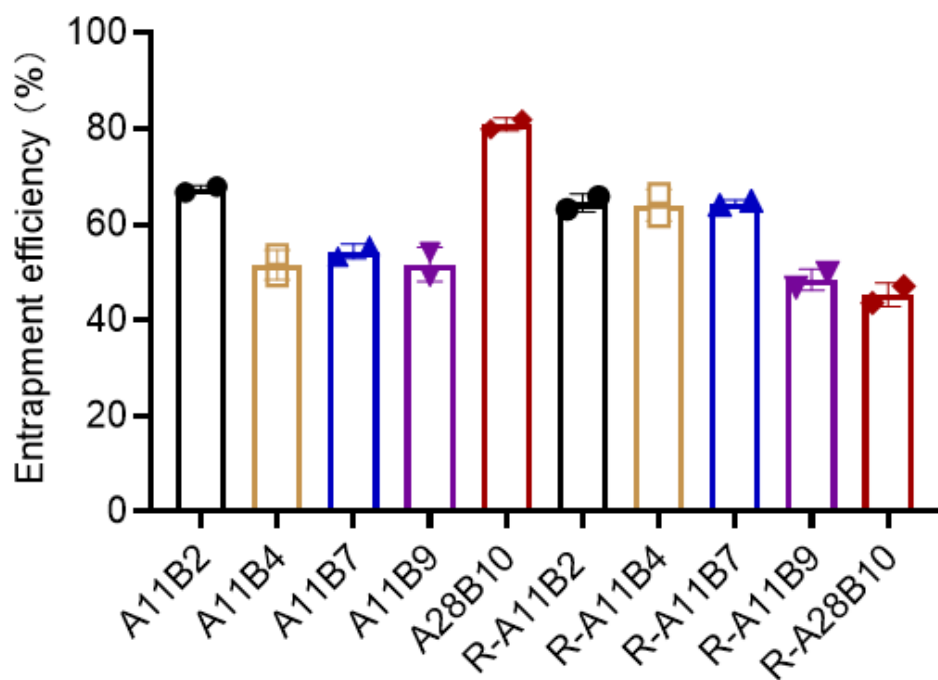
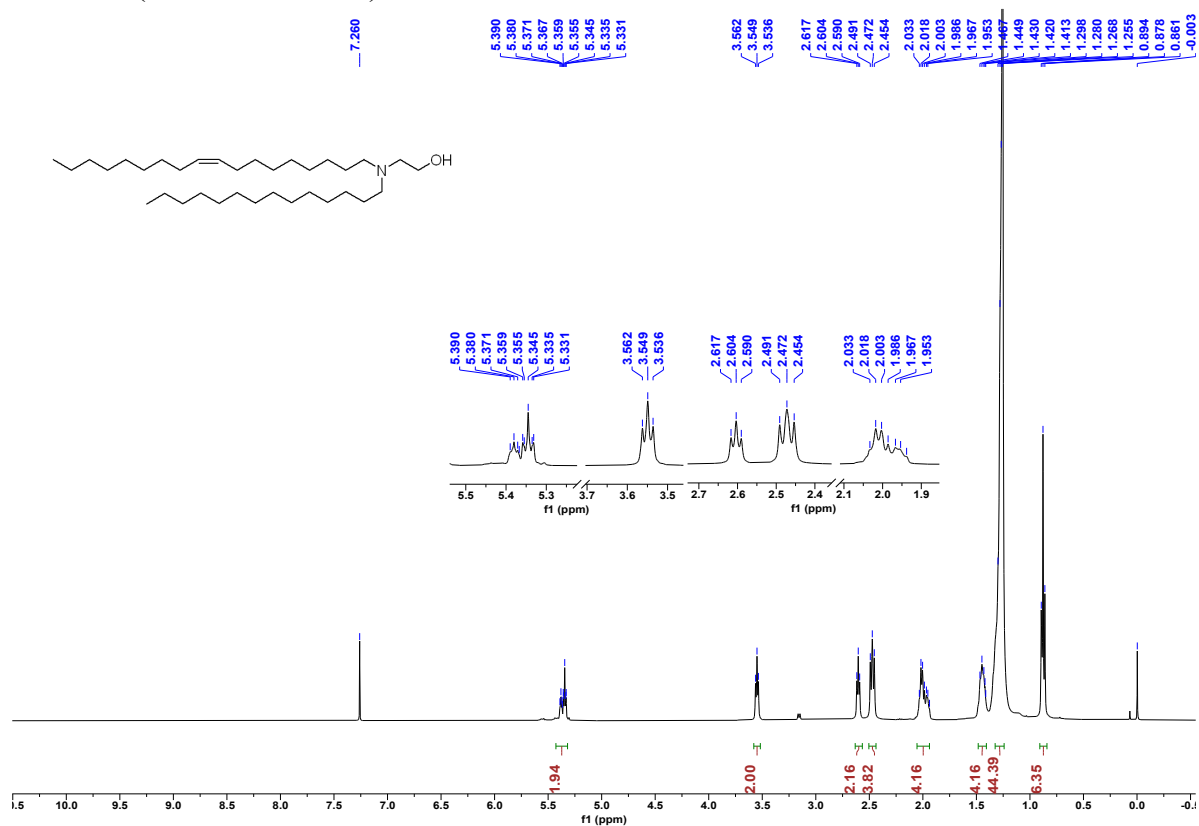
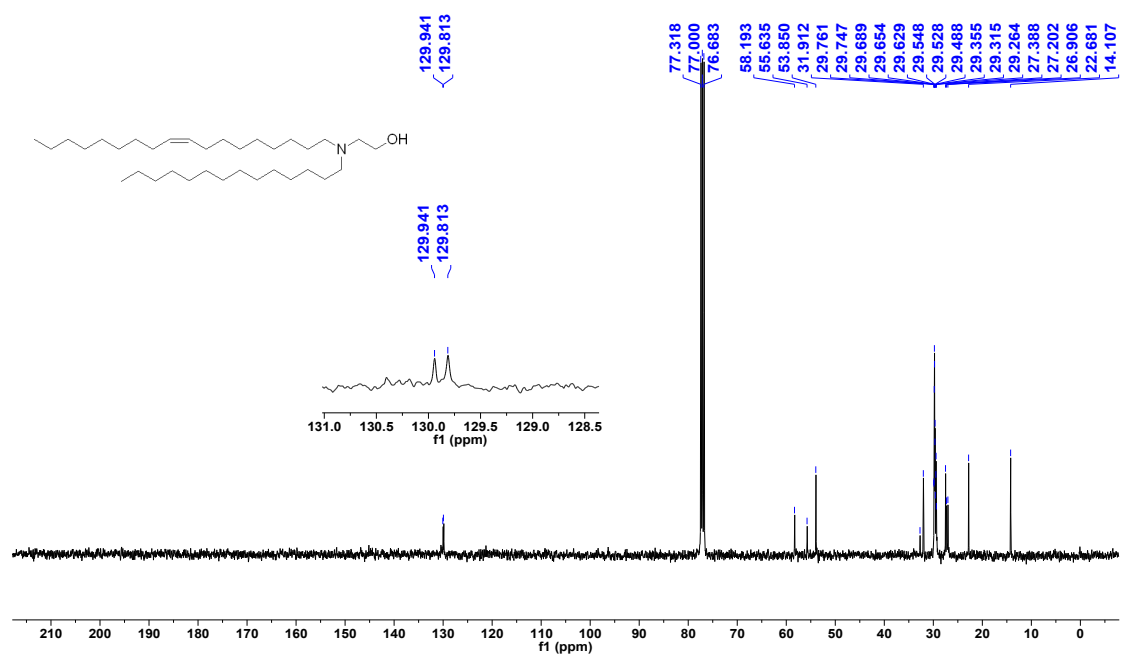


Figure S52. The mRNA encapsulation efficiencies of LNPs formulated with different ILs.

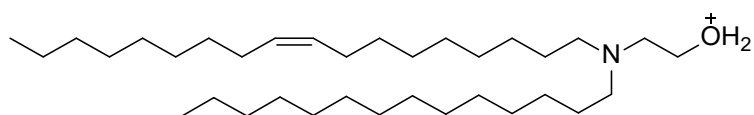
^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)



HRMS (ESI-positive)



Chemical Formula: $C_{34}H_{70}NO^+$

m/z: 508.5452 (100.0%), 509.5486 (36.8%), 510.5520 (6.6%)

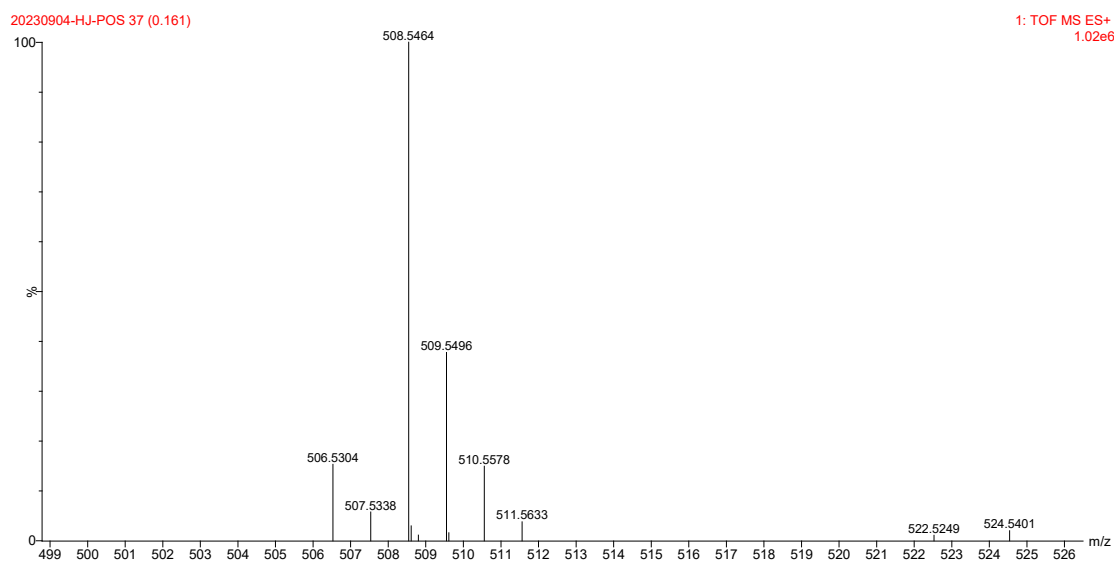
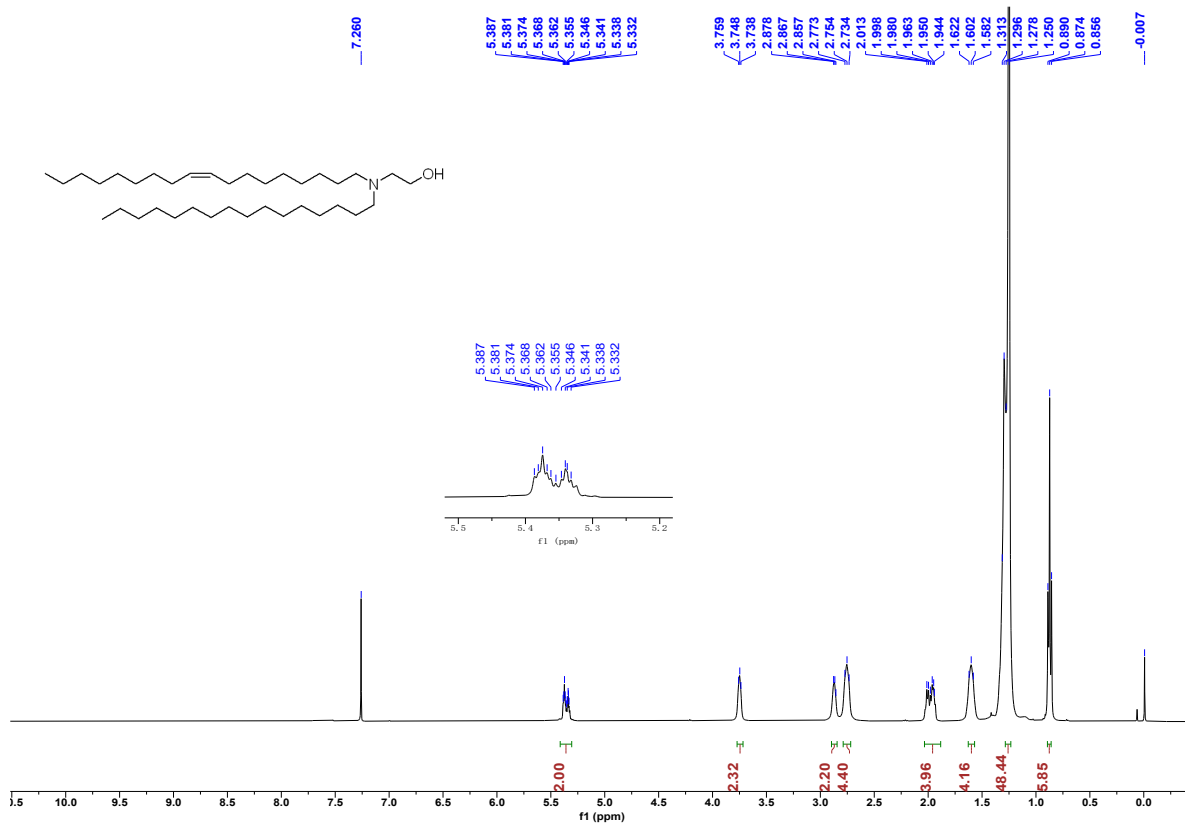


Figure S53. 1H , ^{13}C NMR spectra and HRMS of (Z)-2-(octadec-9-en-1-yl)(tetradecyl)aminoethan-1-ol (*R-AIIB2*).

^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)

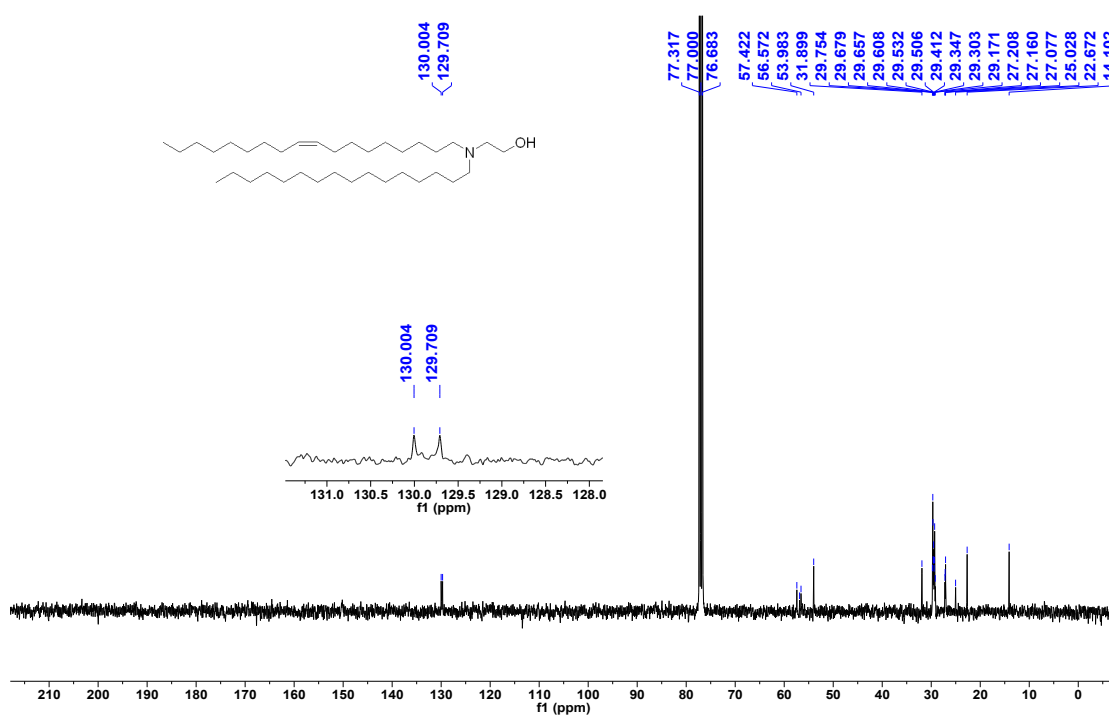
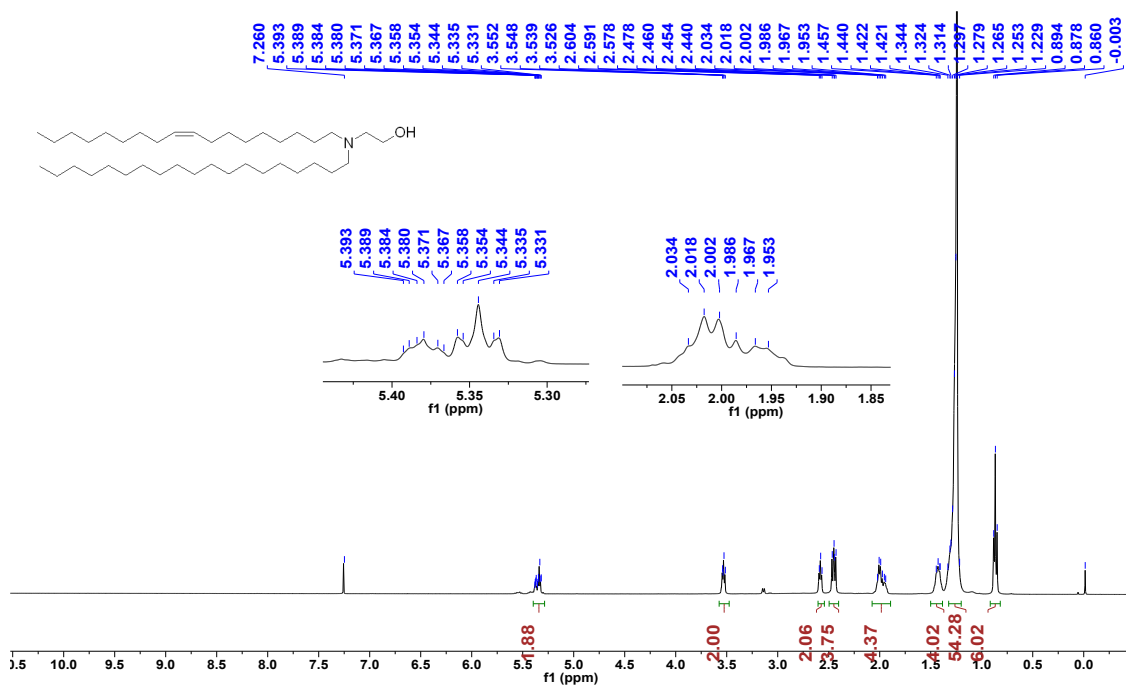


Figure S54. ^1H , ^{13}C NMR spectra and HRMS of (Z)-2-(nonadecyl(octadec-9-en-1-yl)amino)ethan-1-ol (*R-A11B4*)

^1H NMR (400 MHz, CDCl_3)



^{13}C NMR (101 MHz, CDCl_3)

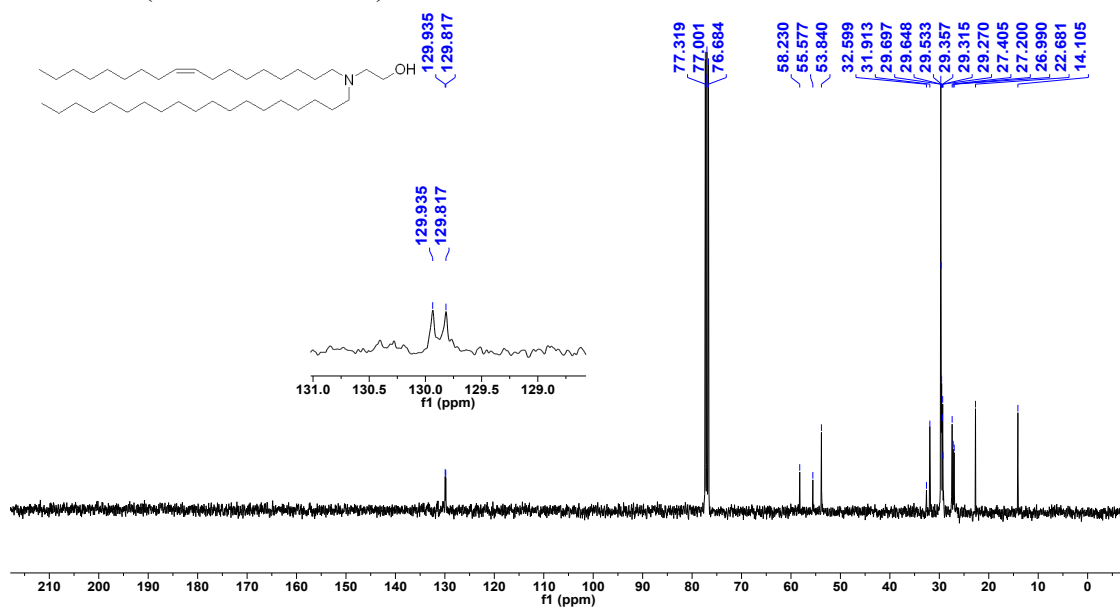


Figure S55. ^1H , ^{13}C NMR spectra and HRMS of (Z)-2-(nonadecyl(octadec-9-en-1-yl)amino)ethan-1-ol (*R-A11B7*)

Chemical structure: CCCCCCCCC=CCCCCCCCCOCN(CCCCCCCCCCCCCO)CO

¹H NMR spectrum (CDCl₃) showing peaks from 0.0 to 7.6 ppm. The inset shows the region from 5.3 to 5.6 ppm.

Integration values (from left to right): 2.74, 1.70, 2.00, 4.14, 2.10, 2.09, 2.16, 2.13, 3.96, 4.44, 37.07, 3.45.

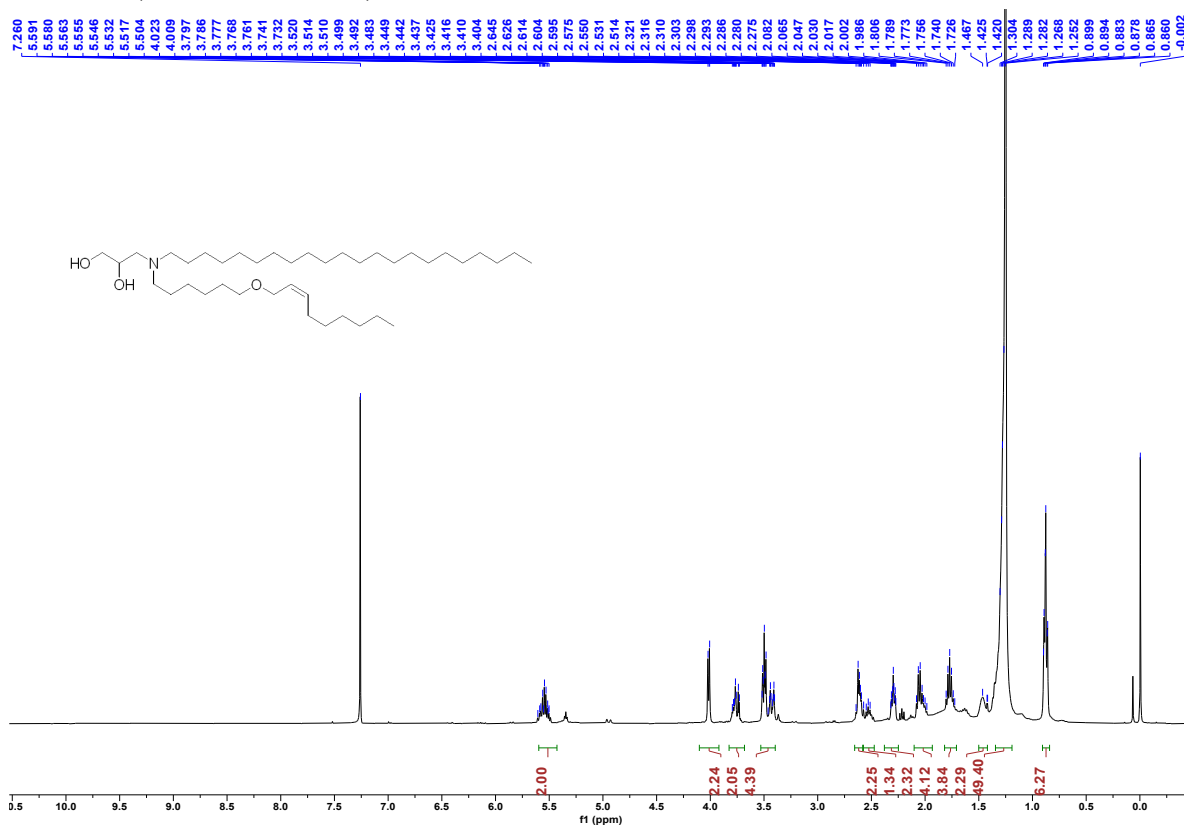
Chemical structure: CCCCCCCCC=CCCCCCCCCOC(C)CCCCCCCCC=CCCCCCCCC

¹³C NMR spectrum (CDCl₃) showing peaks at the following chemical shifts (ppm):

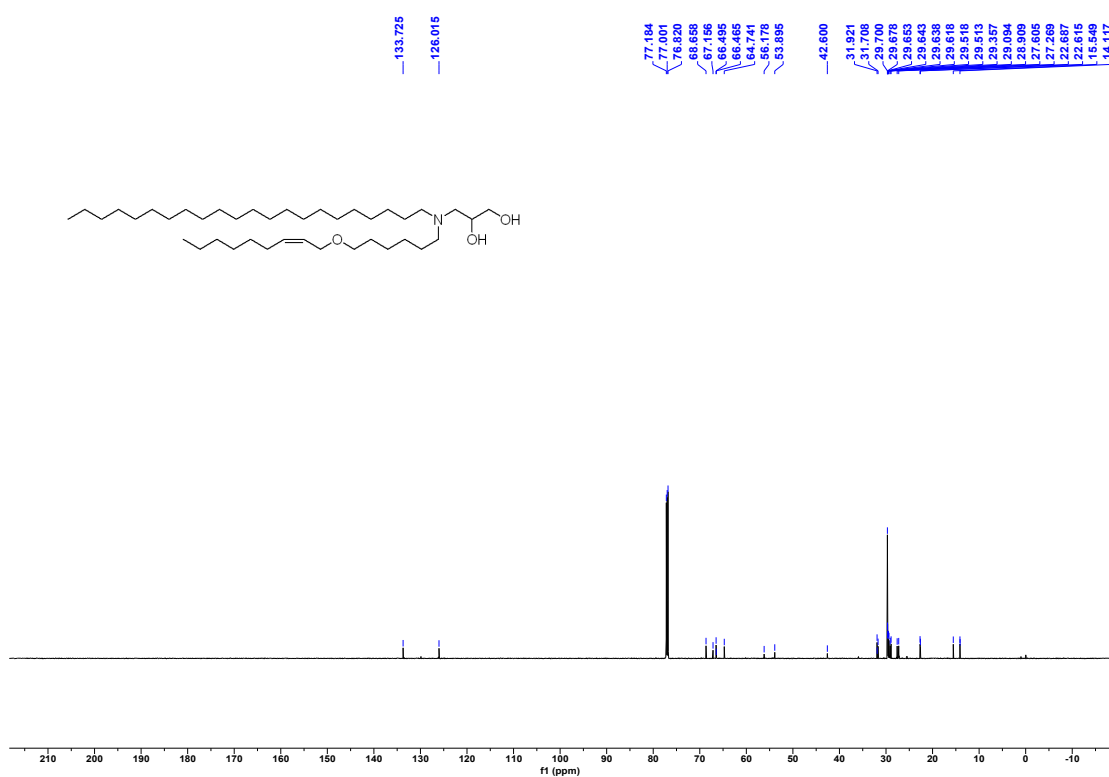
- 132.386
- 129.923
- 129.807
- 128.745
- 77.317
- 77.000
- 76.882
- 71.043
- 69.980
- 58.268
- 58.195
- 55.495
- 50.115
- 31.889
- 29.735
- 29.746
- 29.615
- 29.590
- 29.520
- 29.506
- 29.271
- 27.185
- 26.216
- 24.469
- 14.094

Figure S56. ^1H and ^{13}C NMR spectra of (Z)-2-(((6-(dec-9-en-1-yloxy)hexyl)(octadec-9-en-1-yl)amino)ethan-1-ol (*R-AllB9*)

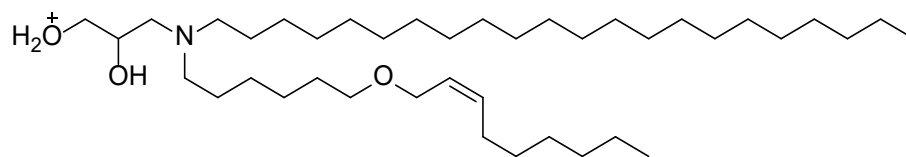
¹H NMR (400 MHz, CDCl₃)



¹³C NMR (176 MHz, CDCl₃)



HRMS (ESI-positive)



Exact Mass: 624.6289

m/z: 624.6290 (100.0%), 625.6323 (43.3%), 626.6357 (9.1%), 627.6390 (1.3%)

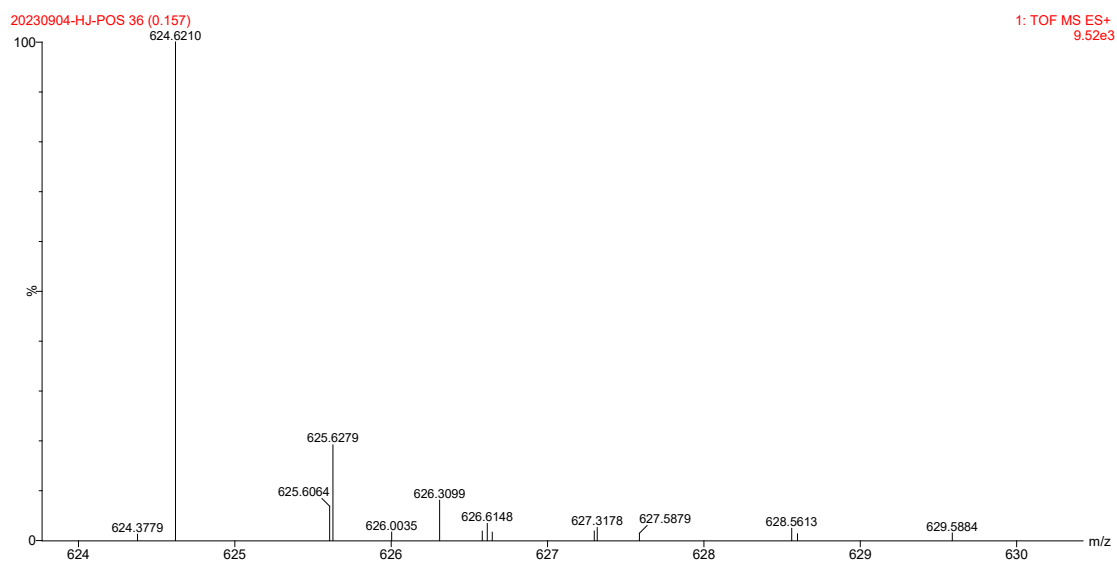


Figure S57. ^1H , ^{13}C NMR spectra and HRMS of (Z)-3-(docosyl(6-(non-2-en-1-yloxy)hexyl)amino)propane-1,2-diol (*R-A28B10*)

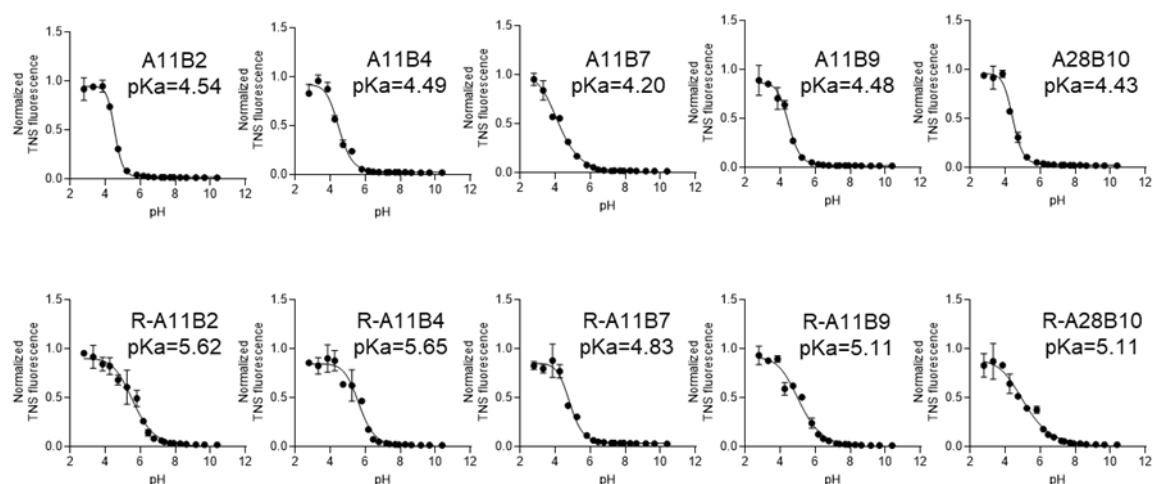


Figure S58. Measurement of pKa of the formulated LNPs by the TNS assay.

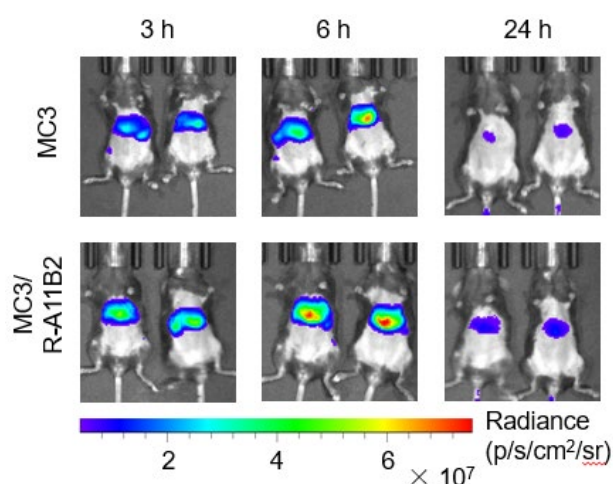


Figure S59. Bioluminescence images of mice at different time points post-intravenous injection of LNPs. Each mouse received 2 μ g of mLuc through intravenous injection.

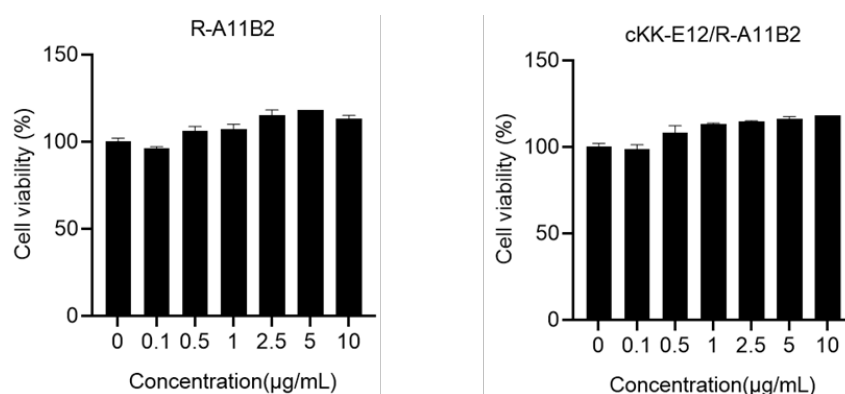


Figure S60. The viability of DC2.4 cells after incubation with different concentrations of R-A11B2 and cKK-E12/R-A11B2 LNPs. Data are shown as mean $\pm$ SD.